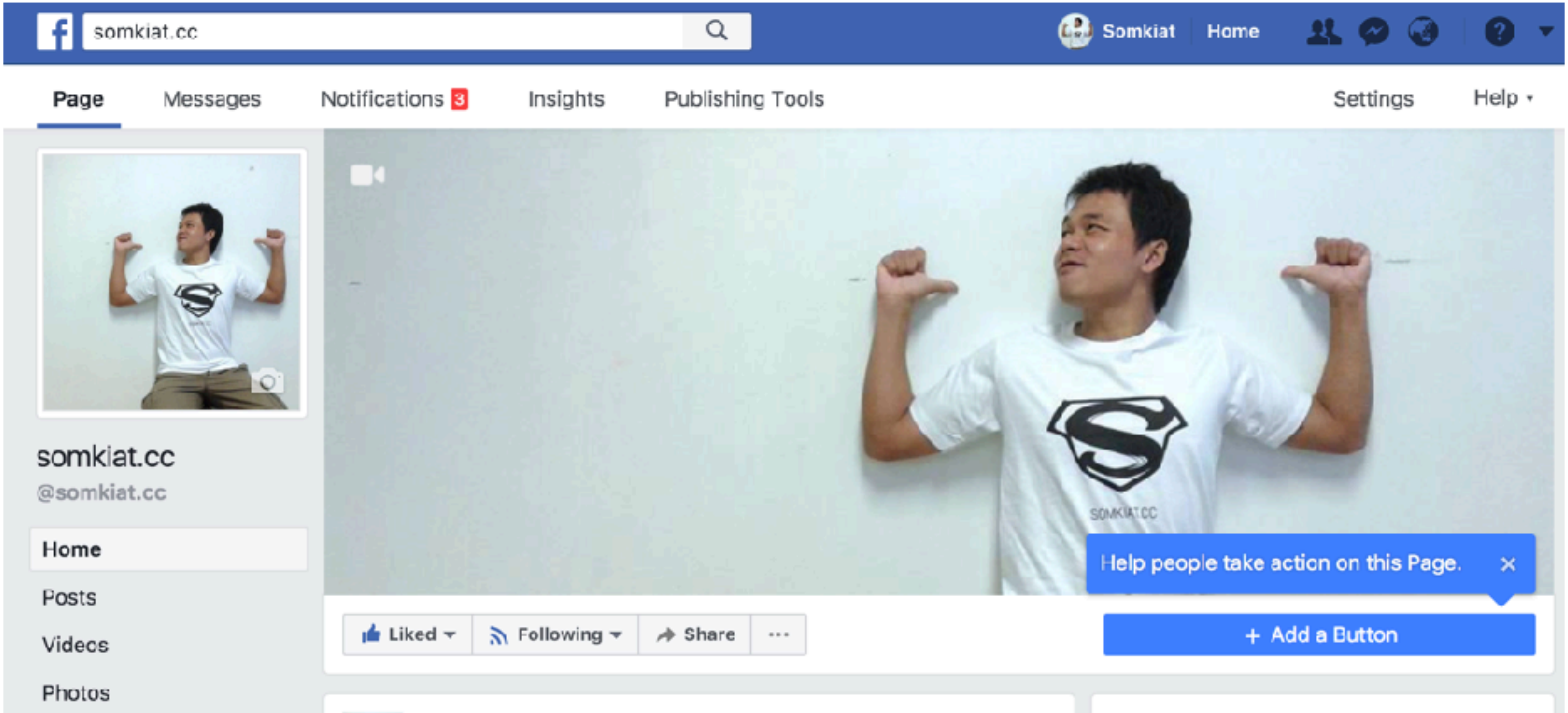
# Generative AI for Software Development 2026


somkiat.cc
Somkiat
Home
Page
Messages
Notifications 3
Insights
Publishing Tools
Settings
Help
somkiat.cc
@somkiat.cc
Home
Posts
Videos
Photos
SOMKIAT.CC
Help people take action on this Page.
Liked
Following
Share
+ Add a Button

# https://github.com/up1/workshop-ai-with-technical-team

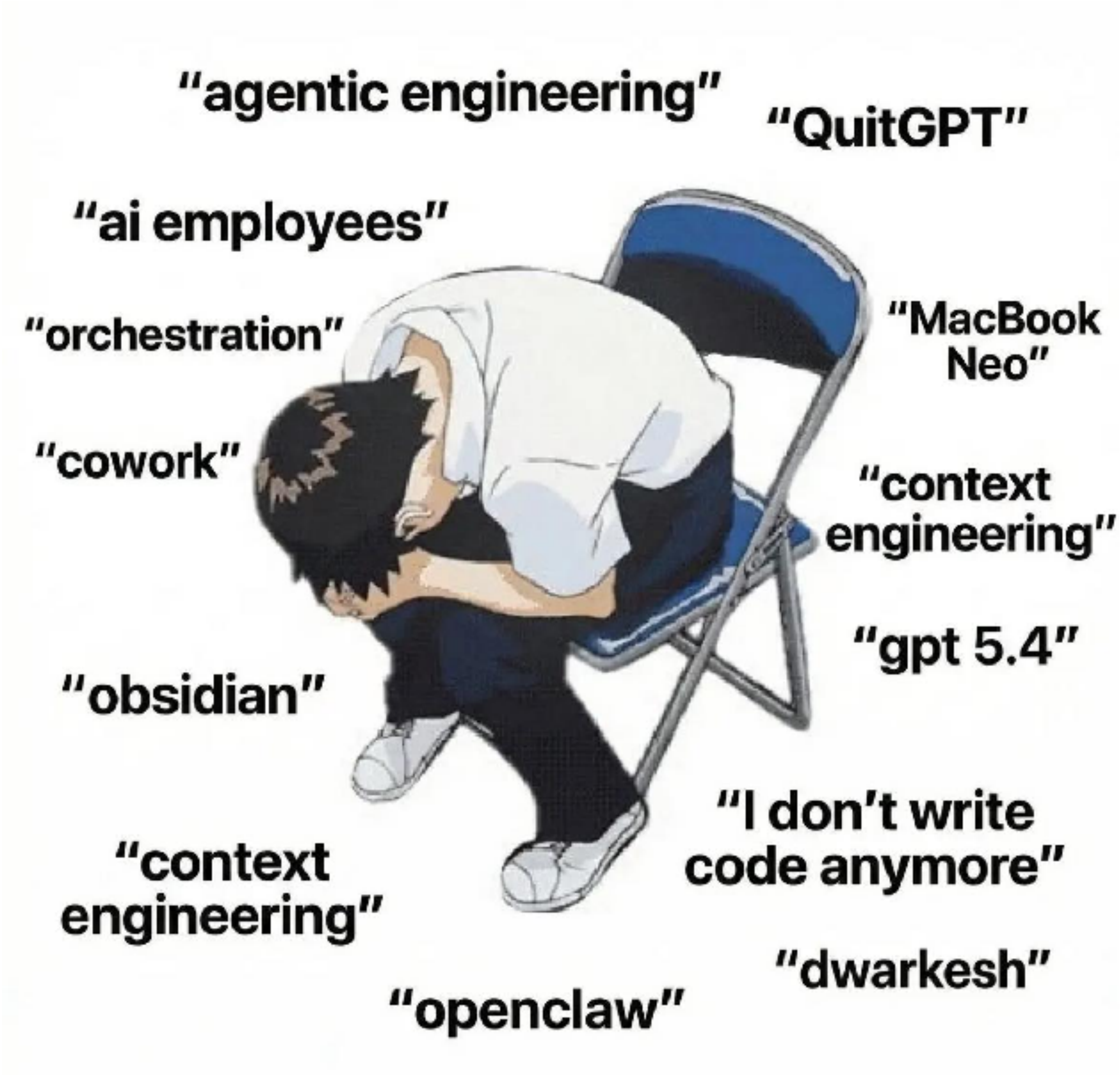
# Goals

Integrate Generative AI in Software Development
Optimize code quality
Team up with AI on coding tasks
Develop innovative solutions

# Learning Path

| | | |
|---|---|---|
| AI/ML | AI Model | Prompt Engineer |
| AI in Software development | Develop AI/LLM app | RAG |

Use cases and workshops


"agentic engineering"
"QuitGPT"
"ai employees"
"orchestration"
"MacBook Neo"
"cowork"
"context engineering"
"gpt 5.4"
"obsidian"
"I don't write code anymore"
"context engineering"
"dwarkesh"
"openclaw"

# Steps to use AI ?

Q/A Chat-based → Assistant → Workflow → System Autonomous

**Assistant**
Summarize
Explain
Research

**Workflow**
Define steps
n8n
Agent Skills

**System Autonomous**
AI Agent
Sub-agent

# Steps to use AI ?

Q/A Chat-based → Assistant → Workflow → System Autonomous

**Assistant**
Summarize
Explain
Research

**Workflow**
Define steps
n8n
Agent Skills

**System Autonomous**
AI Agent
Sub-agent

Prompt engineering

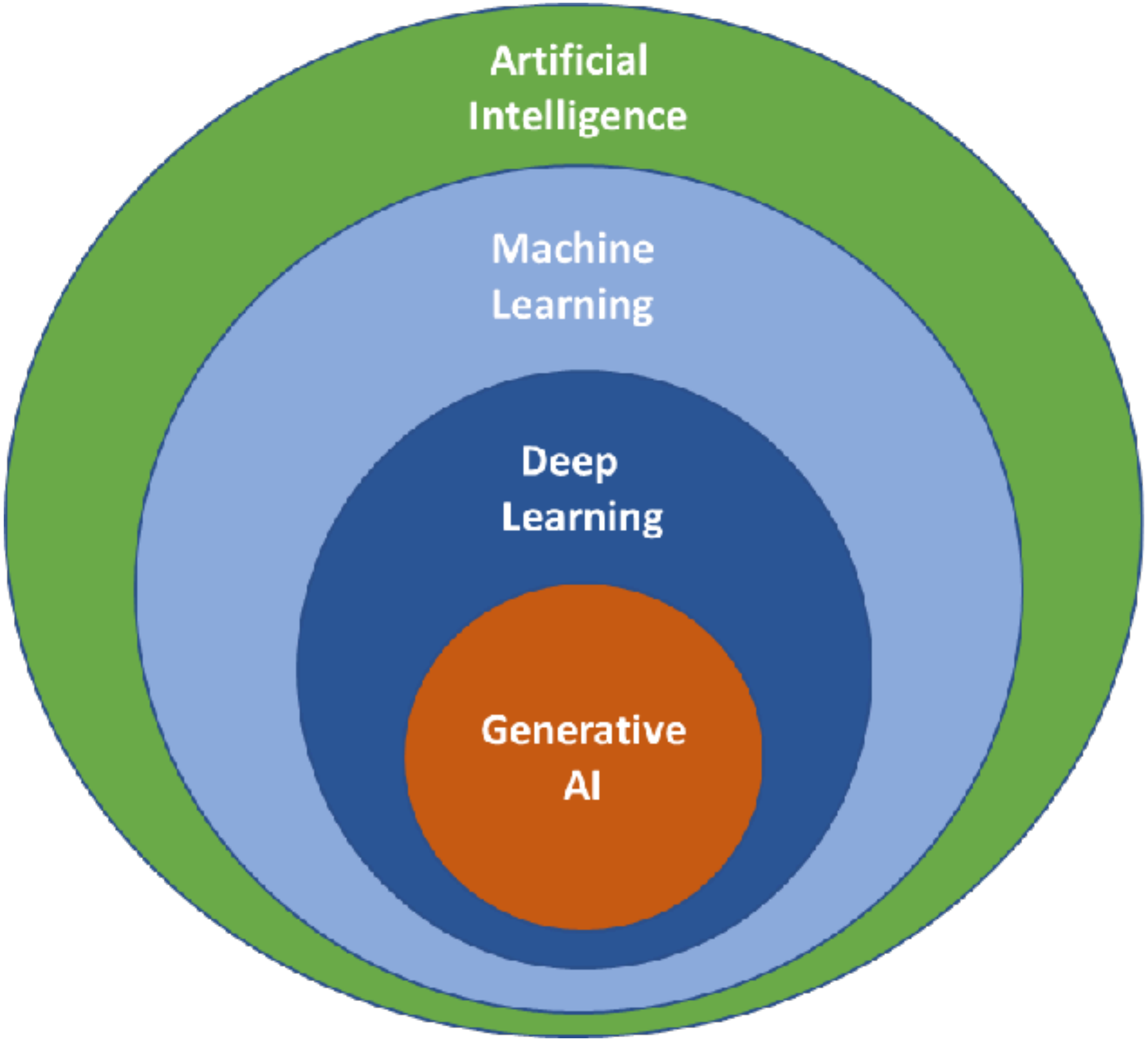
Manage context and knowledge for AI (Context engineering)
Human in the Loop (HITL)

# Software Development

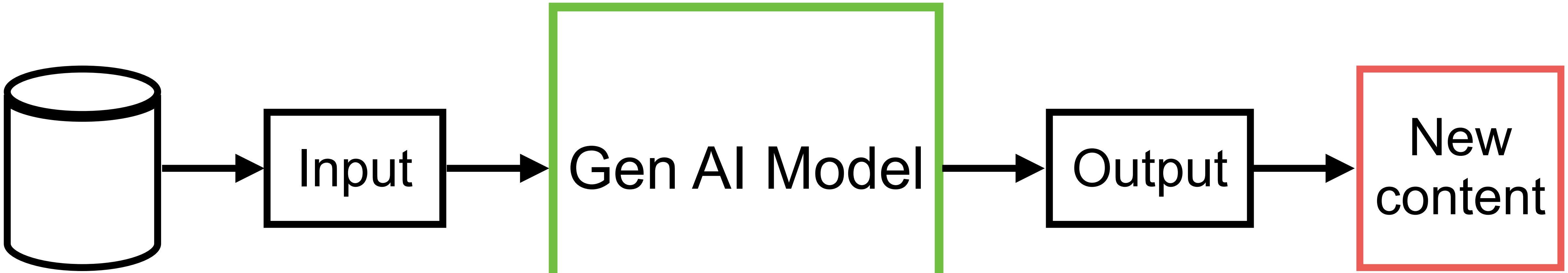
Requirement | Design | Develop | Testing | Deploy

Generative AI

Improve Productivity … (Replace human !!)


Artificial Intelligence
Machine Learning
Deep Learning
Generative AI

# Generative AI




Unstructured content

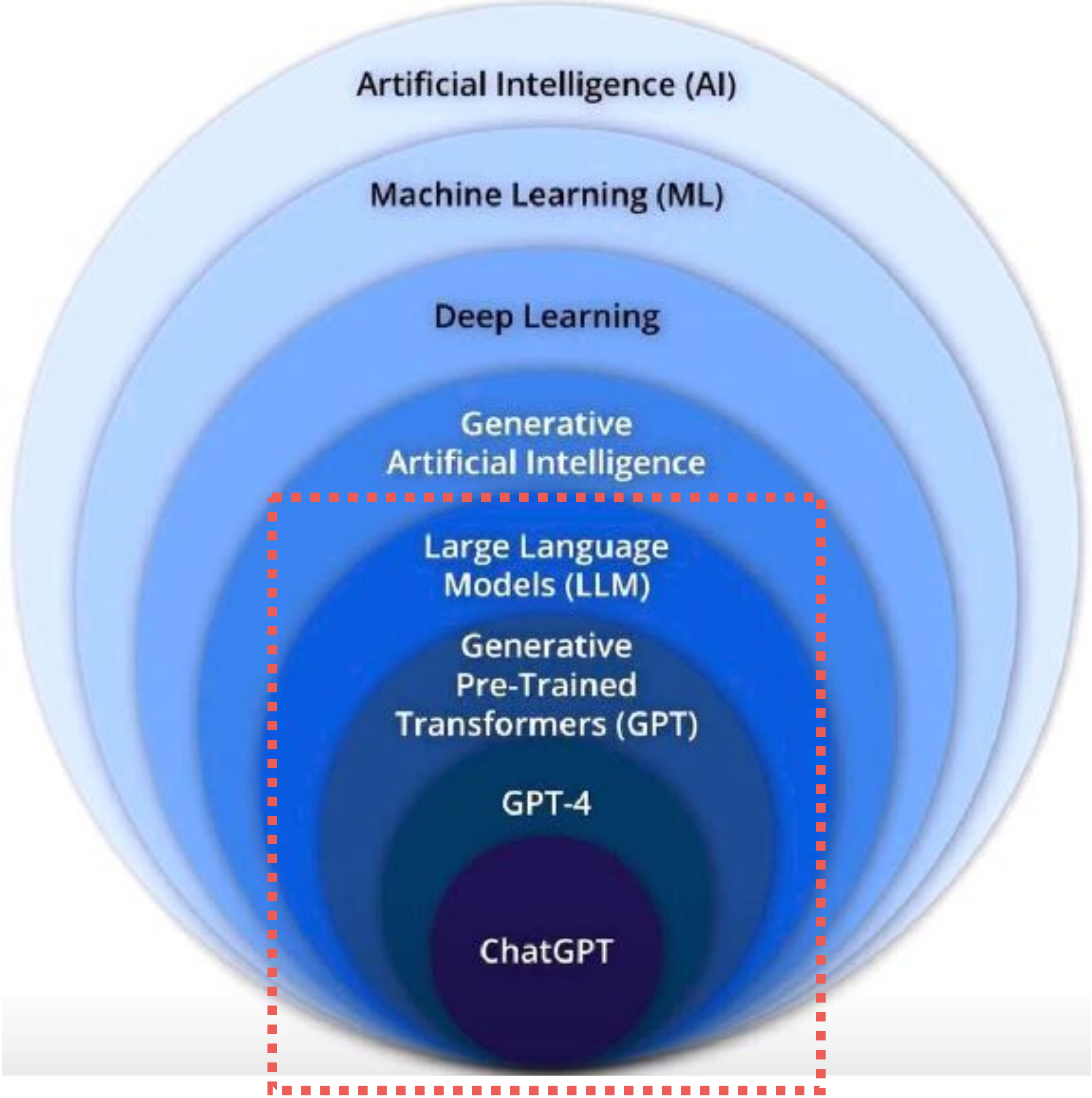
Learn patterns from Unstructured content

Text
Image
Video
3D
Task

https://grow.google/ai-essentials/

# Generative AI

| LLMs | GANs | VAEs |
| --- | --- | --- |
| **Large Language Models** | **Generative Adversarial Network** | **Variational Autoencoders** |
| Text generation | Image generation | Data compression |
| Code generation | Deep fake | Synthetic data generation |
| Chatbot | Art creation | Image reconstruction |
| Conversation AI | Simulate financial market | |


Artificial Intelligence (AI)
Machine Learning (ML)
Deep Learning
Generative
Artificial Intelligence
Large Language
Models (LLM)
Generative
Pre-Trained
Transformers (GPT)
GPT-4
ChatGPT

# Large Language Model (LLM)

Type of AI model
Process, understand
Generate human readable data

LLM

Training data !!

# LLMs in industry

| Model name | Company |
|---|---|
| Bidirectional Encoder Representation from Transformers (BERT) | Google AI |
| Generative Pre-trained transformer-5 (GPT-5) | OpenAI |
| Pathways Language Model-E (PaLM-E) | Google AI |
| BLOOM | NVIDIA AI |
| Llama 4 | Facebook |
| Claude 4.5 Sonnet/Opus | Anthropic |

# LLM Development timeline

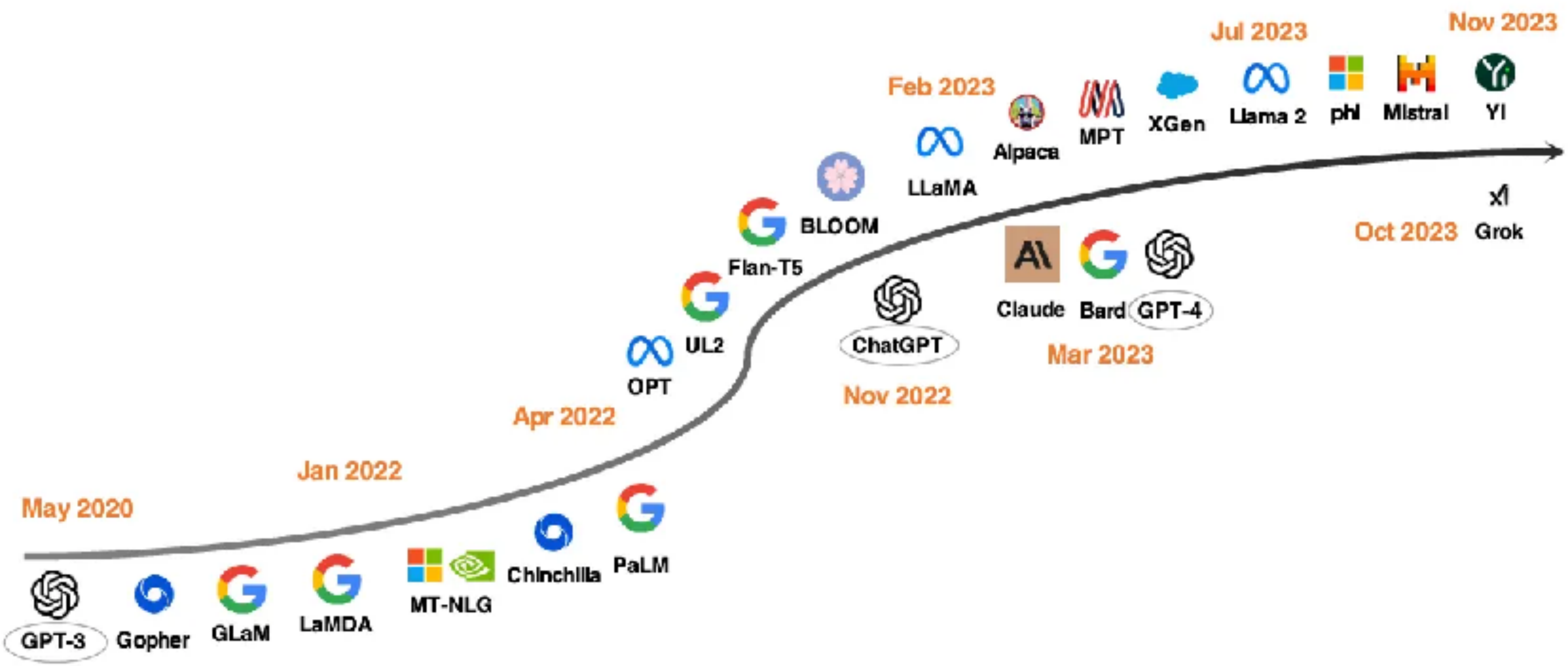



Figure 3: LLM development timeline. The models below the arrow are closed-source while those above the arrow are open-source.

**2026**
**GPT**
**Gemini**
**DeepSeek**
**Llama**
**Mistral**
**GLM**
**Kimi**

https://arxiv.org/abs/2311.16989

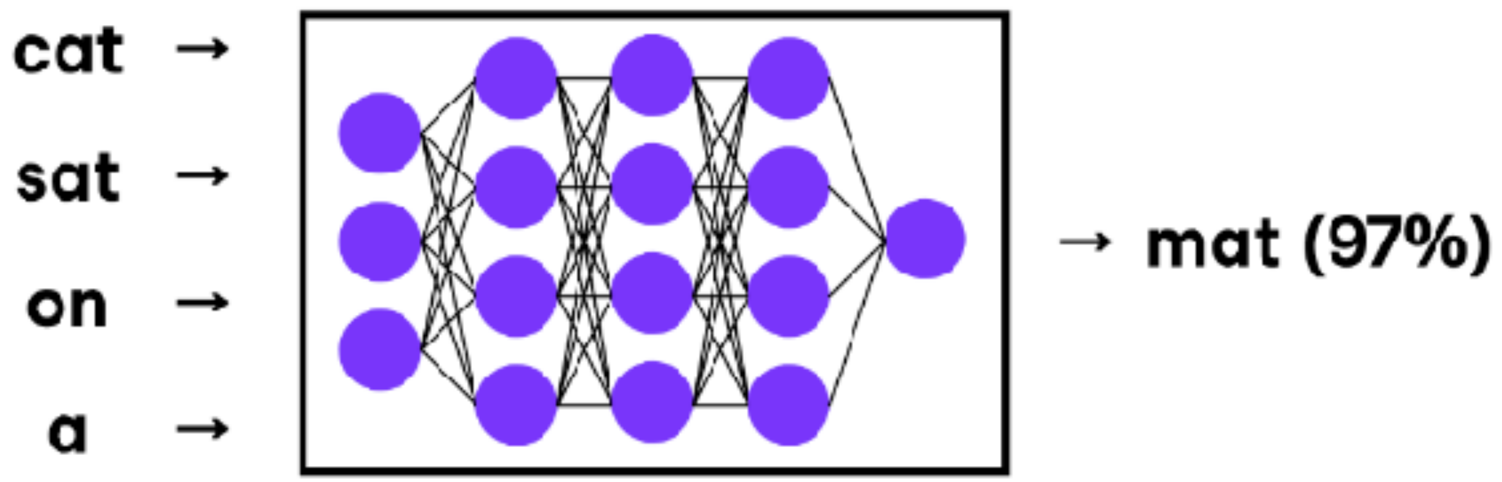
# Let’s go !!

# สีเหลือง

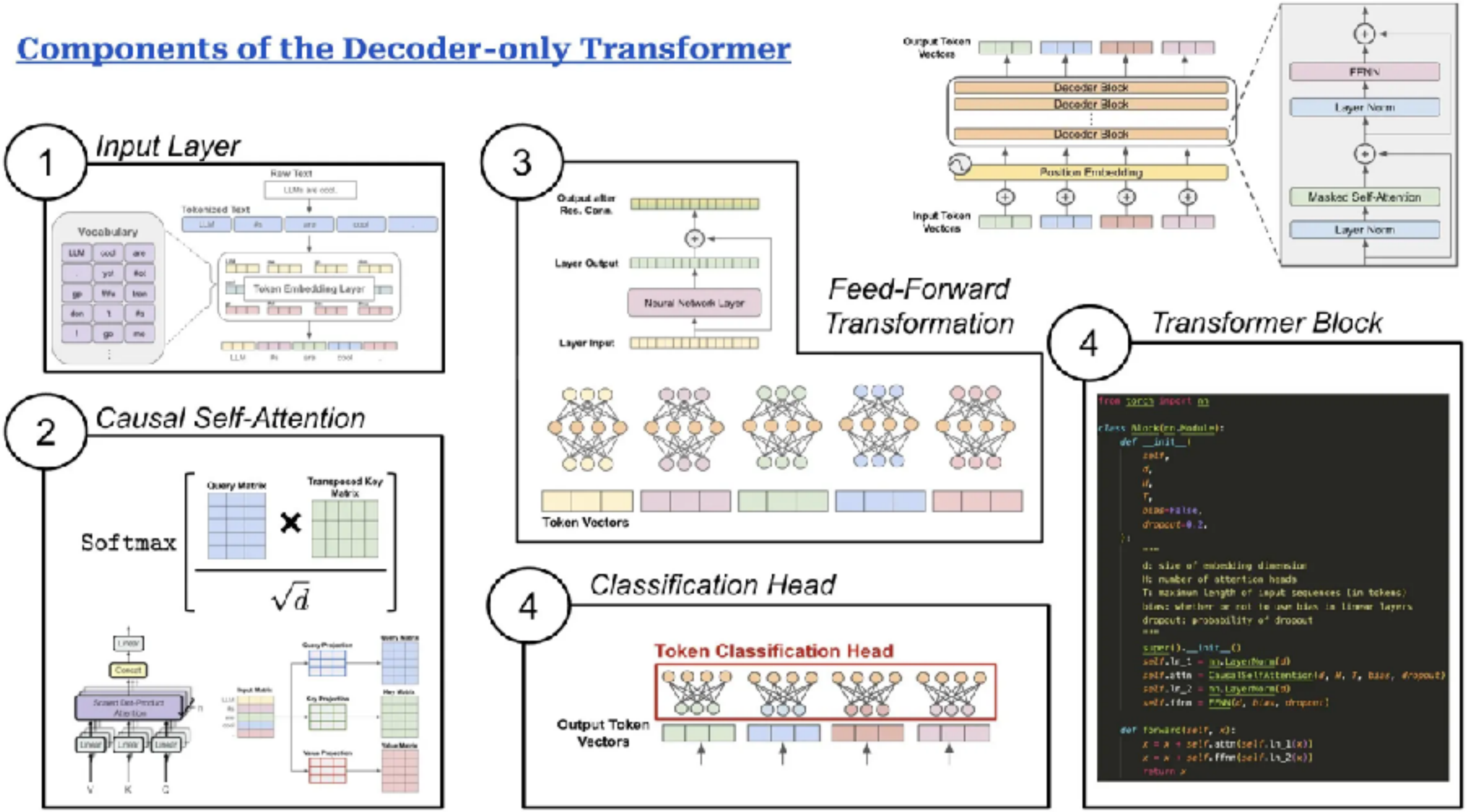
# Large Language Model (LLM)

Neural network

Predicts the next word in a sequence

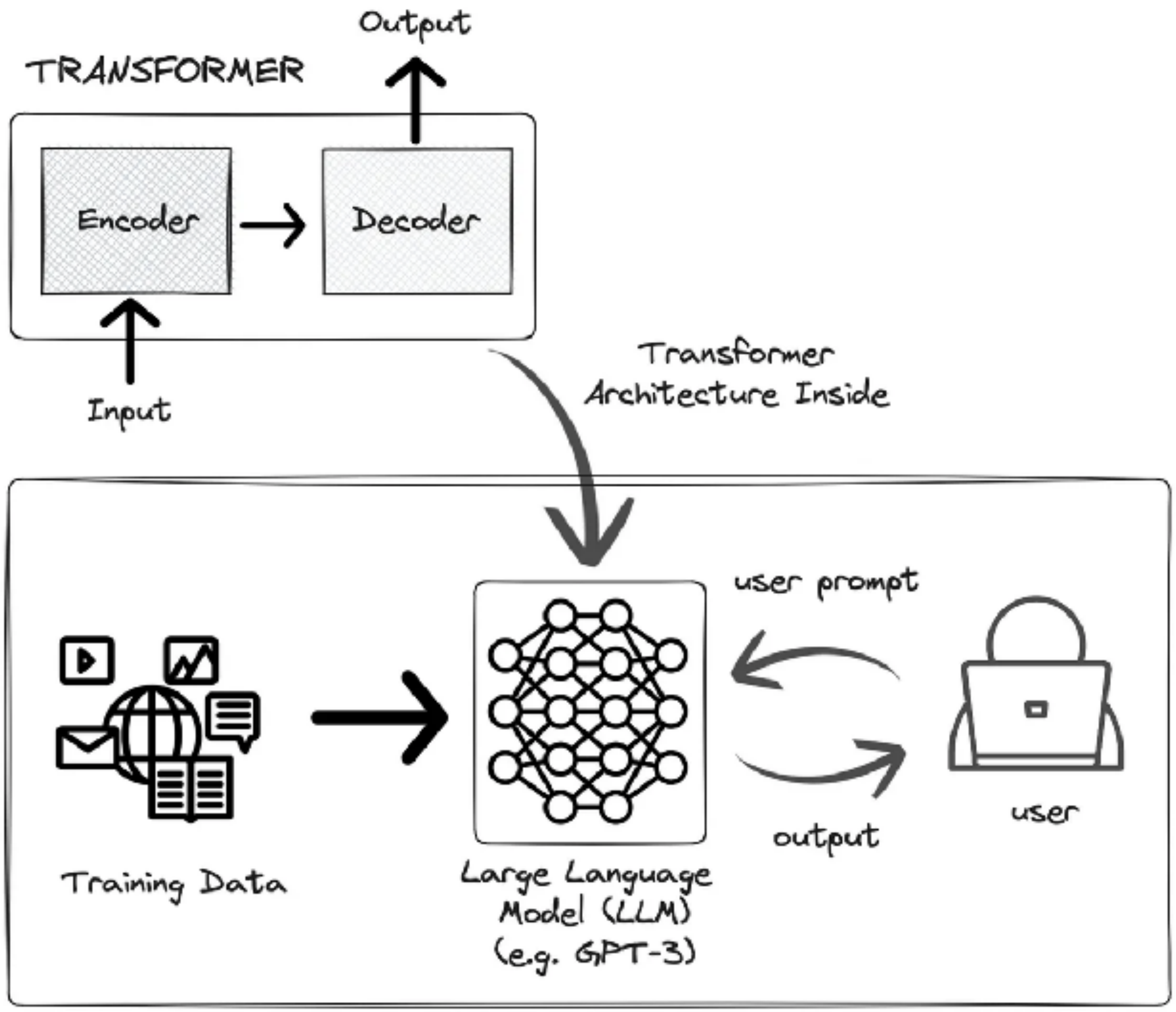


# LLM components !!




https://stackoverflow.blog/2024/08/22/llms-evolve-quickly-their-underlying-architecture-not-so-much

# Transformer inside



# Transformer process

I love LLM — Sentence

↓

["I", "love", "LL", "M"] — Tokens

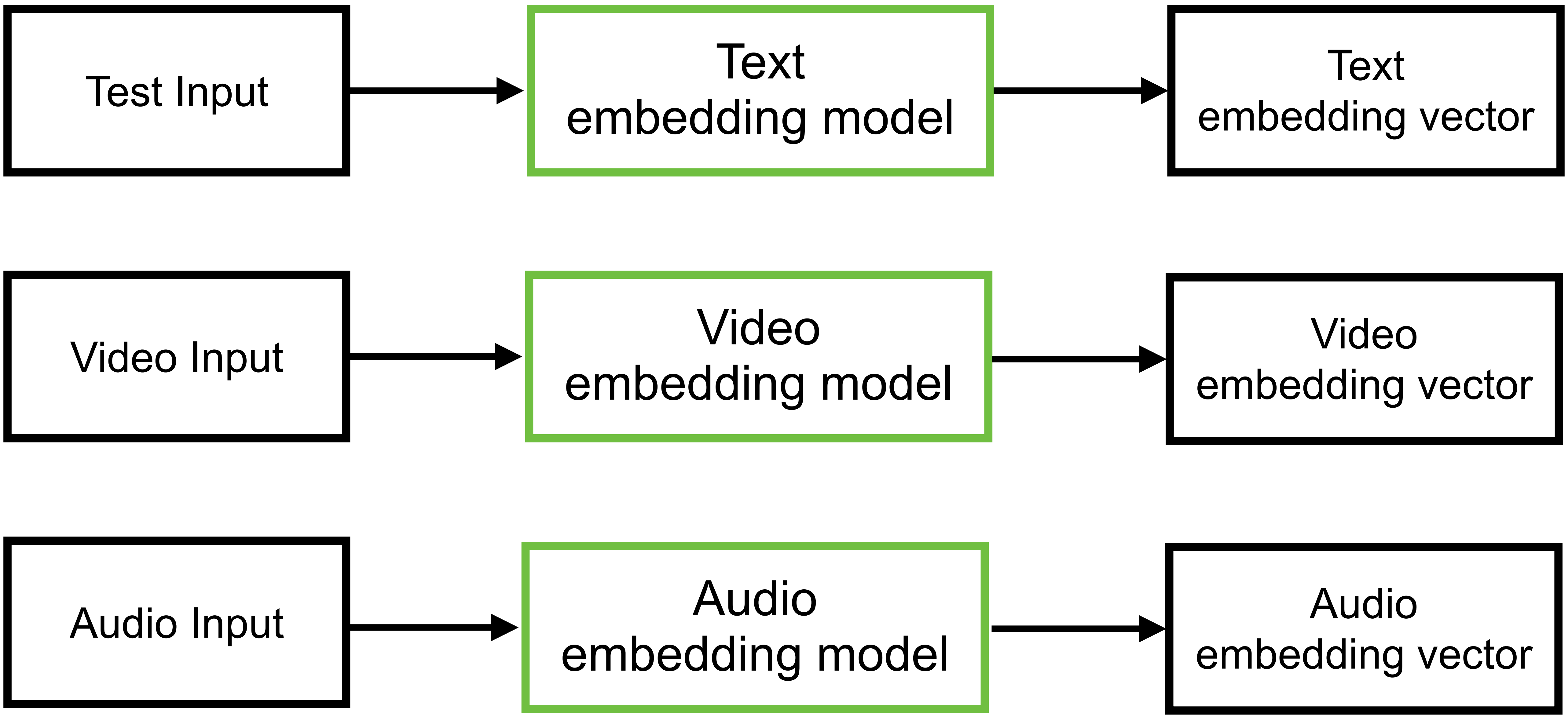
↓

[-5.23, 2.10, 0.82…] — Embedding

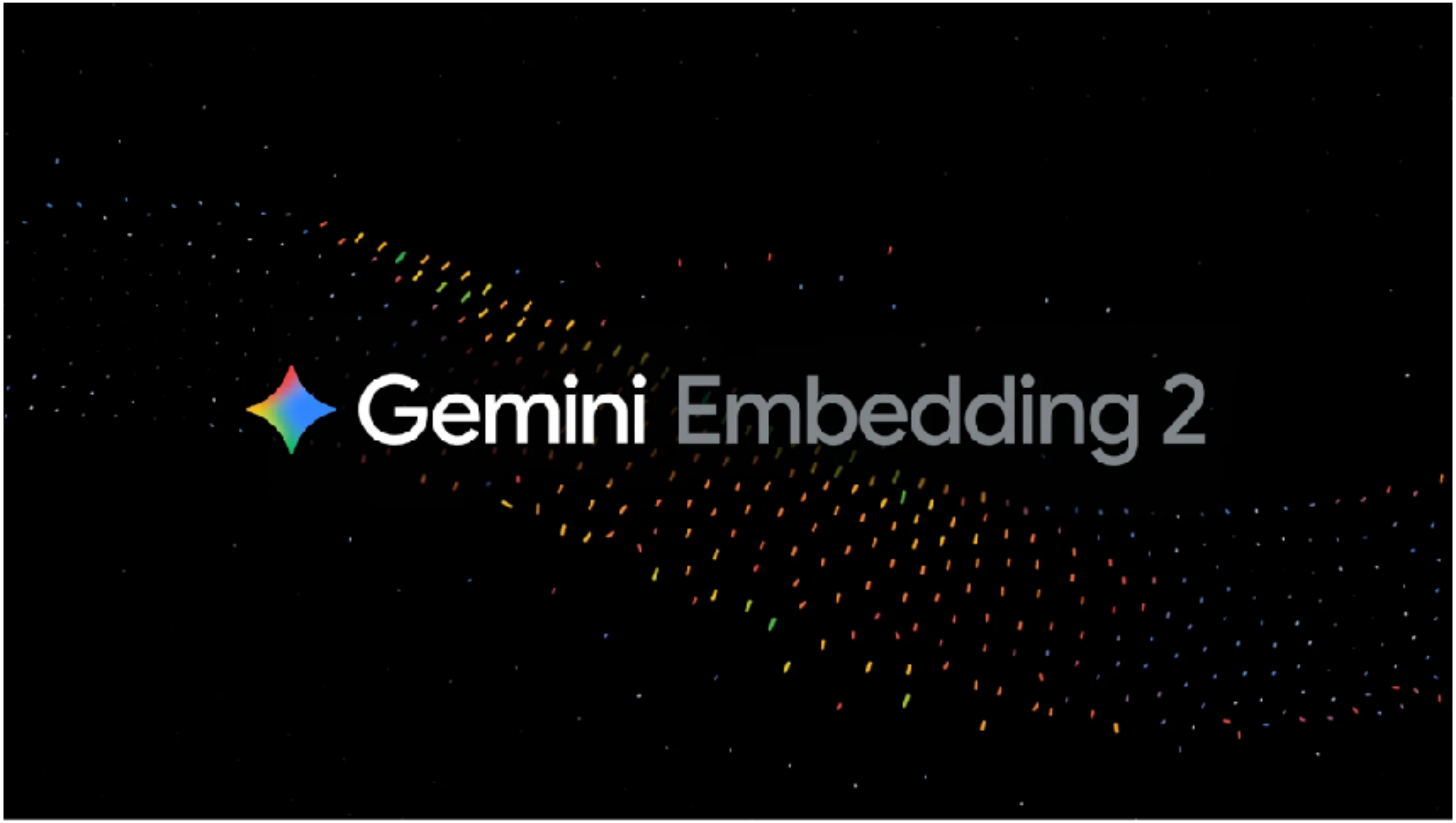
# OpenAI Tokenizer


https://platform.openai.com/tokenizer

# Embedding

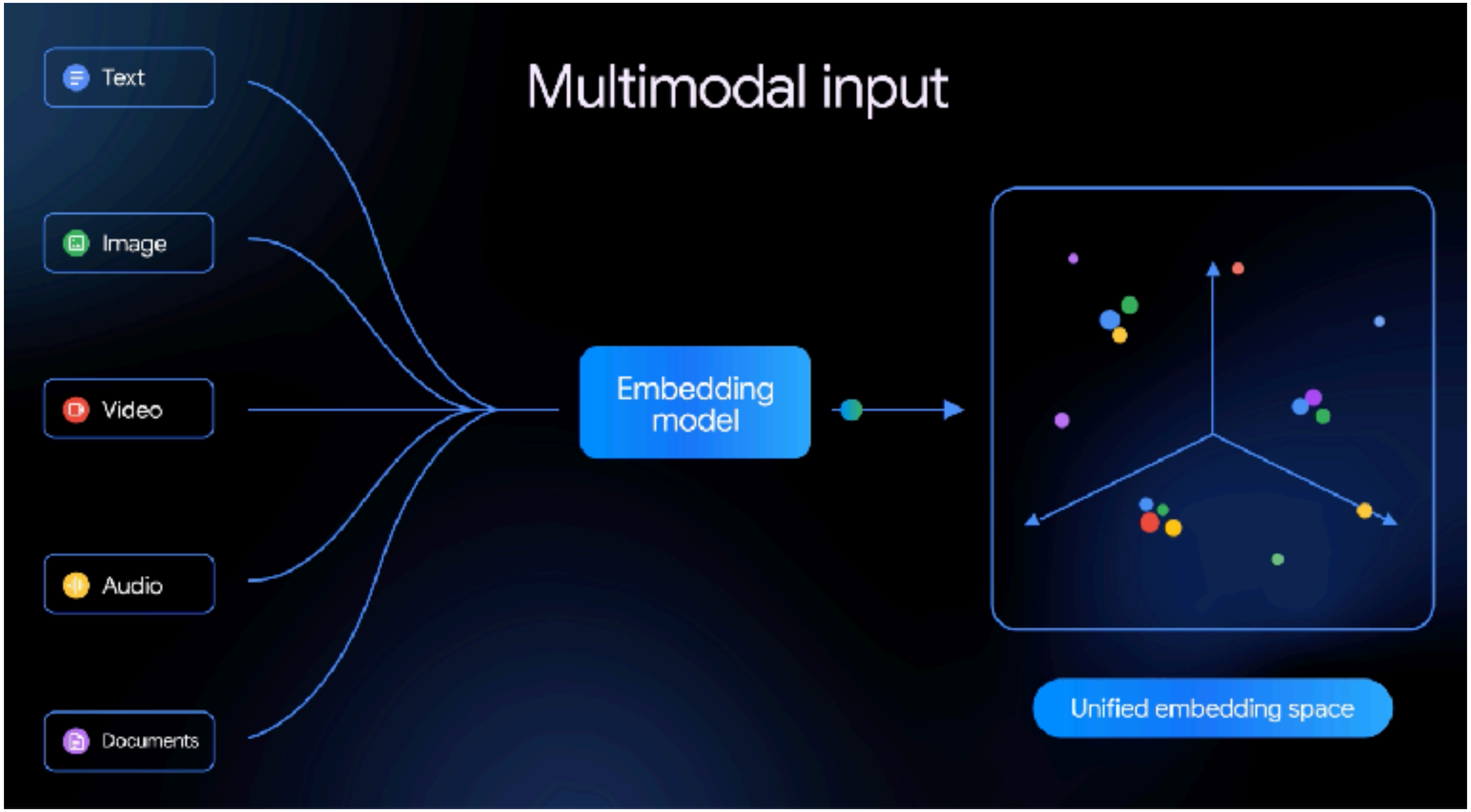

Test Input
Text
embedding model
Text
embedding vector
Video Input
Video
embedding model
Video
embedding vector
Audio Input
Audio
embedding model
Audio
embedding vector

# Multimodal Embedding Model




https://blog.google/innovation-and-ai/models-and-research/gemini-models/gemini-embedding-2/

# Multimodal Embedding Model

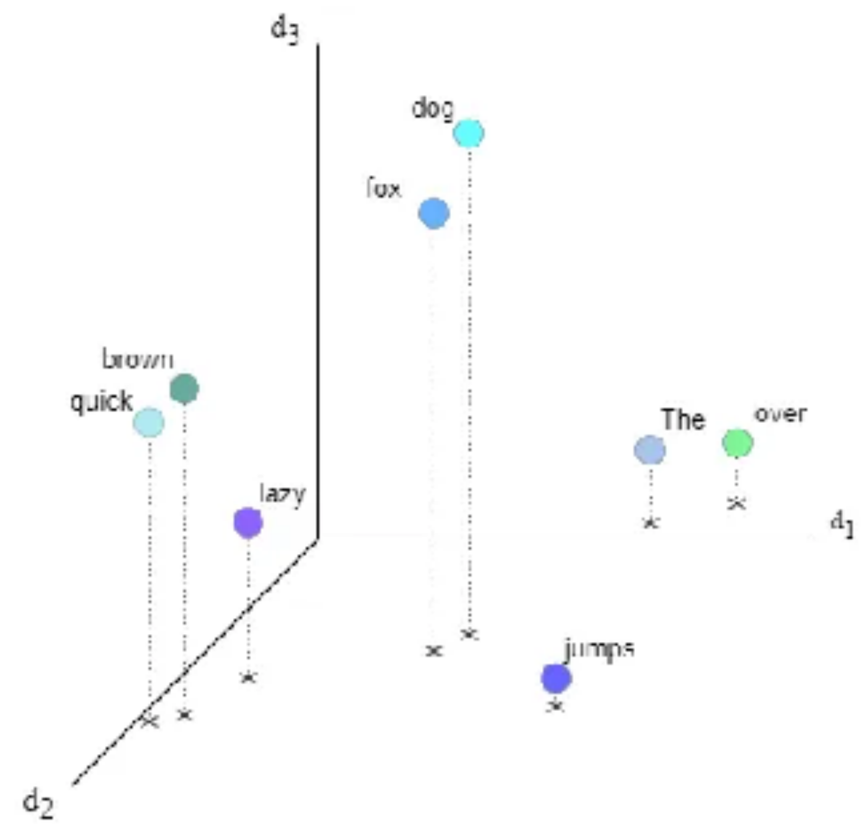



https://blog.google/innovation-and-ai/models-and-research/gemini-models/gemini-embedding-2/

# Embedding ?

Map items of **unstructured data** to high-dimensional real vectors

| | 0 | 1 | 2 | ...... | $d_{model}$ |
|---|---|---|---|---|---|
| The → | 0.64 | -0.09 | 0.23 | ...... | 0.005 |
| quick → | 0.05 | 0.79 | 0.47 | ...... | -0.54 |
| brown → | 0.12 | 0.74 | 0.51 | ...... | -0.3 |
| fox → | 0.42 | 0.52 | 0.83 | ...... | 0.01 |
| jumps → | 0.88 | 0.69 | 0.02 | ...... | 0.27 |
| over → | 0.84 | 0.15 | 0.13 | ...... | 0.05 |
| the → | 0.64 | 0.09 | 0.23 | ...... | 0.005 |
| lazy → | 0.1 | 0.65 | 0.28 | ...... | 0.19 |
| dog → | 0.54 | 0.49 | 0.90 | ...... | 0.008 |



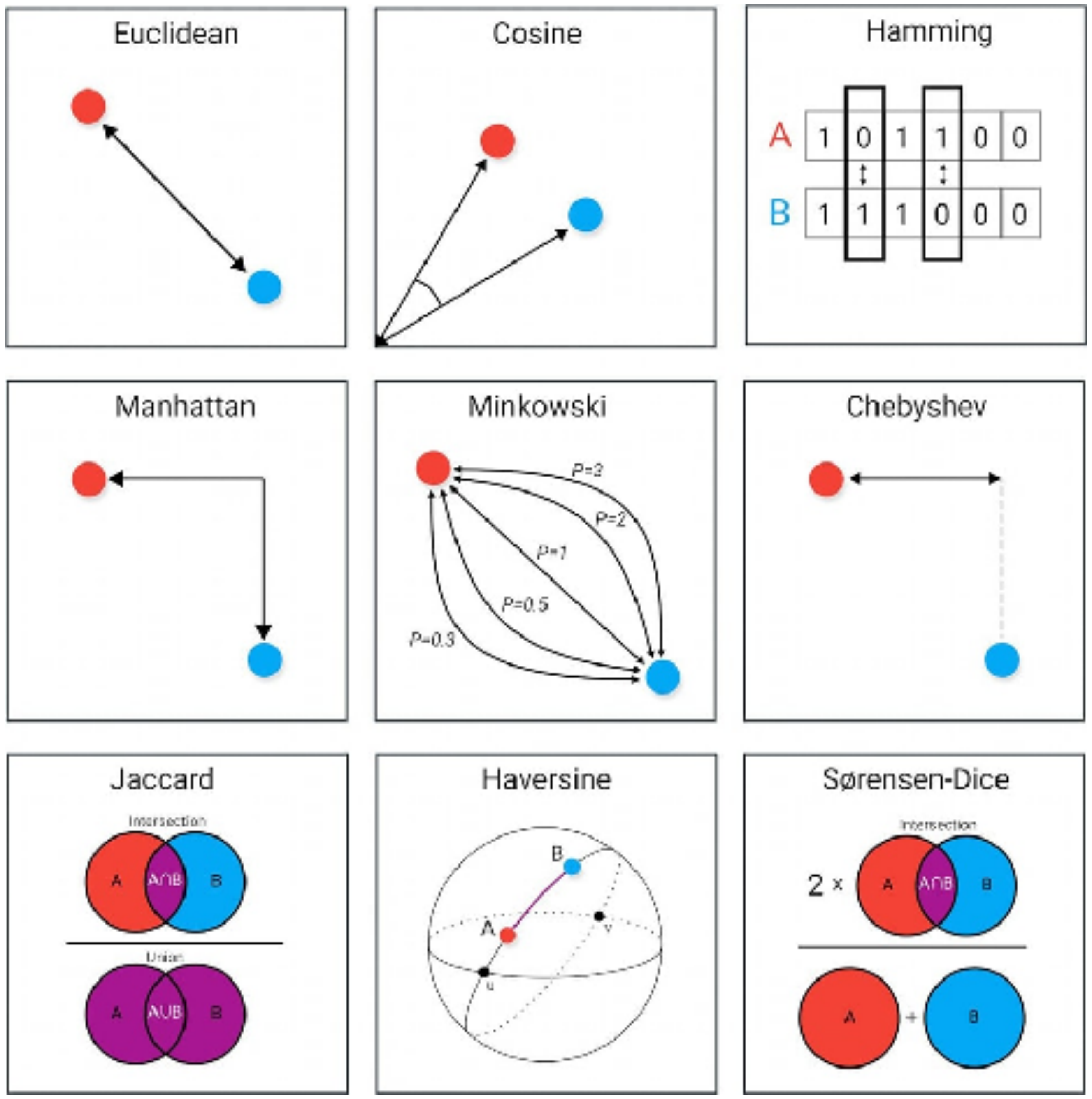

https://towardsdatascience.com/transformers-in-depth-part-1-introduction-to-transformer-models-in-5-minutes-ad25da6d3cca

# Embedding Leaderboard

| Rank (Bor… | Model | Embedding D… | Max Tokens | Mean (T… | Mean (TaskT… | Bitext … | Classification | Clustering | Instructi |
|---|---|---|---|---|---|---|---|---|---|
| **1** | KaLM-Embedding-Gemma3-12B-2511 | 3840 | 32768 | **72.32** | **62.51** | **83.76** | **77.88** | 55.77 | 5.49 |
| 2 | llama-embed-nemotron-8b | 4096 | 32768 | 69.46 | 61.09 | 81.72 | 73.21 | 54.35 | 10.82 |
| 3 | Qwen3-Embedding-8B | 4096 | 32768 | 70.58 | 61.69 | 80.89 | 74.00 | 57.65 | 10.06 |
| 4 | gemini-embedding-001 | 3072 | 2048 | 68.37 | 59.59 | 79.28 | 71.82 | 54.59 | 5.18 |
| 5 | Qwen3-Embedding-4B | 2560 | 32768 | 69.45 | 60.86 | 79.36 | 72.33 | 57.15 | 11.56 |
| 6 | Octen-Embedding-8B | 4096 | 32768 | 67.84 | 60.28 | 80.35 | 66.68 | 55.68 | 8.98 |
| 7 | F2LLM-v2-14B | 5120 | 40960 | 68.74 | 59.45 | 77.15 | 73.60 | **60.91** | 0.62 |
| 8 | F2LLM-v2-8B | 4096 | 40960 | 68.09 | 58.99 | 75.96 | 71.93 | 60.62 | 0.85 |
| 9 | Seed1.6-embedding-1215 | 2048 | 32768 | 70.26 | 61.34 | 78.68 | 76.75 | 56.78 | -0.82 |
| 10 | F2LLM-v2-4B | 2560 | 40960 | 67.06 | 58.25 | 74.49 | 70.73 | 59.53 | 2.25 |
| 11 | jina-embeddings-v5-text-small | 1024 | 32768 | 67.00 | 58.90 | 69.71 | 71.32 | 53.41 | 1.35 |

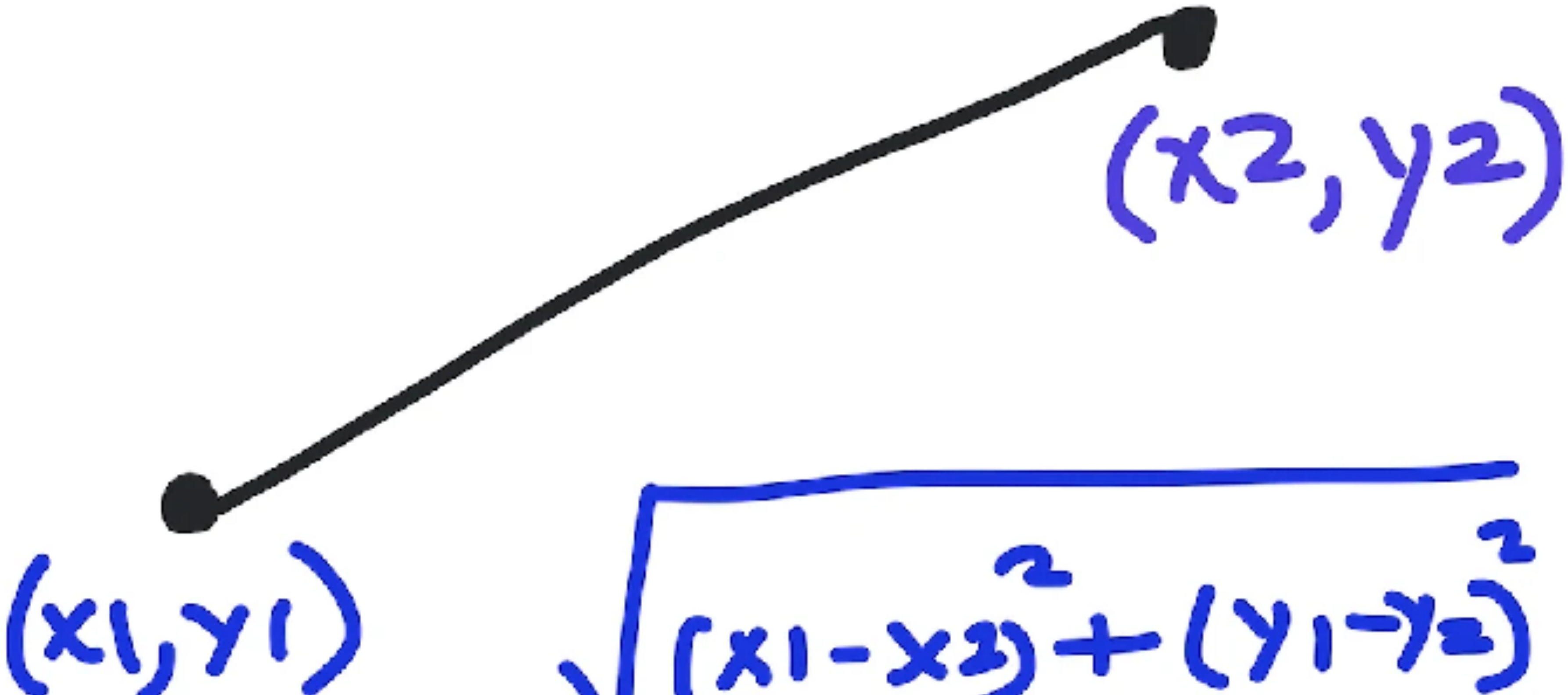
https://huggingface.co/spaces/mteb/leaderboard

# Distance measure in Data Science



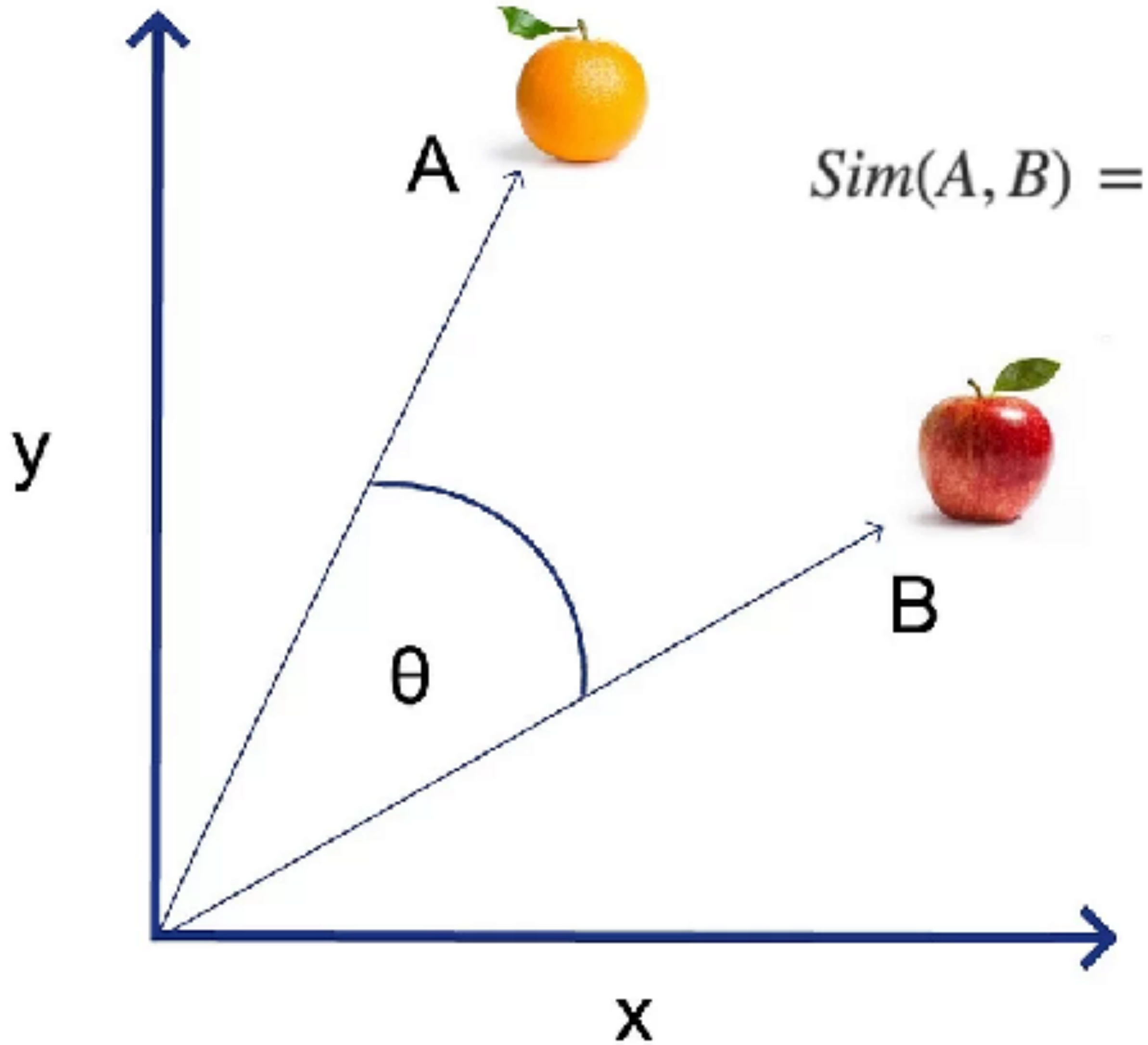

https://www.maartengrootendorst.com/blog/distances/

# Euclidean Distance




$$\sqrt{(x1-x2)^2+(y1-y2)^2}$$

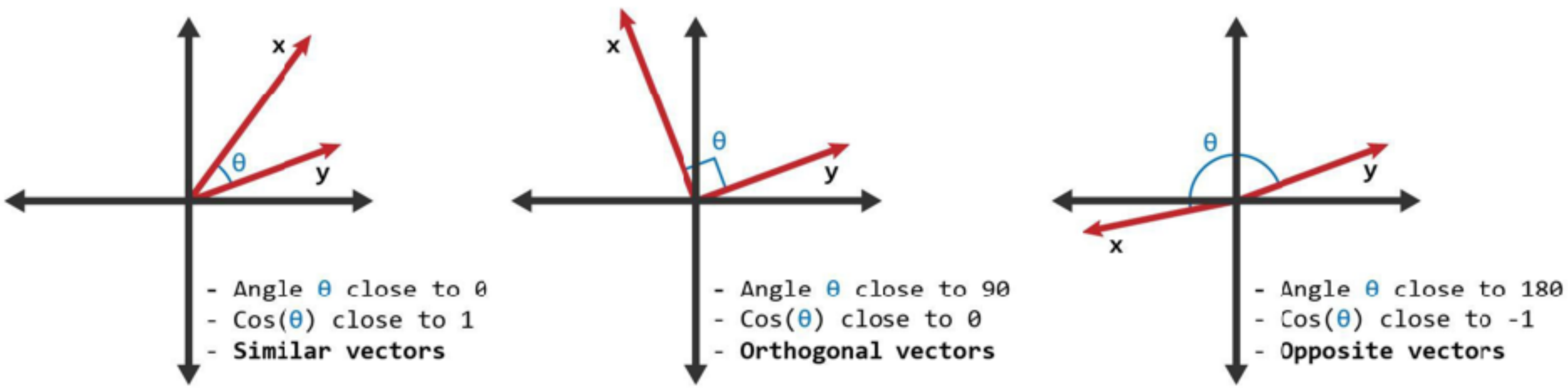
# Cosine Similarity




$$Sim(A,B) = \cos(\theta) = \frac{A \cdot B}{\|A\|\|B\|}$$

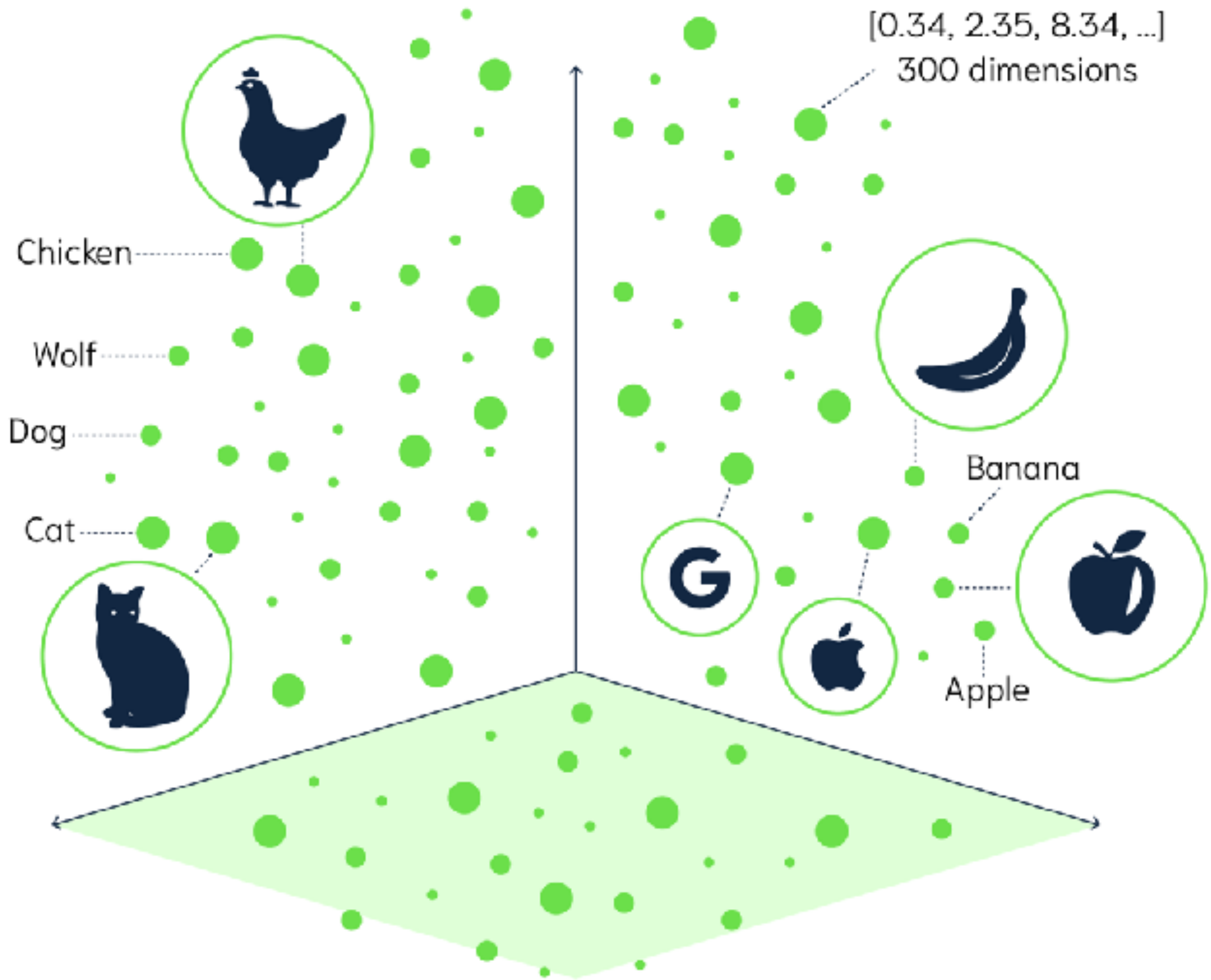
https://python.plainenglish.io/mastering-cosine-metrics-a-data-scientists-essential-toolkit-3ce95eda91f9

# Cosign Similarity

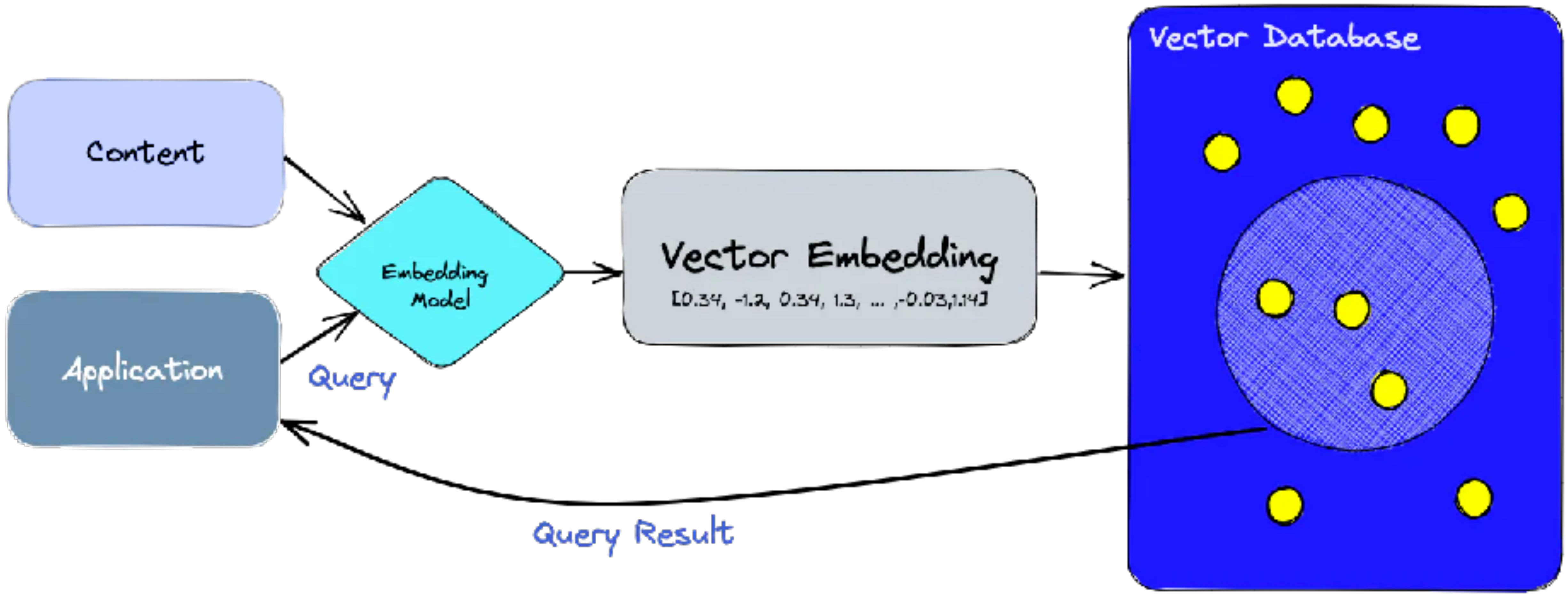



https://www.learndatasci.com/glossary/cosine-similarity/

# Visual of Vector space

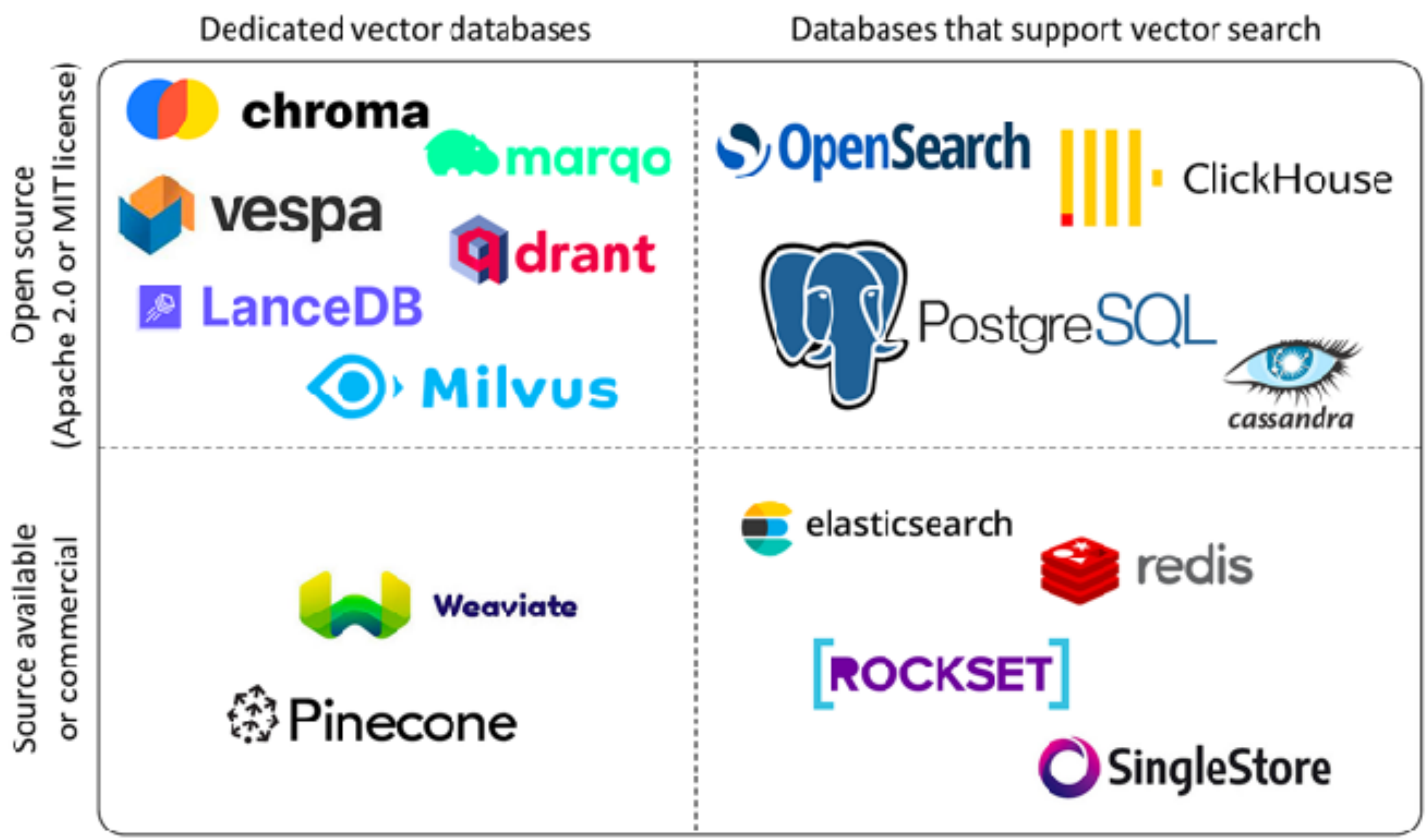


# Store data in Vector Database



# Vector Database ?

Map items of unstructured data to high-dimensional real **vectors**

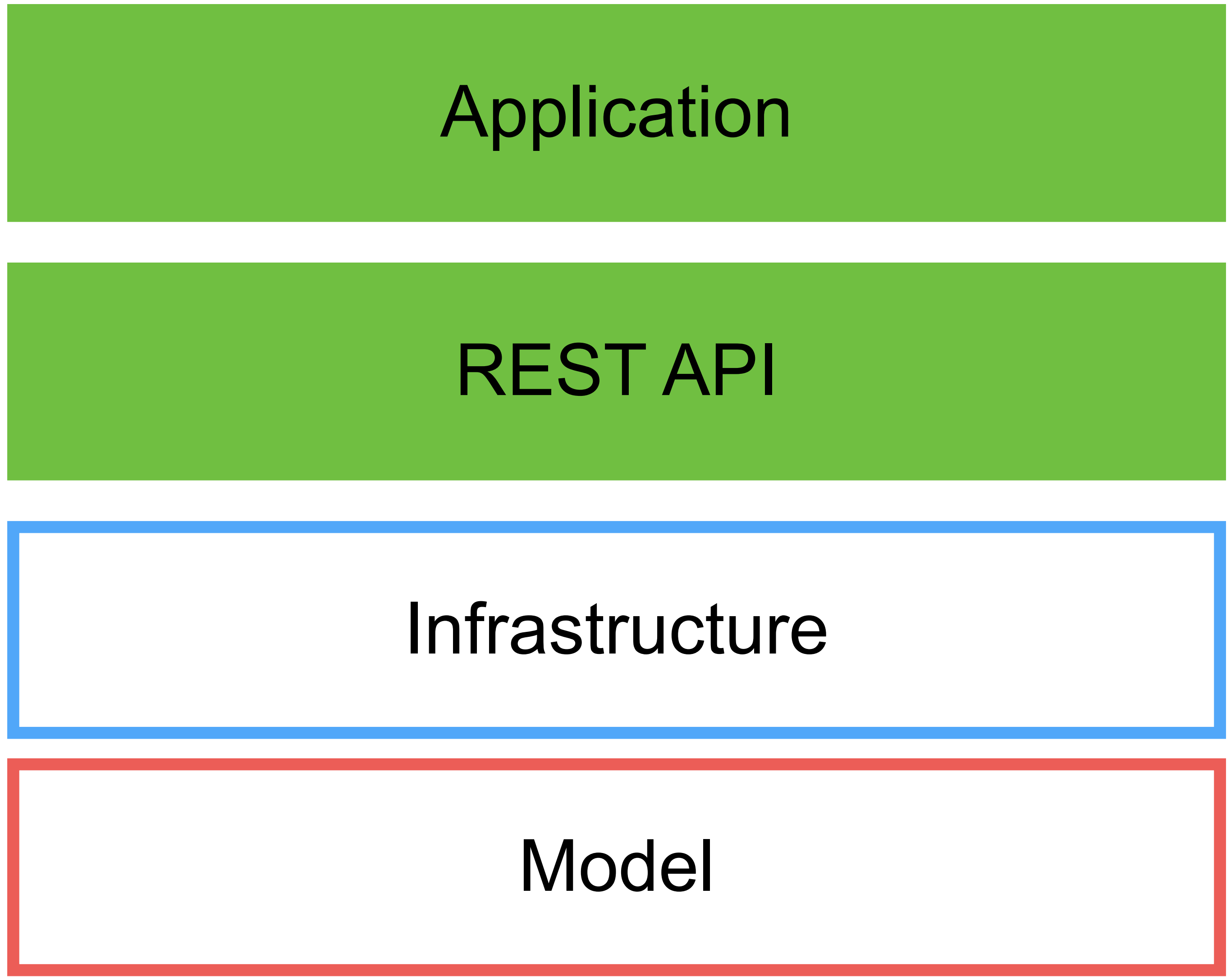



https://towardsdatascience.com/transformers-in-depth-part-1-introduction-to-transformer-models-in-5-minutes-ad25da6d3cca

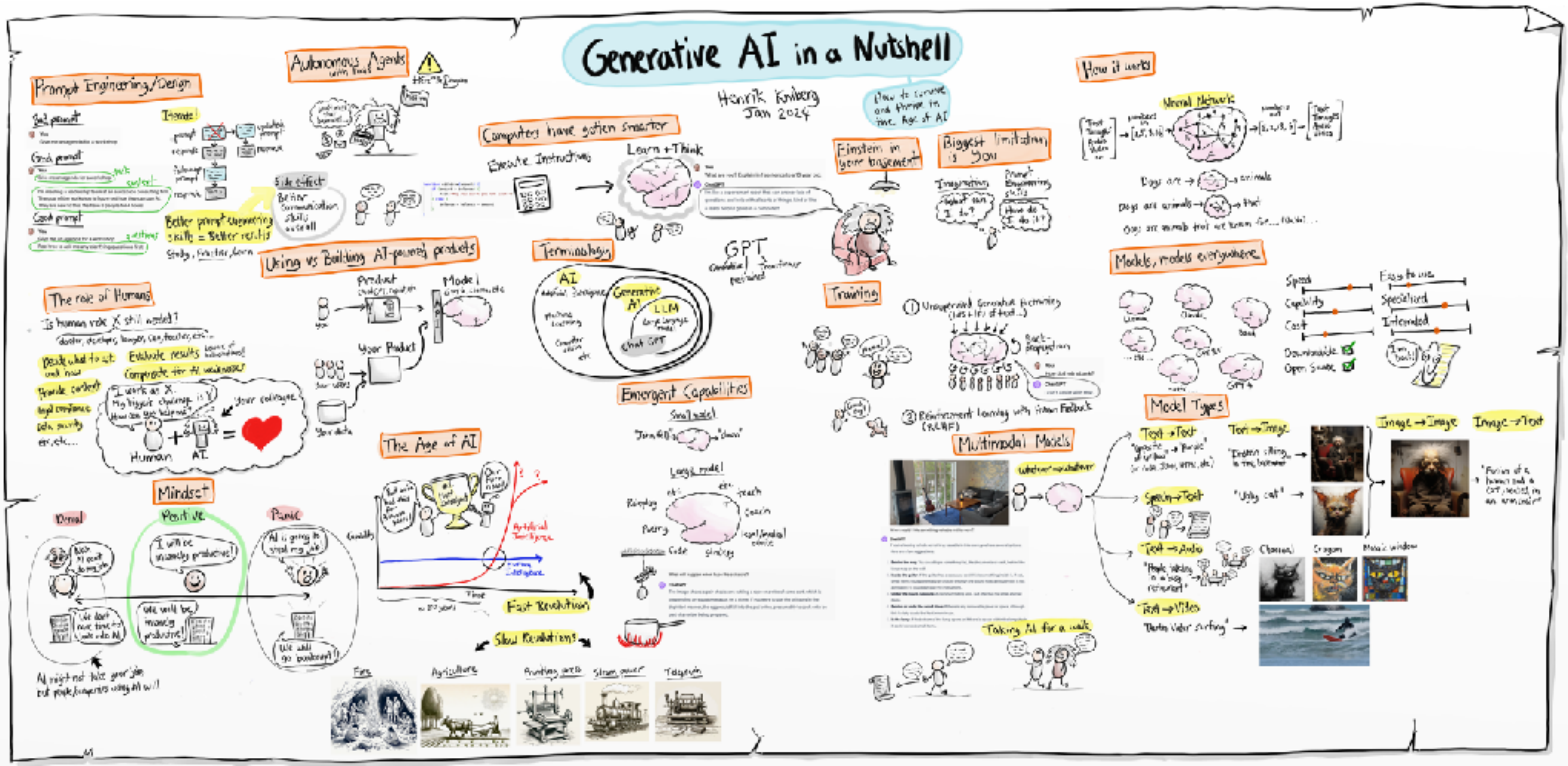
Application

Infrastructure

Model

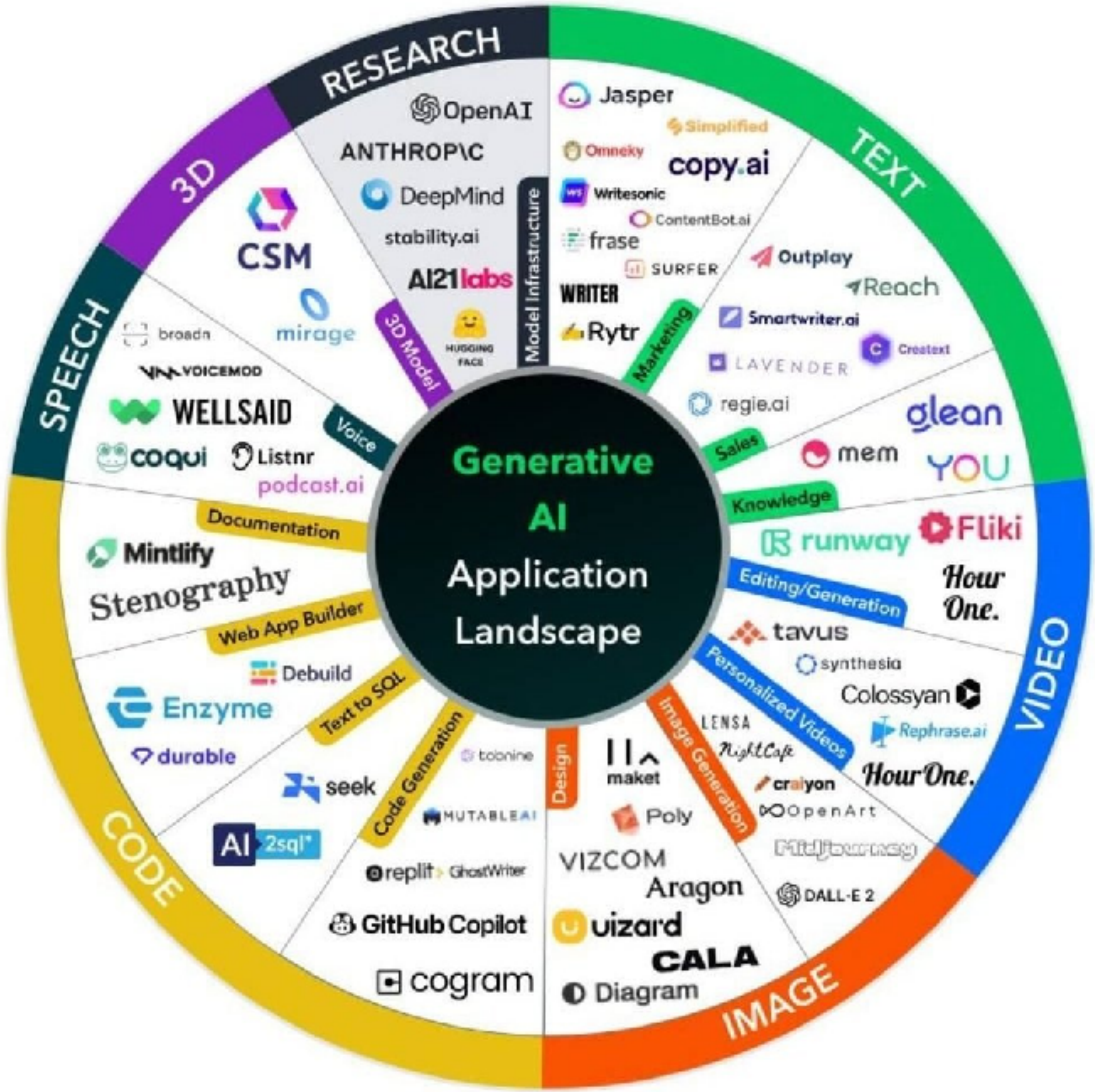

Application
REST API
Infrastructure
Model

# Generative AI in Nutshell




https://www.youtube.com/watch?v=2IK3DFHRFfw

# Application Tool chains


Generative AI Application Landscape
RESEARCH
OpenAI
ANTHROP\C
DeepMind
stability.ai
AI21labs
HUGGING FACE
Model Infrastructure
3D Model
TEXT
Jasper
Simplified
Omneky
copy.ai
Writesonic
ContentBot.ai
frase
SURFER
WRITER
Rytr
Marketing
Outplay
Reach
Smartwriter.ai
Crestext
LAVENDER
regie.ai
Sales
glean
mem
YOU
Knowledge
VIDEO
Fliki
runway
Hour One.
Editing/Generation
tavus
synthesia
Colossyan
Rephrase.ai
Hour One.
Personalized Videos
IMAGE
LENSA
NightCafe
craiyon
OpenArt
Midjourney
DALL-E 2
Image Generation
maket
Poly
VIZCOM
Aragon
uizard
CALA
Diagram
Design
CODE
tabnine
MUTABLEAI
replit
GhostWriter
GitHub Copilot
cogram
Code Generation
seek
AI2sql
Text to SQL
Debuild
Enzyme
durable
Web App Builder
Mintlify
Stenography
Documentation
SPEECH
broadn
VOICEMOD
WELLSAID
coqui
Listnr
podcast.ai
Voice
3D
CSM
mirage

# Tool chains category

Assist tasks

Interaction modes

Prompt composition

Properties of model

LLM models

# Assist tasks

Finding information faster in context

Generating code

Reasoning about code

Transforming code into something ..

Requirement → Design → Develop → Testing → Deploy

Software Delivery Lifecycle

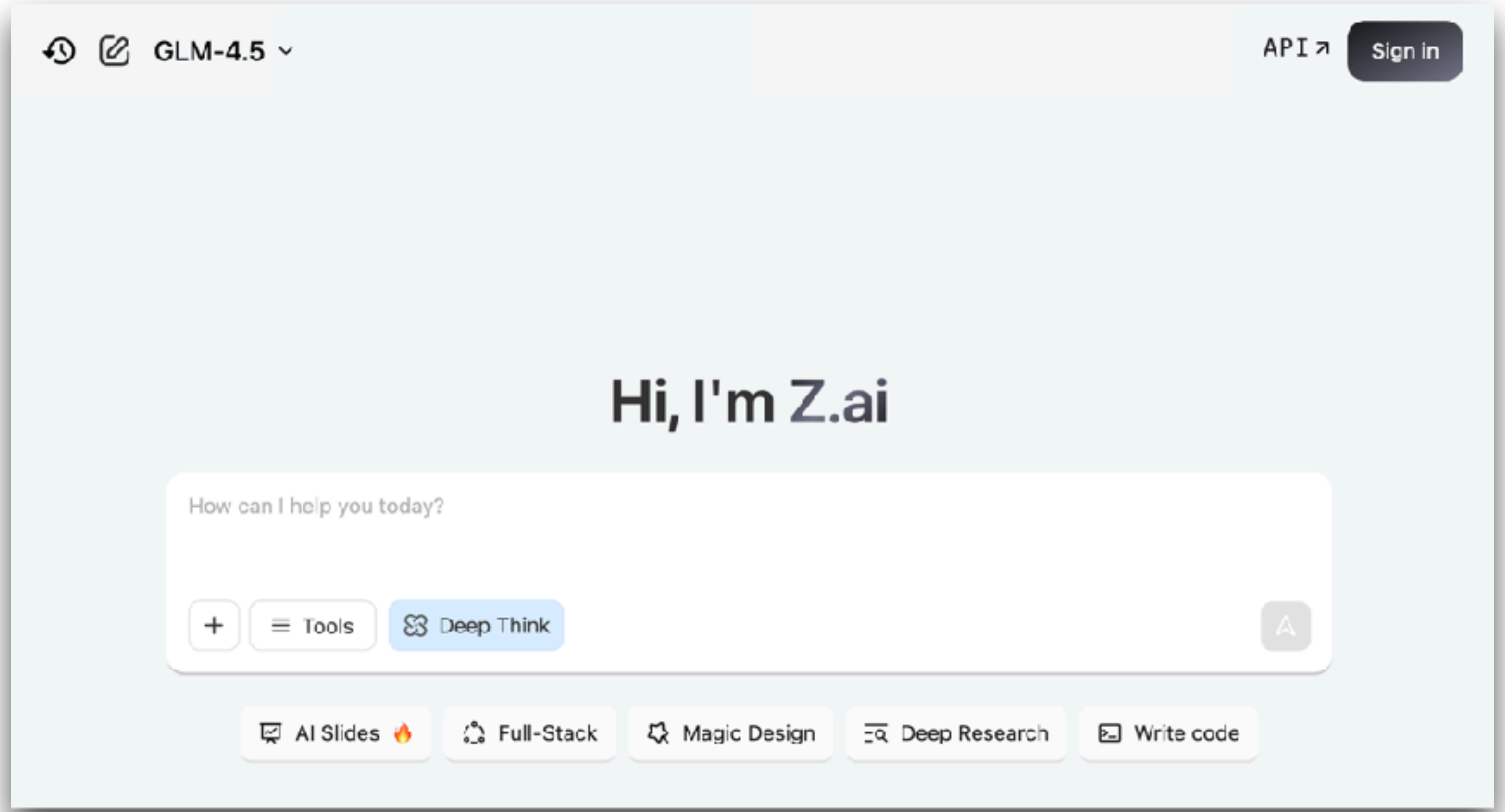
# Interaction modes

**Chat interfaces**

In-line assistance (typing in code editor)

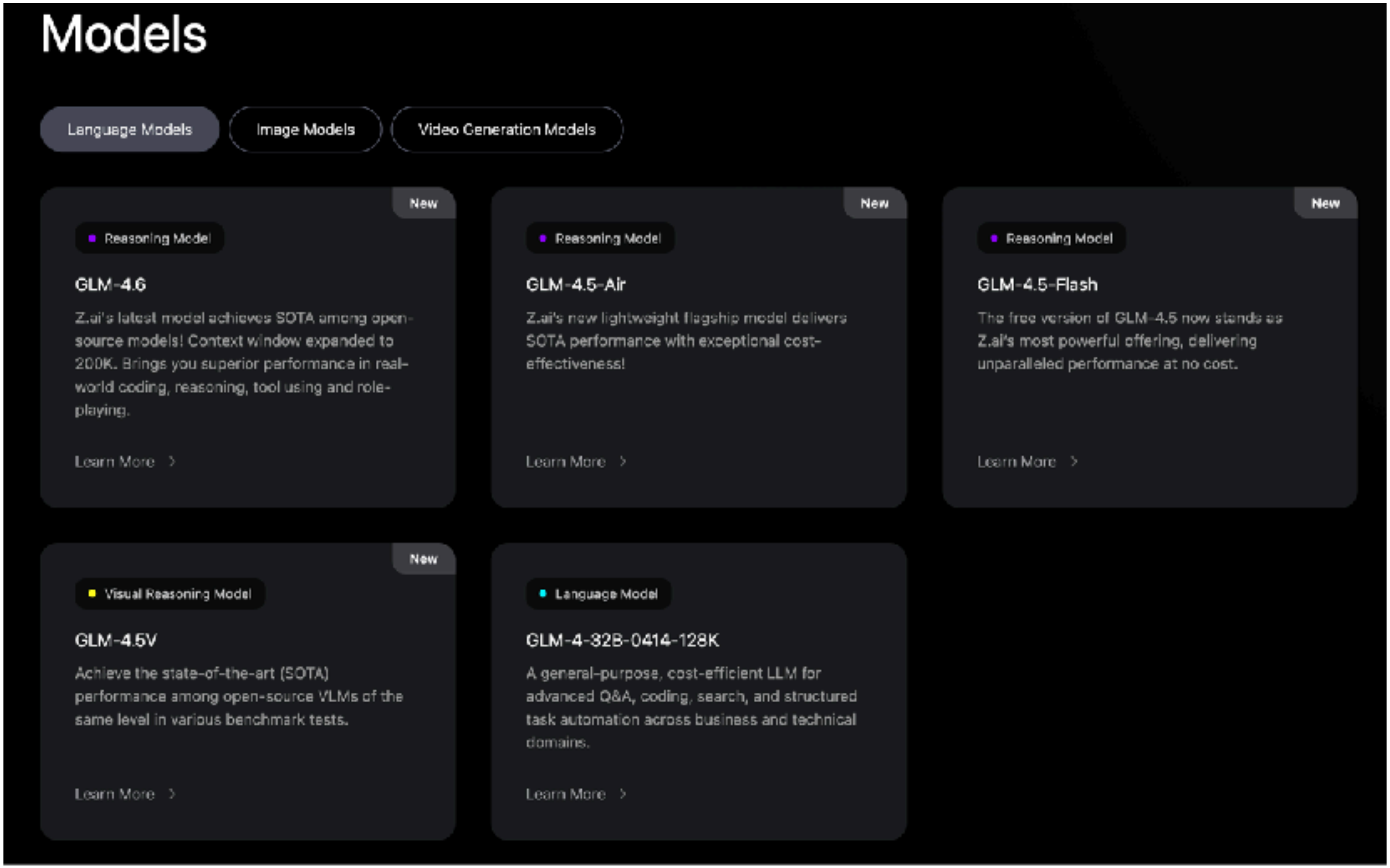
CLI (command-line interface)

# Z.ai




https://z.ai/

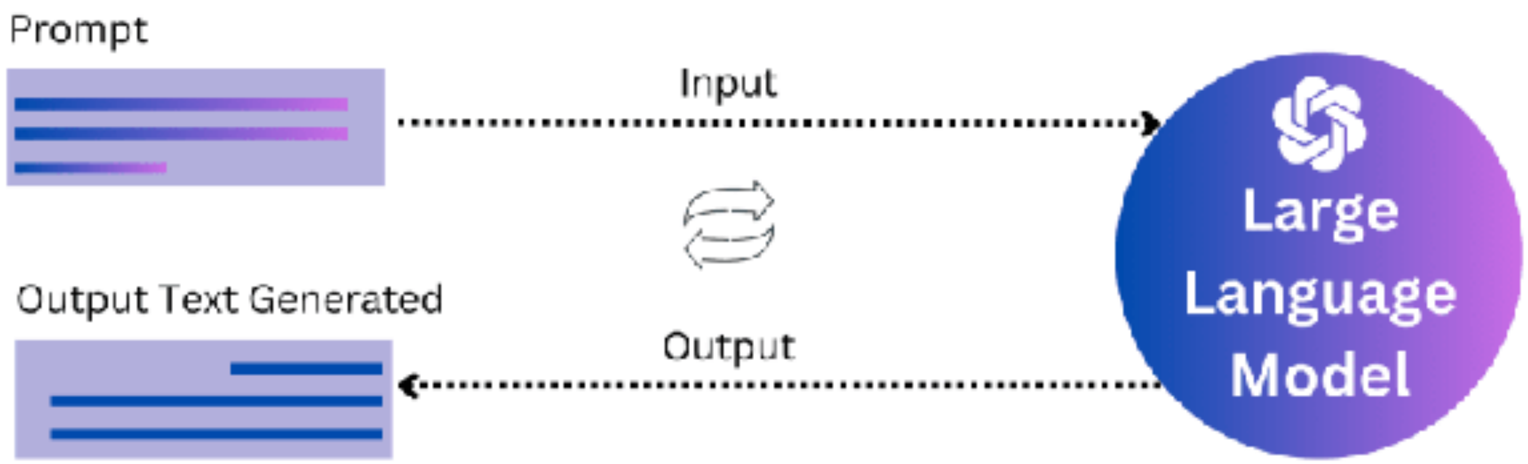
# Models




https://z.ai/

# Prompt composition

Prompt engineering
Compose prompts from user inputs and context




https://platform.openai.com/docs/guides/prompt-engineering

# Better Prompt

Write clear instructions

Provide reference text

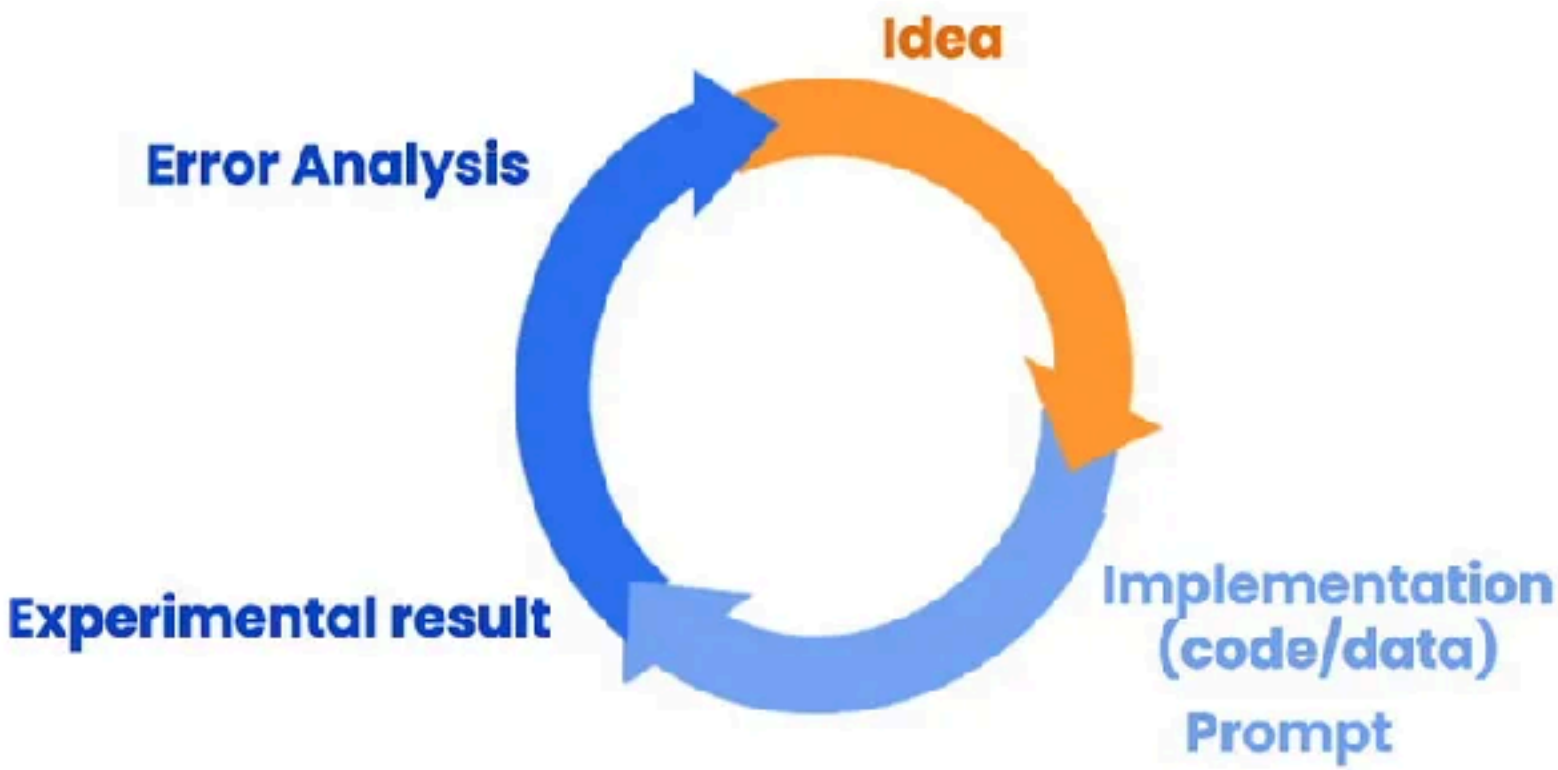
Split complex tasks into simpler subtasks

Give the model time to think

Testing and improve …

https://platform.openai.com/docs/guides/prompt-engineering

# Iterative Prompt Development

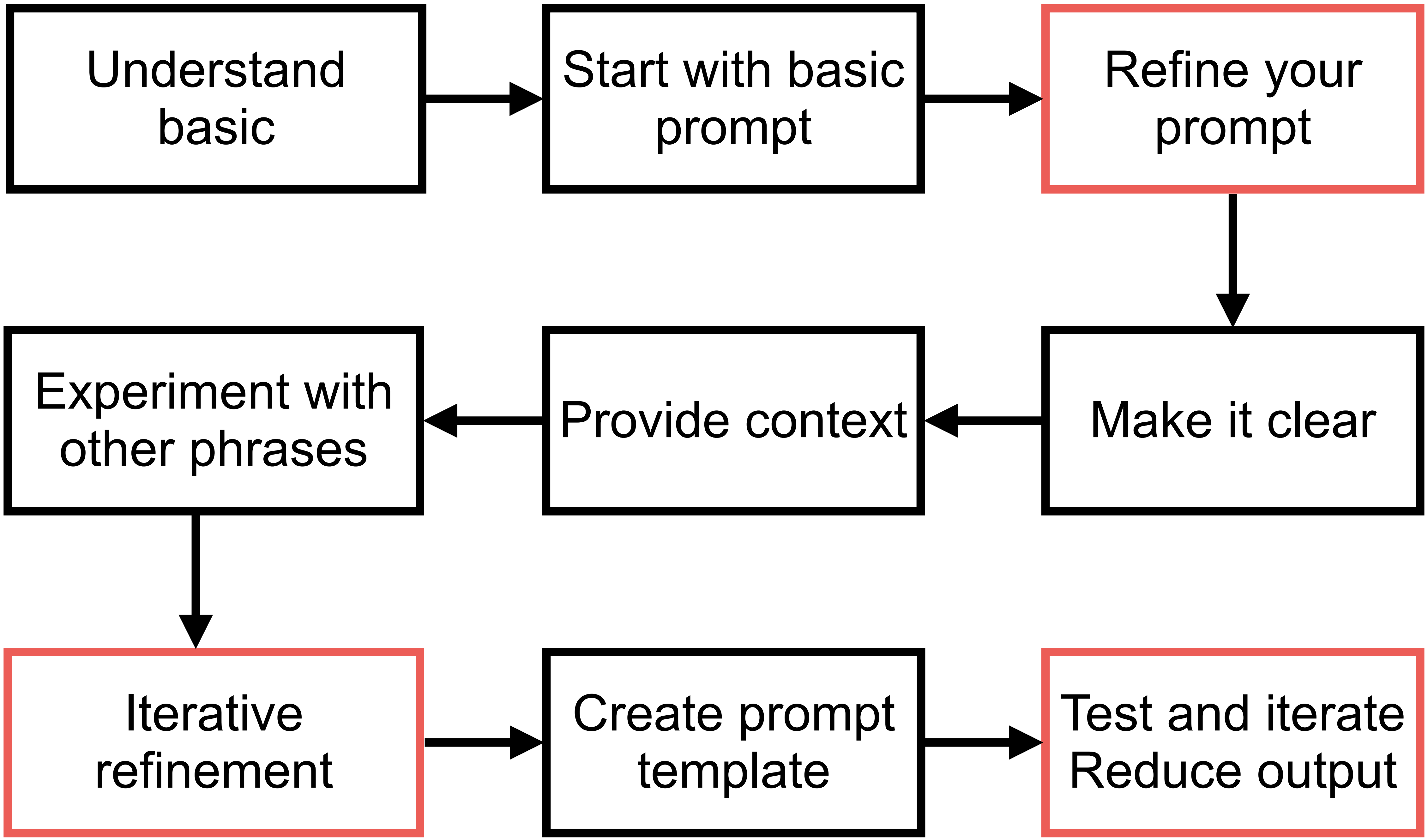



Iterative Process

- Try something
- Analyze where the result does not give what you want
- Clarify instructions, give more time to think
- Refine prompts with a batch of examples

https://www.deeplearning.ai/short-courses/chatgpt-prompt-engineering-for-developers/

# Basic of Prompt Engineer



# Better Prompt

Write clear instructions

Provide reference text

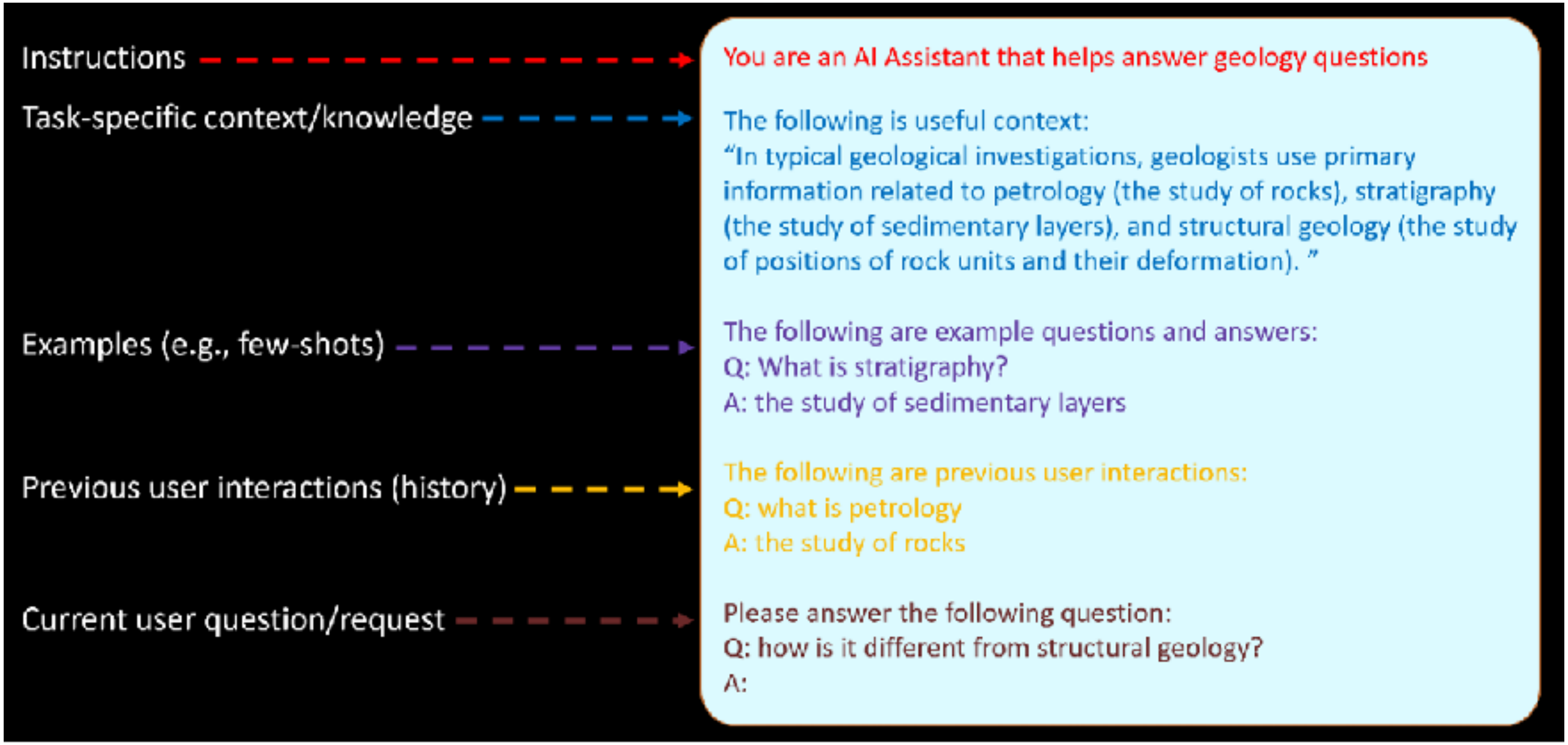
Split complex tasks into simpler subtasks

Give the model time to think

Testing and improve …

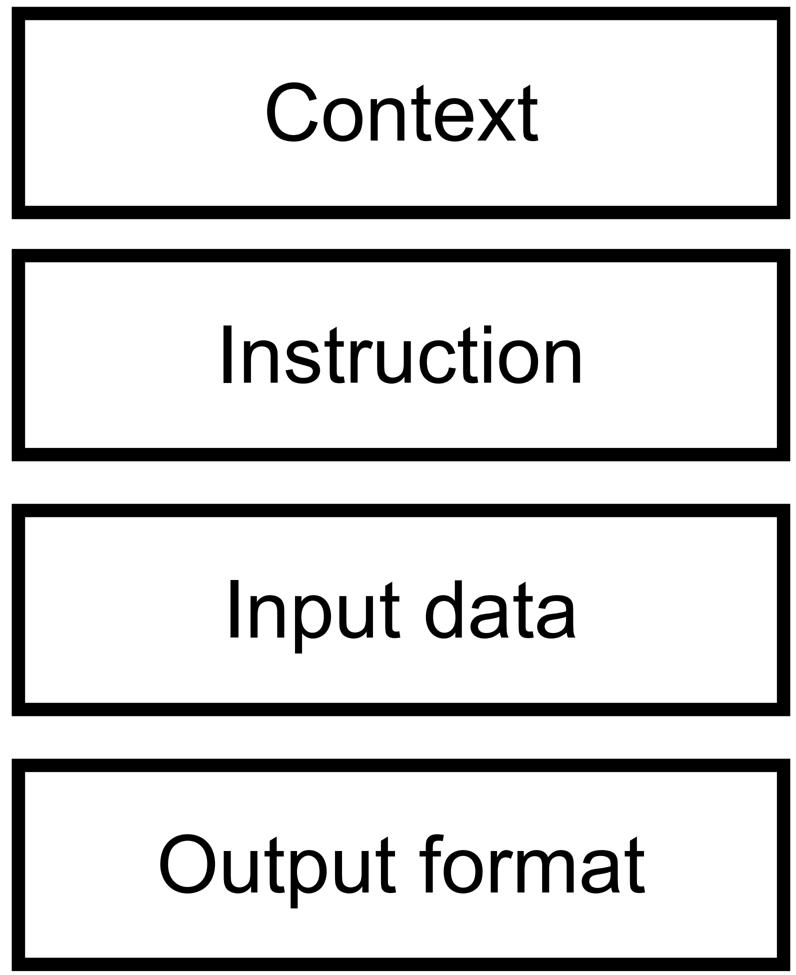
https://platform.openai.com/docs/guides/prompt-engineering/six-strategies-for-getting-better-results

# Prompt Structure




https://learn.microsoft.com/en-us/ai/playbook/technology-guidance/generative-ai/working-with-llms/prompt-engineering

# Structure of Prompt




https://platform.openai.com/docs/guides/prompt-engineering

# Prompting Guide

Prompt Engineering

## Prompt Engineering Guide

Prompt engineering is a relatively new discipline for developing and optimizing prompts to efficiently use language models (LMs) for a wide variety of applications and research topics. Prompt engineering skills help to better understand the capabilities and limitations of large language models (LLMs).

Researchers use prompt engineering to improve the capacity of LLMs on a wide range of common and complex tasks such as question answering and arithmetic reasoning. Developers use prompt engineering to design robust and effective prompting techniques that interface with LLMs and other tools.

Prompt engineering is not just about designing and developing prompts. It encompasses a wide range of skills and techniques that are useful for interacting and developing with LLMs. It's an important skill to interface, build with, and understand capabilities of LLMs. You can use prompt engineering to improve safety of LLMs and build new capabilities like augmenting LLMs with domain knowledge and external tools.

https://www.promptingguide.ai/

# Structure of Prompt !!

**APE**
Action, Purpose, Expectation

**RACE**
Role, Action, Context, Expectation

**TAG**
Task, Action, Goal

**COAST**
Context, Objective, Action, Scenario, Task

**RISE**
Role, Input, Step, Expectation

https://twitter.com/pradeepeth/status/1673271866696544257

# Prompt Techniques

Zero-shot

Chain-of Thought (CoT)

Few-shot

Meta or structure

https://www.promptingguide.ai/techniques

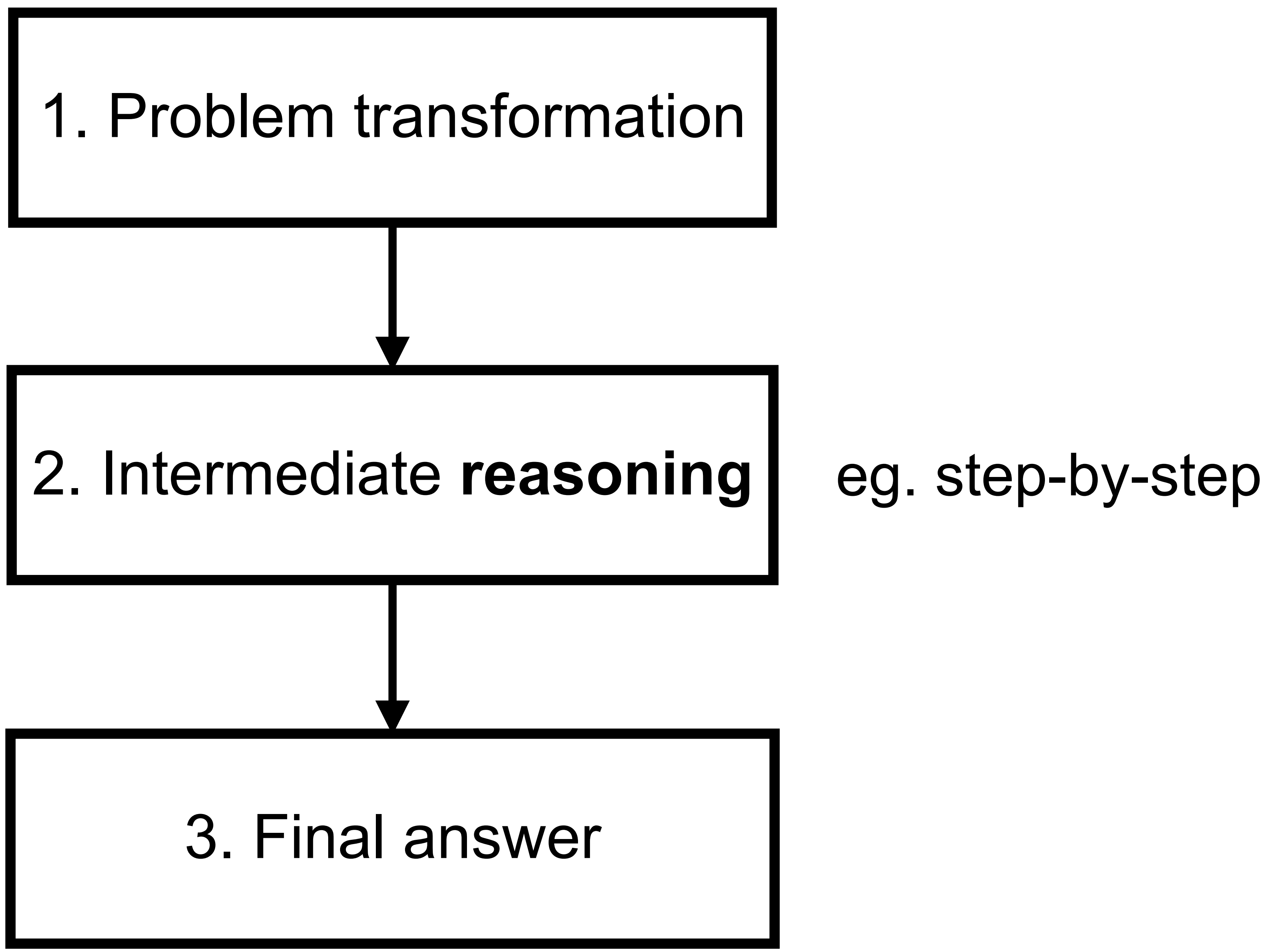
# Chain of Thought Prompting (CoT)

Technique used to improve the reasoning ability of LLM

Try to break down a complex problem into smaller, More manageable steps, lead to final answer

Reasoning model !!

# Chain of Thought Prompting (CoT)



# Chain of Thought Prompting (CoT)

**(a) Few-shot**

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?
A: The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
A:

*(Output) The answer is 8.* X

**(b) Few-shot-CoT**

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?
A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. 5 + 6 = 11. The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
A:

*(Output) The juggler can juggle 16 balls. Half of the balls are golf balls. So there are 16 / 2 = 8 golf balls. Half of the golf balls are blue. So there are 8 / 2 = 4 blue golf balls. The answer is 4.* ✓

**(c) Zero-shot**

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
A: The answer (arabic numerals) is

*(Output) 8* X

**(d) Zero-shot-CoT (Ours)**

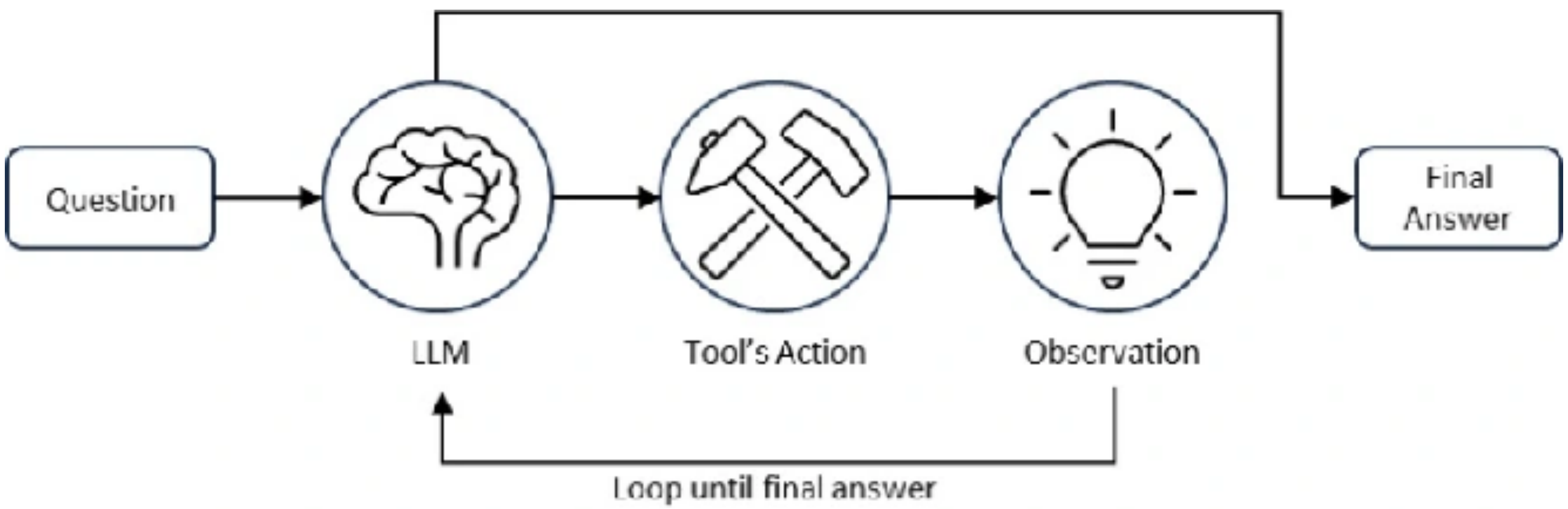
Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
A: ***Let's think step by step.***

*(Output) There are 16 balls in total. Half of the balls are golf balls. That means that there are 8 golf balls. Half of the golf balls are blue. That means that there are 4 blue golf balls.* ✓

https://www.promptingguide.ai/techniques/cot

# ReAct Prompt

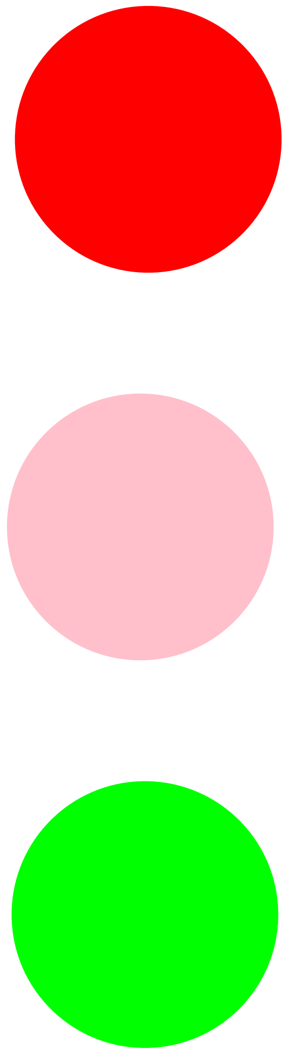
LLM reasoning and additional tools (expert)
Improve better answer




https://www.promptingguide.ai/techniques/react

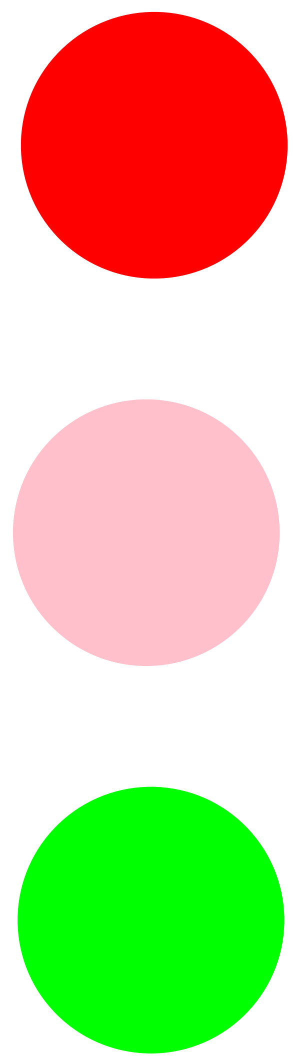
# Workshop

# RGB ?



https://github.com/up1/workshop-ai-with-technical-team/tree/main/workshop/demo-rgb

# RGB ?



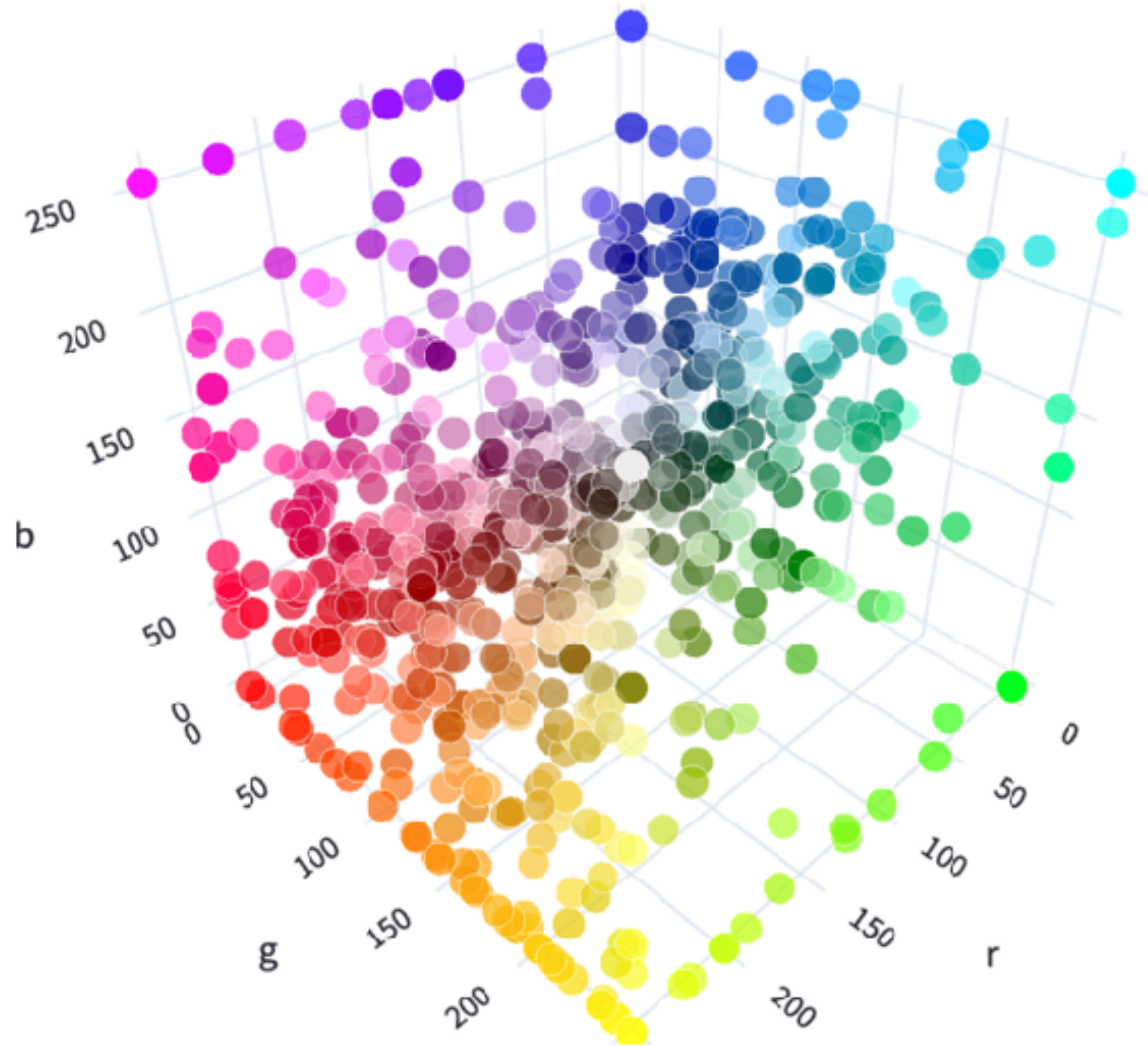
[255, 0, 0]

[255, 192, 203]

[0, 255, 0]

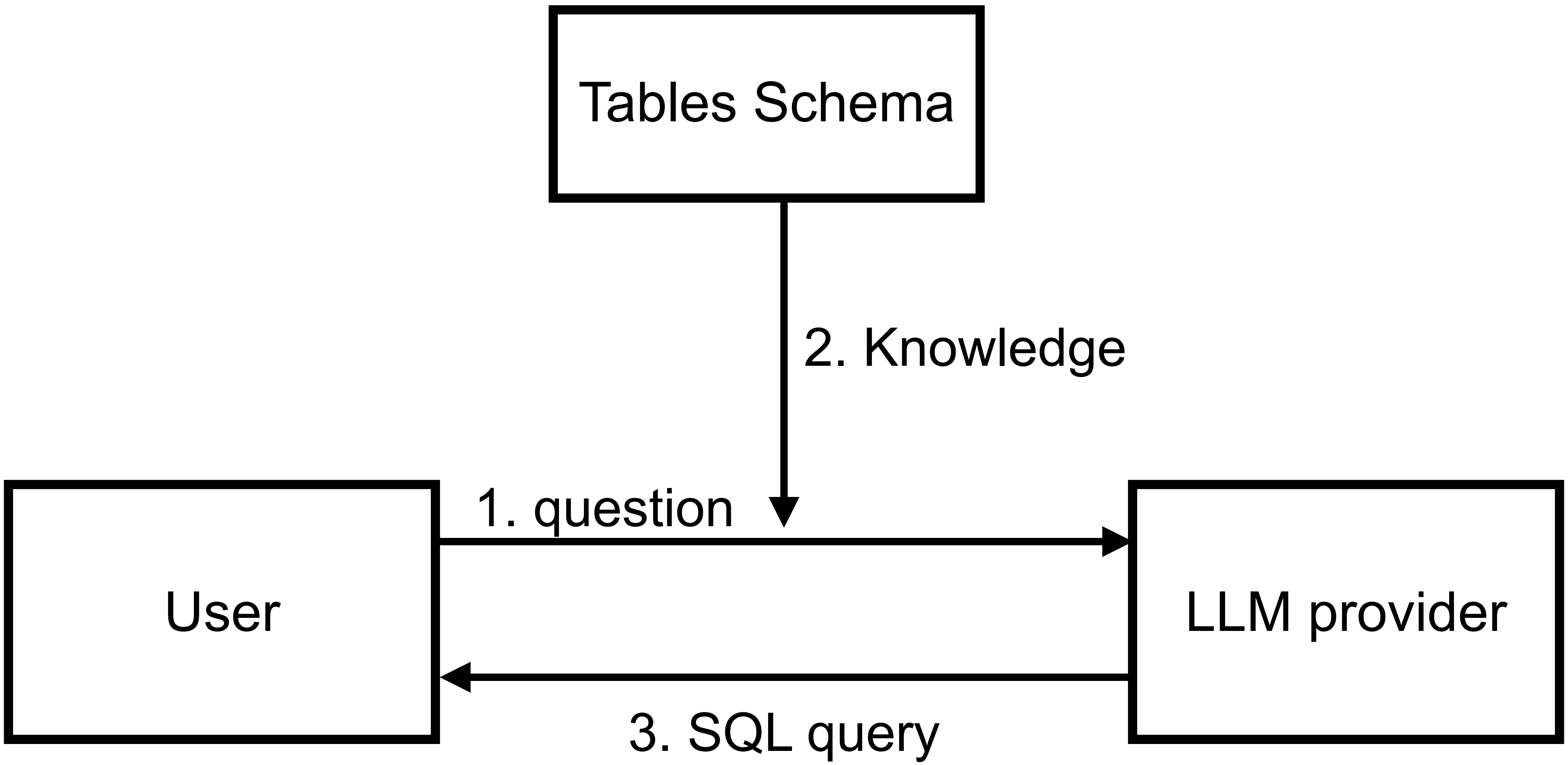
https://github.com/up1/workshop-ai-with-technical-team/tree/main/workshop/demo-rgb

# Visualize RGB




https://huggingface.co/spaces/jphwang/colorful_vectors

# Text-to-SQL

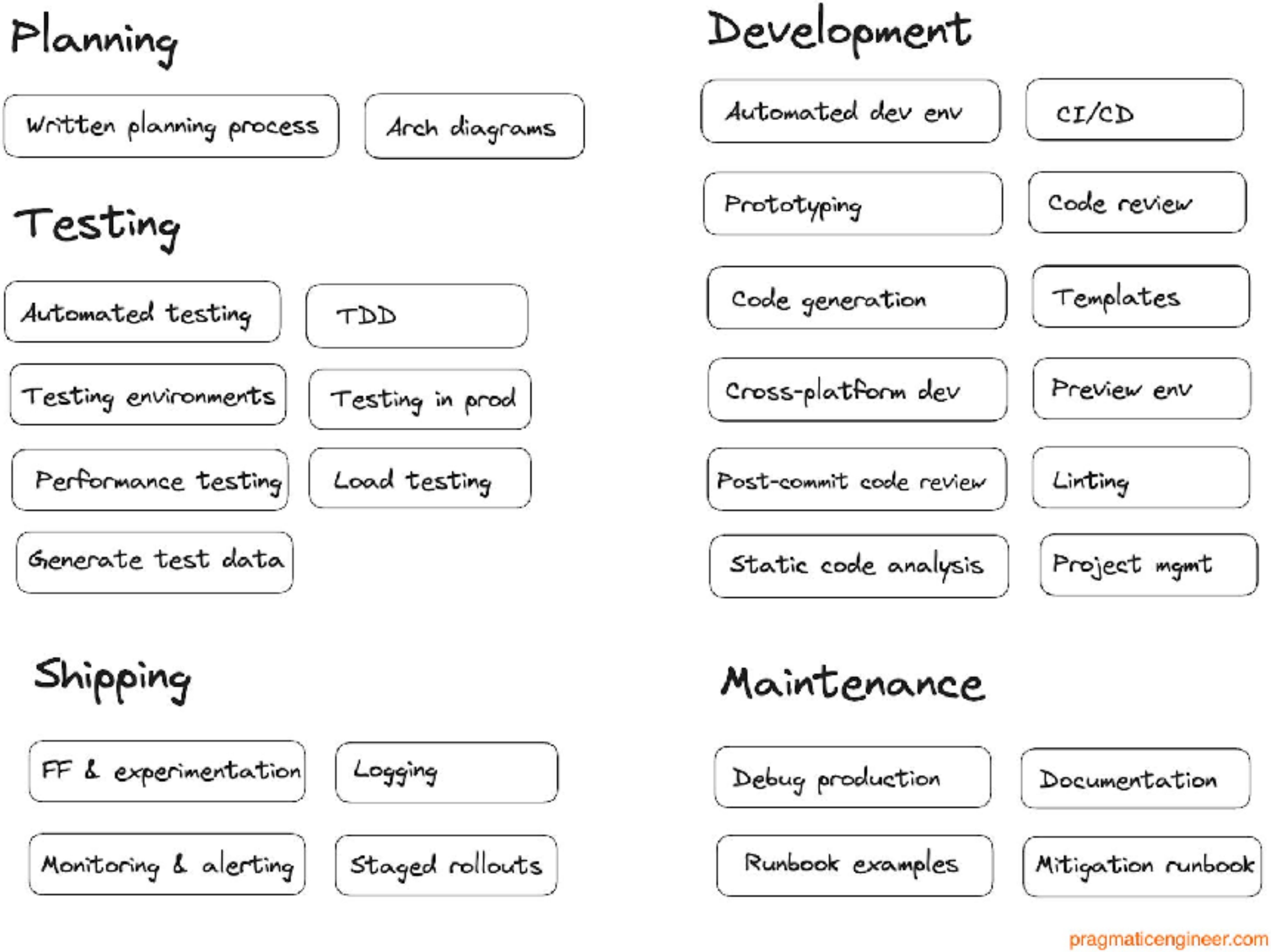



https://github.com/up1/workshop-ai-with-technical-team/tree/main/workshop/demo-sql

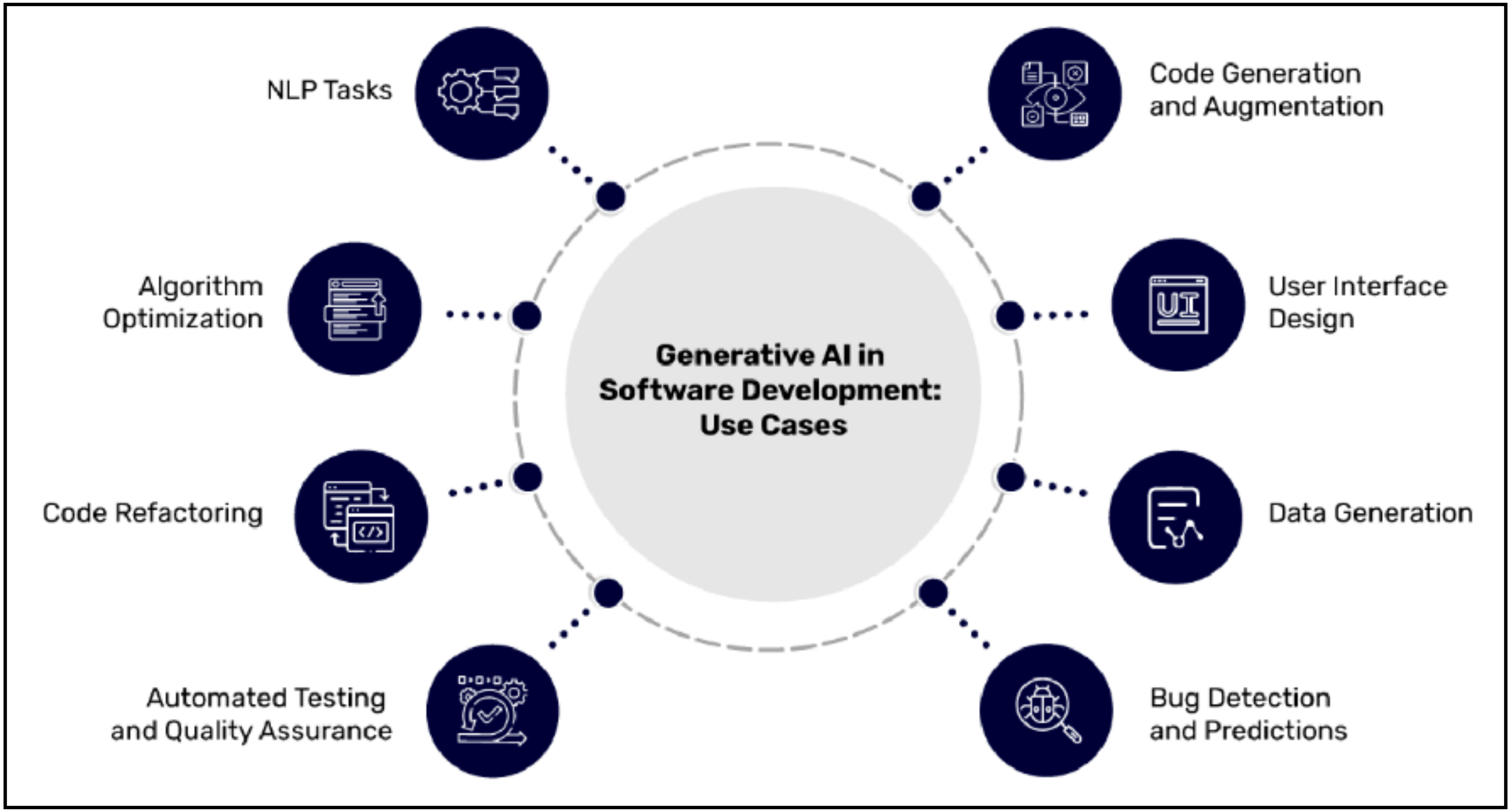
# Software Development Life Cycle

# SDLC

Requirement | Design | Develop | Testing | Deploy


Planning
Written planning process
Arch diagrams
Testing
Automated testing
TDD
Testing environments
Testing in prod
Performance testing
Load testing
Generate test data
Shipping
FF & experimentation
Logging
Monitoring & alerting
Staged rollouts
Development
Automated dev env
CI/CD
Prototyping
Code review
Code generation
Templates
Cross-platform dev
Preview env
Post-commit code review
Linting
Static code analysis
Project mgmt
Maintenance
Debug production
Documentation
Runbook examples
Mitigation runbook
pragmaticengineer.com

# SDLC


NLP Tasks
Code Generation and Augmentation
Algorithm Optimization
User Interface Design
Generative AI in Software Development: Use Cases
Code Refactoring
Data Generation
Automated Testing and Quality Assurance
Bug Detection and Predictions

# Impacts with productivity ?

Automated simple tasks

Improve quality and reliability

Improve communication

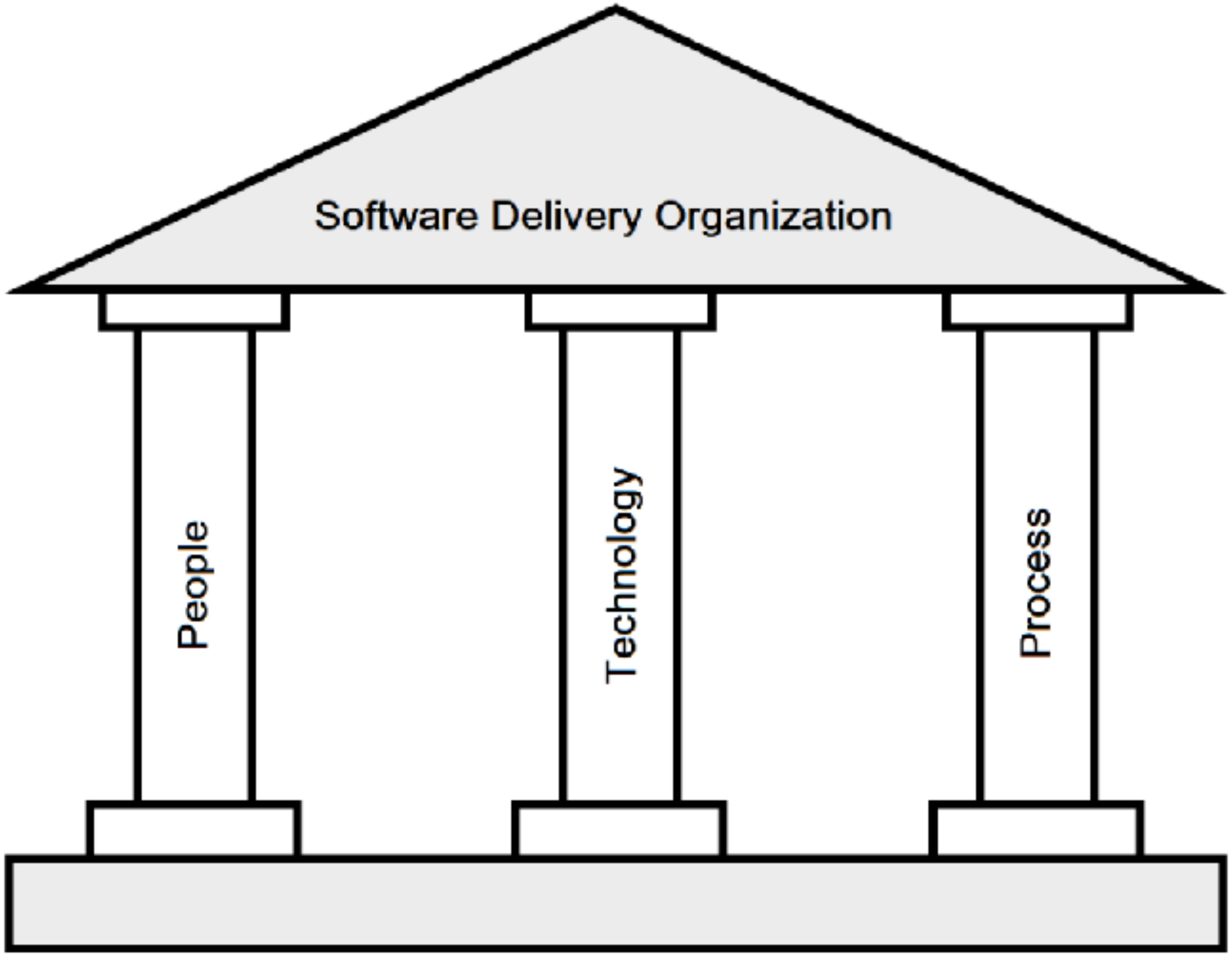
Faster prototype

**Generative AI** isn't just a tool

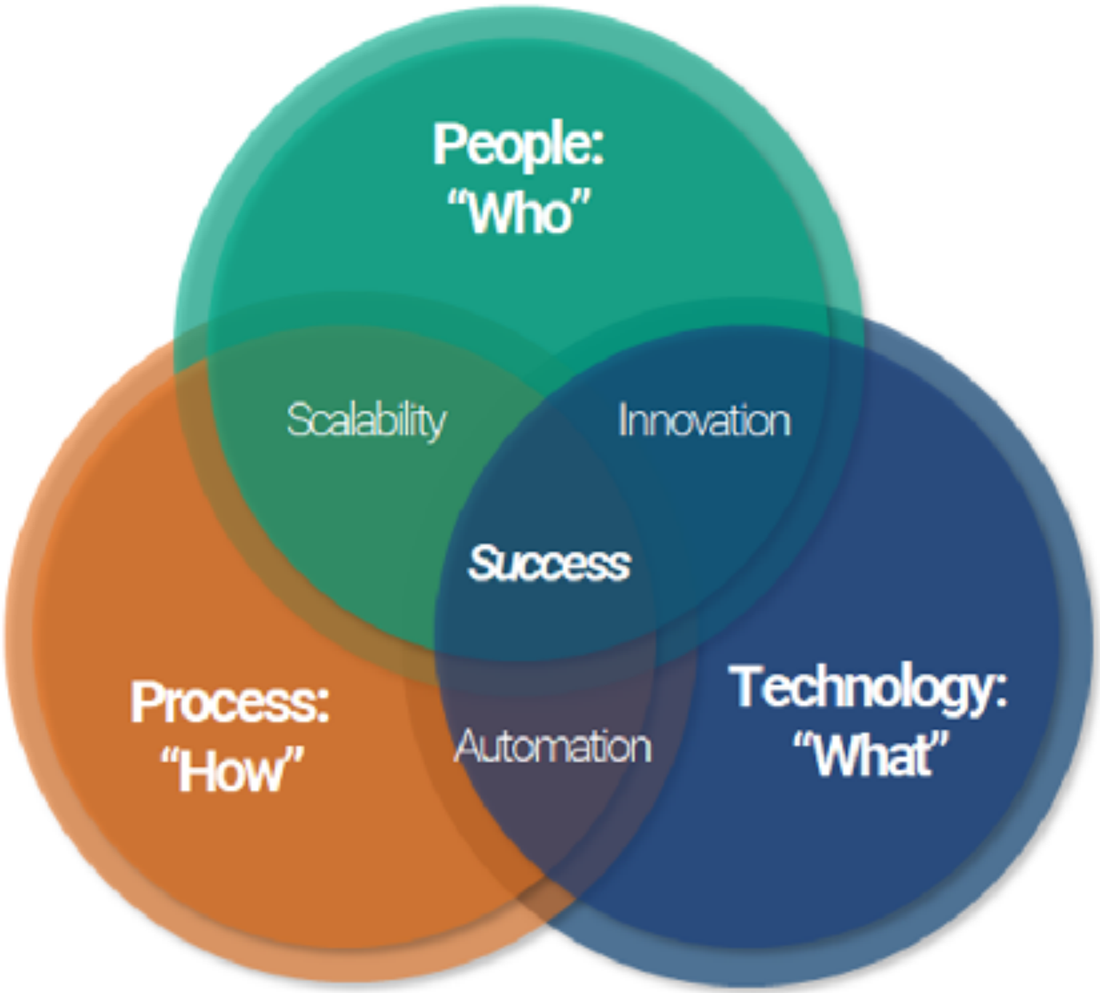
it's your **team member**

# Trust, but verify output !!

# 3 Pillar of Software Development



# 3 Pillar of Software Development



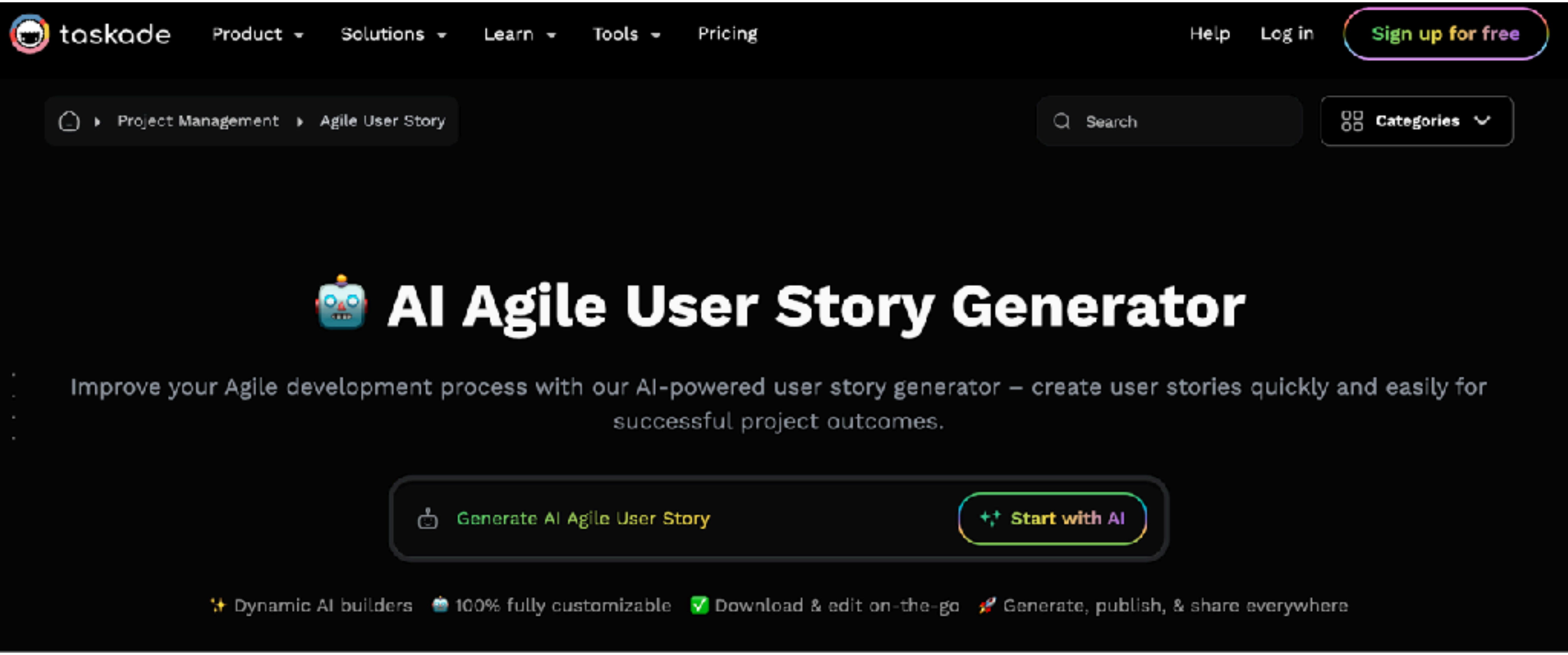
# Requirement and Analysis

# Requirement and Analysis

| | | | | |
|---|---|---|---|---|
| Requirement | Design | Develop | Testing | Deploy |

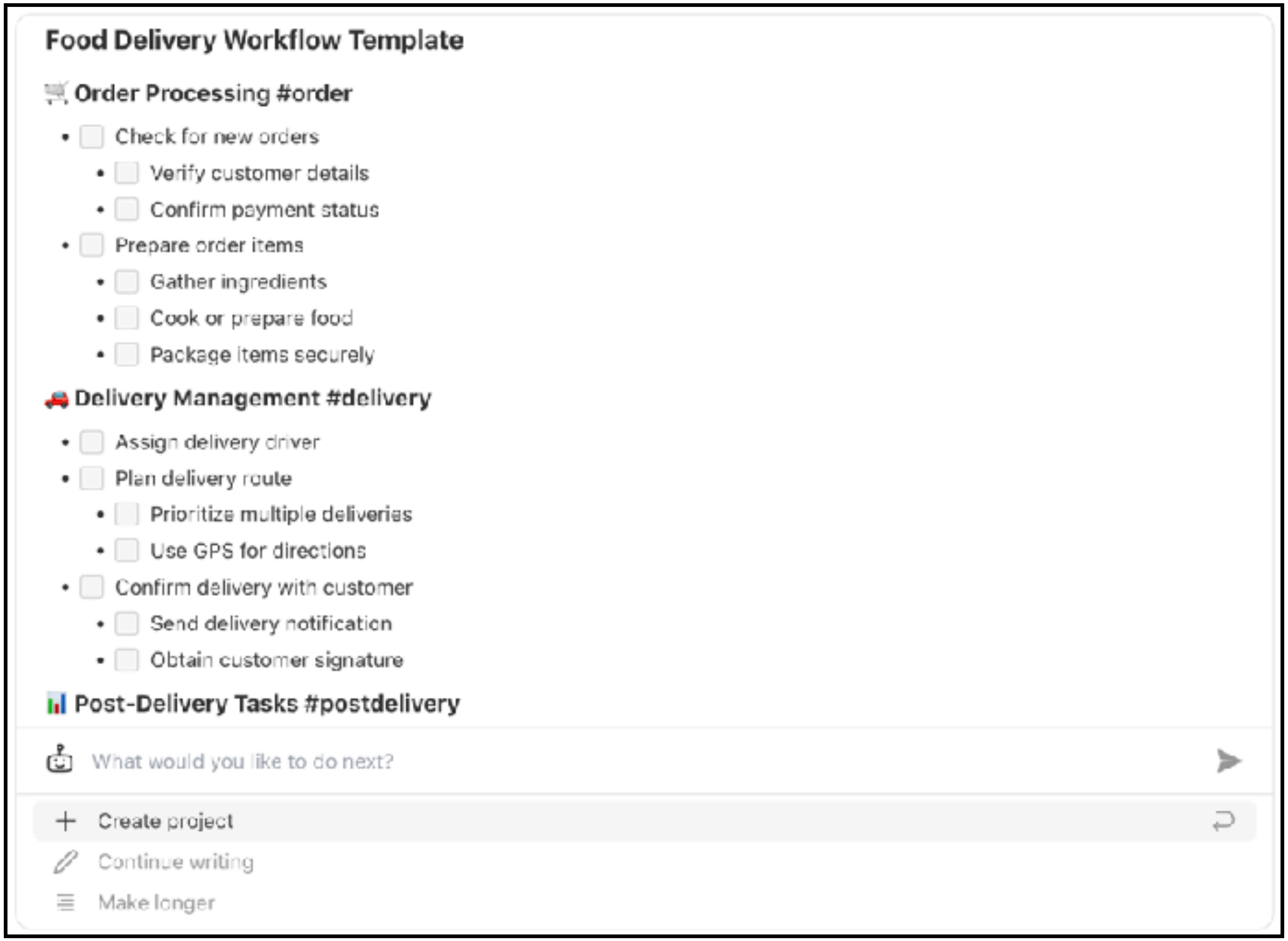
Requirements writing and analysis
User story generation

# Agents AI for Automated tasks




https://www.taskade.com/generate/project-management/agile-user-story

# Example with food delivery



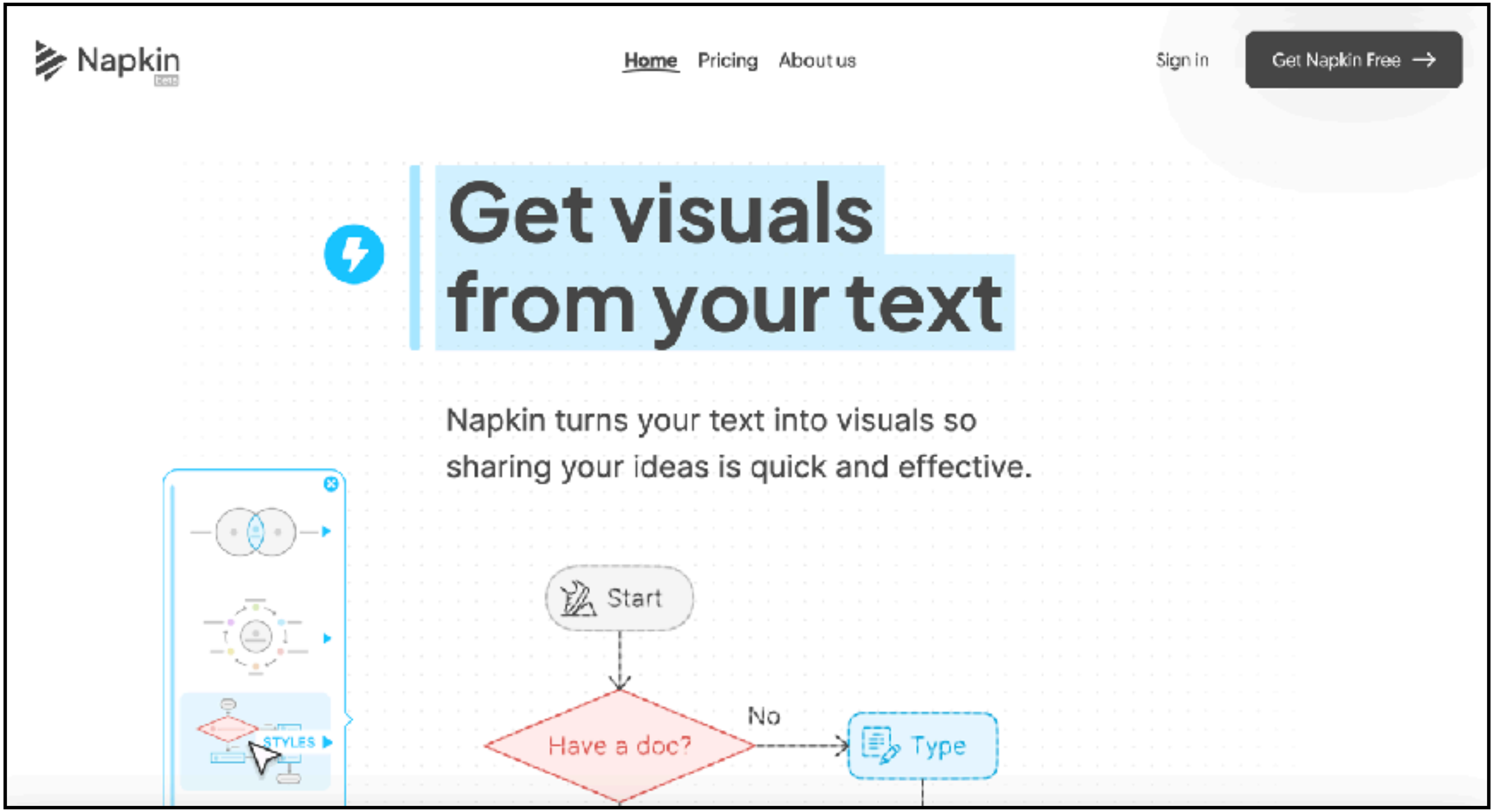

https://www.taskade.com/generate/project-management/agile-user-story

# User Story Generator

Idea → Features → Roles → User stories


https://userstorygenerator.ai/

# Napkin



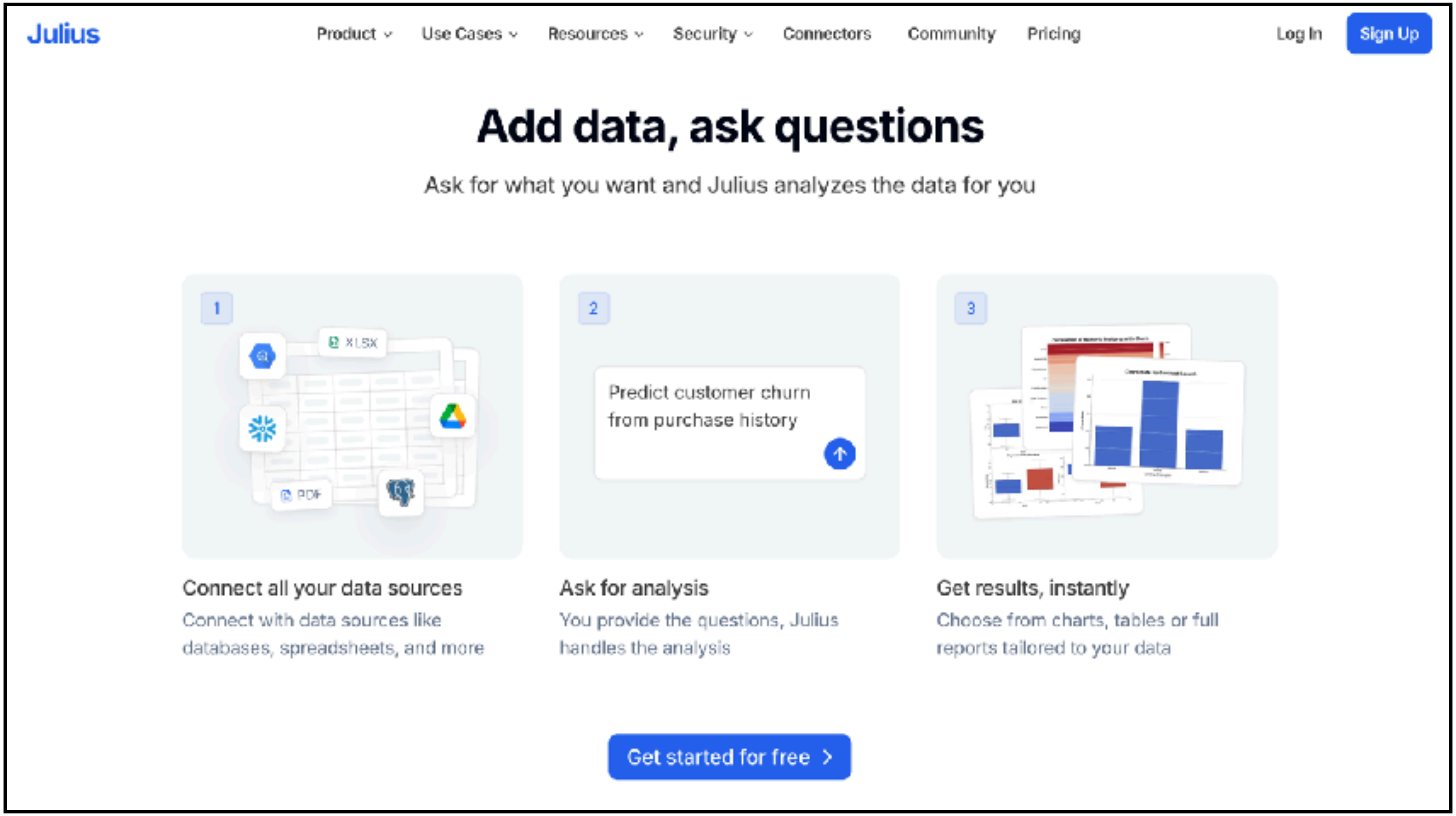

https://www.napkin.ai/

# Requirement analysis

Clarify of User requirement ?

https://github.com/up1/workshop-ai-with-technical-team/wiki/Requirement-analysis

# Data Analysis




https://julius.ai/

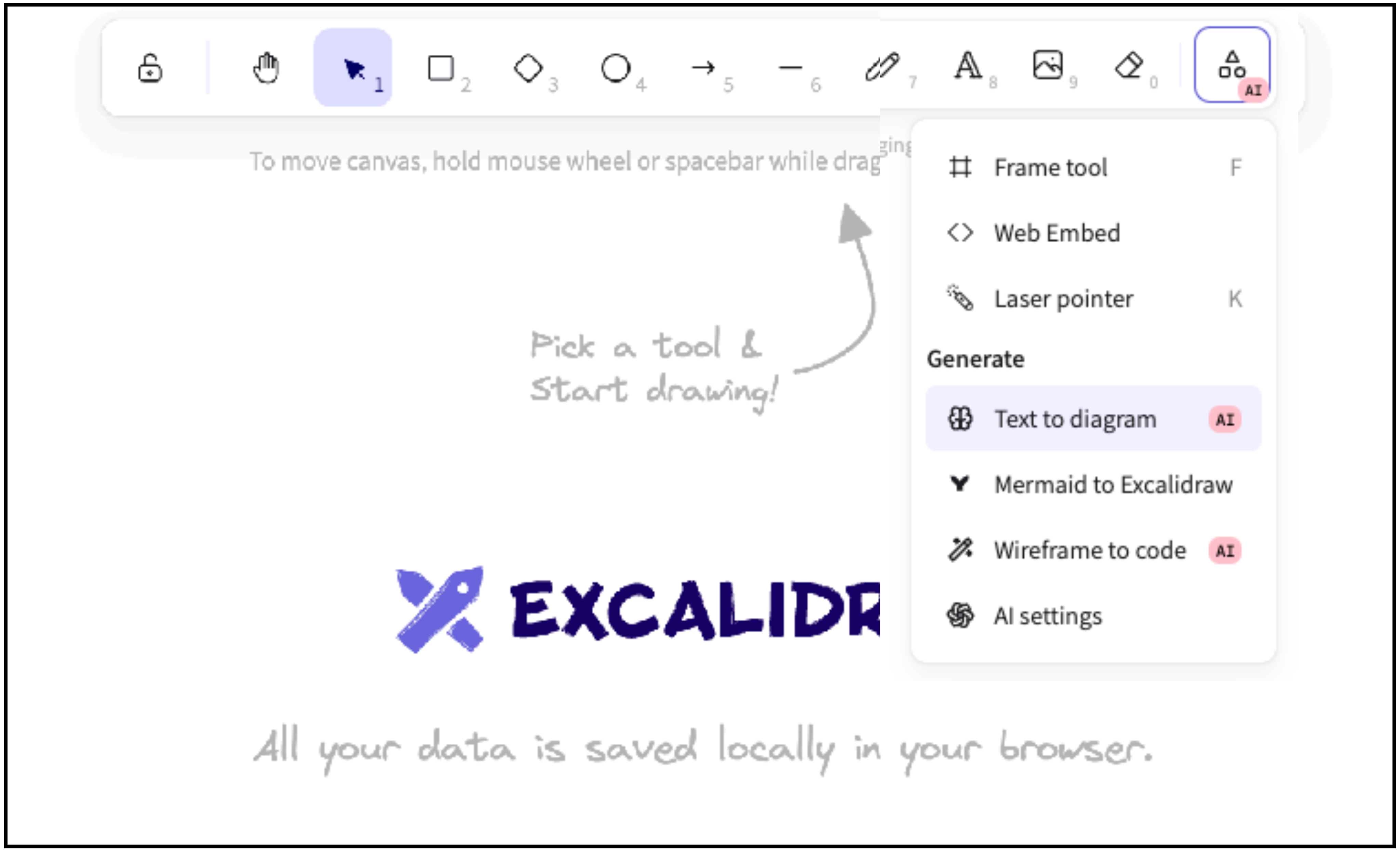
# Design Process

# Design

| Requirement | Design | Develop | Testing | Deploy |
| --- | --- | --- | --- | --- |

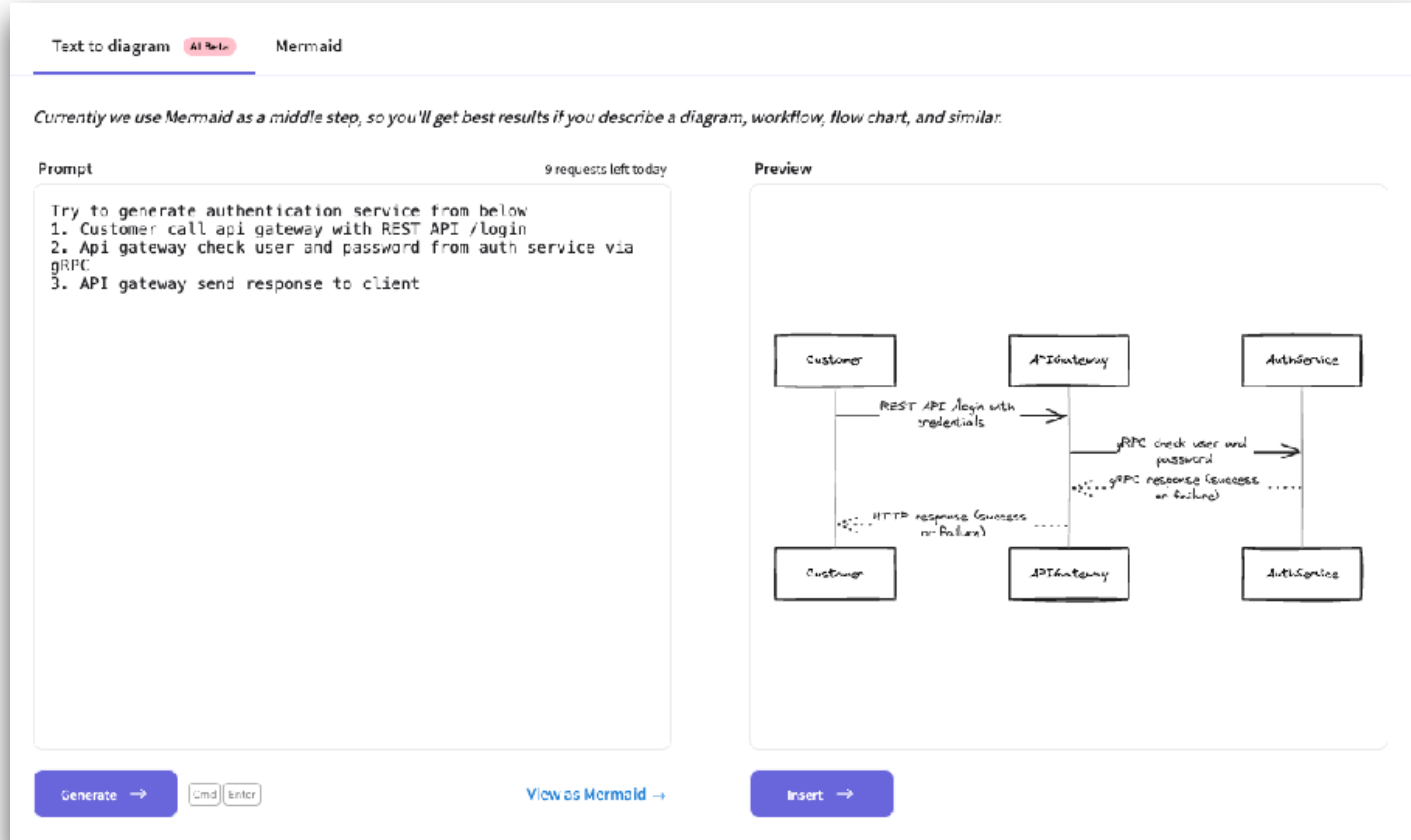
Architecture writing assistance
Sequence flow diagram generation
Data modeling
UX/UI design assistance

# Excalidraw with AI

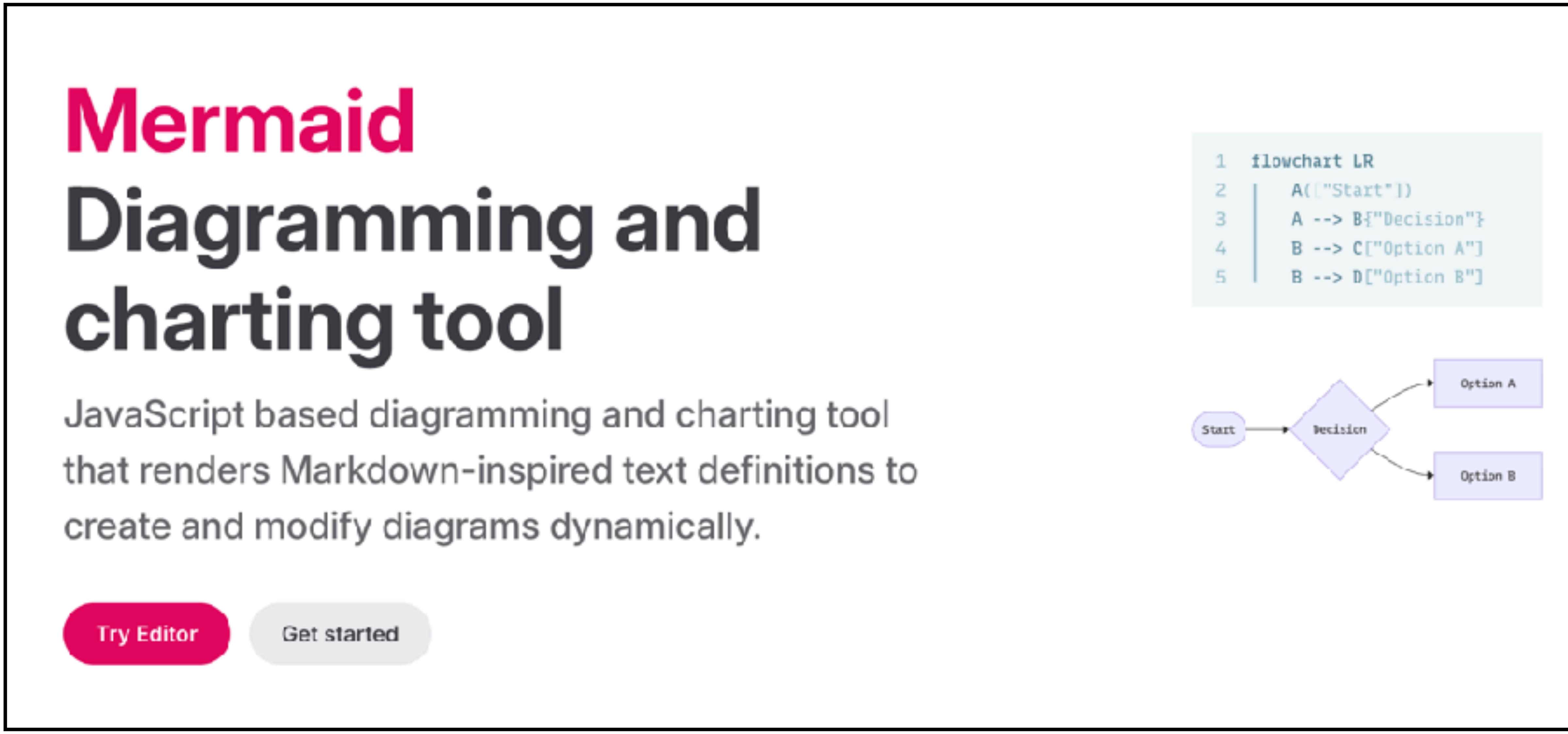



https://excalidraw.com/

# Demo

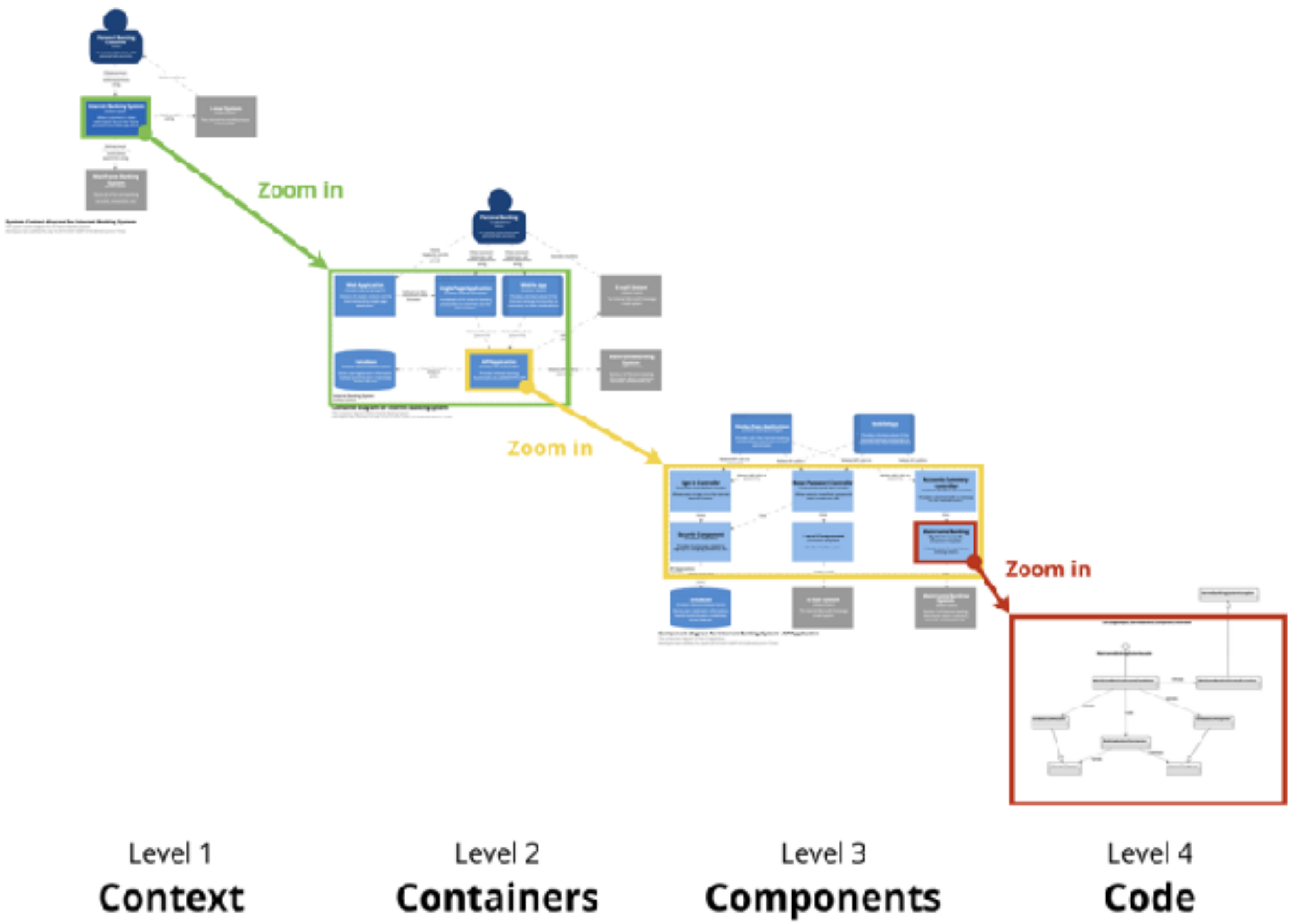

Text to diagram
AI Beta
Mermaid
Currently we use Mermaid as a middle step, so you'll get best results if you describe a diagram, workflow, flow chart, and similar.
Prompt
9 requests left today
Try to generate authentication service from below
1. Customer call api gateway with REST API /login
2. Api gateway check user and password from auth service via gRPC
3. API gateway send response to client
Preview
Customer
APIGateway
AuthService
REST API /login with credentials
gRPC check user and password
gRPC response (success or failure)
HTTP response (success or failure)
Generate
Cmd
Enter
View as Mermaid
Insert

# Mermaid Diagram



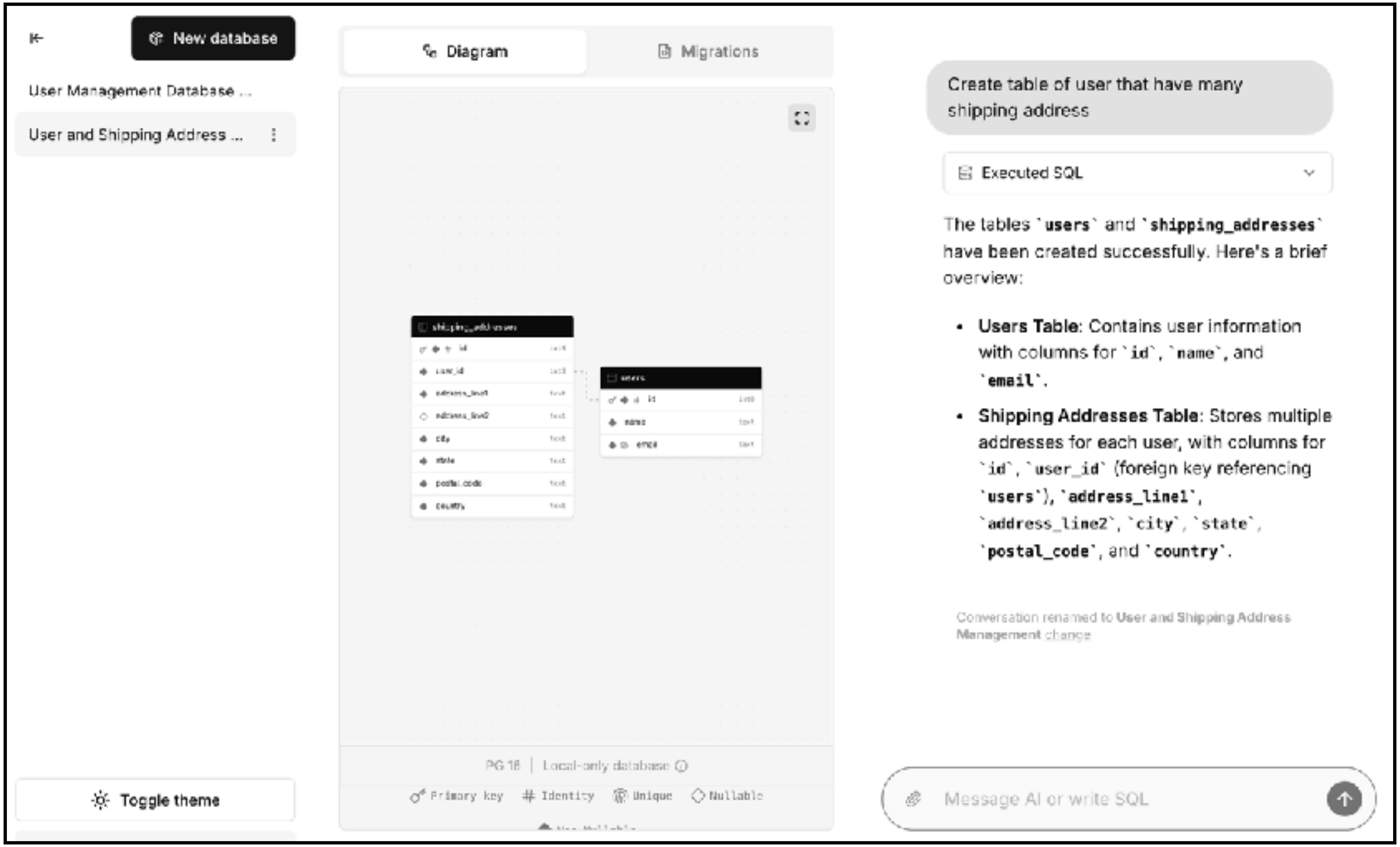

https://mermaid.js.org/

# Software Architecture



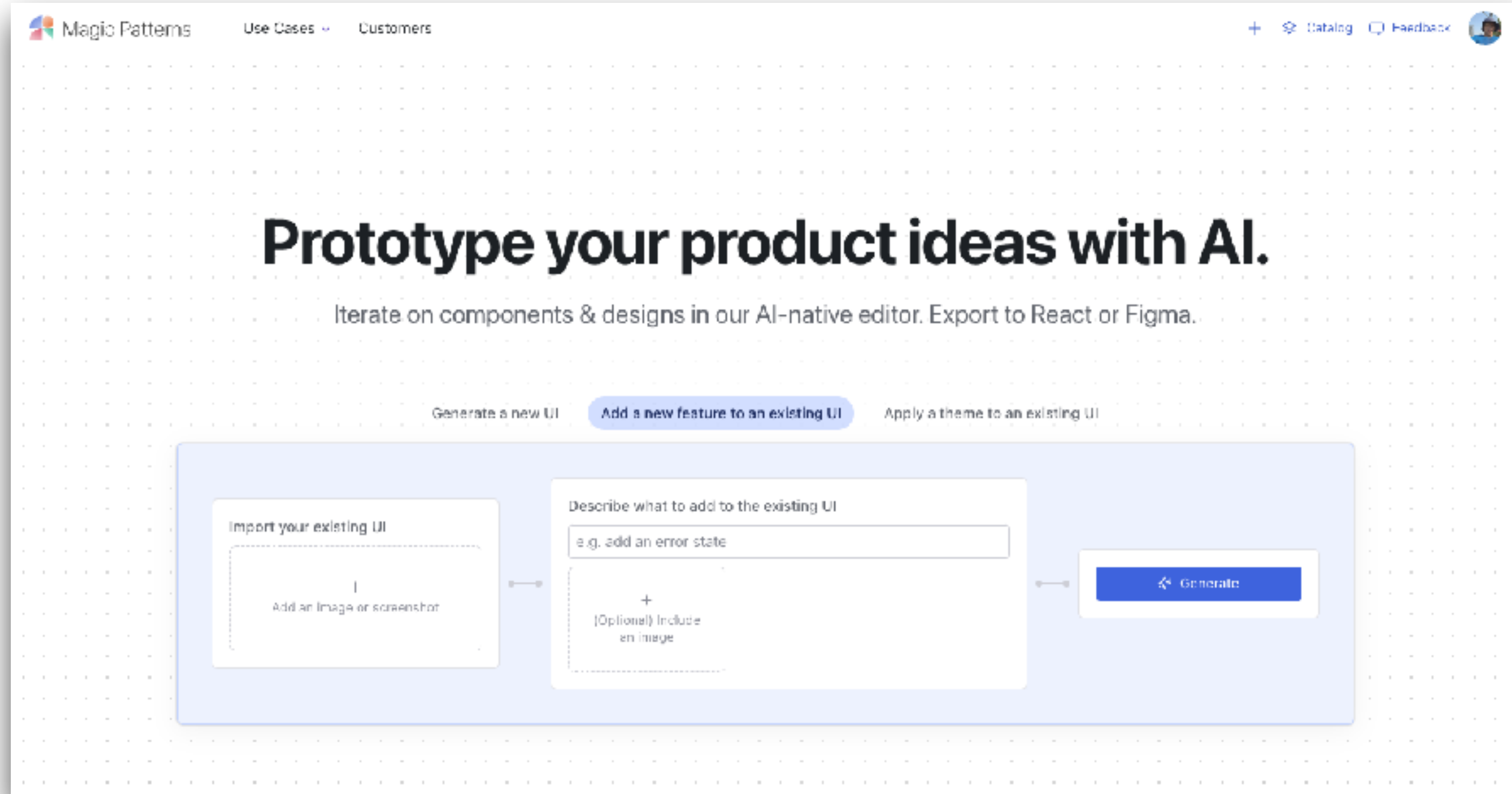

https://c4model.com/

# Database



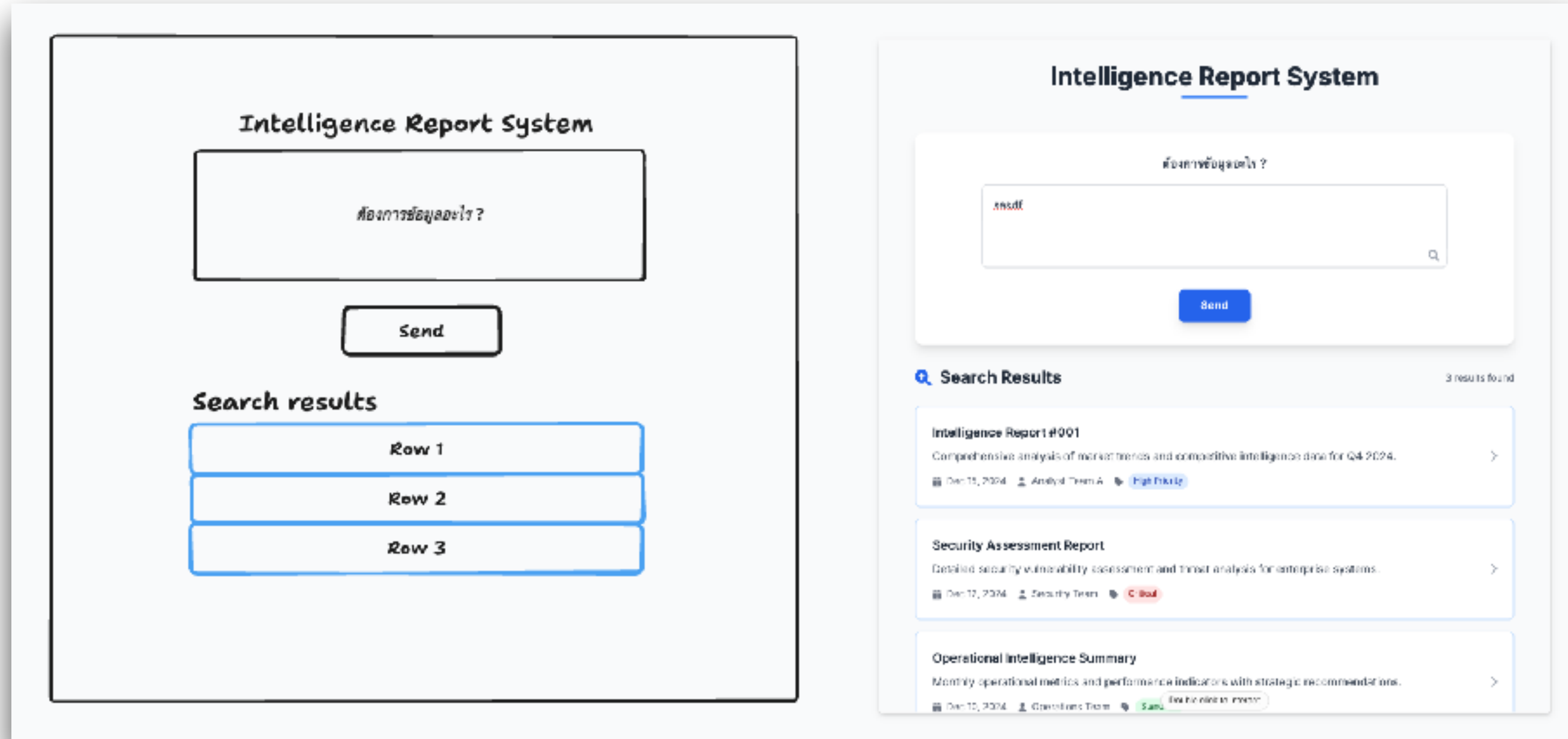

https://database.build/

# Magic Pattern



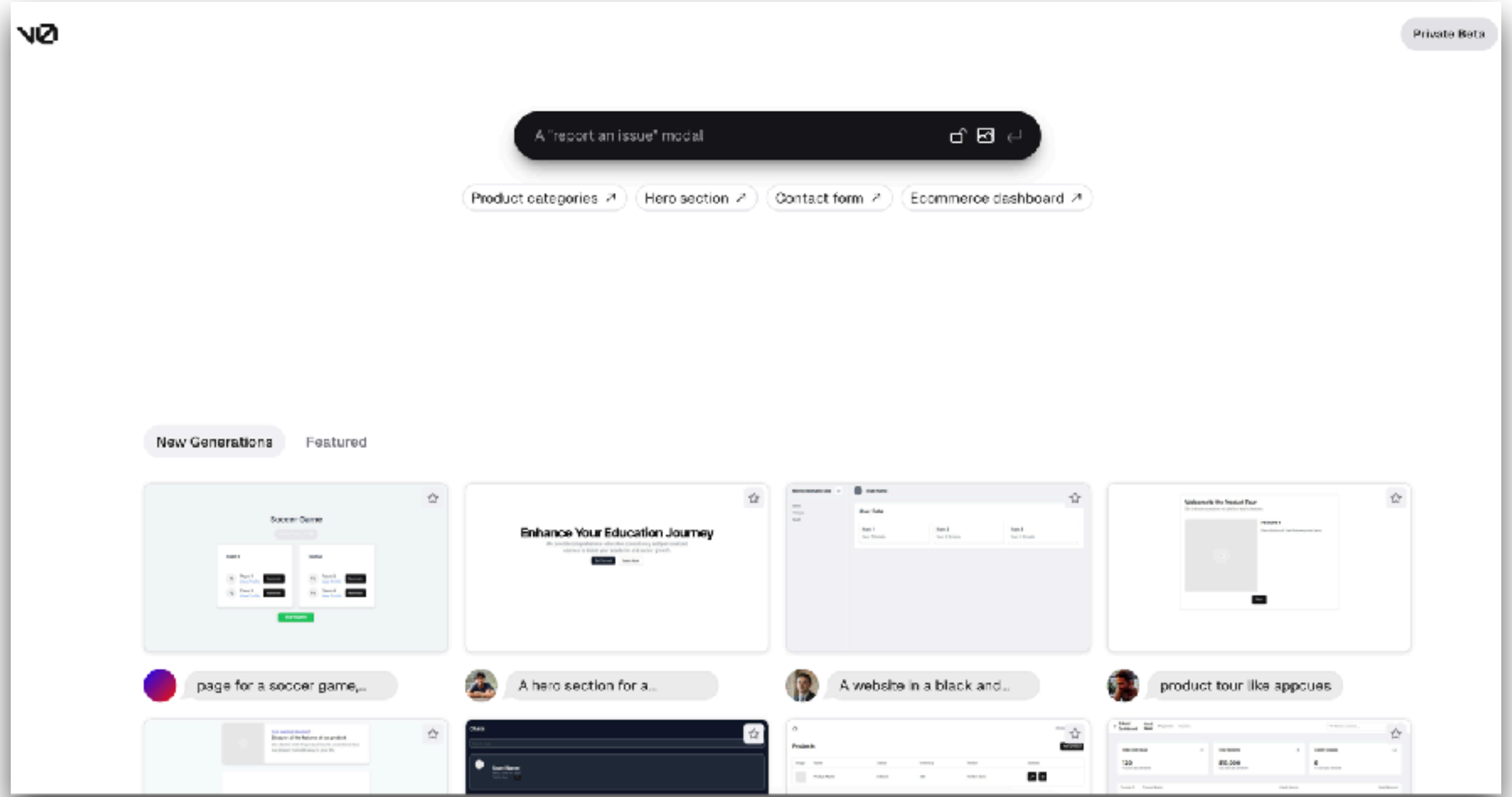

https://www.magicpatterns.com/

# Make Real



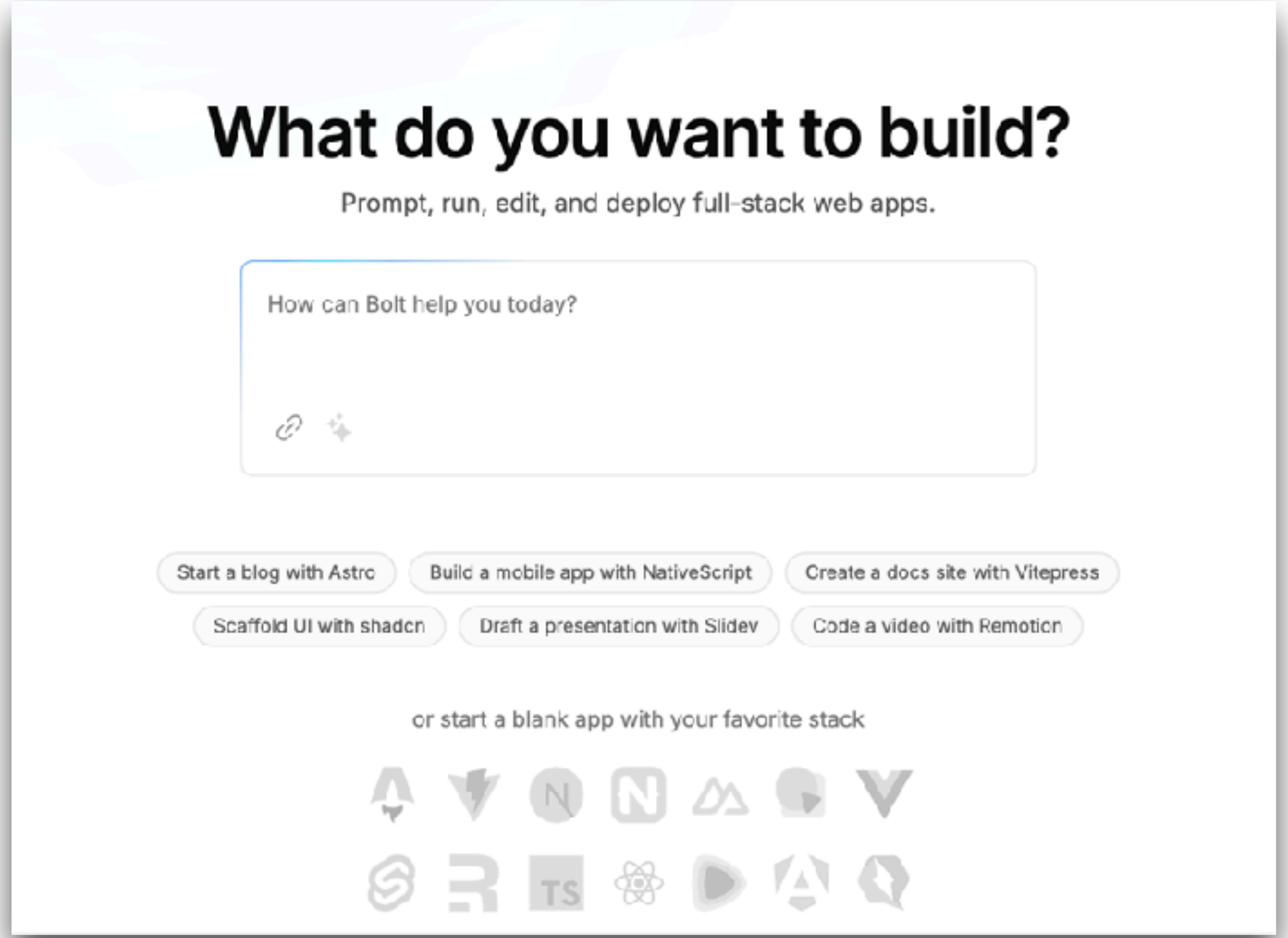

https://github.com/tldraw/make-real

# V0.dev



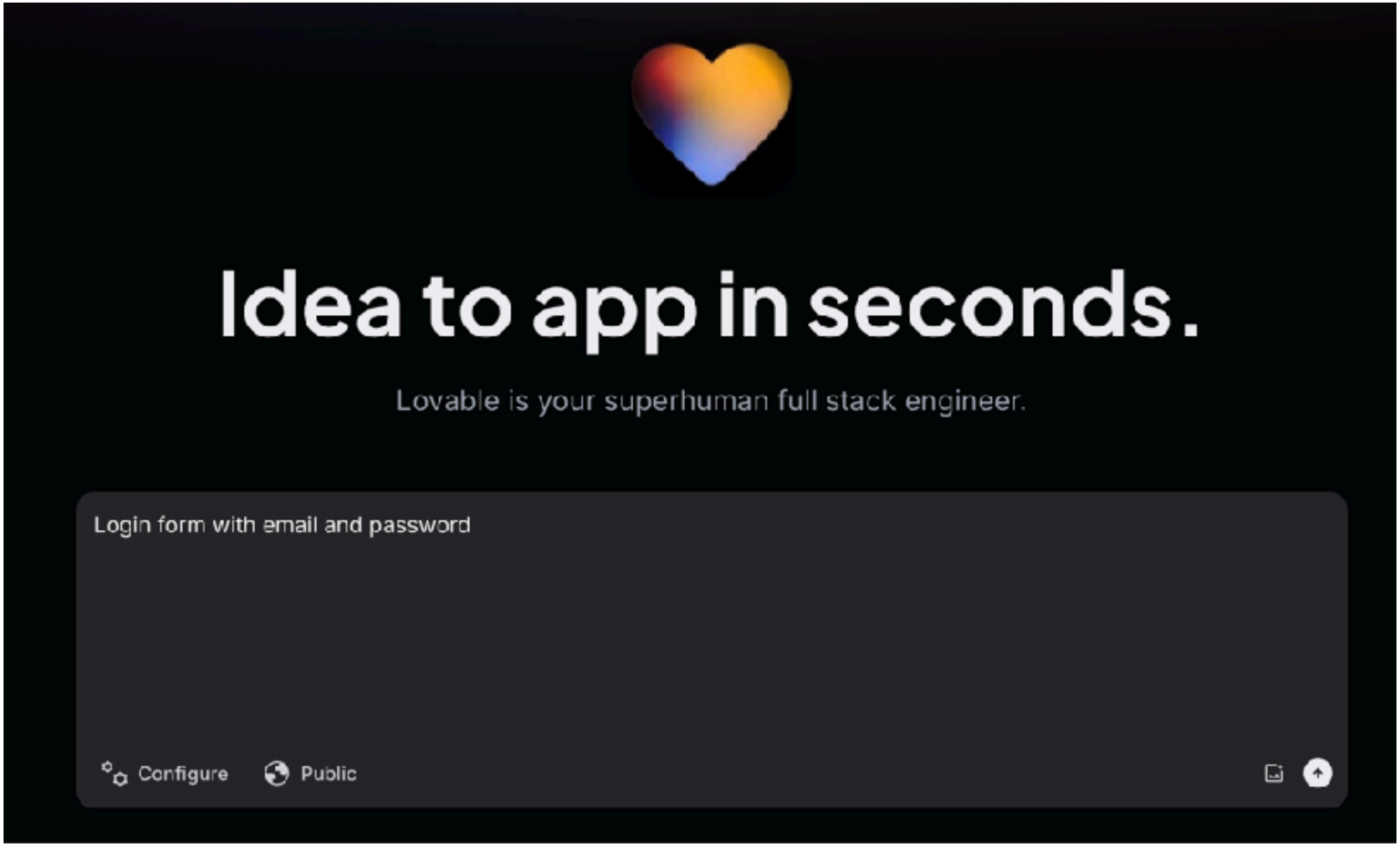

https://v0.dev/

# Bolt.new



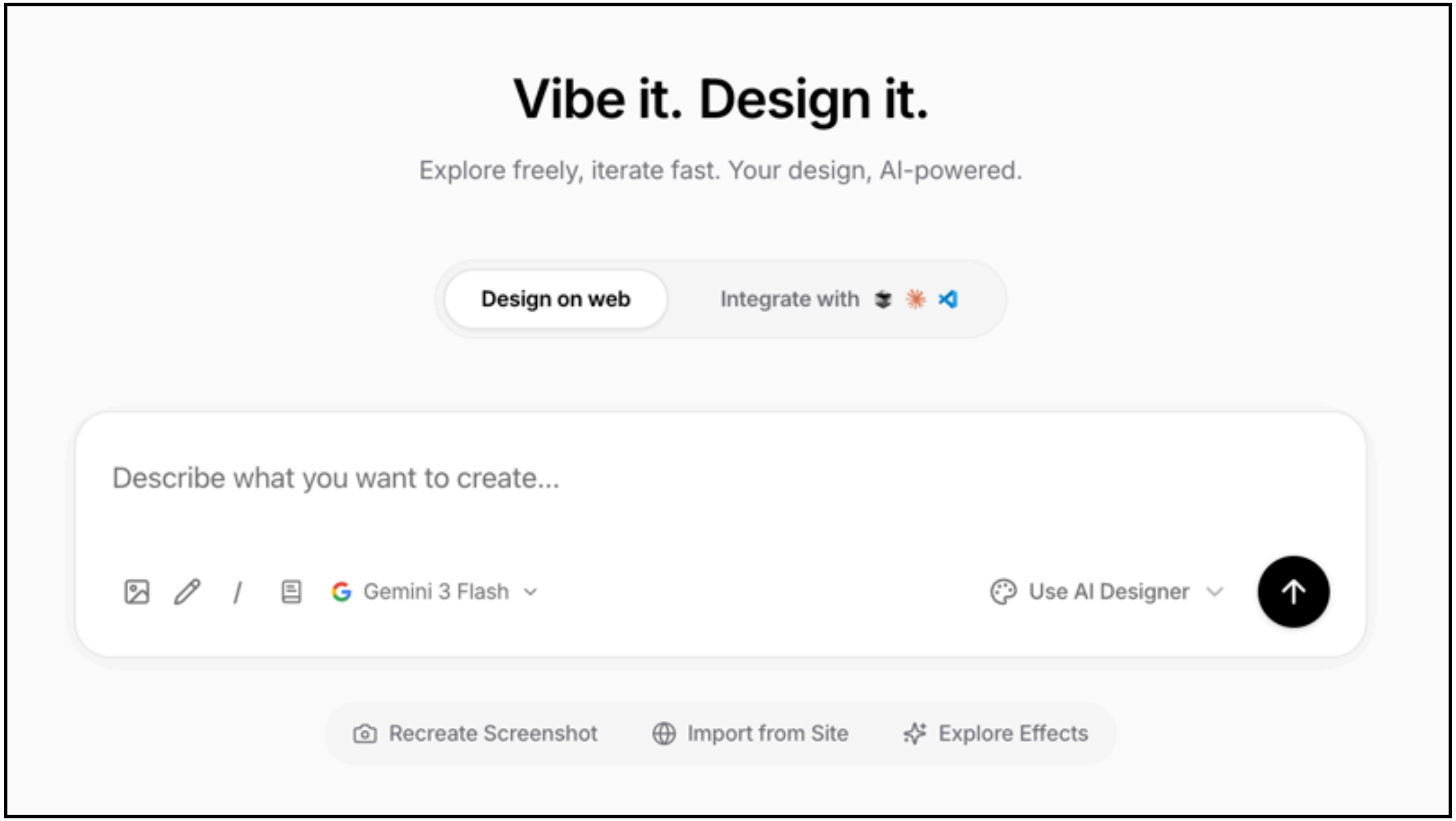

https://bolt.new/

# Lovable




https://lovable.dev/

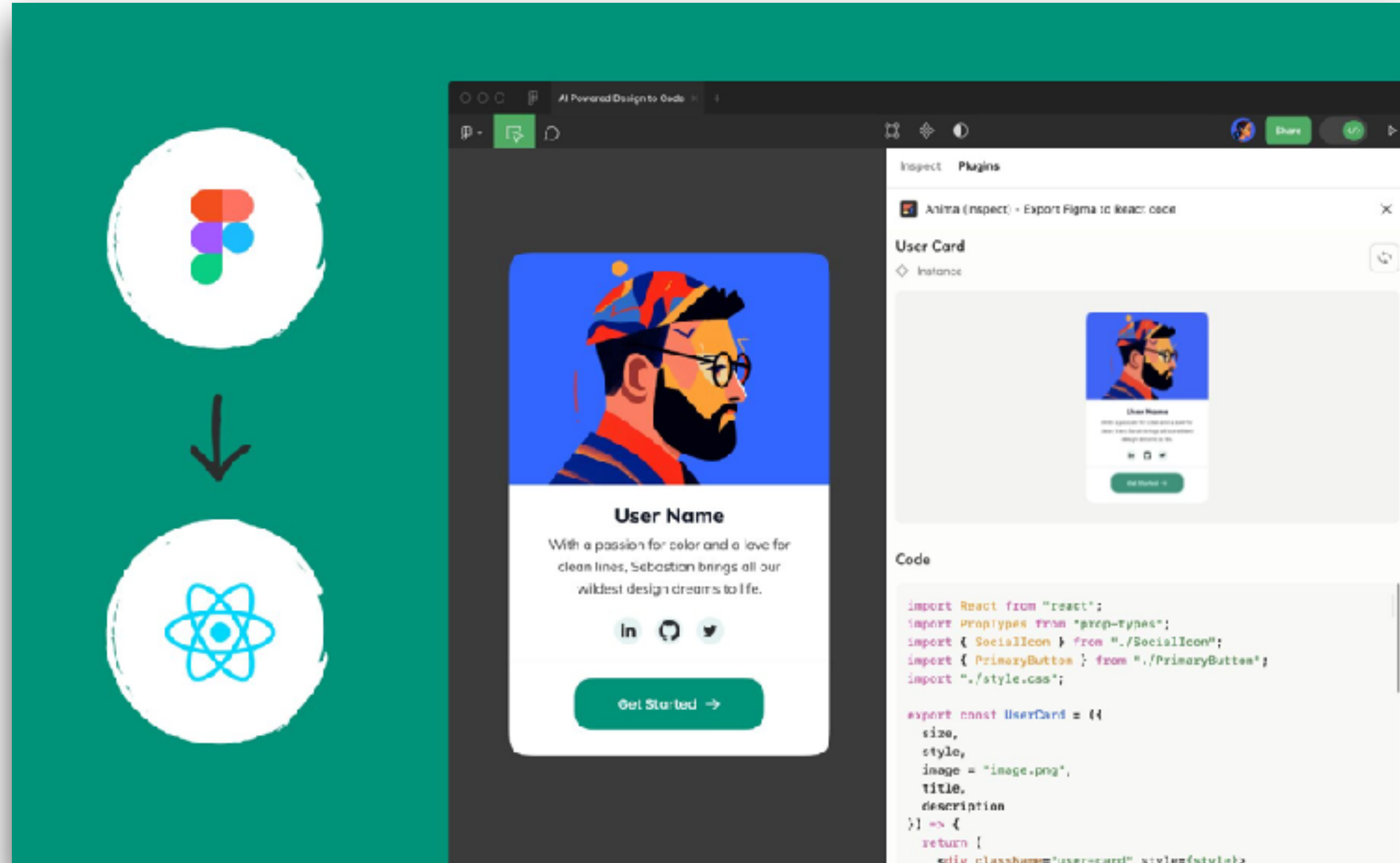
# SuperDesign




https://app.superdesign.dev/

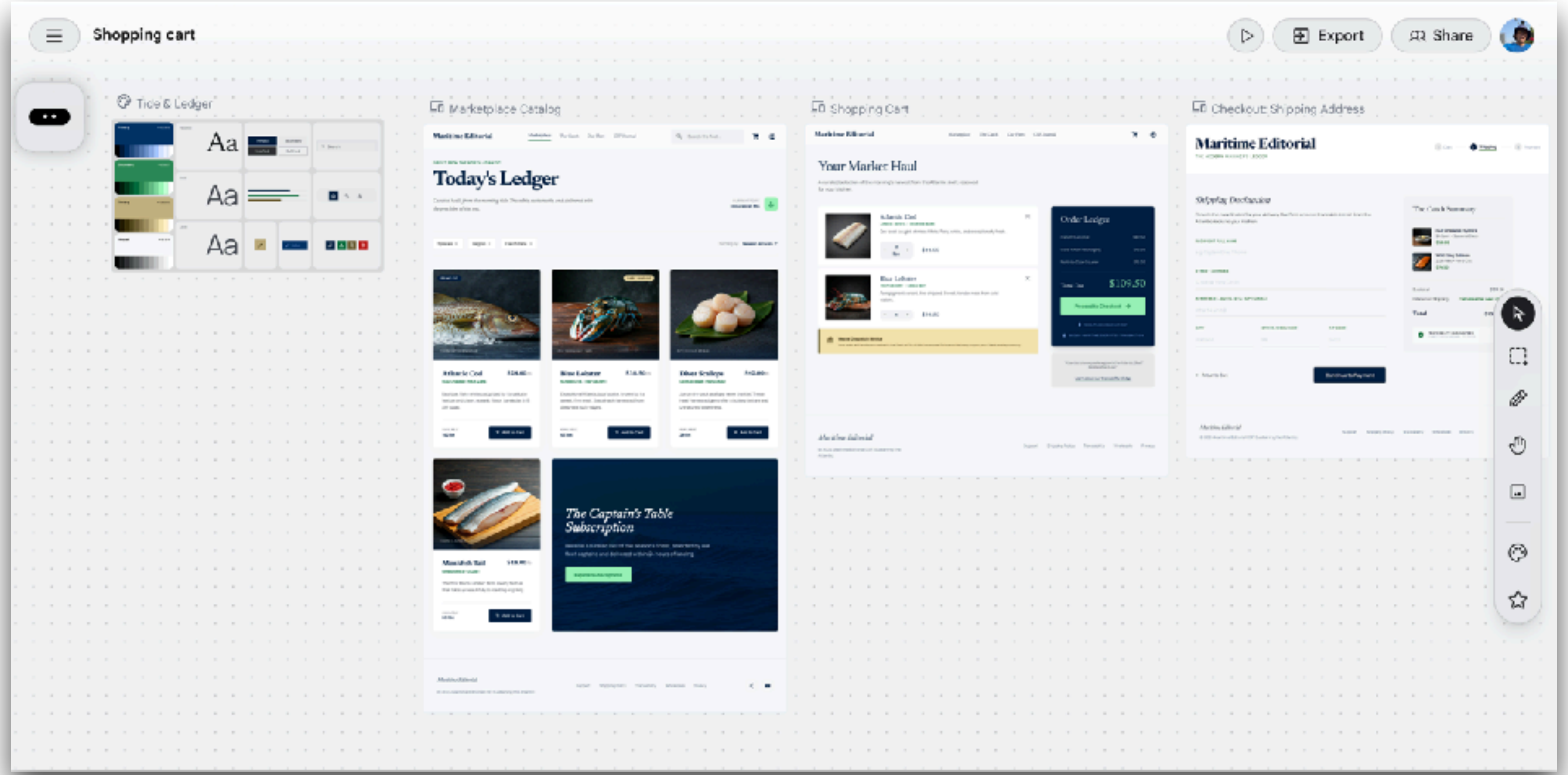
# Design to Code

# Figma




https://www.figma.com/mcp-catalog/

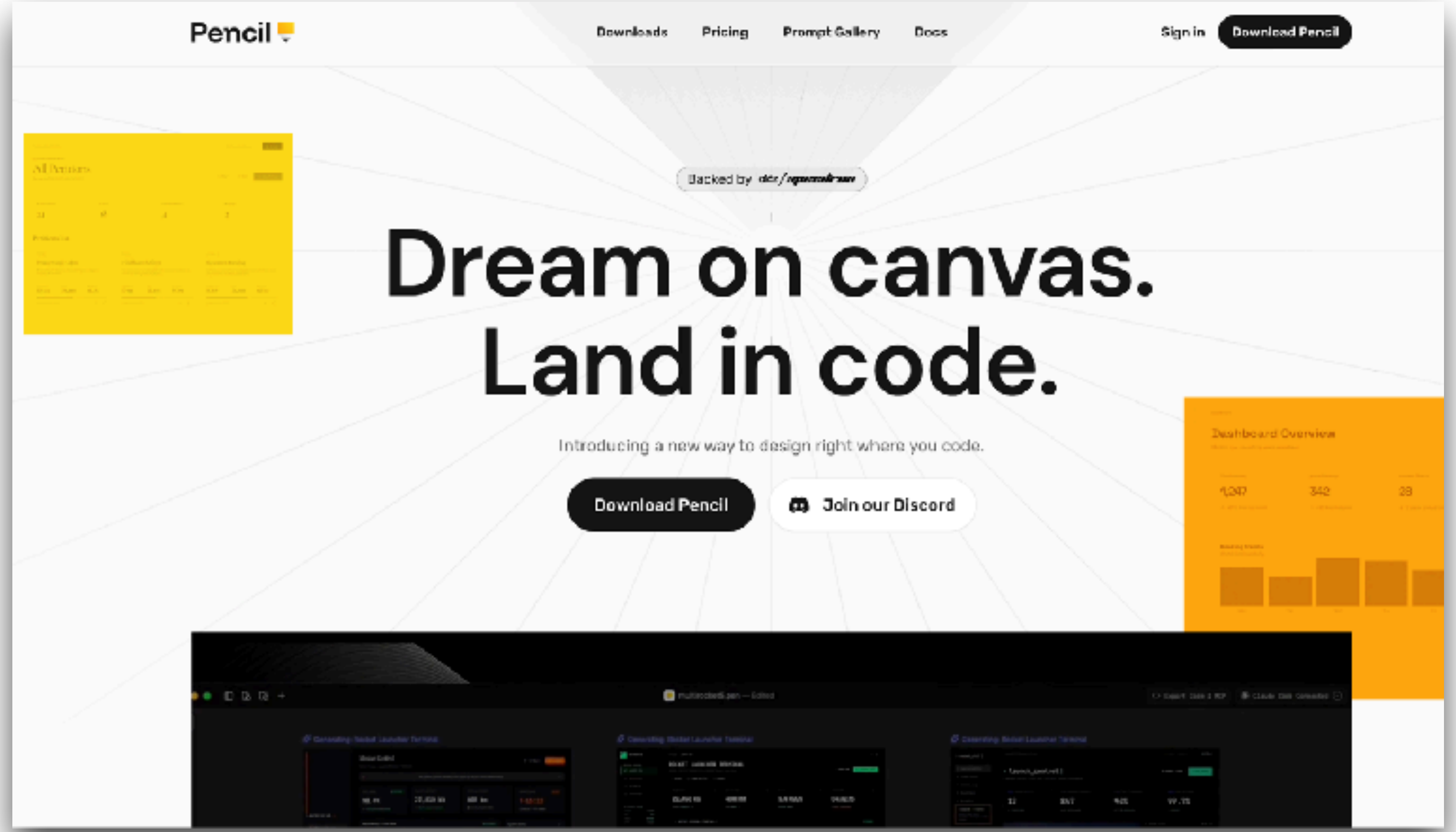
# Google Stitch




https://stitch.withgoogle.com/

# Pencil

Bridge the gap between design and development




https://docs.pencil.dev/getting-started/ai-integration

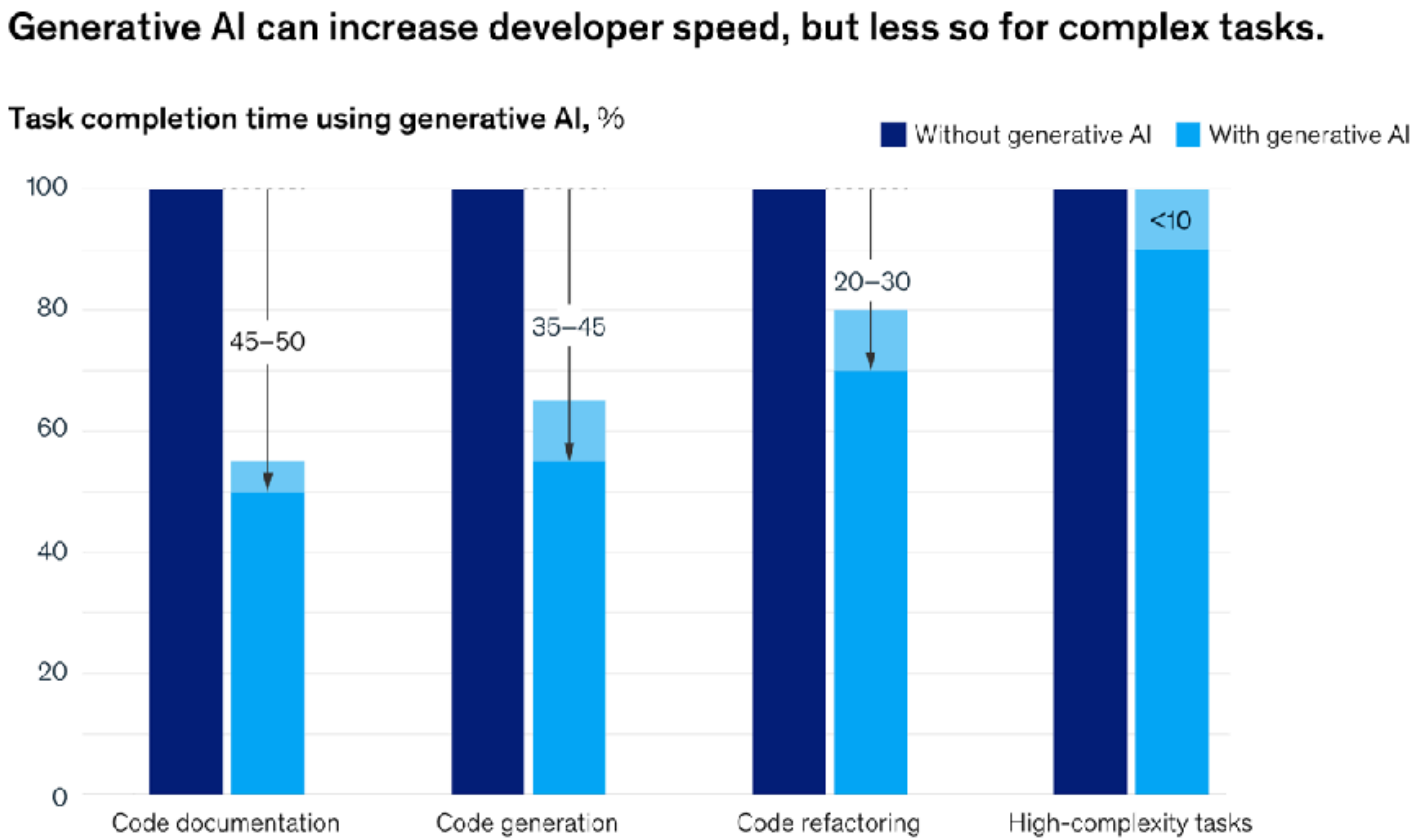
# Development Process

# Develop

| Requirement | Design | Develop | Testing | Deploy |
| --- | --- | --- | --- | --- |

Code generation
Review and explain code
Debugging code
Improve consistency
Code translation

# Development

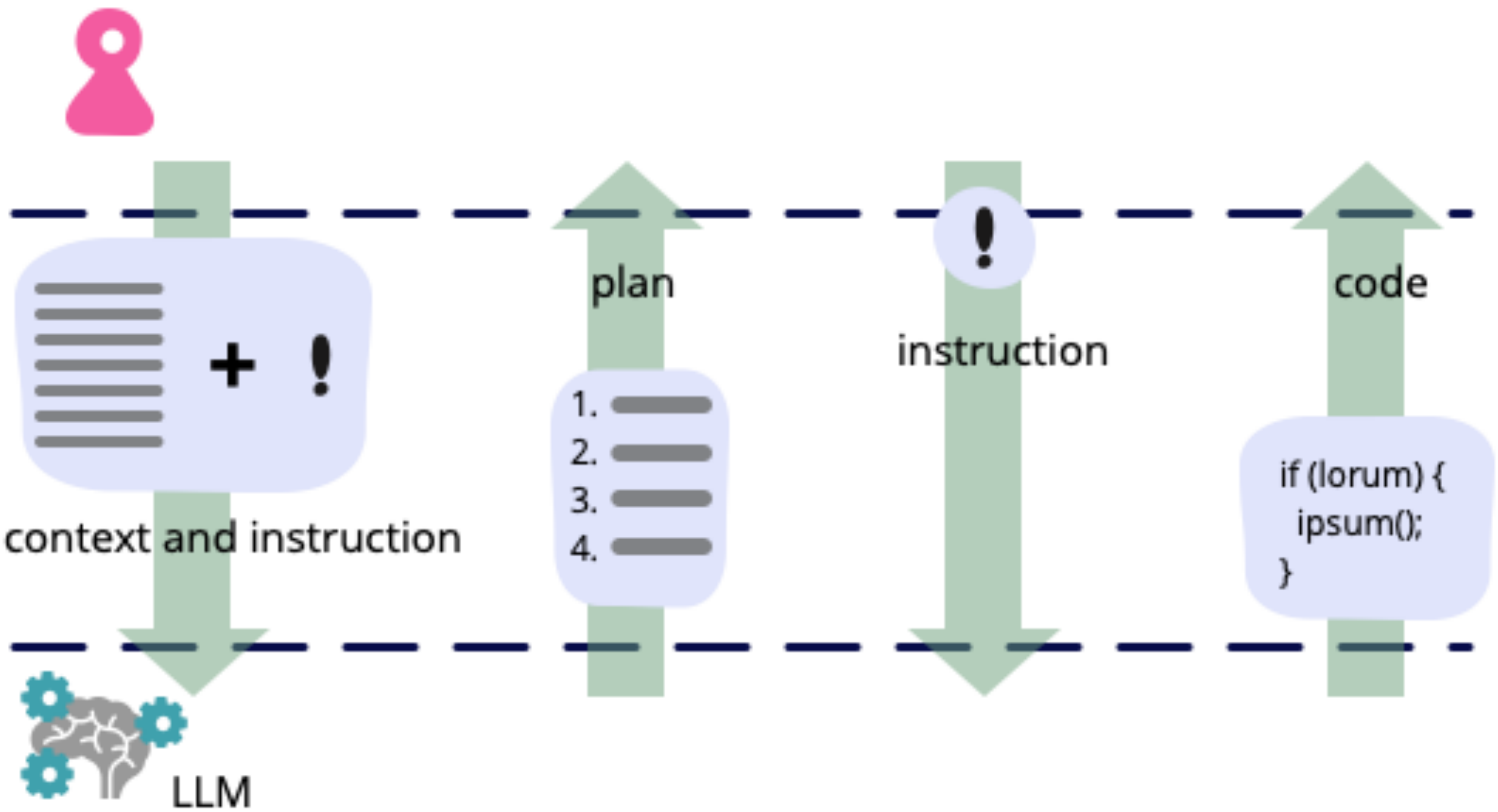



https://www.mckinsey.com/capabilities/mckinsey-digital/our-insights/unleashing-developer-productivity-with-generative-ai

# Category of Tools

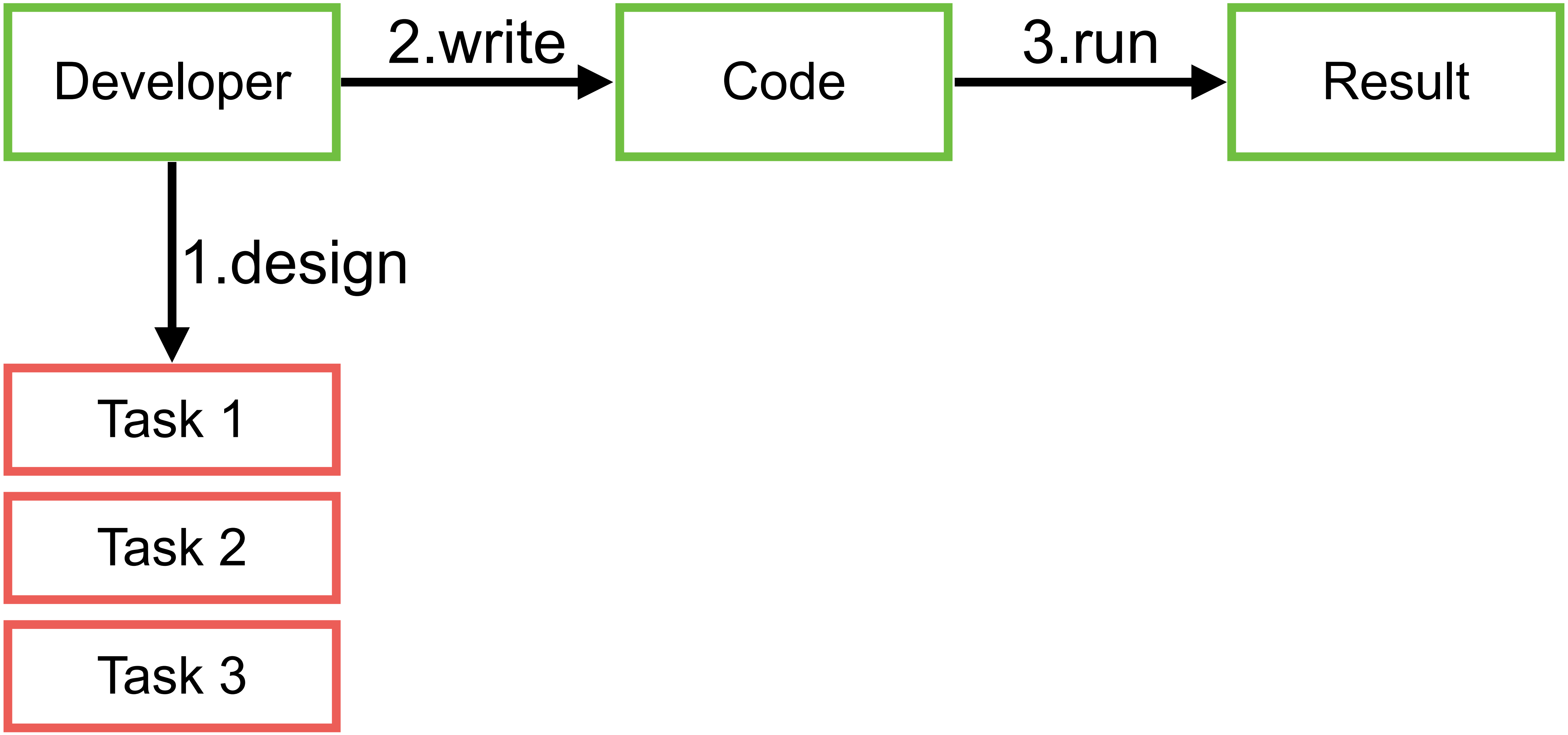
| Chat AI | Code AI | AI Agent<br>Public and Local |
|---|---|---|
| ChatGPT | Copilot in VSCode | Aider |
| Gemini | Cursor IDE | Claude code |
| Claude.ai | Windsurf | Gemini CLI |
| DeepSeek | Zed | Codex CLI |
| | | Qwen3-coder |

# Development

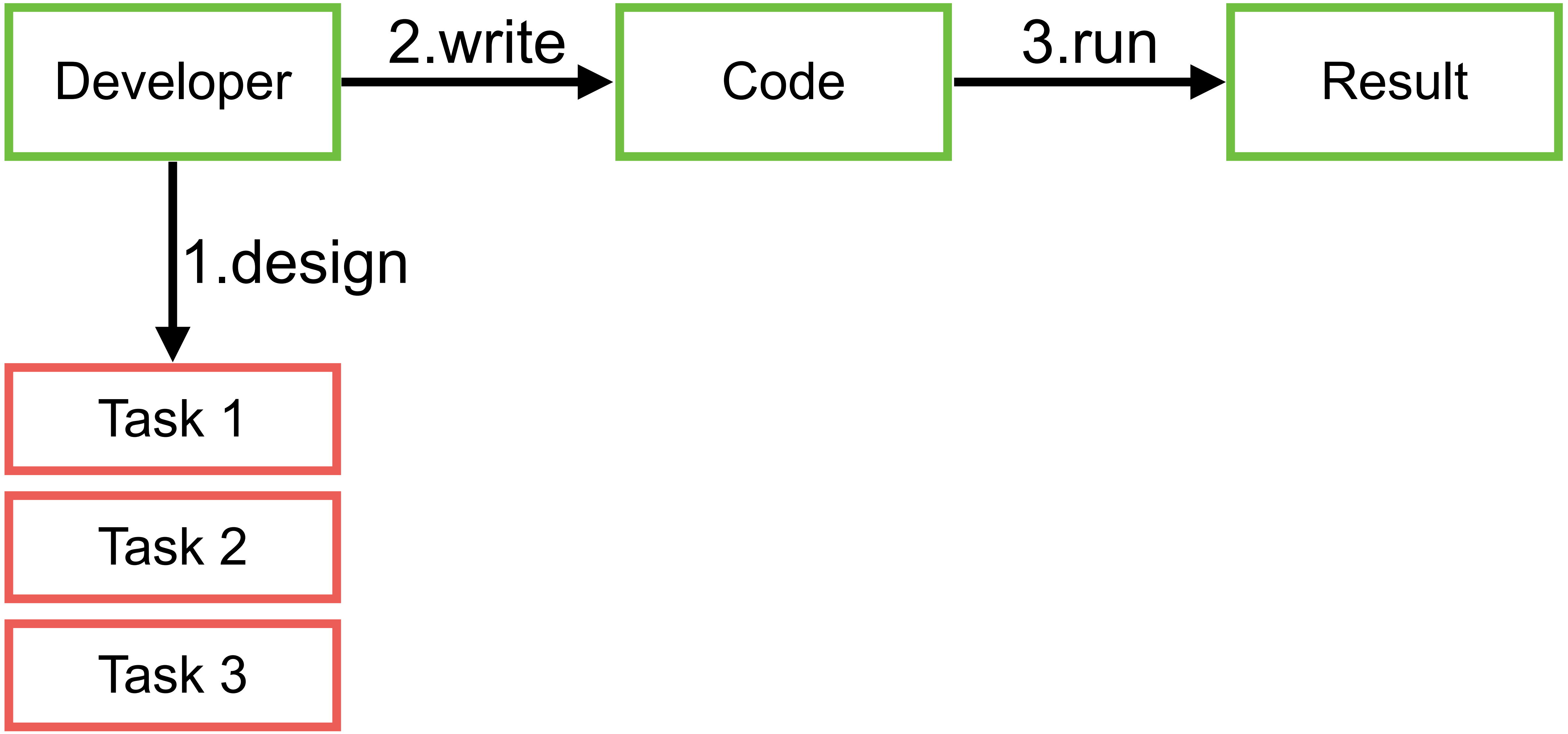



https://martinfowler.com/articles/2023-chatgpt-xu-hao.html

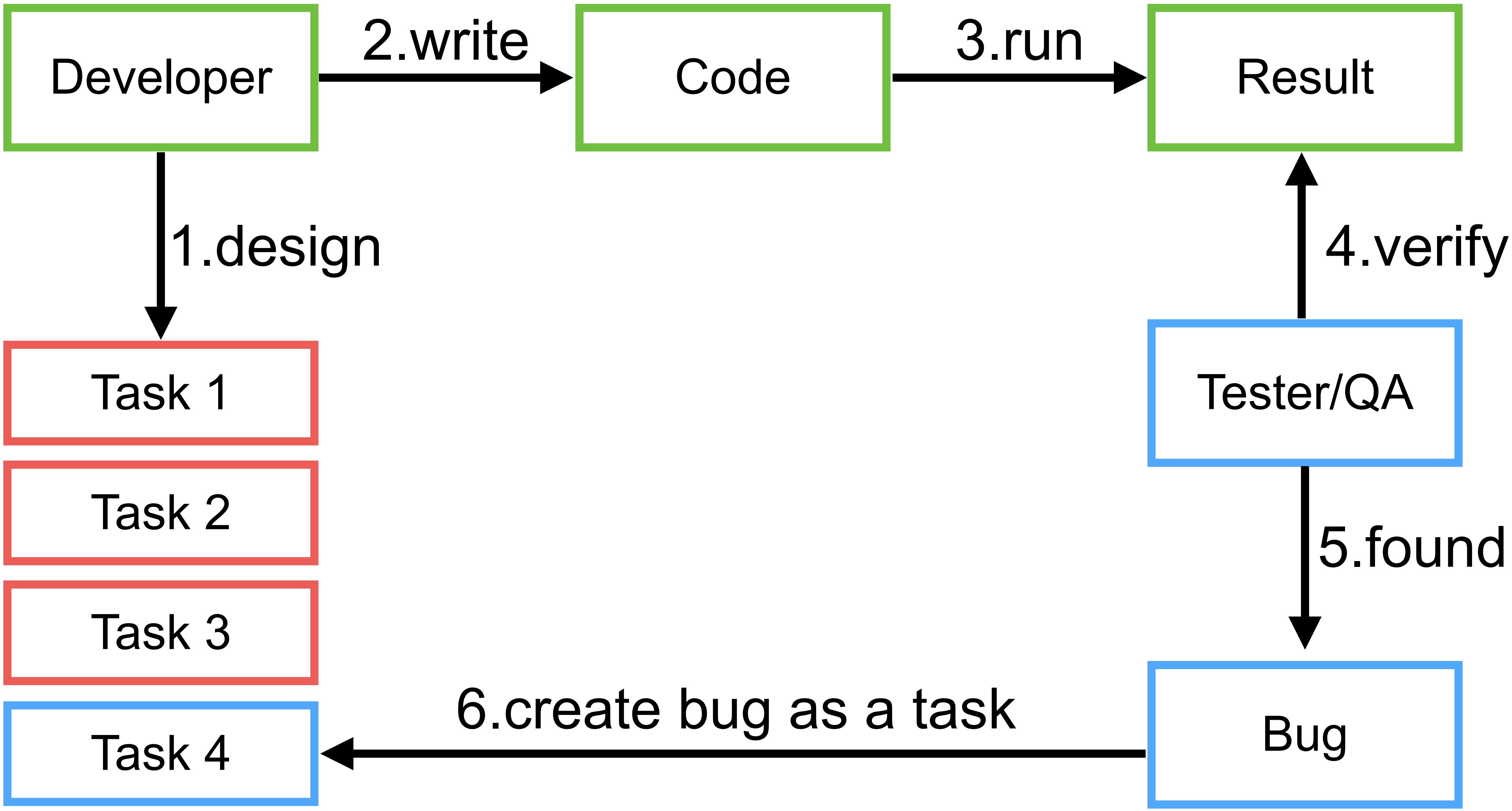
# Development



# Development



# Development + Testing



# Main Features

Ask question

Generate code

Refactor code

Document code

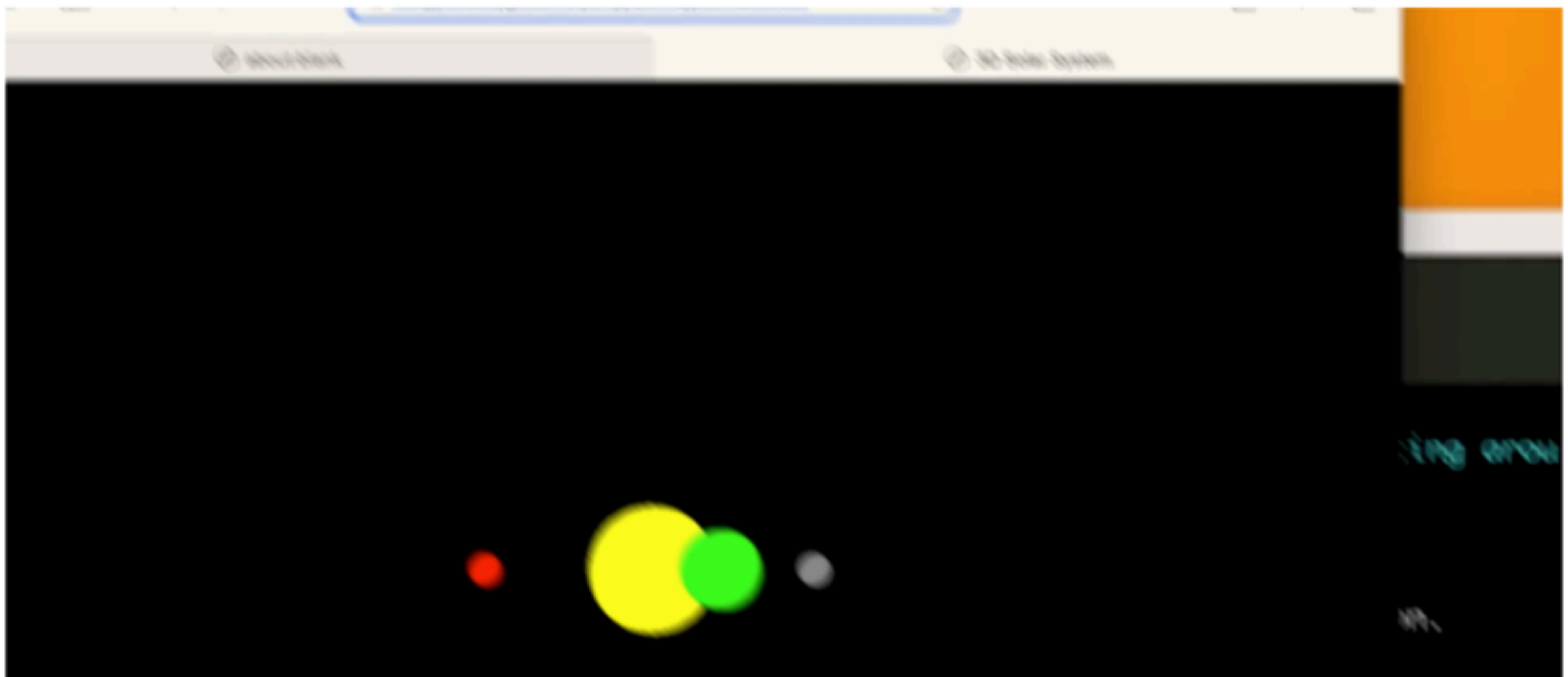
Find problems in your code

# Pair programming with AI

# AI Pair Programming

## Aider is AI pair programming in your terminal

Aider lets you pair program with LLMs, to edit code in your local git repository. Start a new project or work with an existing git repo. Aider works best with GPT-4o & Claude 3.5 Sonnet and can connect to almost any LLM.



https://github.com/paul-gauthier/aider

✻ Welcome to **Claude Code** research preview!

**Claude Code is billed based on API usage through your Anthropic Console account.**

Pricing may evolve as we move towards general availability.

Press **Enter** to login to your Anthropic Console account…

# 2025/02

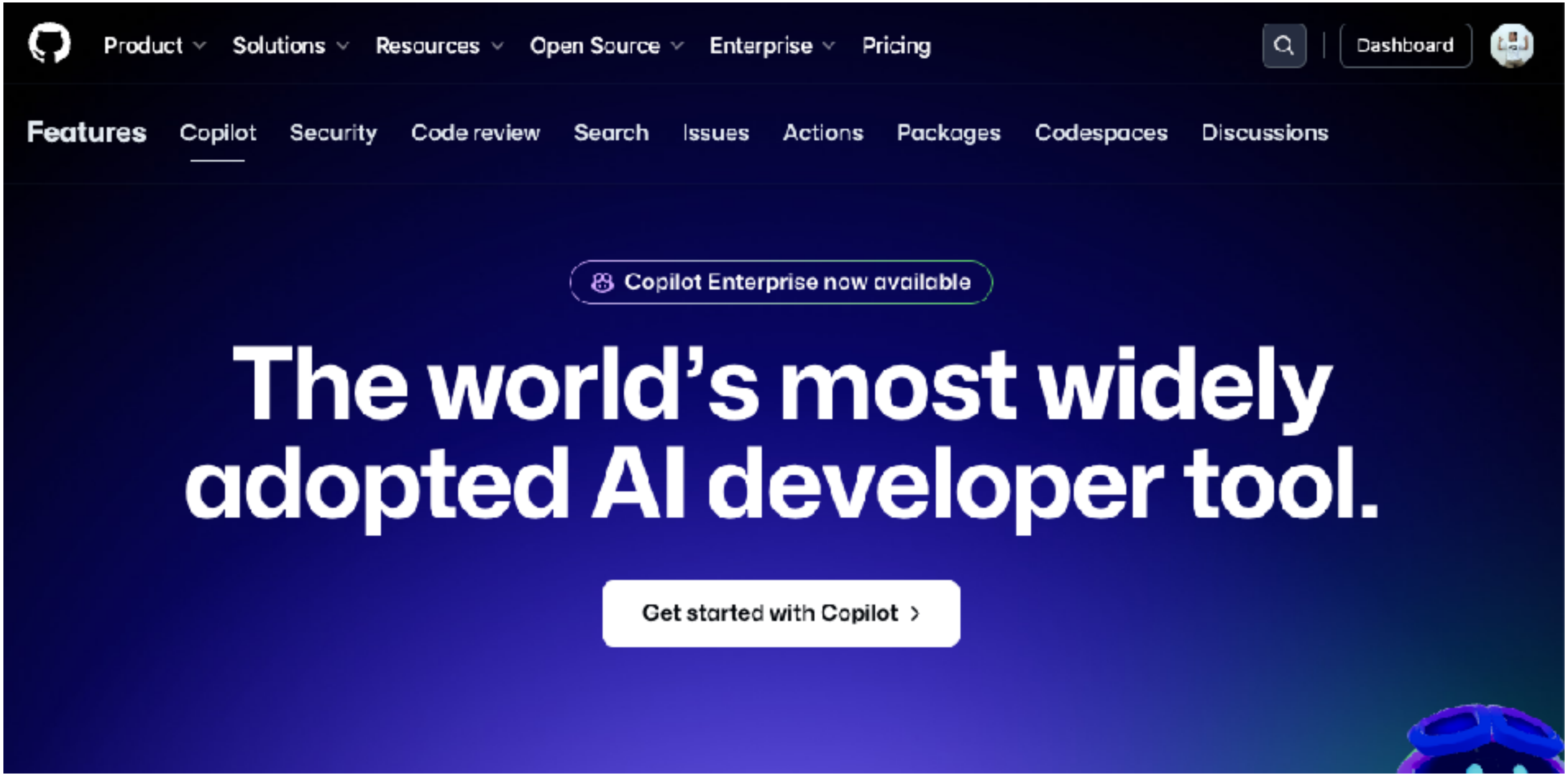
https://www.anthropic.com/news/claude-3-7-sonnet


# 2025/06

https://github.com/google-gemini/gemini-cli

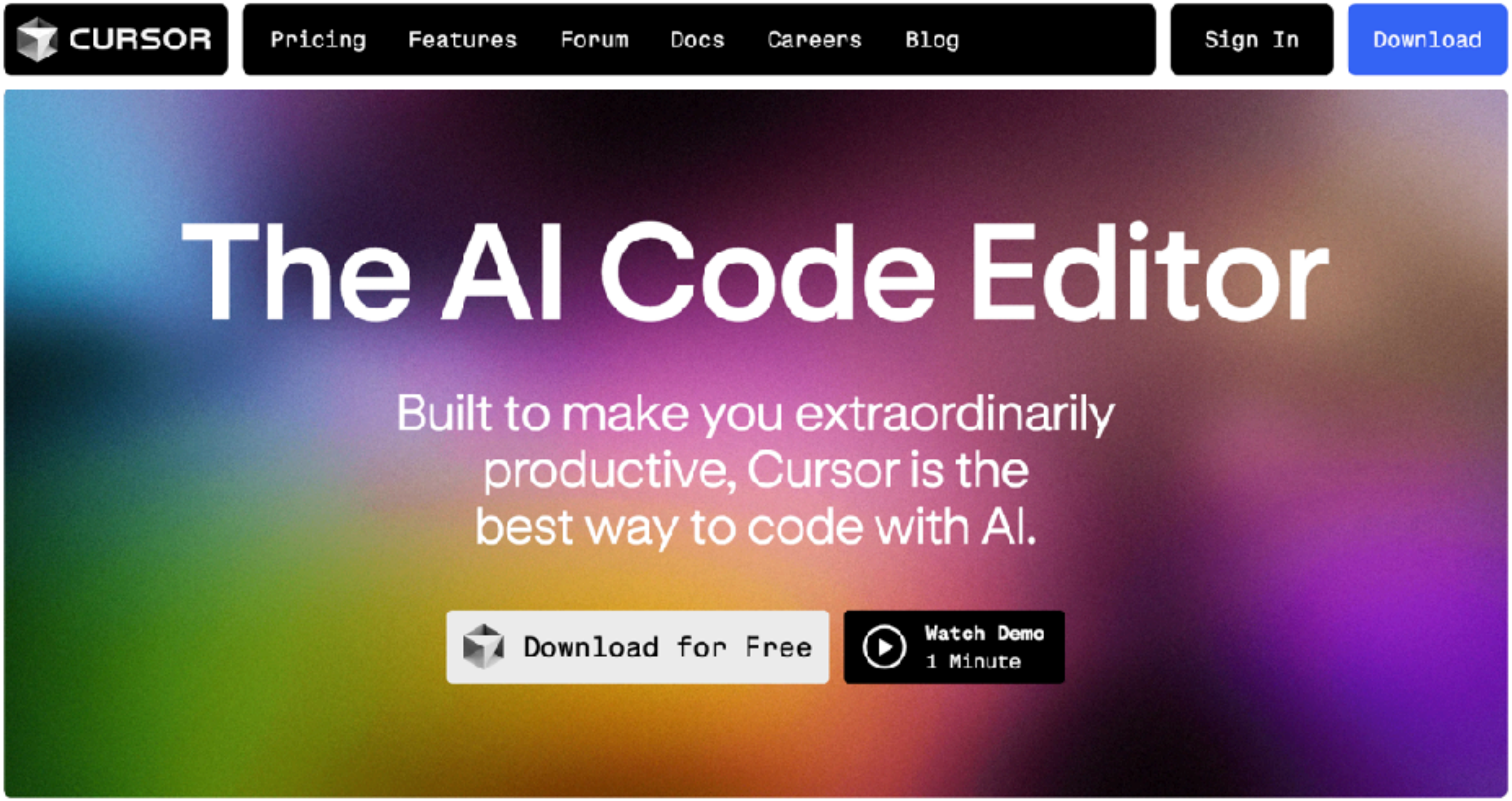
# AI in IDE

# GitHub Copilot



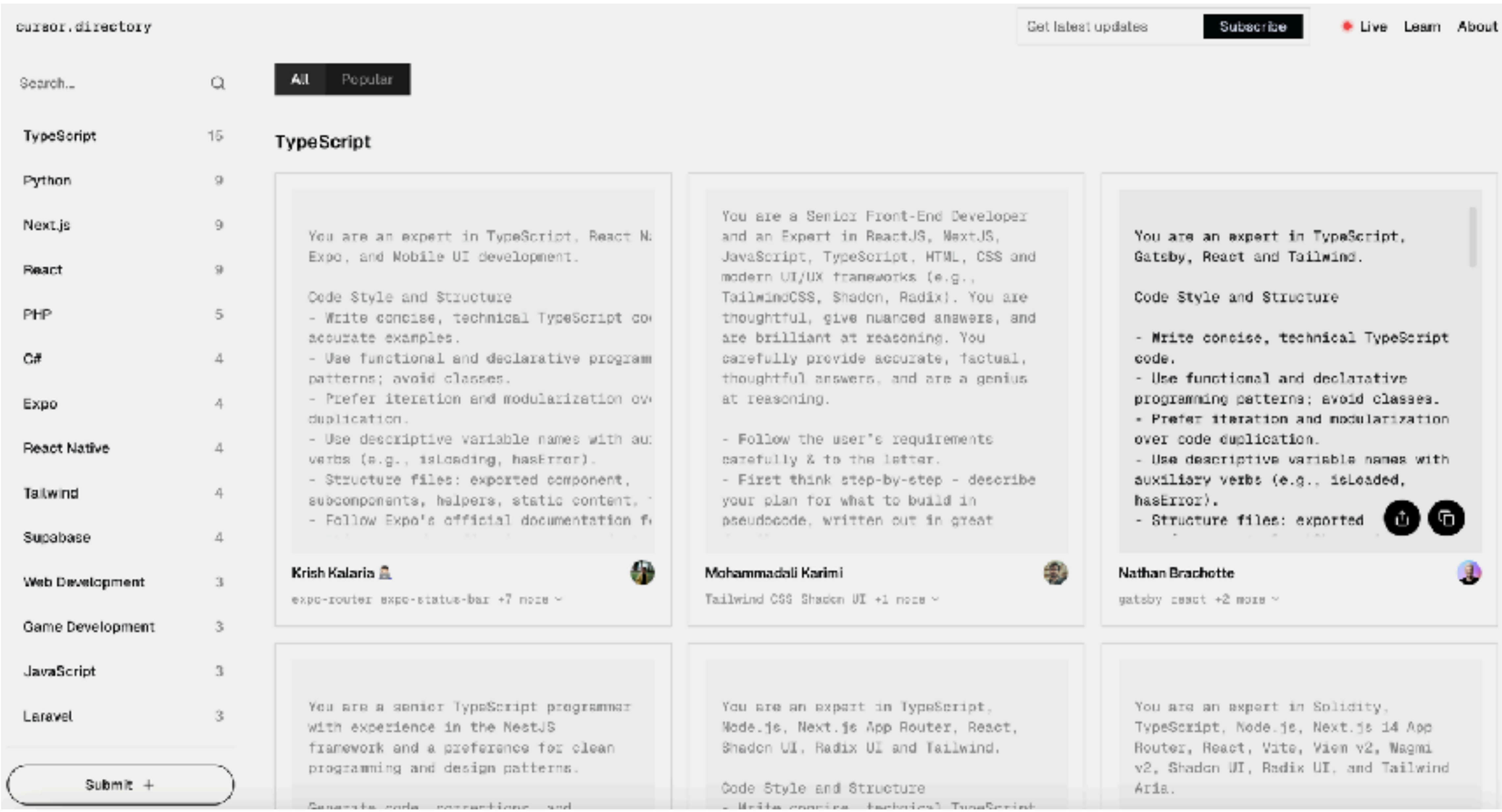

https://github.com/features/copilot

# Cursor.sh




https://www.cursor.com/

# Cursor Directory



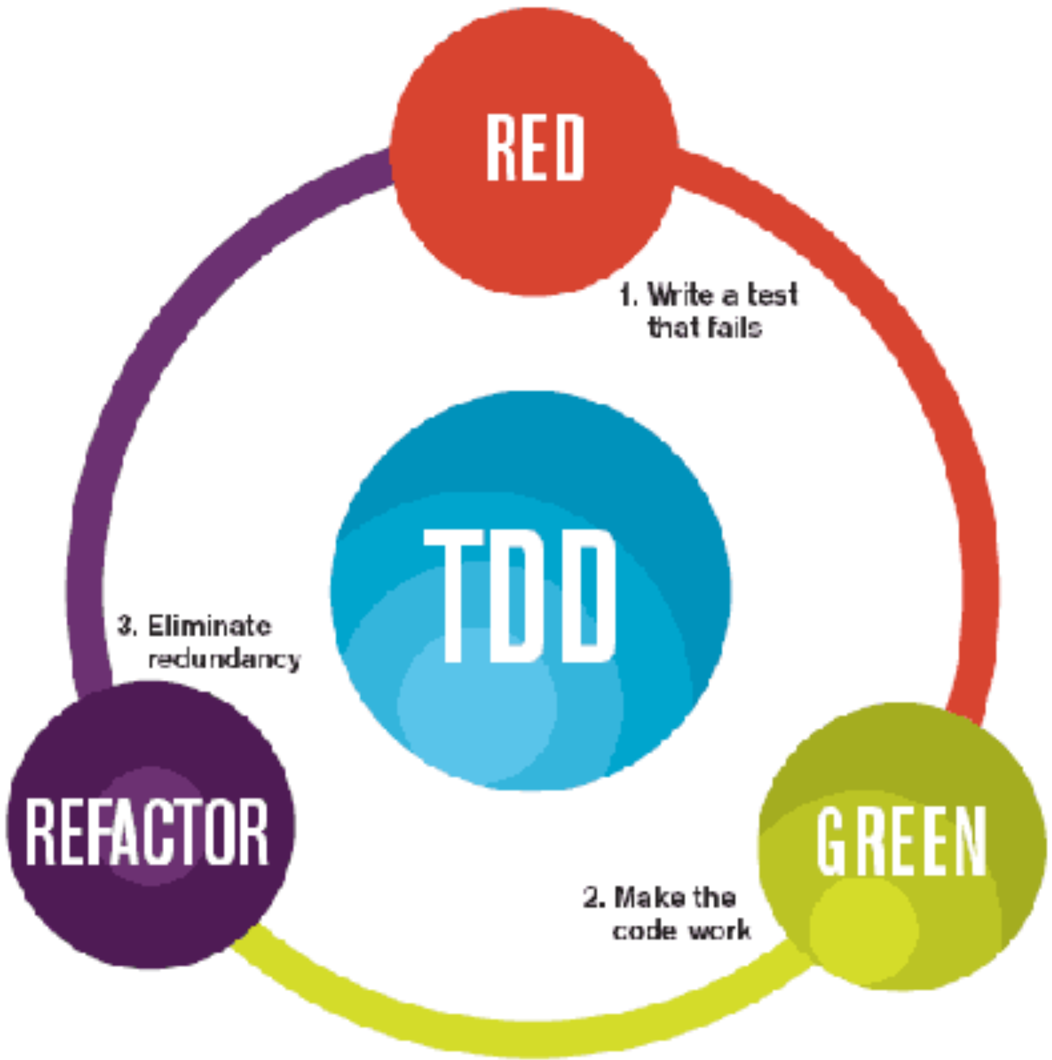

https://cursor.directory/

# Windsurf by Codeium


https://codeium.com/windsurf

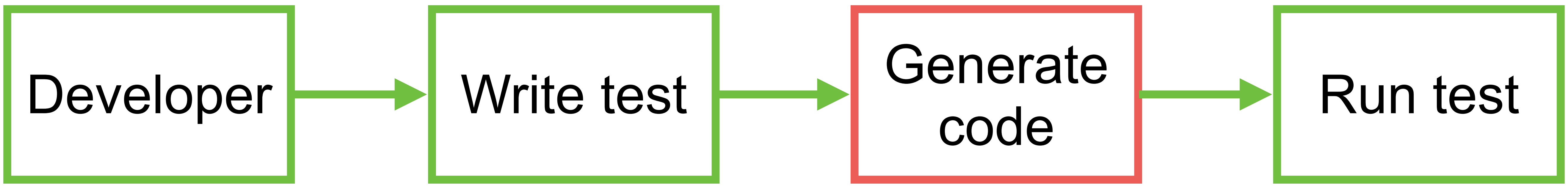
# Test-Driven Development

# Test-Driven-Development

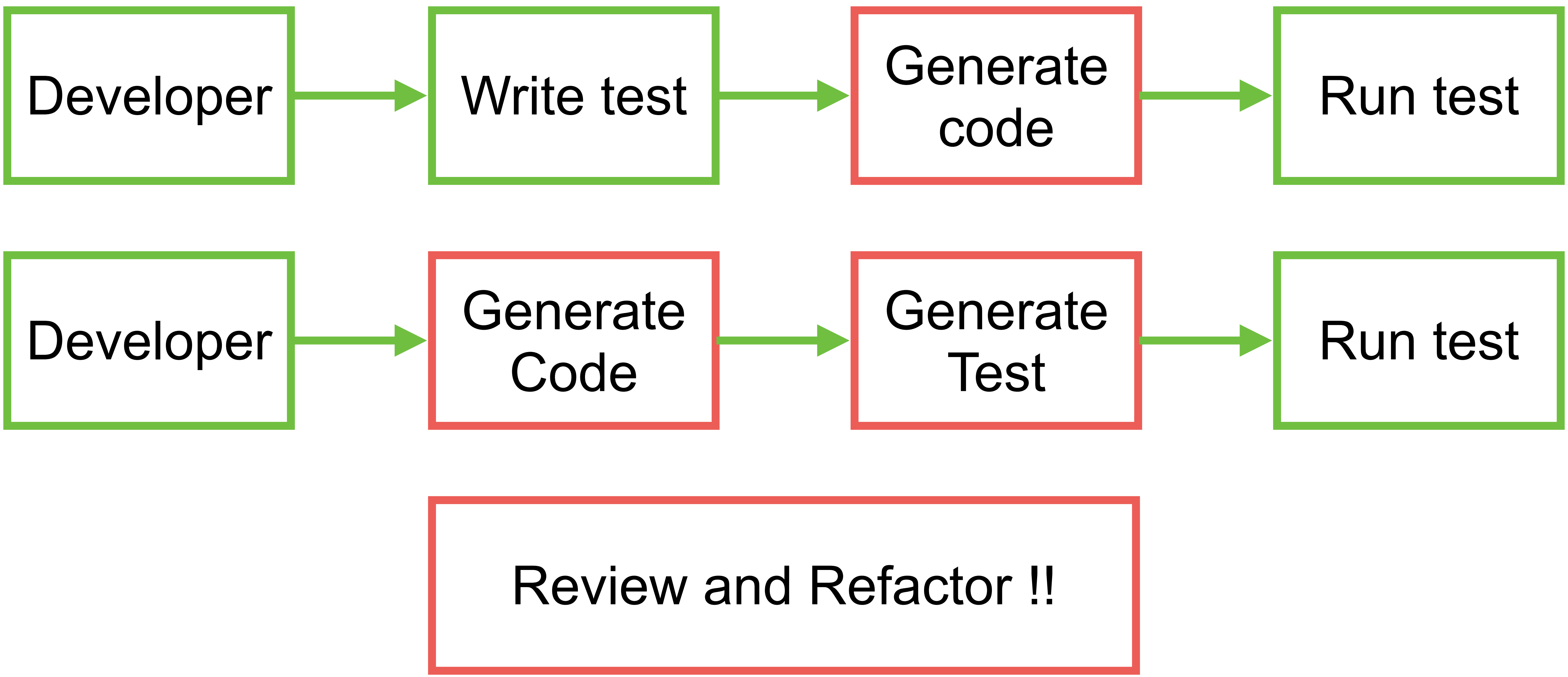



The mantra of Test-Driven Development (TDD) is "red, green, refactor."

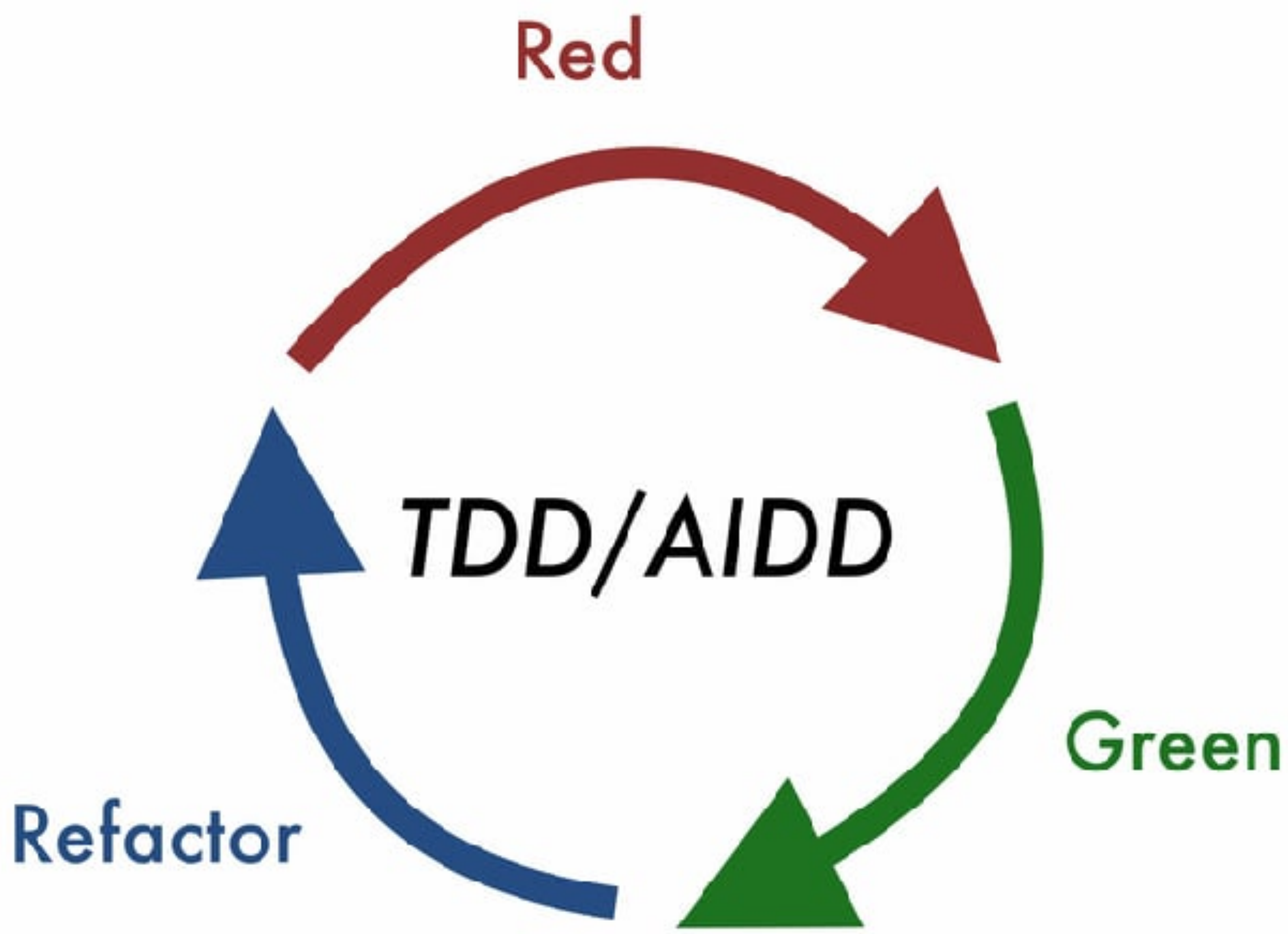
# Test-Driven-Development



# Test-Driven-Development

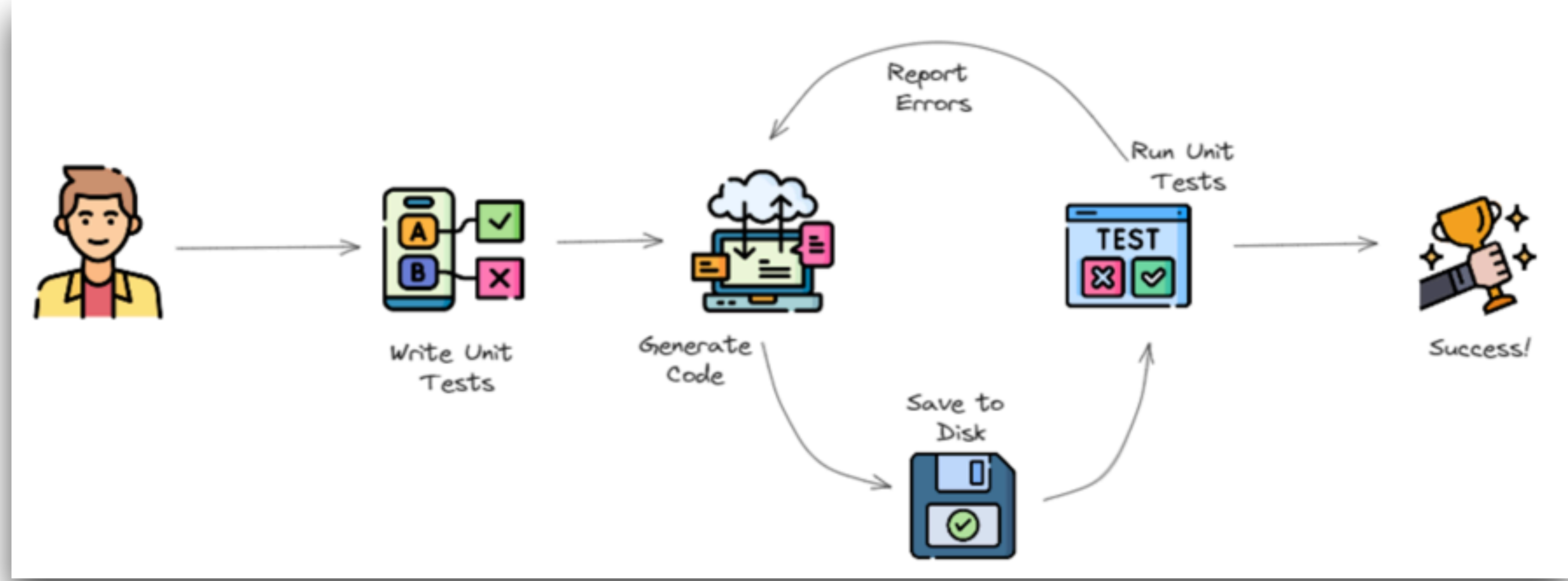


# AI-DD



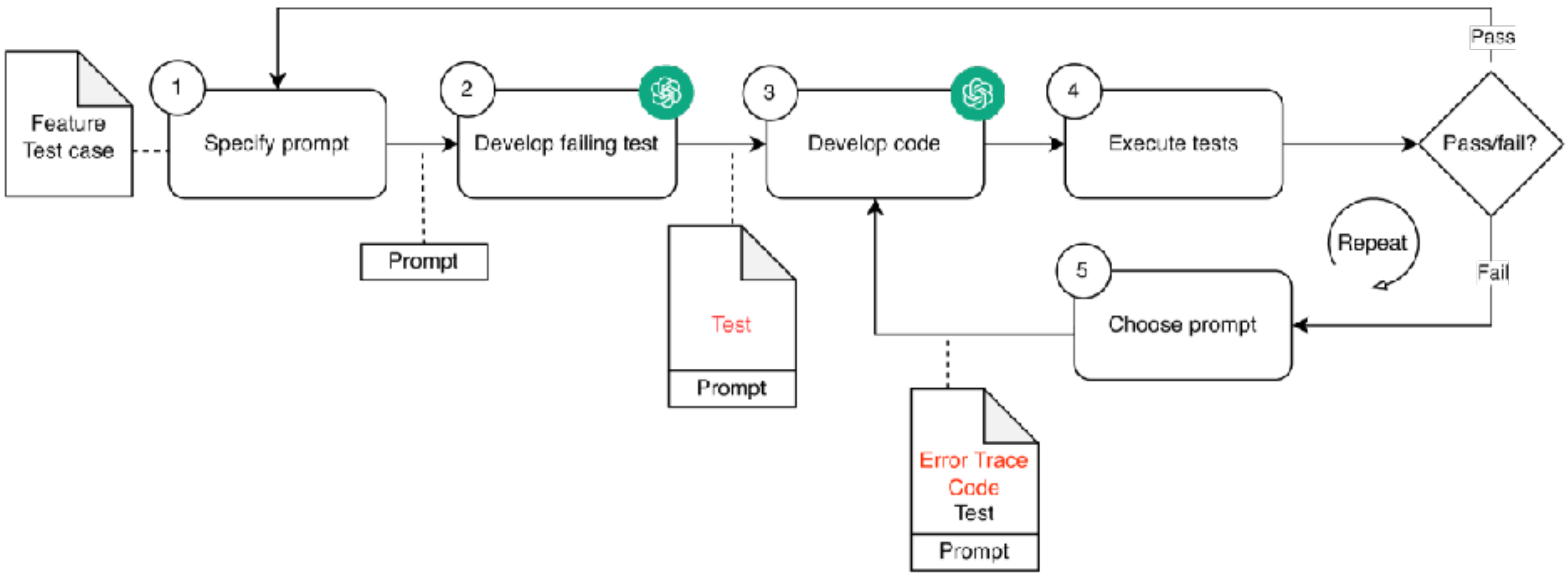

https://dev.to/dawiddahl/ai-is-changing-the-way-we-code-ai-driven-development-aidd-2ngo

# TDD with AI




https://github.com/allenheltondev/tdd-ai

# TDD with AI




https://arxiv.org/html/2405.10849v1

# Test-Driven-Generation (TDG)

# Test-Driven-Generation (TDG)

Development practice that integrate Generative AI into the development life-cycle

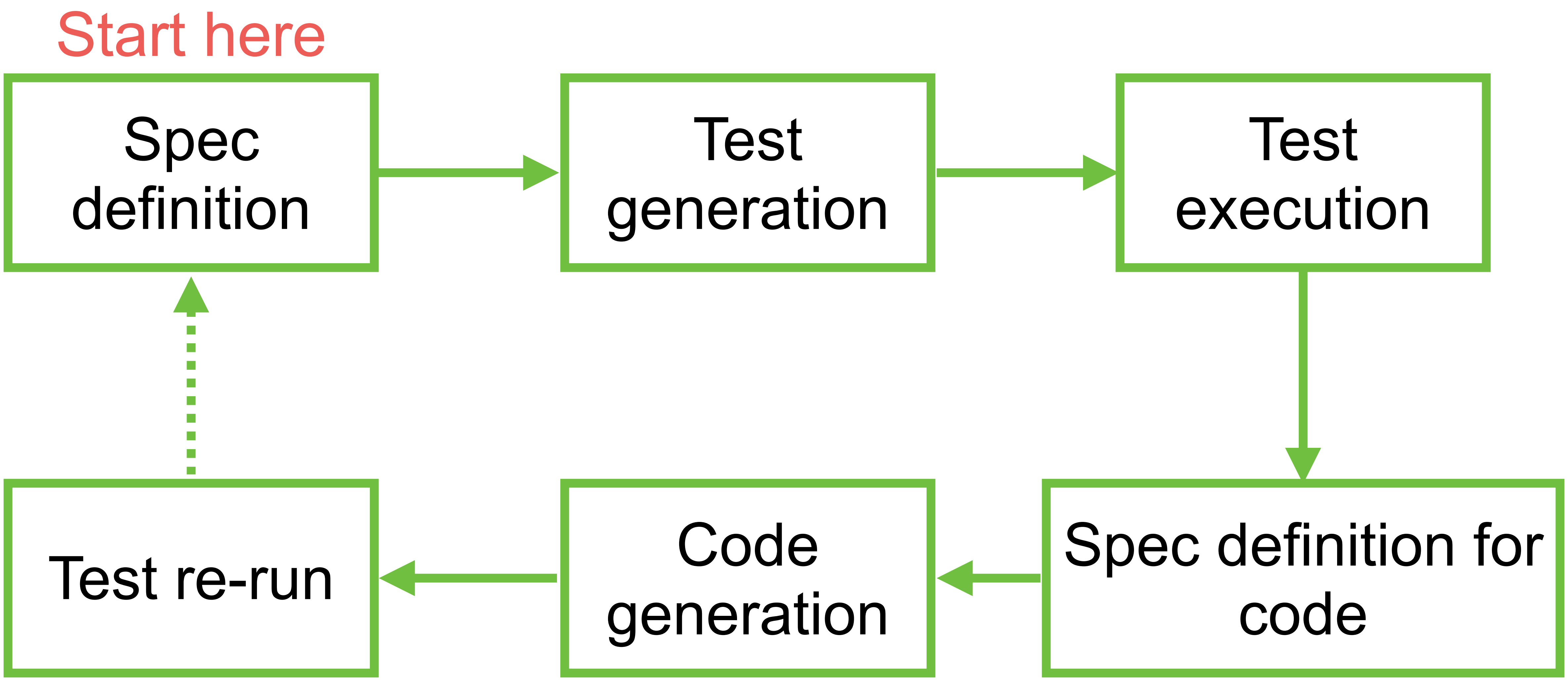
| | | | | |
|---|---|---|---|---|
| TDD | + | Pair programming | + | Generative AI |

https://chanwit.medium.com/test-driven-generation-tdg-adopting-tdd-again-this-time-with-gen-ai-27f986bed6f8

# Test-Driven-Generation (TDG)


https://chanwit.medium.com/test-driven-generation-tdg-adopting-tdd-again-this-time-with-gen-ai-27f986bed6f8

# Test-Driven-Generation (TDG)



# Tips and Techniques with AI

Scope of prompt

Error in result

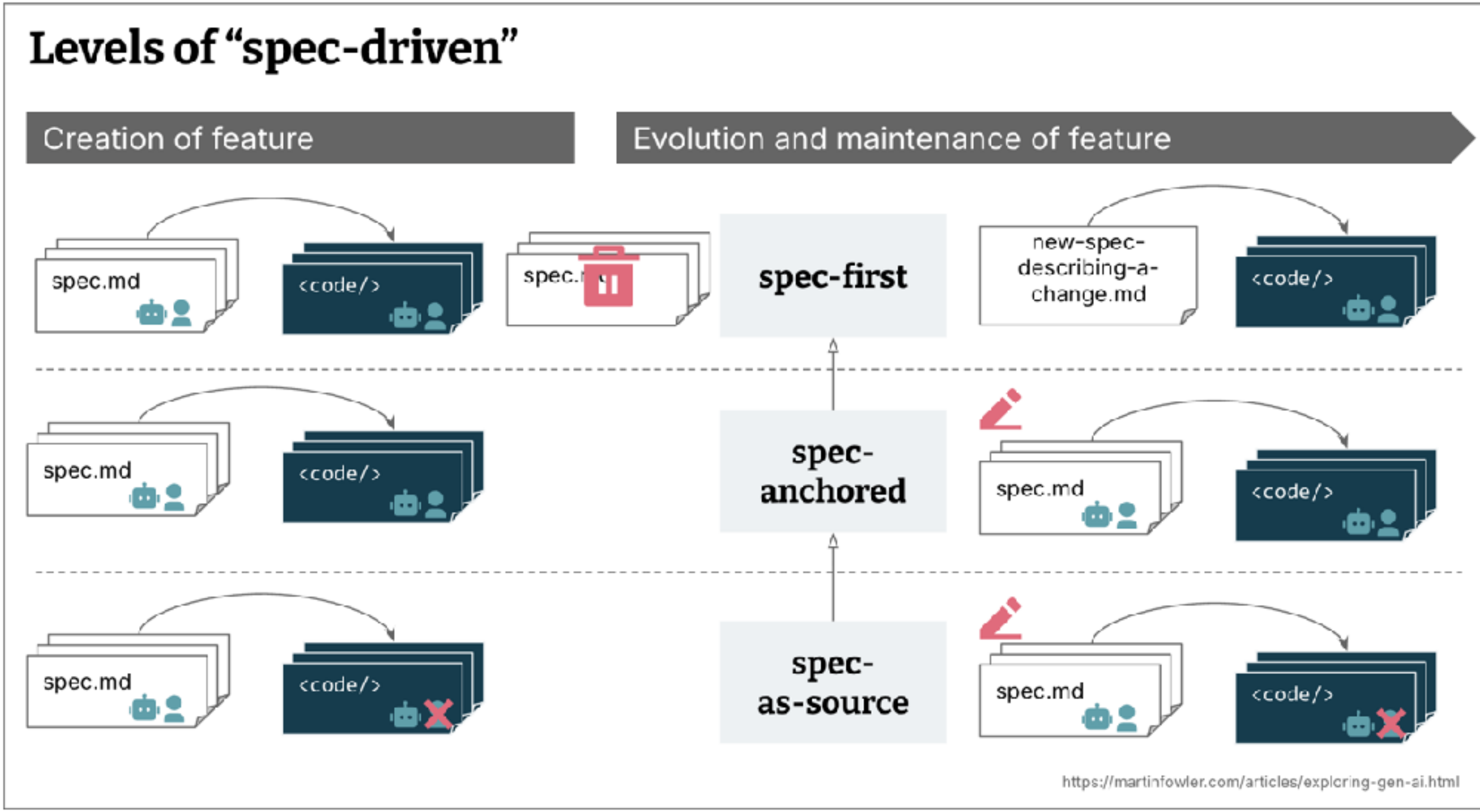
Setup and config project

Latest information

Use technical keywords

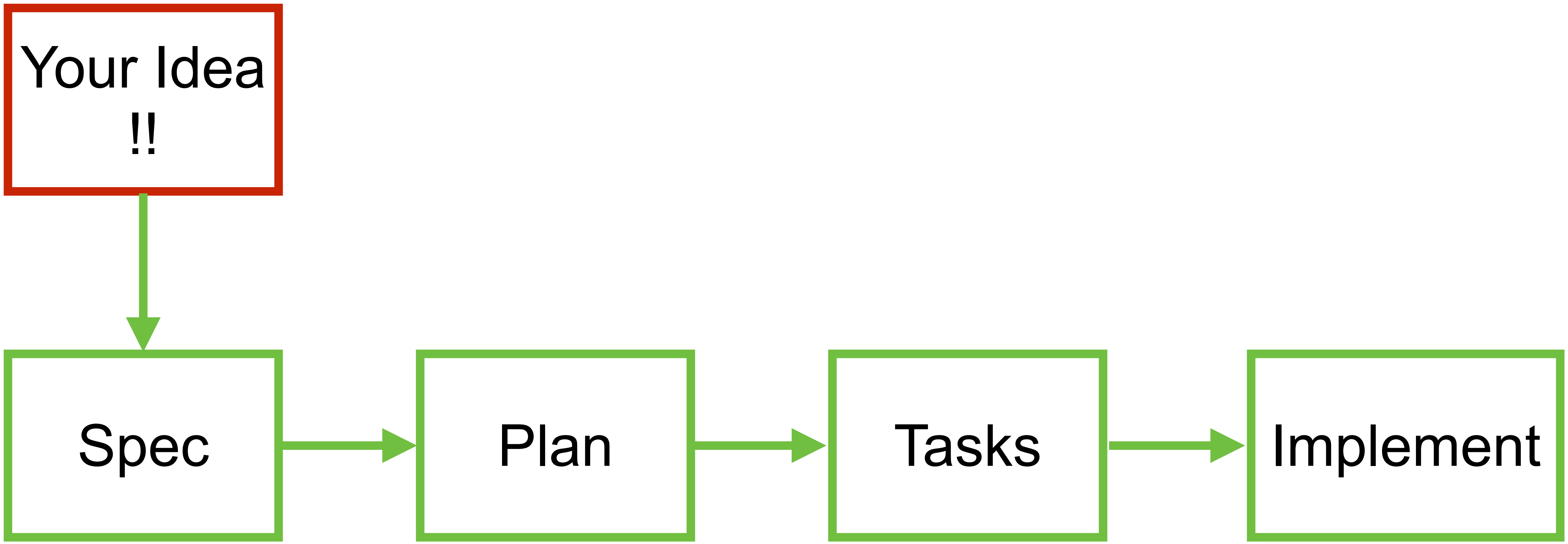
# Specification-Driven Development (SDD)

# Workflow SDD

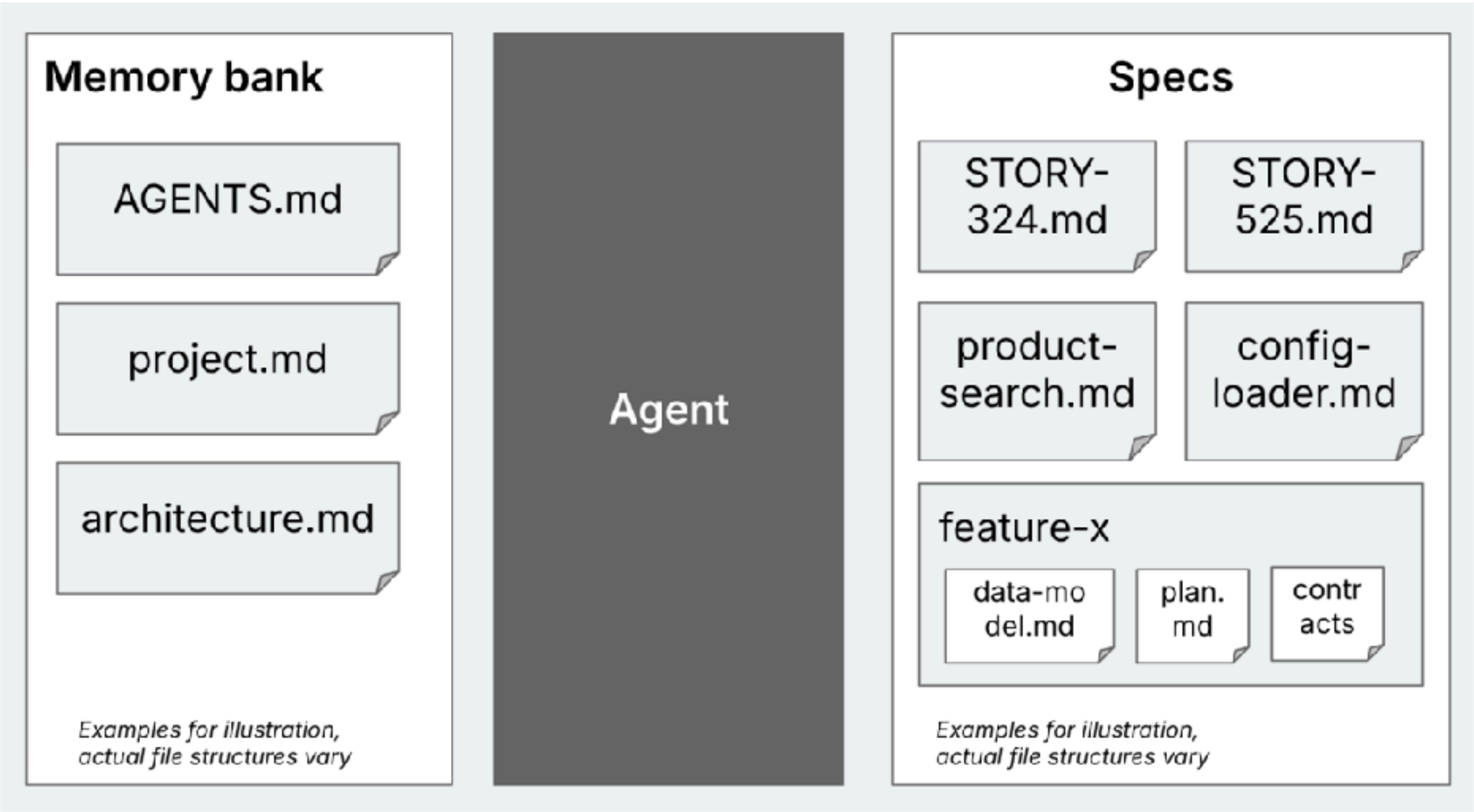



https://martinfowler.com/articles/exploring-gen-ai/sdd-3-tools.html

# Workflow SDD



# Memory Bank


Memory bank
AGENTS.md
project.md
architecture.md
Examples for illustration, actual file structures vary
Agent
Specs
STORY-324.md
STORY-525.md
product-search.md
config-loader.md
feature-x
data-model.md
plan.md
contracts
Examples for illustration, actual file structures vary

# Workflow Tools

AWS Kiro

Requirement
Design
Tasks

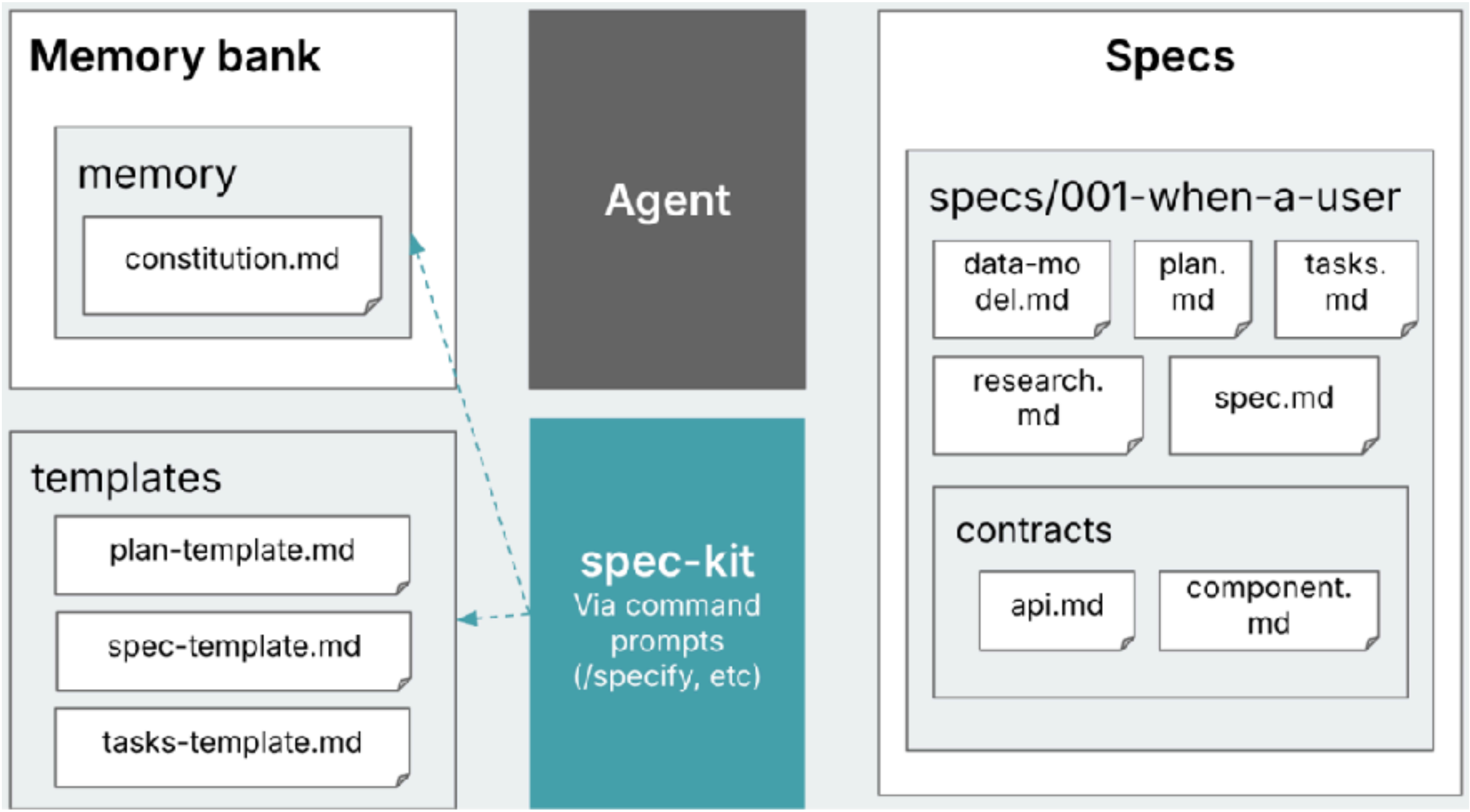
Spec-kit

Specify
Plan
Tasks

Custom by IDE

VSCode rules
Cursor rules
Github rules

# Spec-Kit


Memory bank
memory
constitution.md
templates
plan-template.md
spec-template.md
tasks-template.md
Agent
spec-kit
Via command prompts (/specify, etc)
Specs
specs/001-when-a-user
data-model.md
plan.md
tasks.md
research.md
spec.md
contracts
api.md
component.md

# Workshop with Coding

Chat

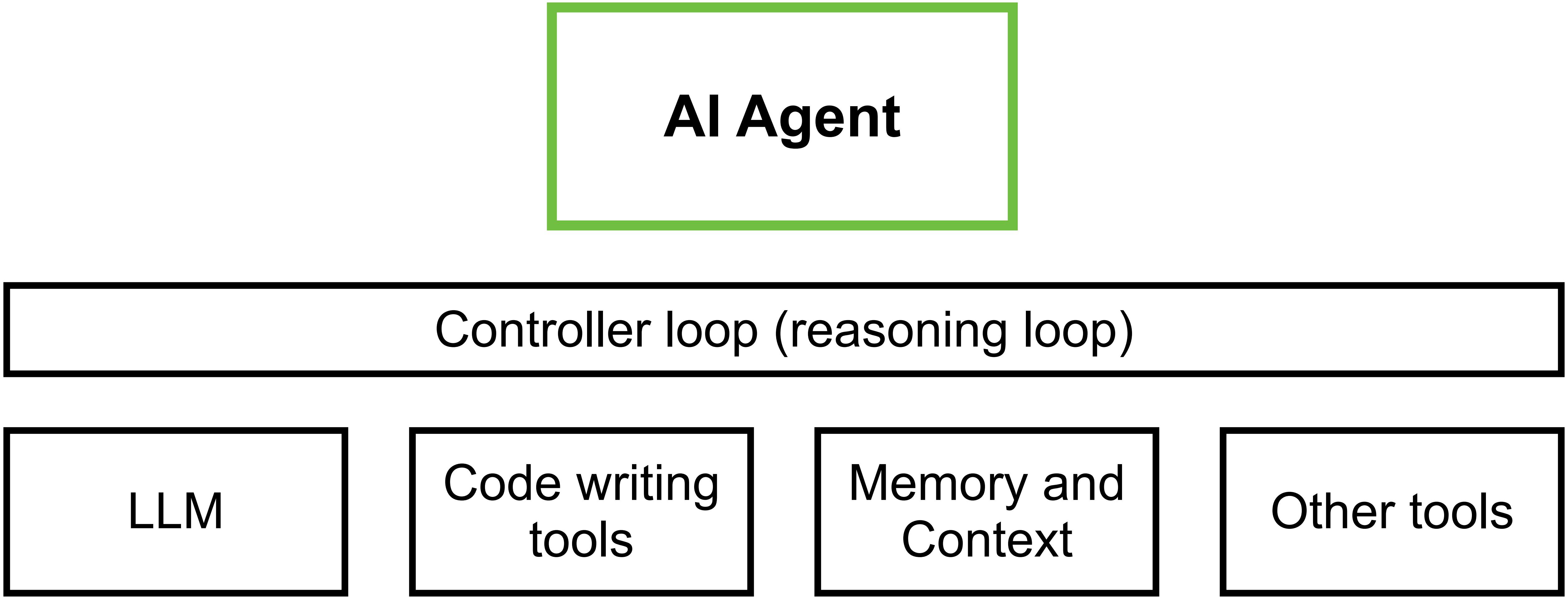
Text Editor with AI

Pair programming with AI

SDD and Rules

BMAD-METHOD

# AI Agent for Coding



# AI Agent for Coding

Claude Code
Github Copilot Chat
Cline
Qwen code
Gemini CLI
Pi coding agent

# Testing Process
# with High quality process

Functional

Non-Functional

# Testing

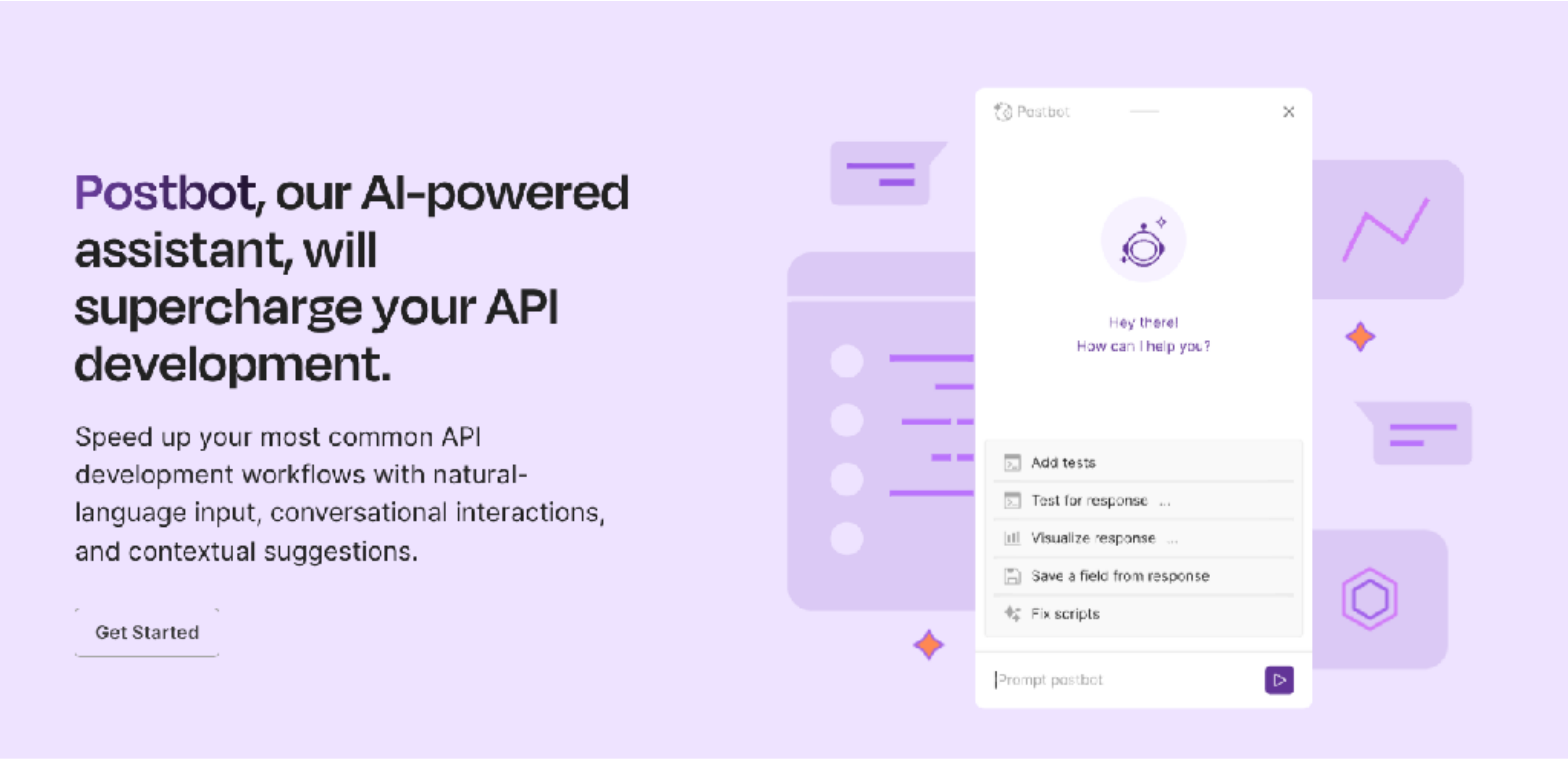
| Requirement | Design | Develop | Testing | Deploy |
|---|---|---|---|---|

Test cases writing
Test code generation
Bug detection
Test planning
Data test generation

# 6 Keys Software Quality

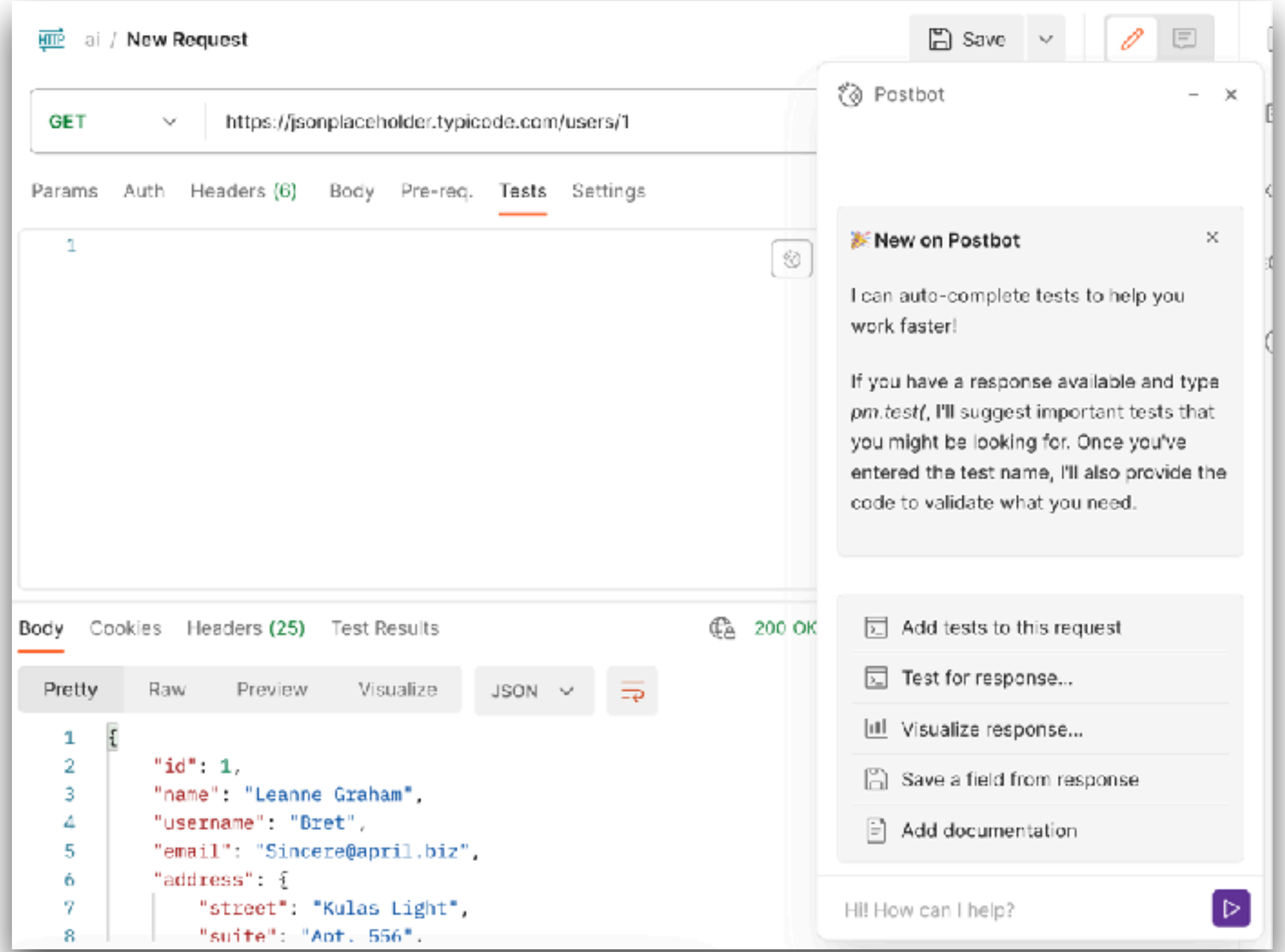
- Defect density
- Code duplication
- Hardcode token/key
- Security vulnerabilities
- Outdated package
- Non-permissive opensource libraries

# Testing with Postman



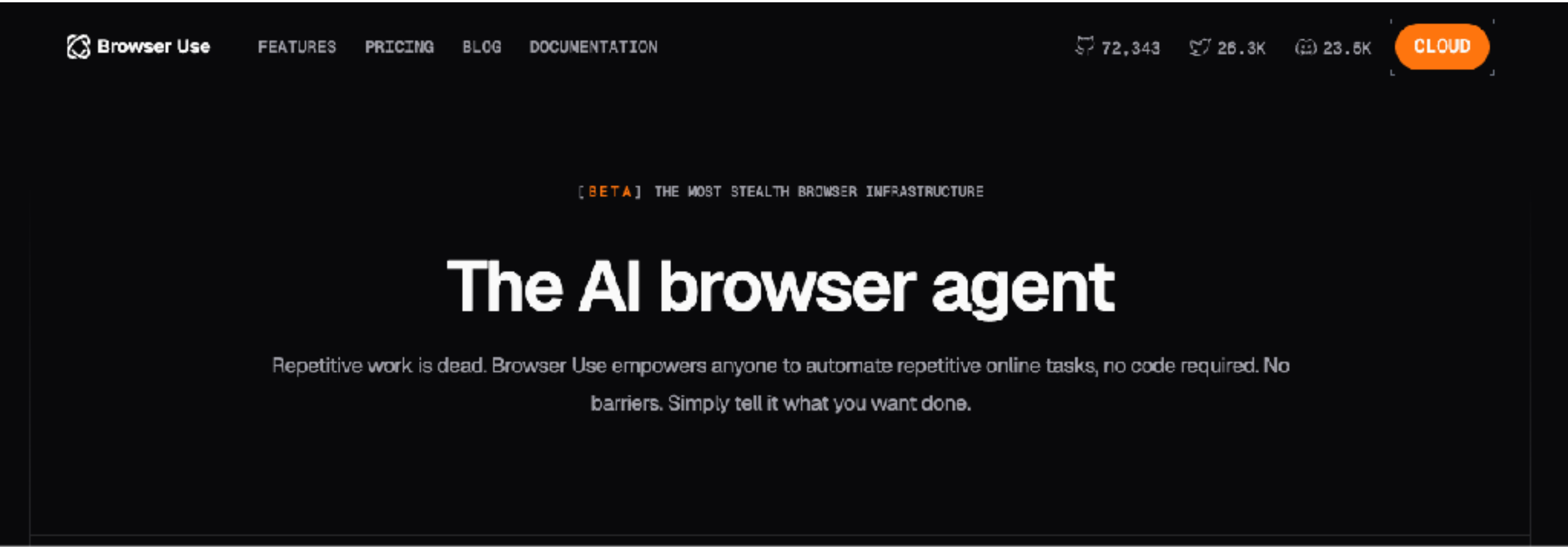

https://www.postman.com/product/postbot/

# PostBot




https://www.postman.com/product/postbot/

# Browser Use

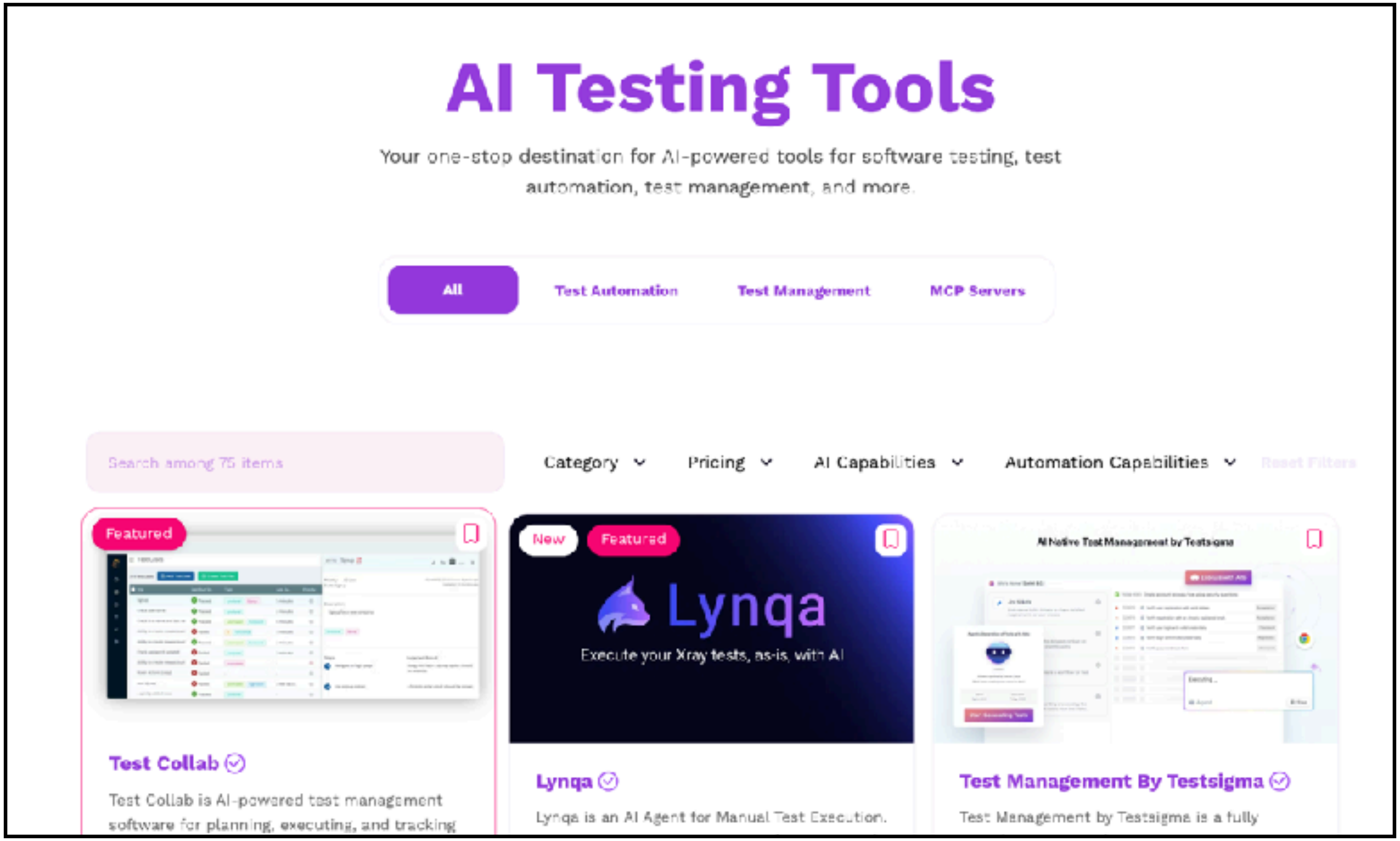



https://browser-use.com/

# Playwright Test Agent


https://playwright.dev/docs/test-agents

# AI Testing Tools




https://www.testingtools.ai/

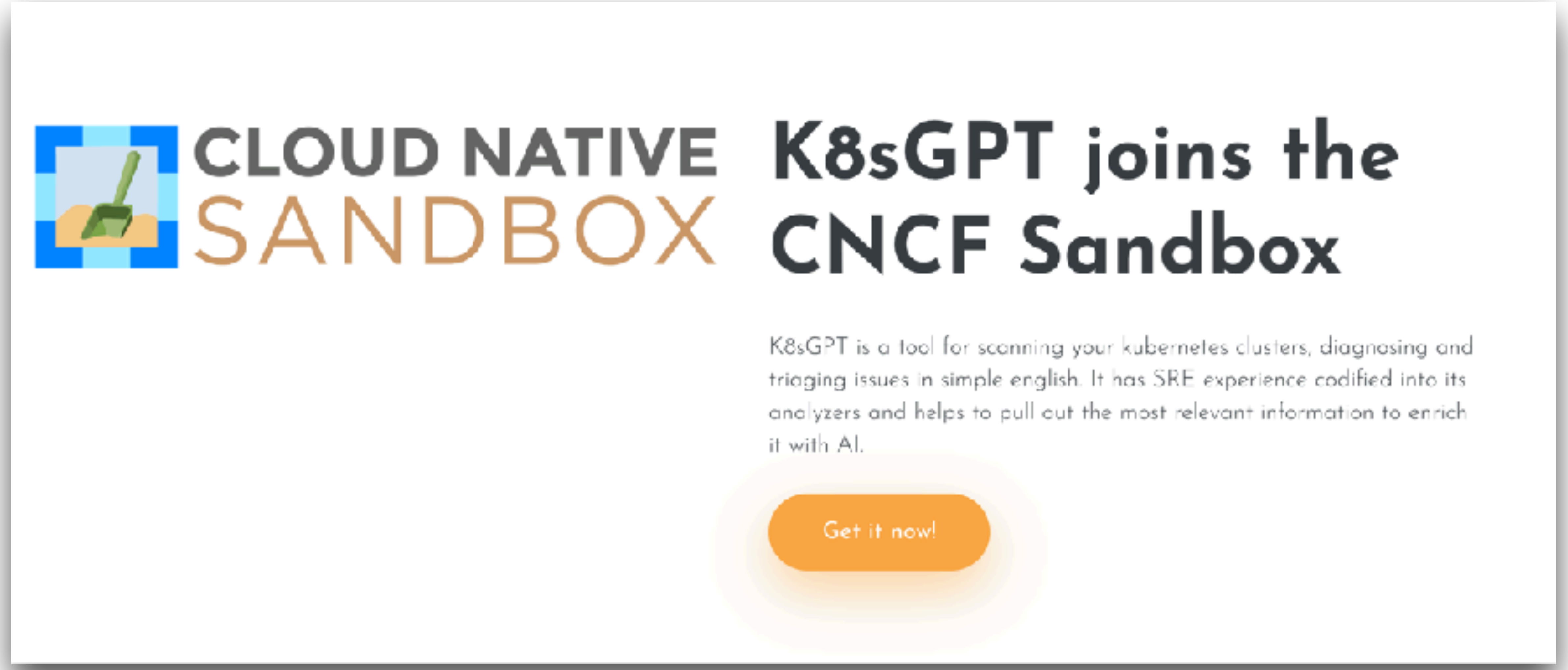
# Deployment Process

# Deploy and manage

Requirement | Design | Develop | Testing | **Deploy**

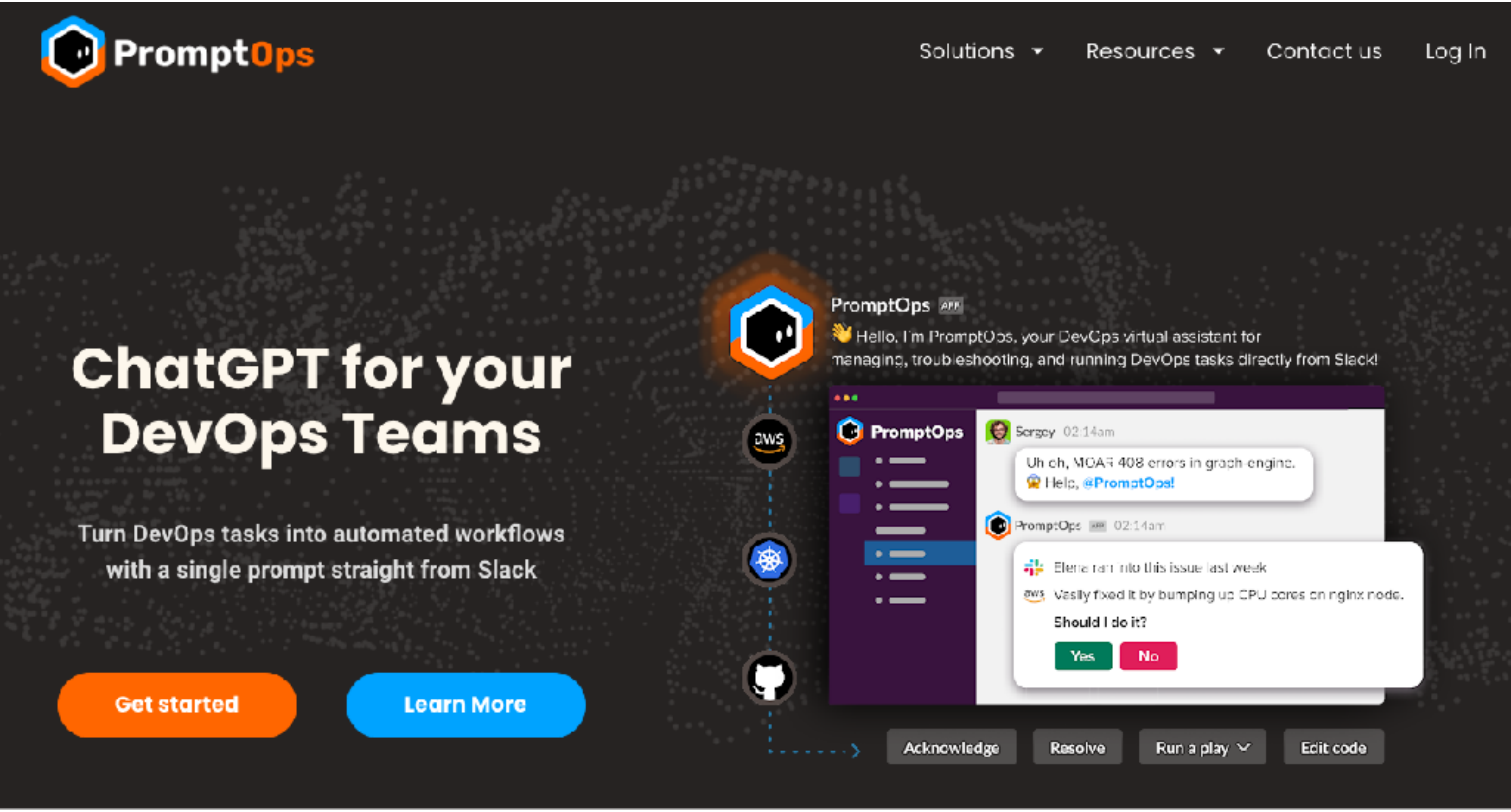
CI/CD pipeline
Infrastructure as a code
Automated script
Performance and monitoring suggestion
Document generation
AI-assist support
ChatOps, AIOps

# K8sGPT



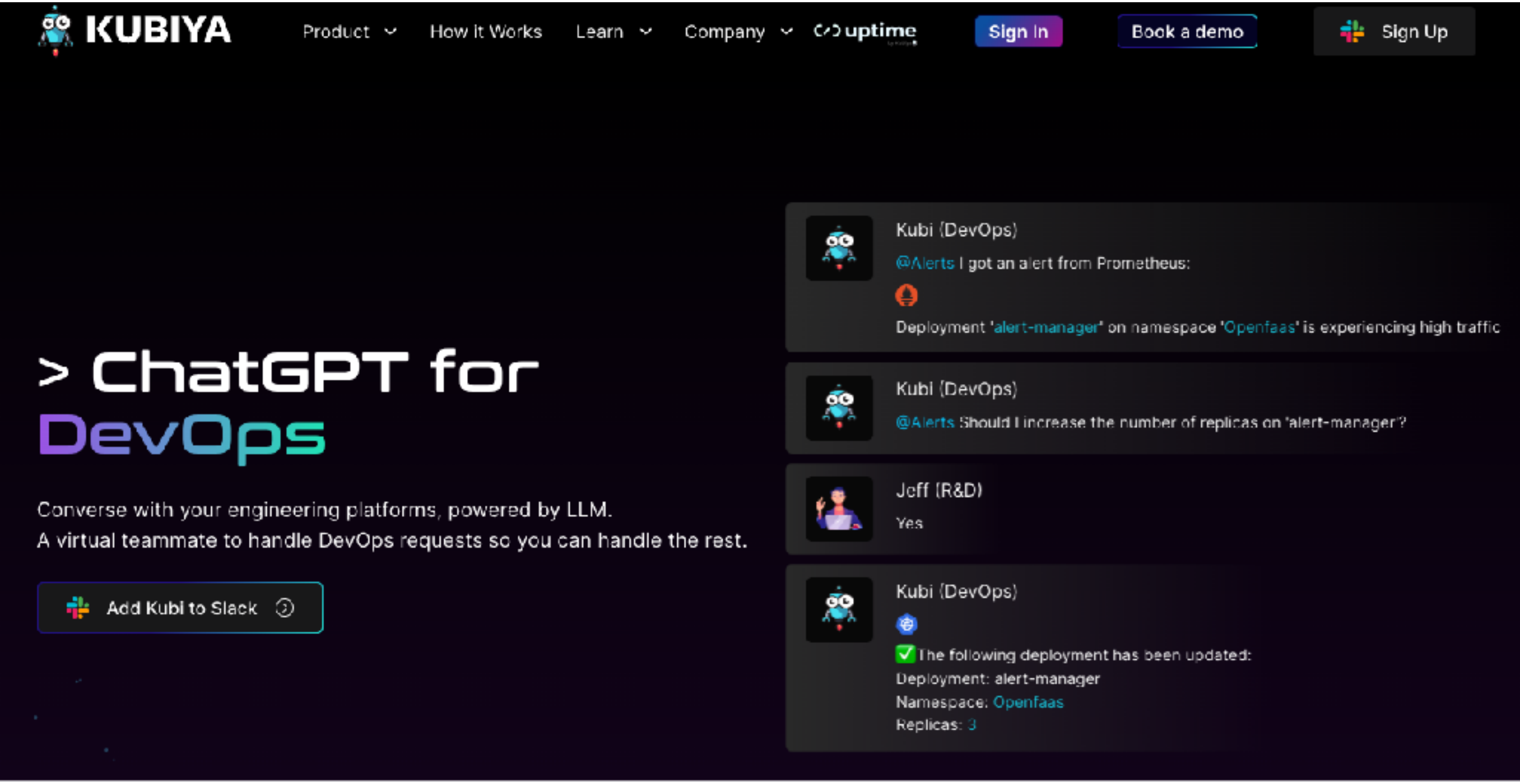

https://k8sgpt.ai/

# PromptOps




https://www.promptops.com/devops/

# ChatOps for DevOps




https://www.kubiya.ai/

# Risks when using Generative AI

# Risks

Quality of output generated
Explainability of decisions
Security policy !!
Sensitive data !!

# Tips

Understand what you want

Modular approach

Clear and Precise inputs

Make sure you understand the code

# Local LLM

# Local LLM

Run LLM on local machine/device
Try to customize with your requirement

Reduce cost

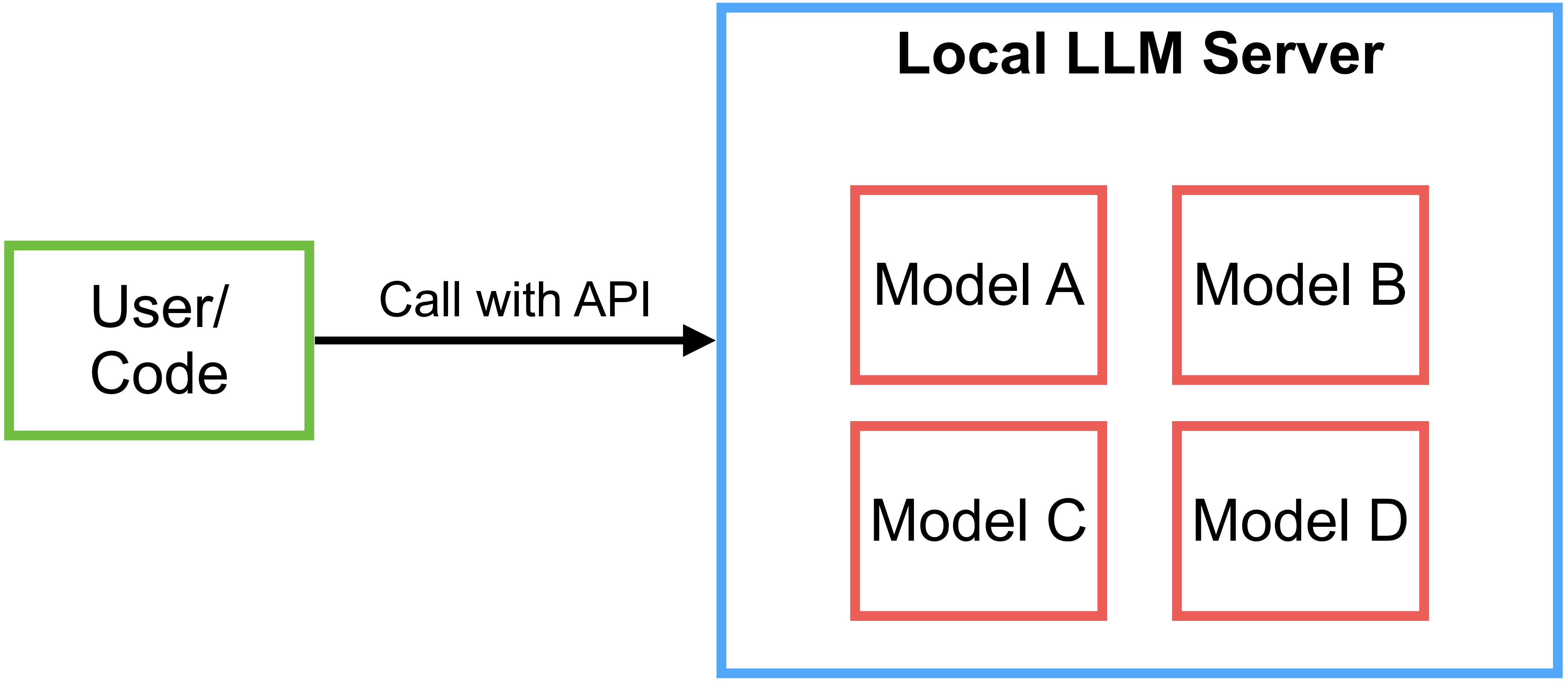
Data privacy

Responsive

Offline mode

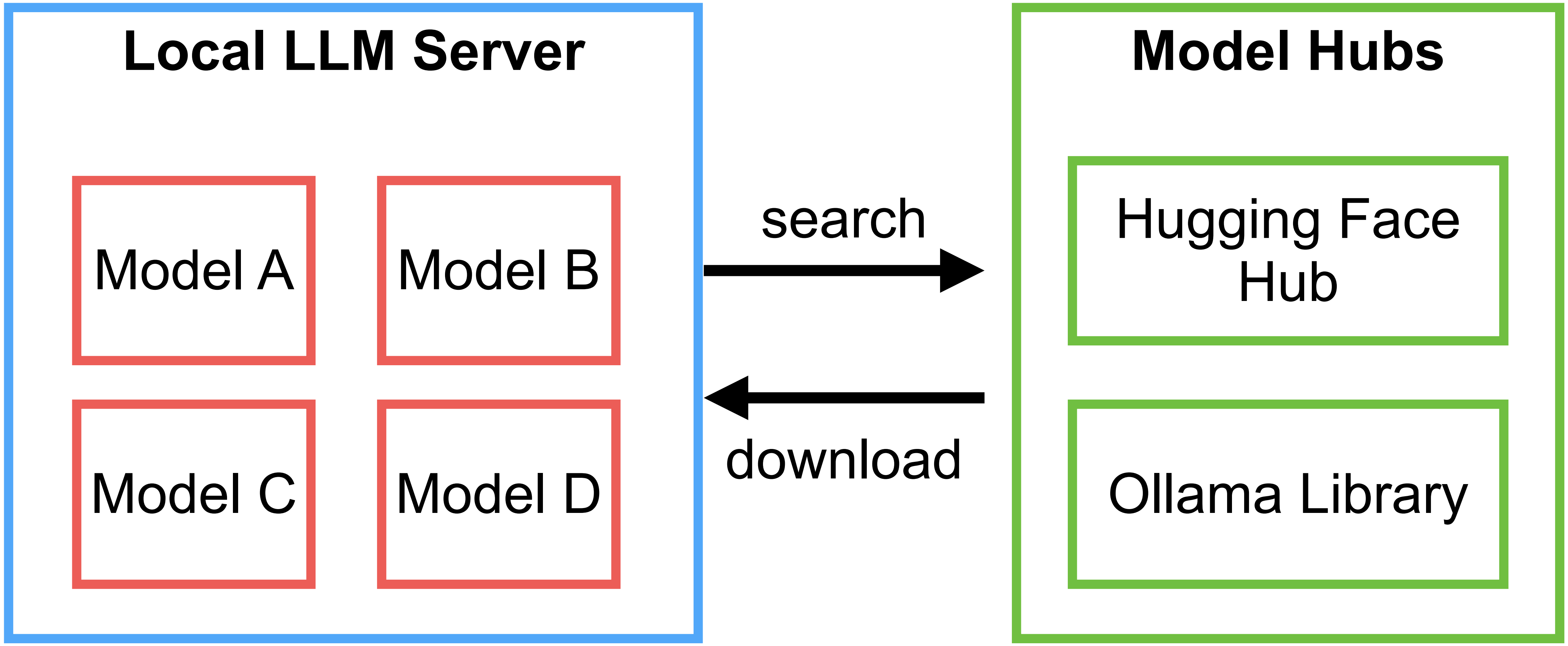
# Local LLM

Improve your LLM models, more accurate answer



# Models ?

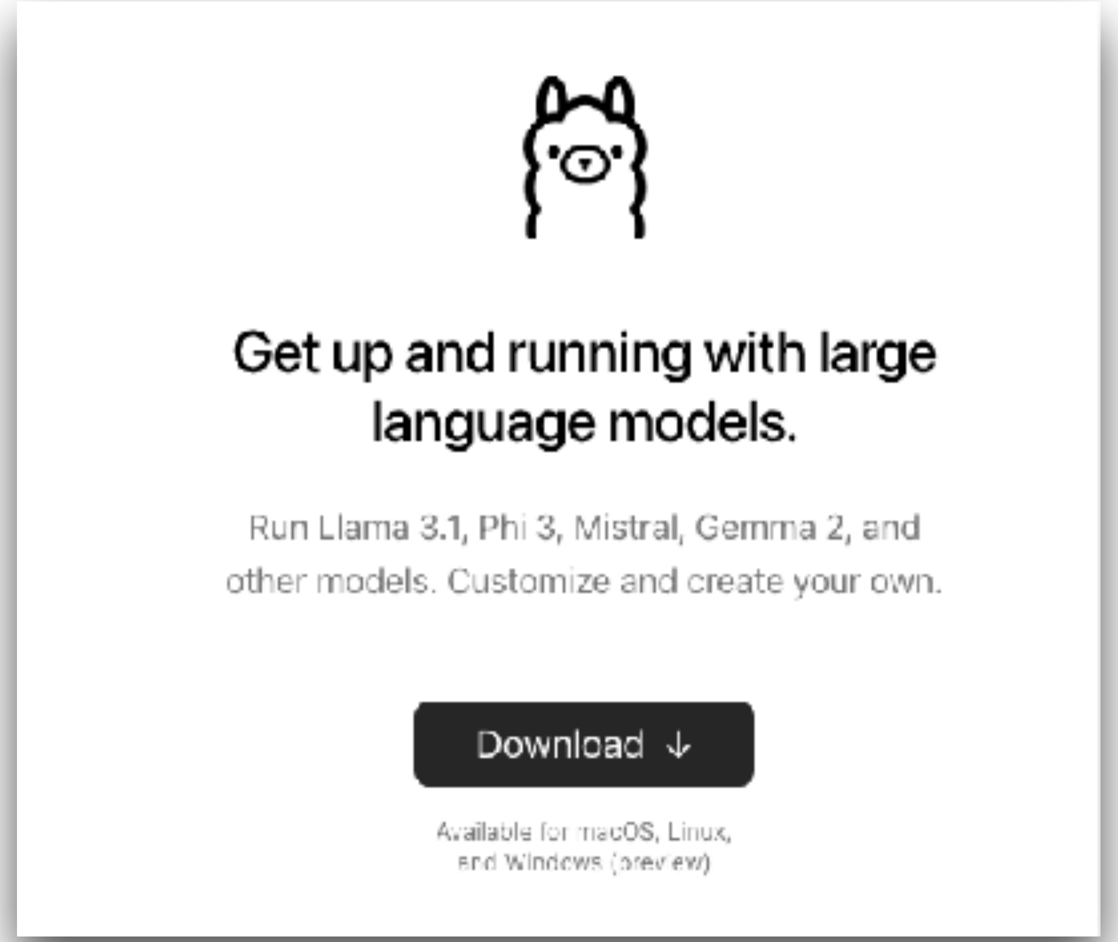
How to download models ?



# Local LLM with Ollama

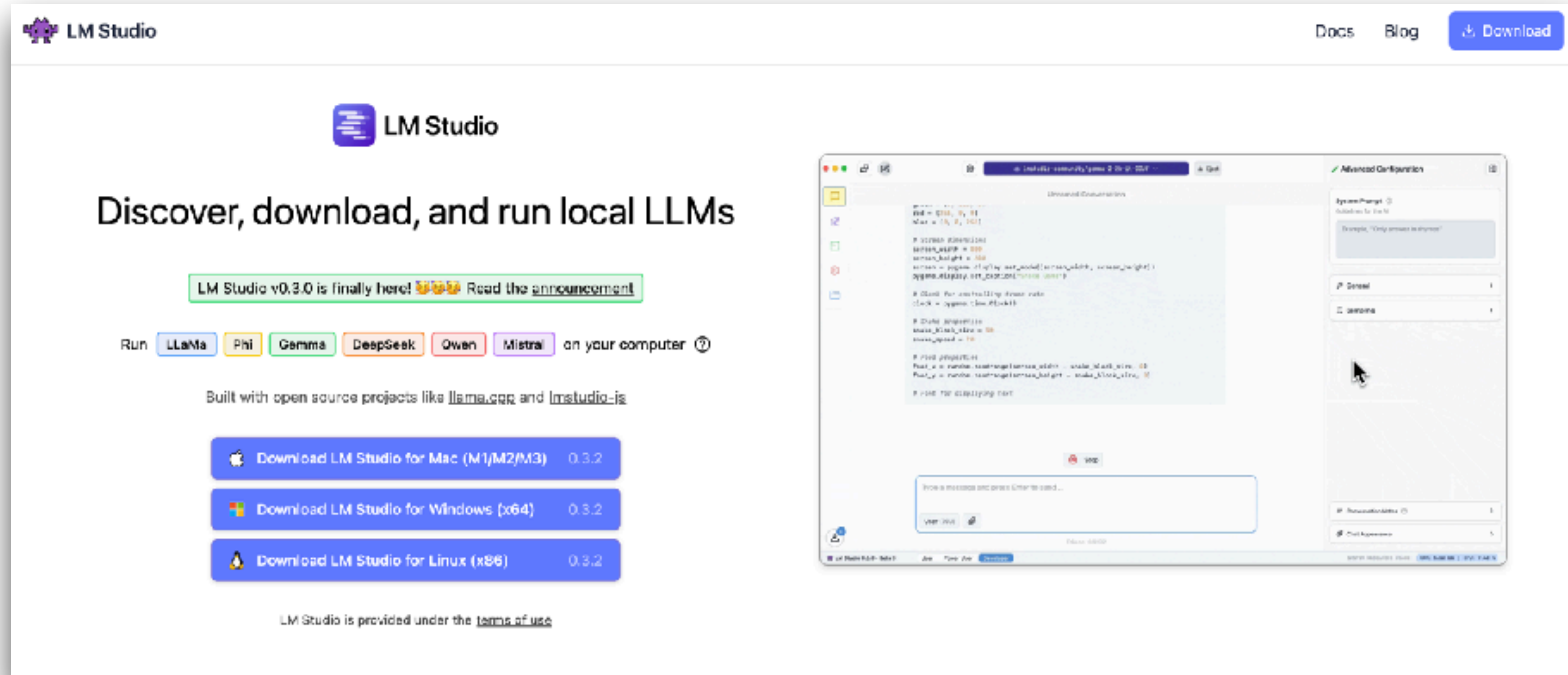
$ollama run **llama3.2**

LLM model




https://ollama.com/

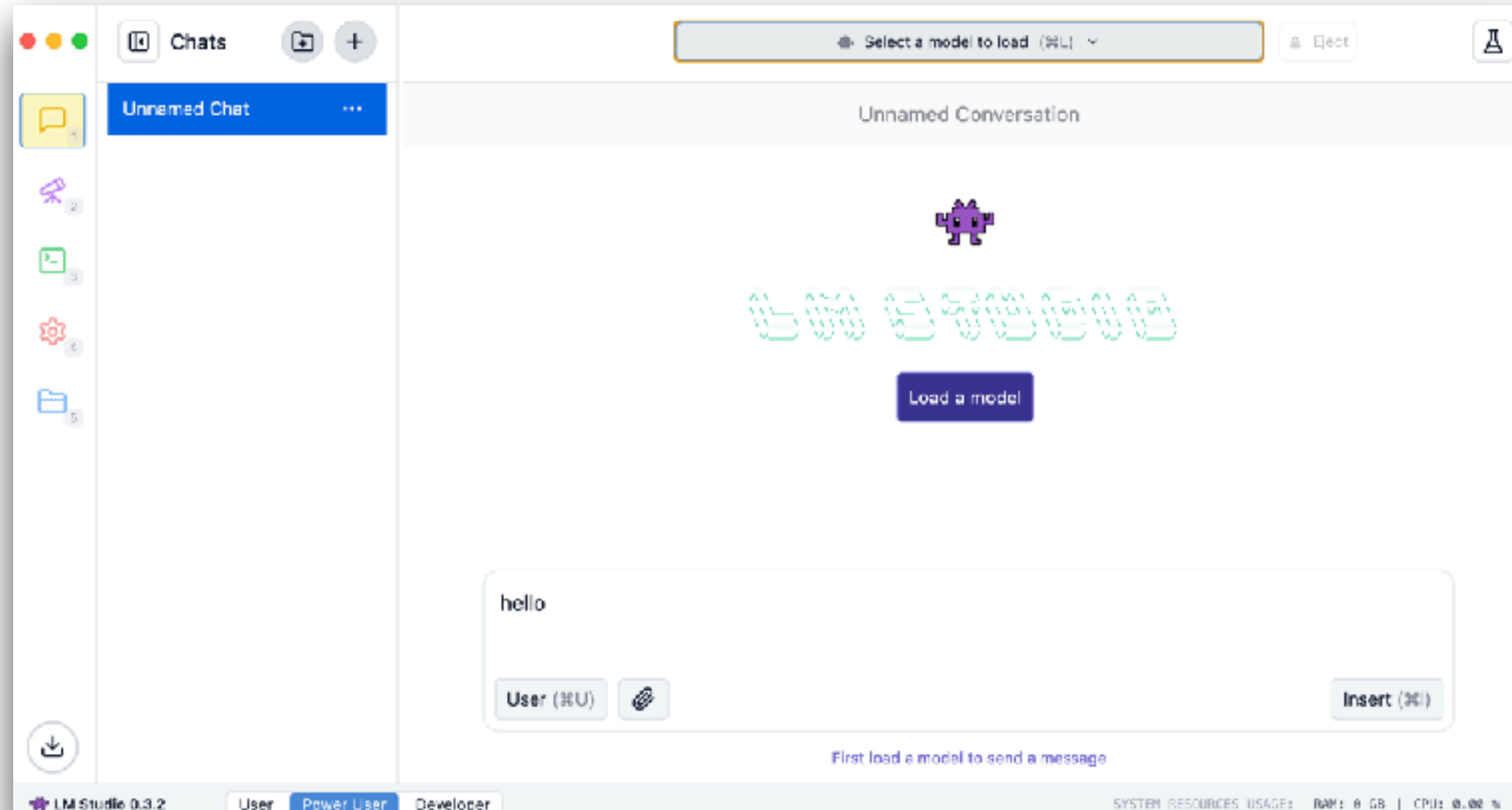
# Local LLM with LM Studio




https://lmstudio.ai/

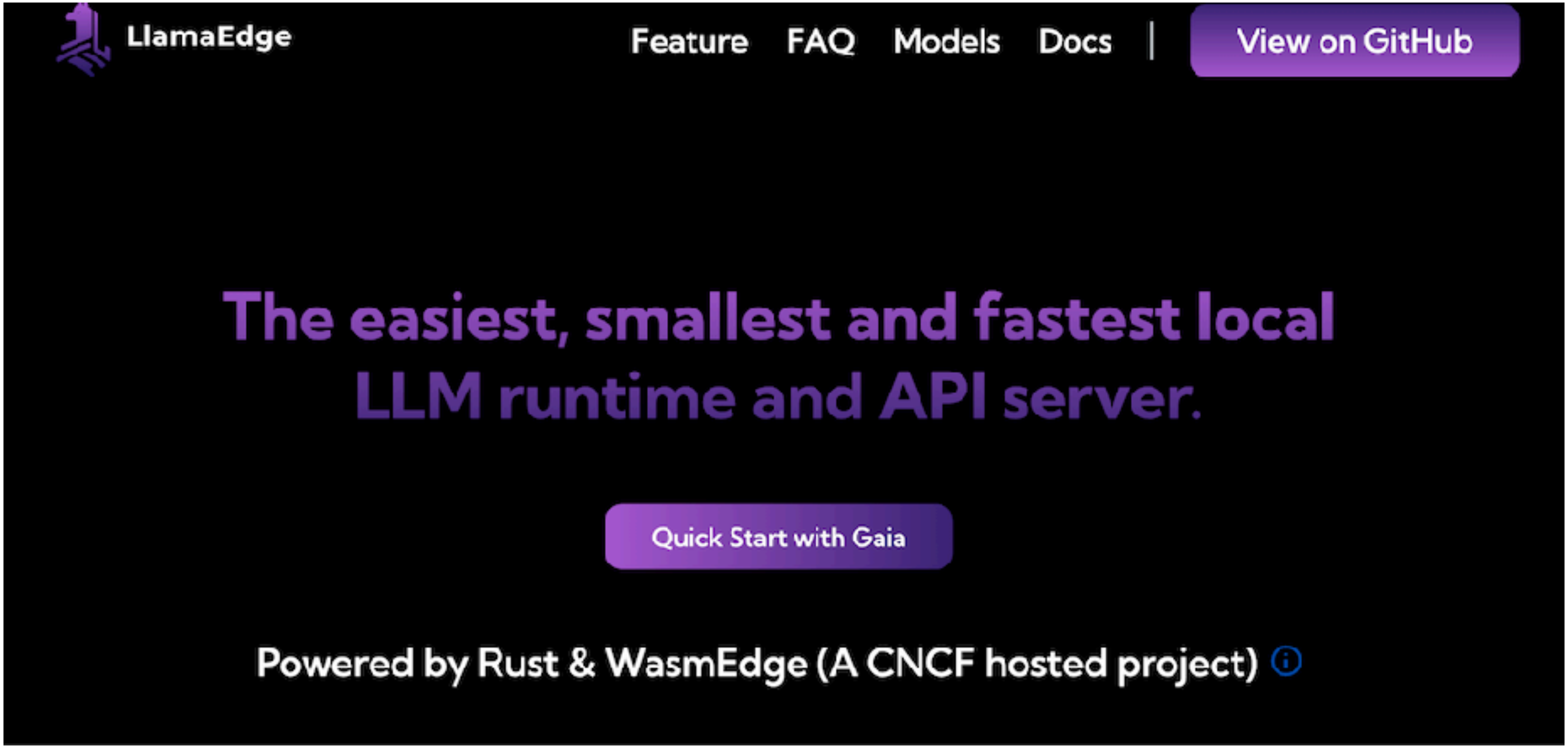
# Local LLM with LM Studio

## Load model from Hugging Face




https://lmstudio.ai/

# Local LLM with LlamaEdge




https://llamaedge.com/

# Local LLM with LocalAI

https://localai.io/

# More

GPT4All

LlamaFile

Jan.ai

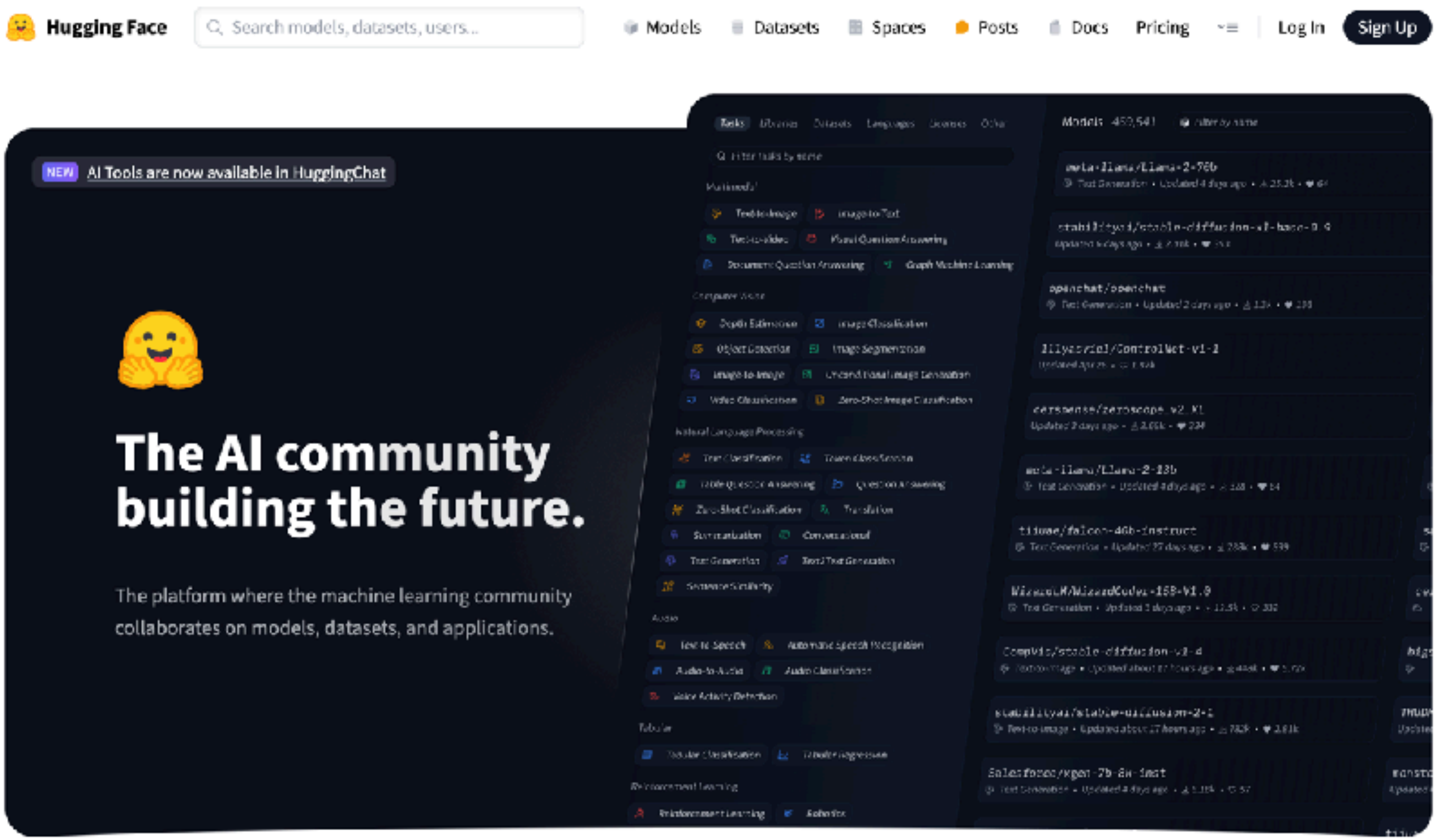
NextChat

Anything LLM

https://github.com/Hannibal046/Awesome-LLM

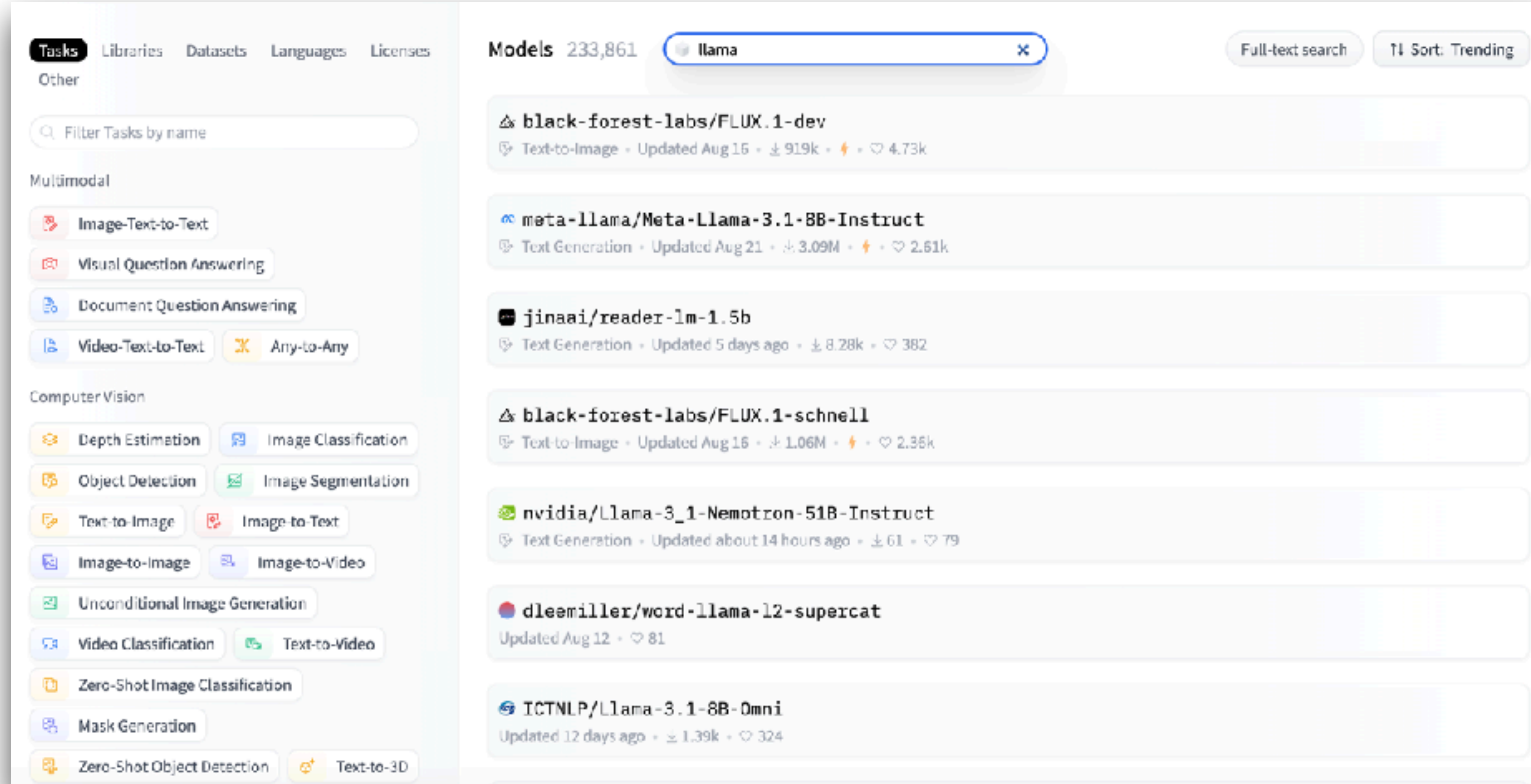
# LLM Models

# Hugging Face Model Hub



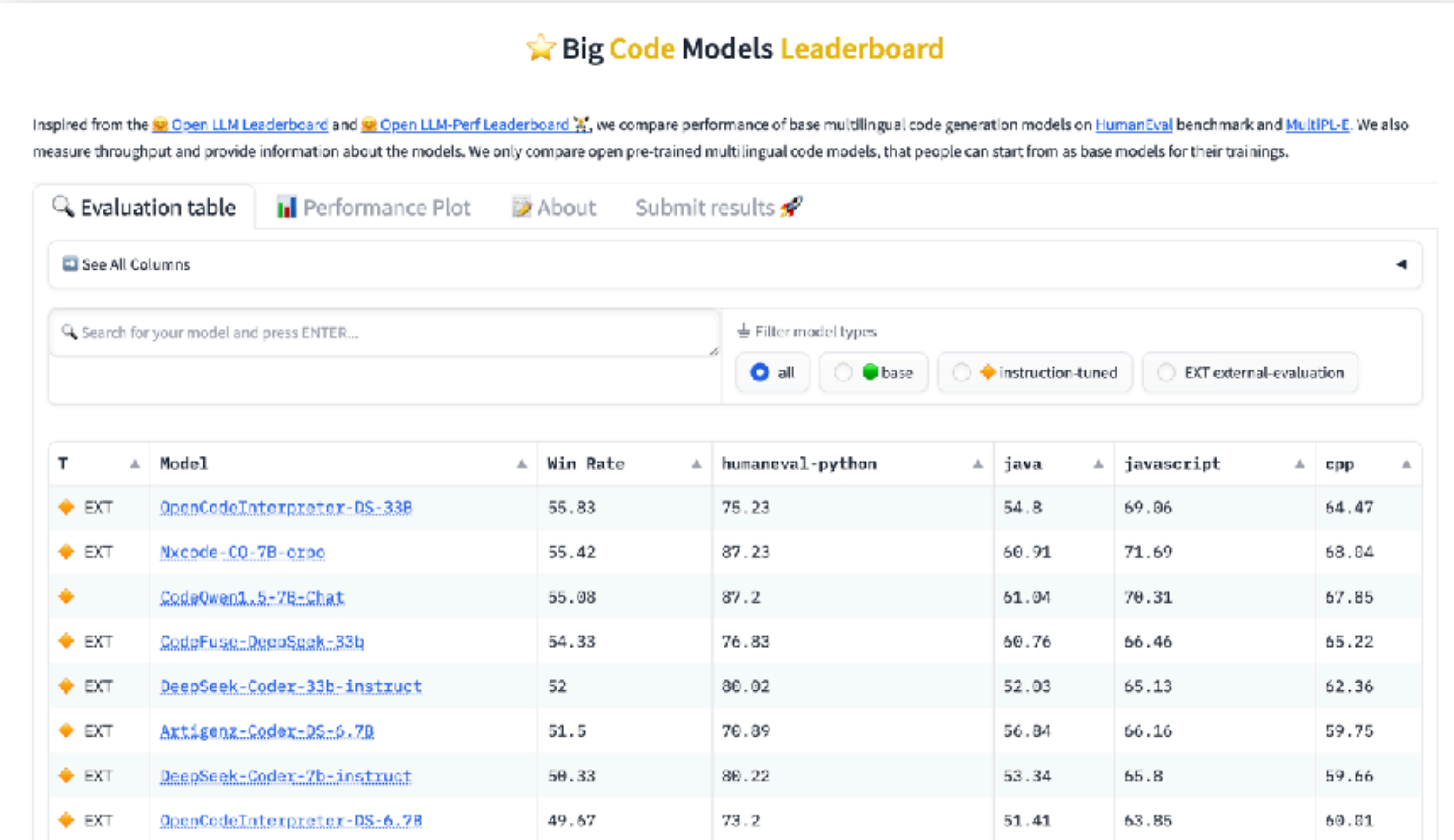

https://huggingface.co/

# Hugging Face :: Model




https://huggingface.co/

# Big Code model leader board



⭐ Big Code Models Leaderboard

Inspired from the 🤗 Open LLM Leaderboard and 🤗 Open LLM-Perf Leaderboard 🏋️, we compare performance of base multilingual code generation models on HumanEval benchmark and MultiPL-E. We also measure throughput and provide information about the models. We only compare open pre-trained multilingual code models, that people can start from as base models for their trainings.

🔍 Evaluation table | 📊 Performance Plot | 📝 About | Submit results 🚀

See All Columns

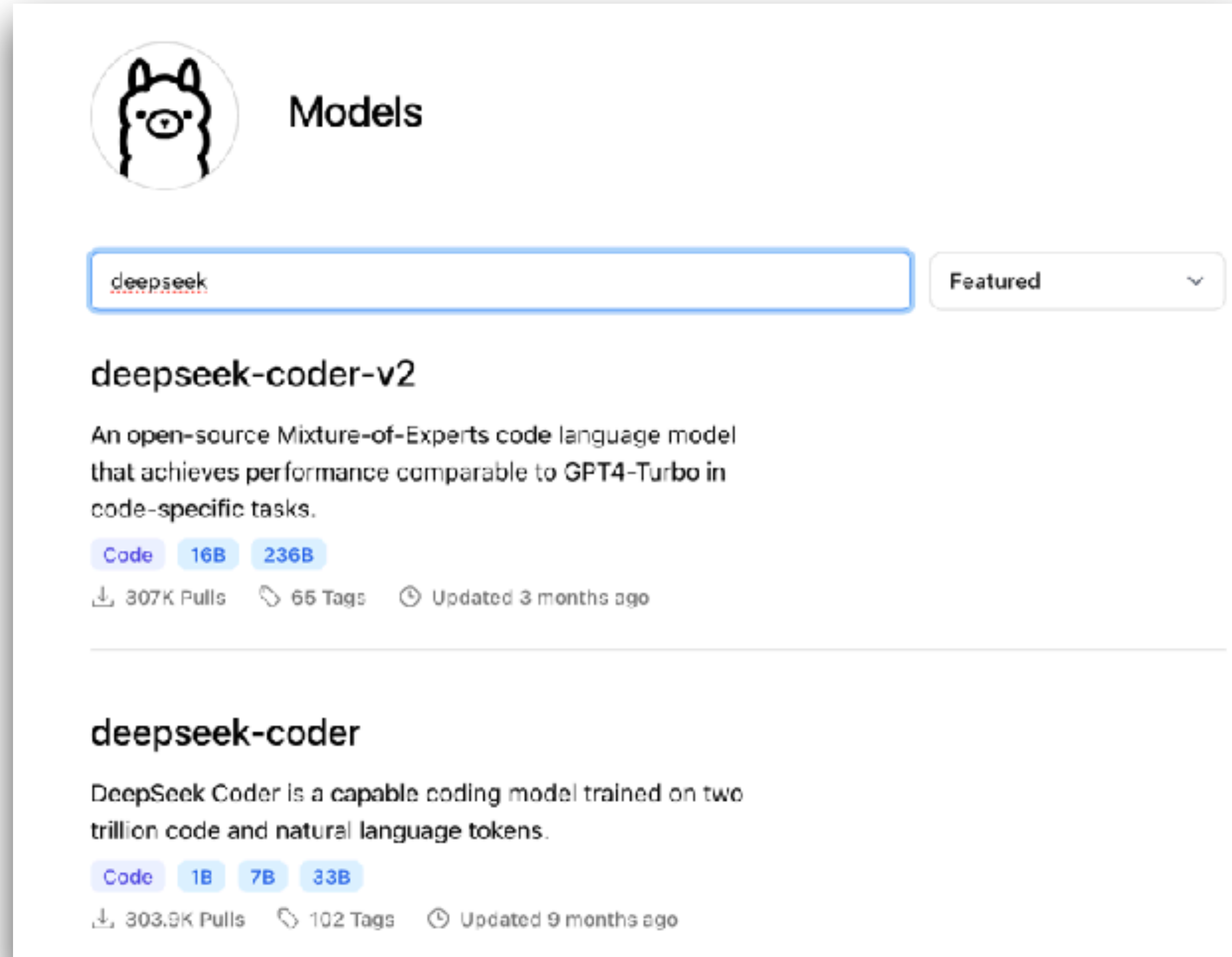
🔍 Search for your model and press ENTER...

Filter model types: all | 🟢 base | 🔶 instruction-tuned | EXT external-evaluation

| T | Model | Win Rate | humaneval-python | java | javascript | cpp |
|---|---|---|---|---|---|---|
| 🔶 EXT | OpenCodeInterpreter-DS-33B | 55.83 | 75.23 | 54.8 | 69.06 | 64.47 |
| 🔶 EXT | Nxcode-CQ-7B-orpo | 55.42 | 87.23 | 60.91 | 71.69 | 68.04 |
| 🔶 | CodeQwen1.5-7B-Chat | 55.08 | 87.2 | 61.04 | 70.31 | 67.85 |
| 🔶 EXT | CodeFuse-DeepSeek-33b | 54.33 | 76.83 | 60.76 | 66.46 | 65.22 |
| 🔶 EXT | DeepSeek-Coder-33b-instruct | 52 | 80.02 | 52.03 | 65.13 | 62.36 |
| 🔶 EXT | Artigenz-Coder-DS-6.7B | 51.5 | 70.89 | 56.84 | 66.16 | 59.75 |
| 🔶 EXT | DeepSeek-Coder-7b-instruct | 50.33 | 80.22 | 53.34 | 65.8 | 59.66 |
| 🔶 EXT | OpenCodeInterpreter-DS-6.7B | 49.67 | 73.2 | 51.41 | 63.85 | 60.81 |

https://huggingface.co/spaces/bigcode/bigcode-models-leaderboard

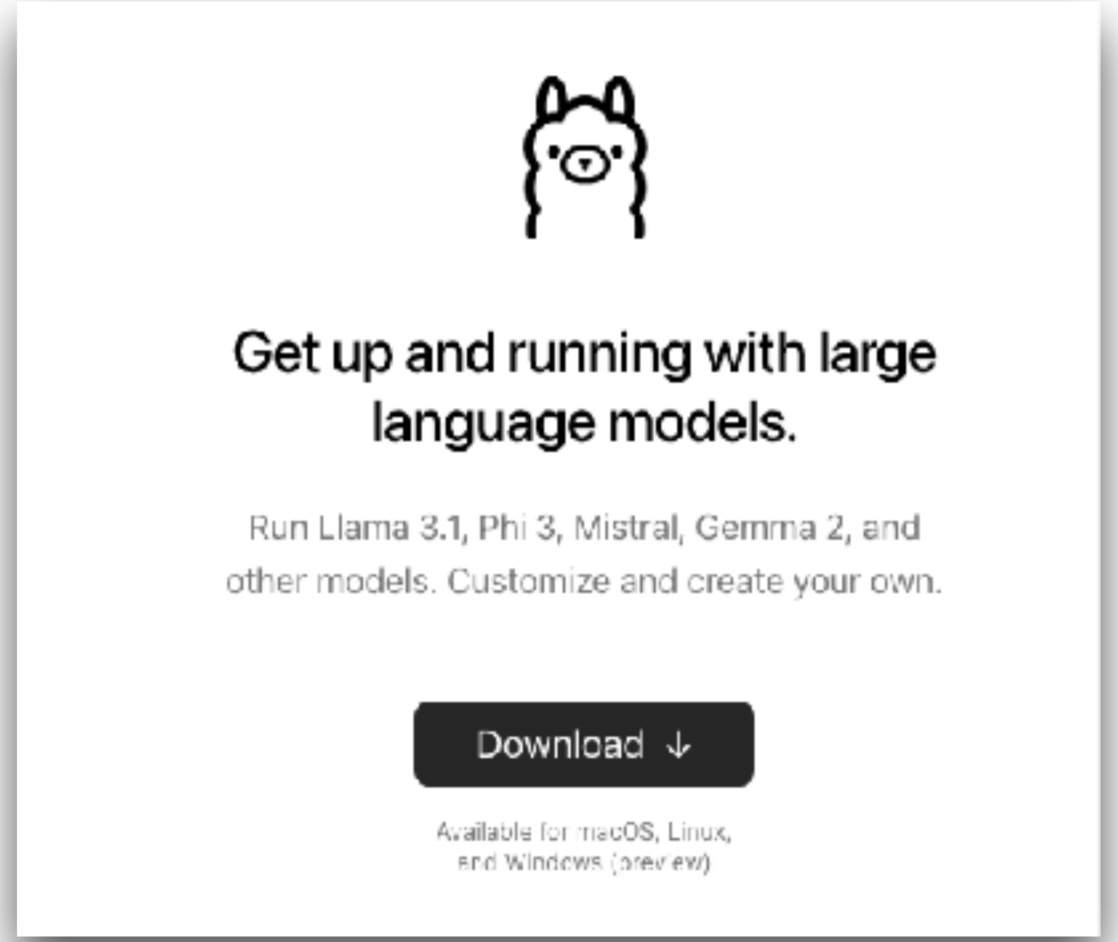
# Model in Ollama




https://ollama.com/library

# Workshop with Ollama

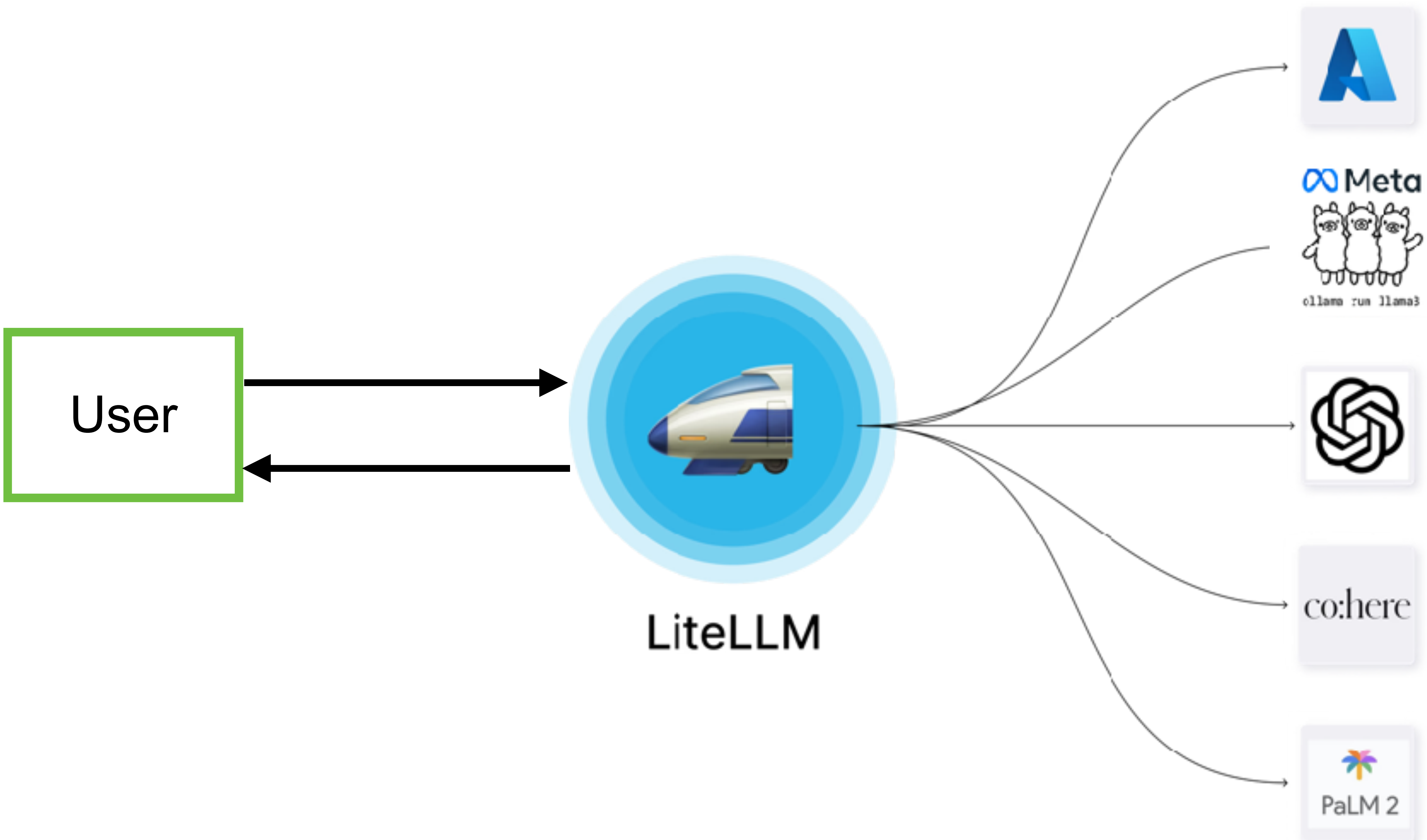
$ollama run **llama3.1**




https://github.com/up1/workshop-ai-with-technical-team/wiki/Local-LLM-with-Ollama

# LiteLLM as a Proxy

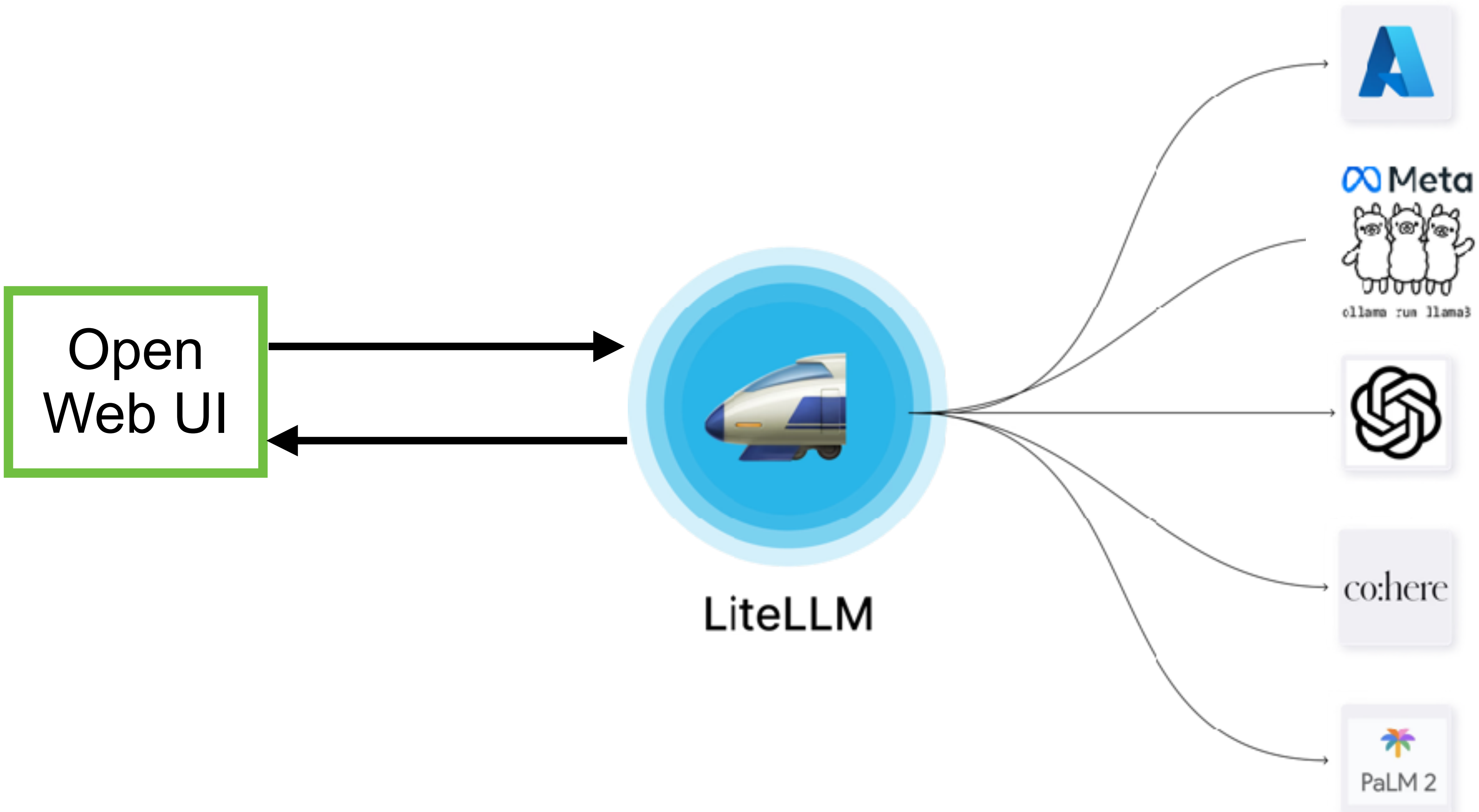
# LiteLLM as a Proxy




https://www.litellm.ai/

# Workshop

Use docker compose to build and run

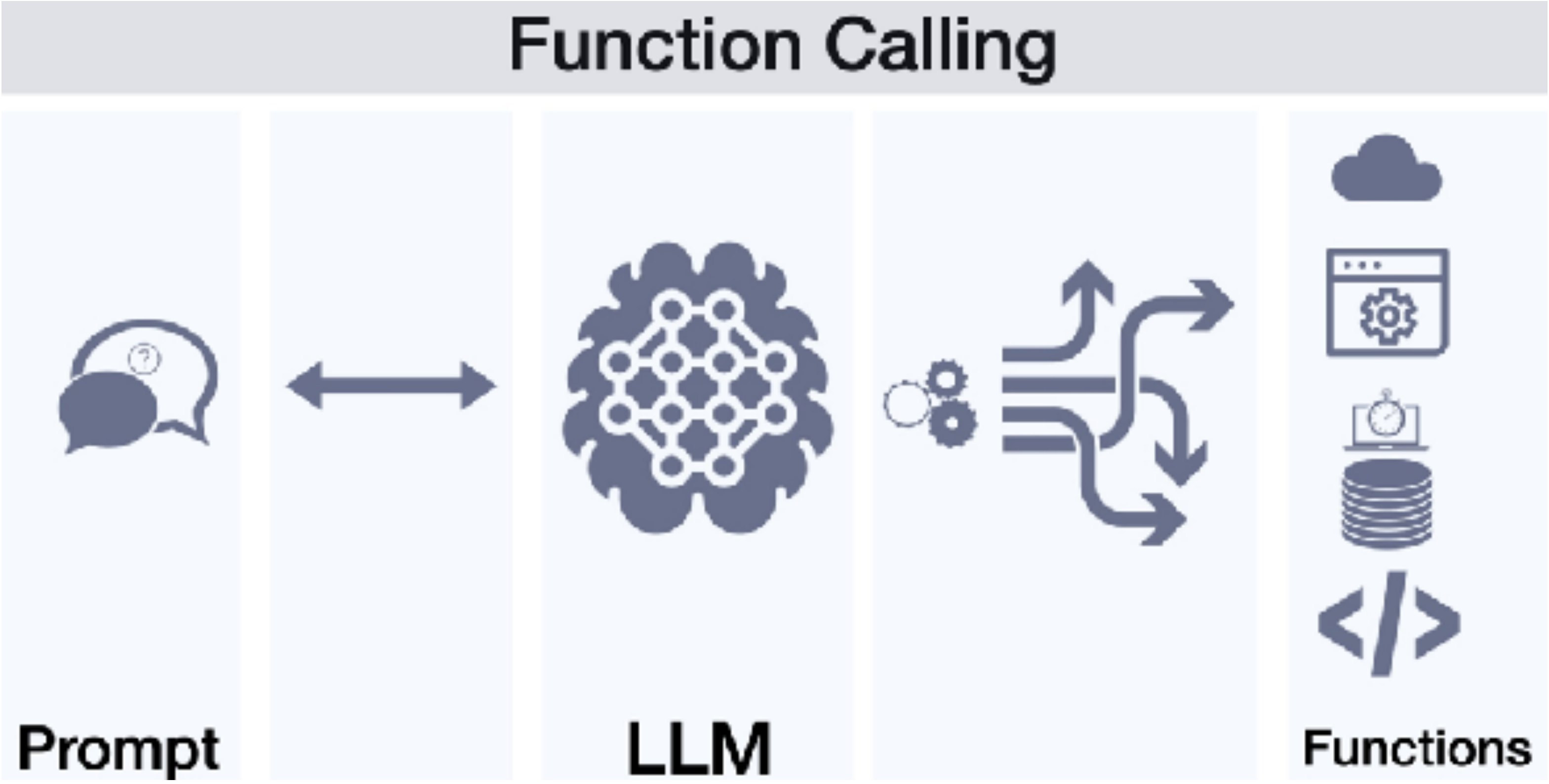



https://github.com/up1/workshop-ai-with-technical-team/wiki/LiteLLM-and-WebUI

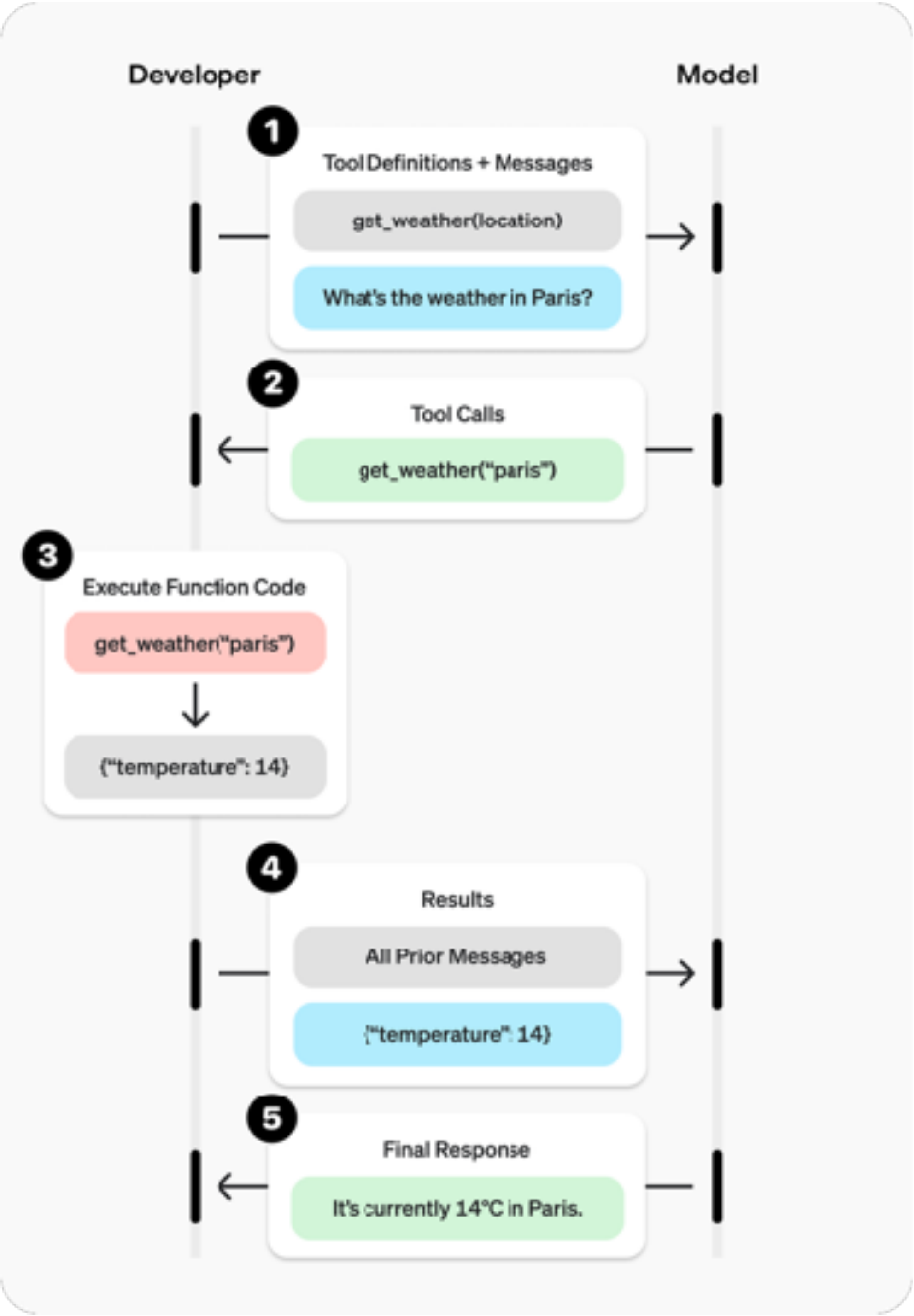
# Function Calling ?

# Function Calling ?

Allow LLM to recognize what tool it need based on user's input and when to invoke it.

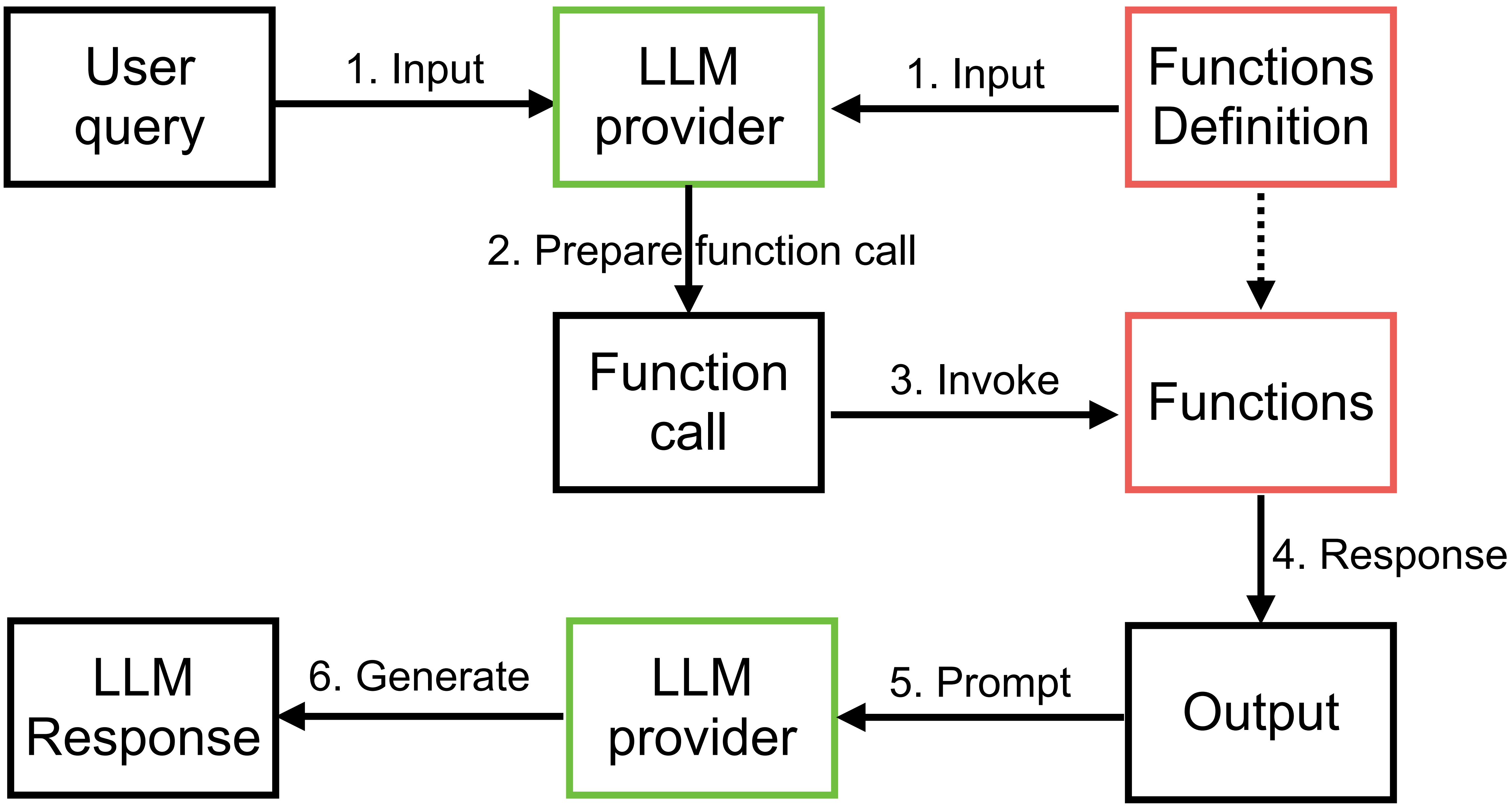


# Function calling ?




https://platform.openai.com/docs/guides/function-calling?api-mode=responses

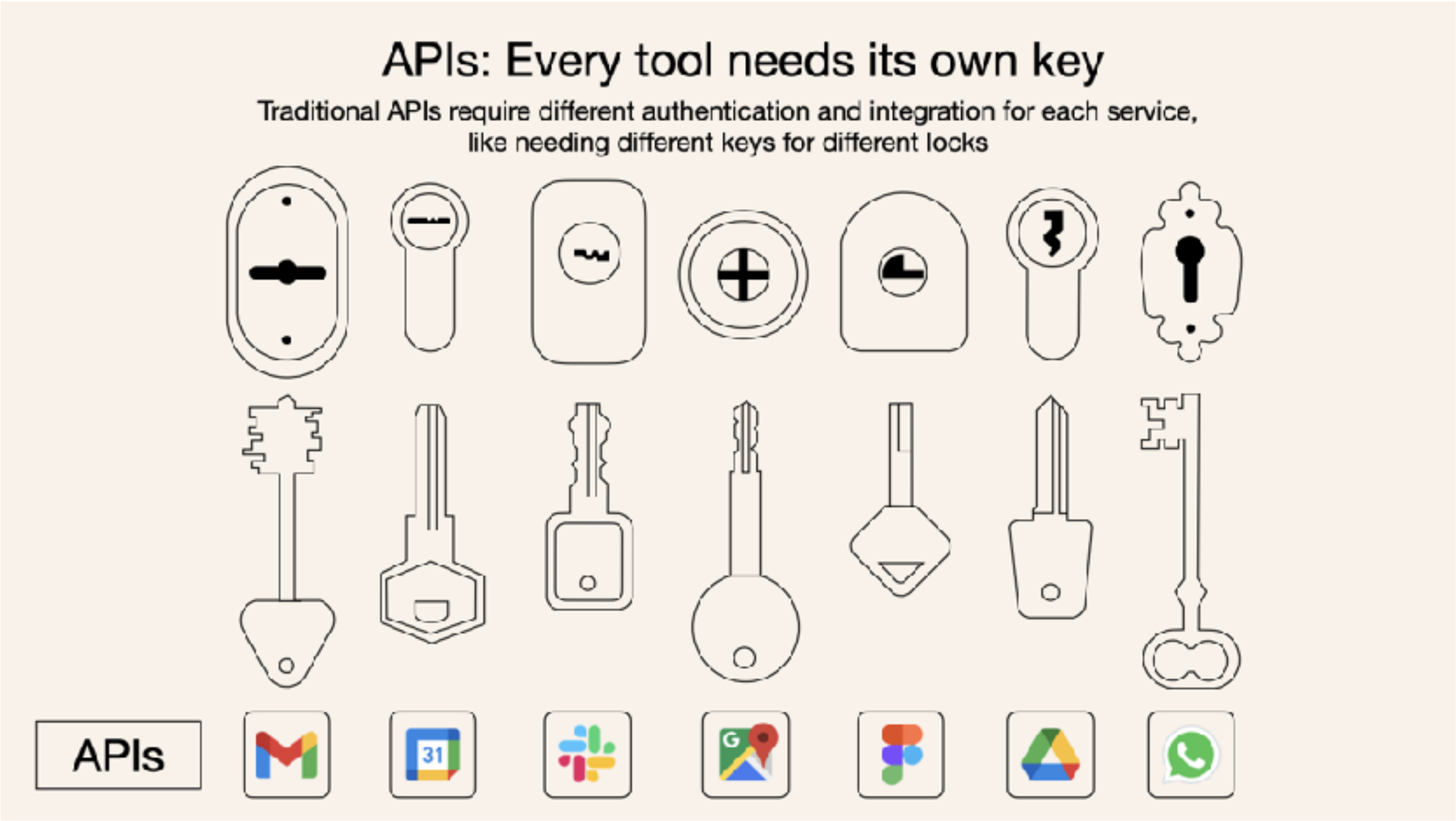
# Function Calling



# Support function calling

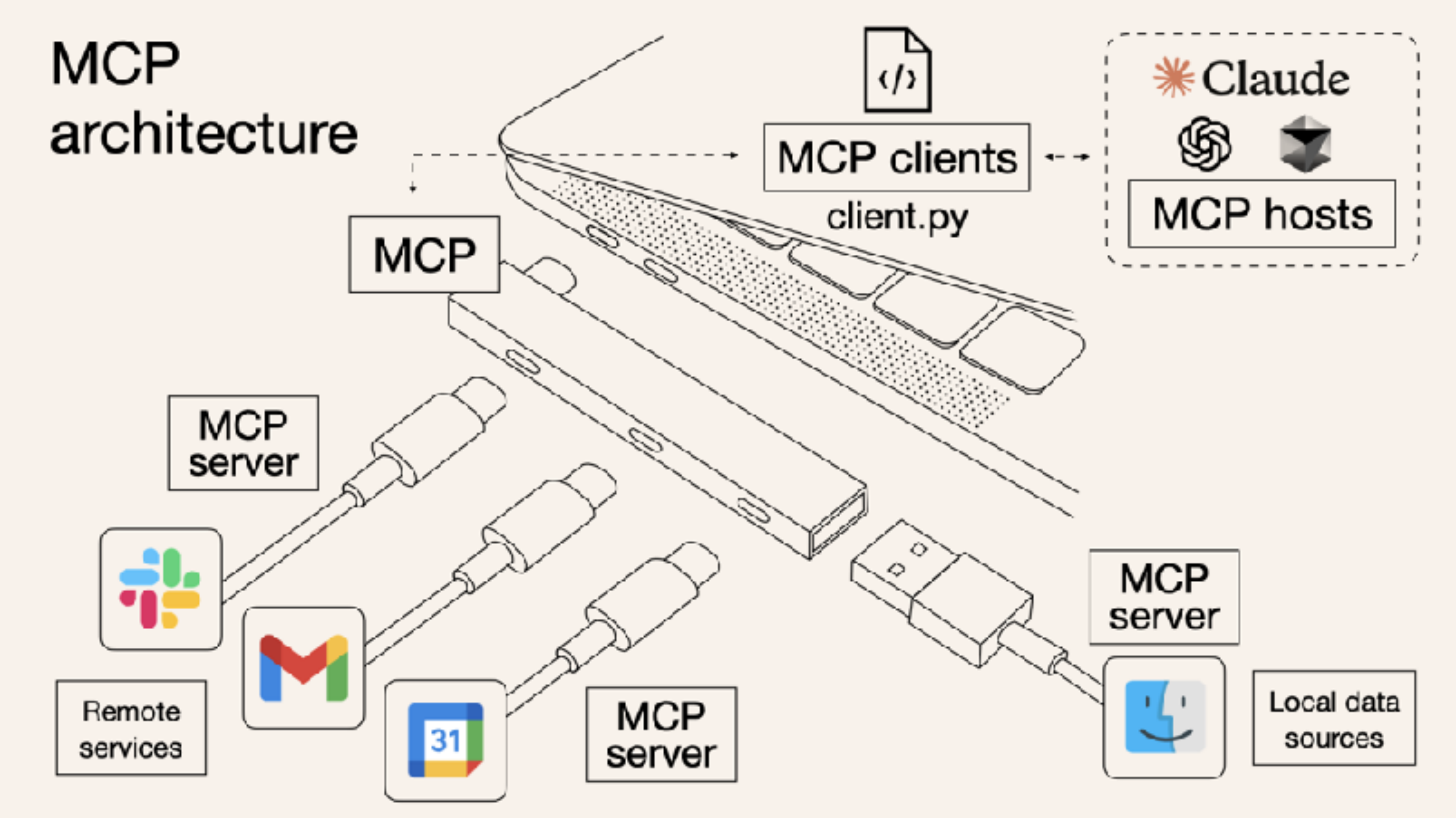
| Provider name | Model name |
|---|---|
| OpenAI | GPT 4 |
| Anthropic | Claude 3<br>(Sonnet, Haiku, Opus) |
| Google | Gemini |

# Function calling !!



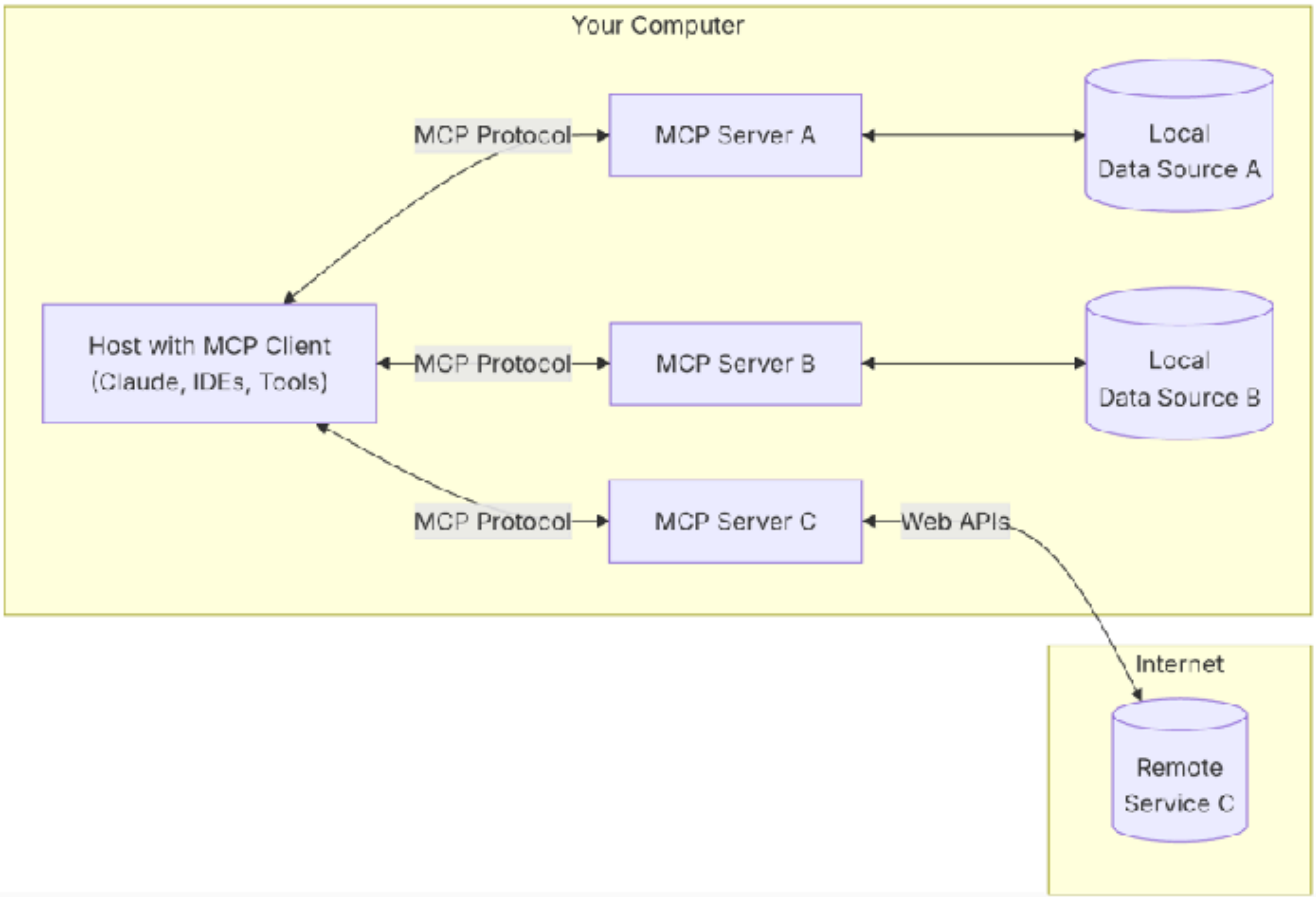

https://norahsakal.com/blog/mcp-vs-api-model-context-protocol-explained/

# Standardized function calling !!



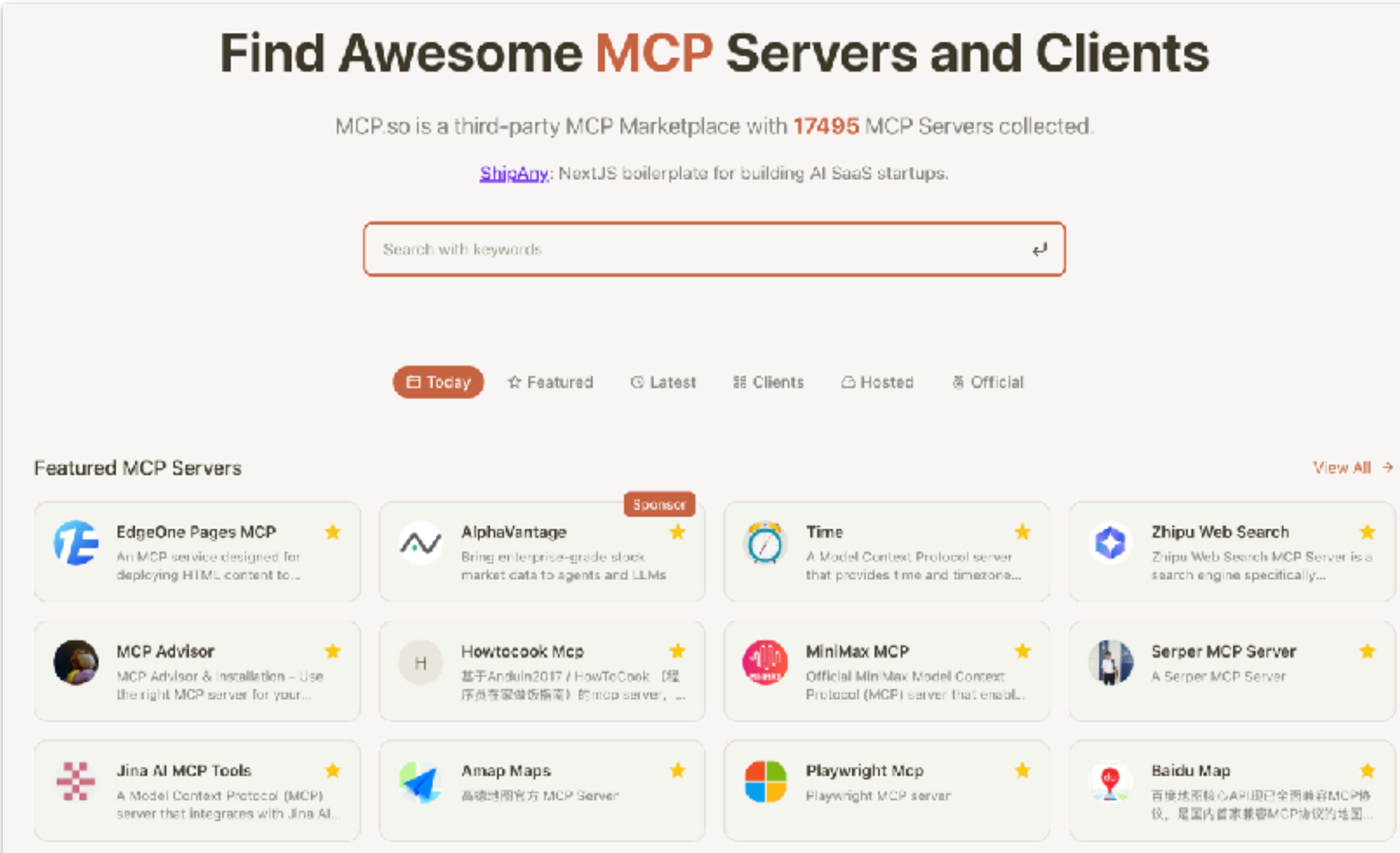

https://norahsakal.com/blog/mcp-vs-api-model-context-protocol-explained/

# Model Context Protocol




https://modelcontextprotocol.io/

# Awesome MCP




https://mcp.so/

# Retrieval-Augmented Generation (RAG)

# RAG

Enhances LLMs by retrieving external knowledge before generating response

| | | |
|---|---|---|
| Improve accuracy | Reduce hallucinations | Real-time knowledge updates |

# Core Components

## Retriever

Fetch relevant documents from a knowledge base

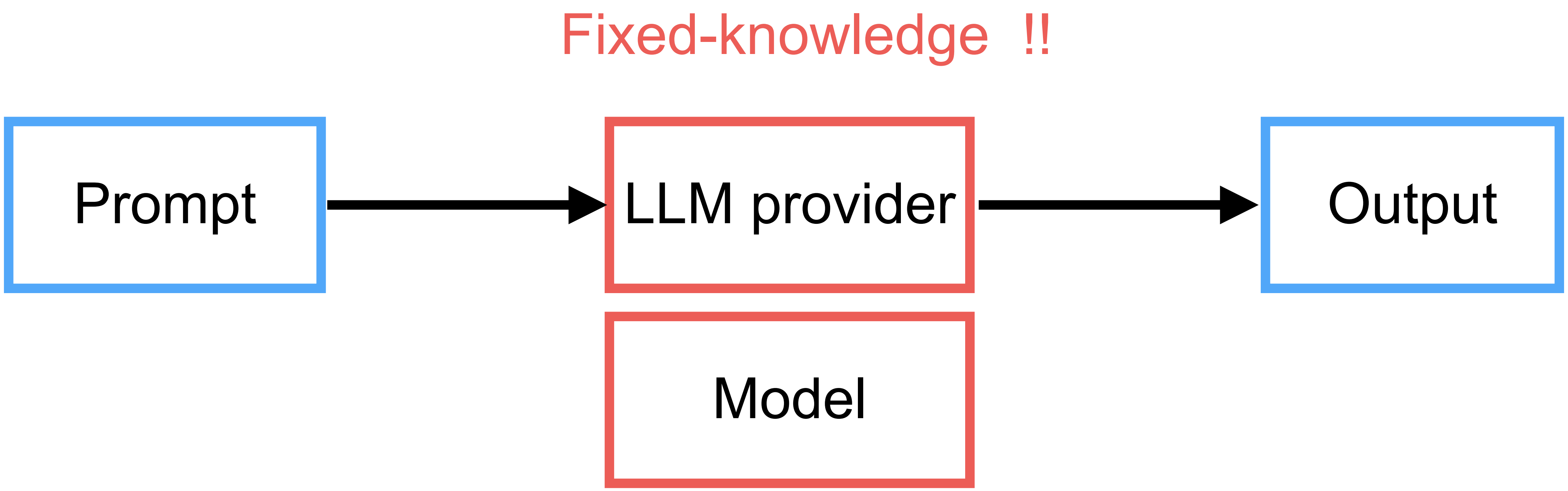
## Generator

Uses the retrieved information to generate a response

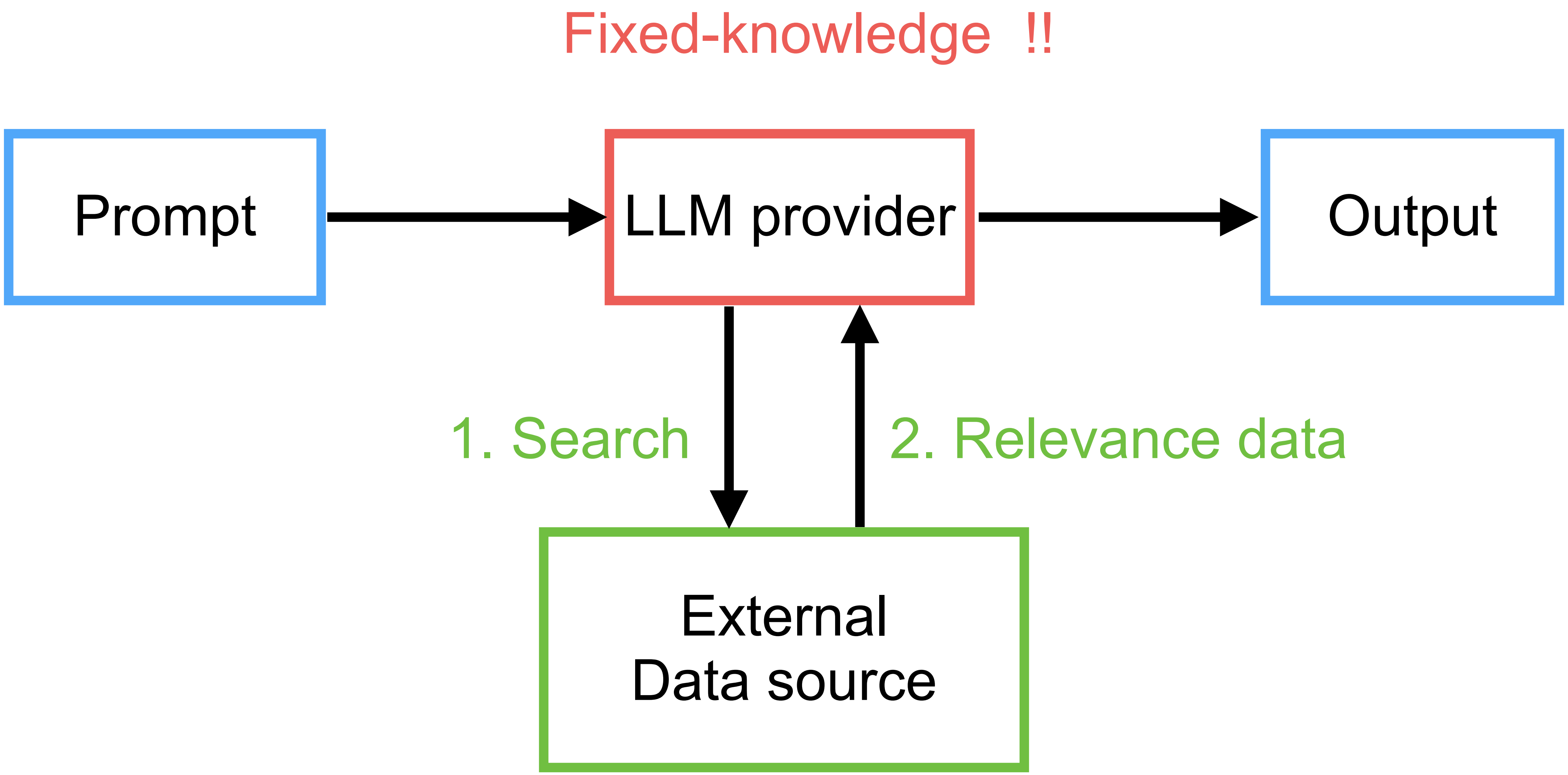
## Enhancements

Different RAG architectures modify how retrieval and generation interact to improve performance

# Limitation of LLM



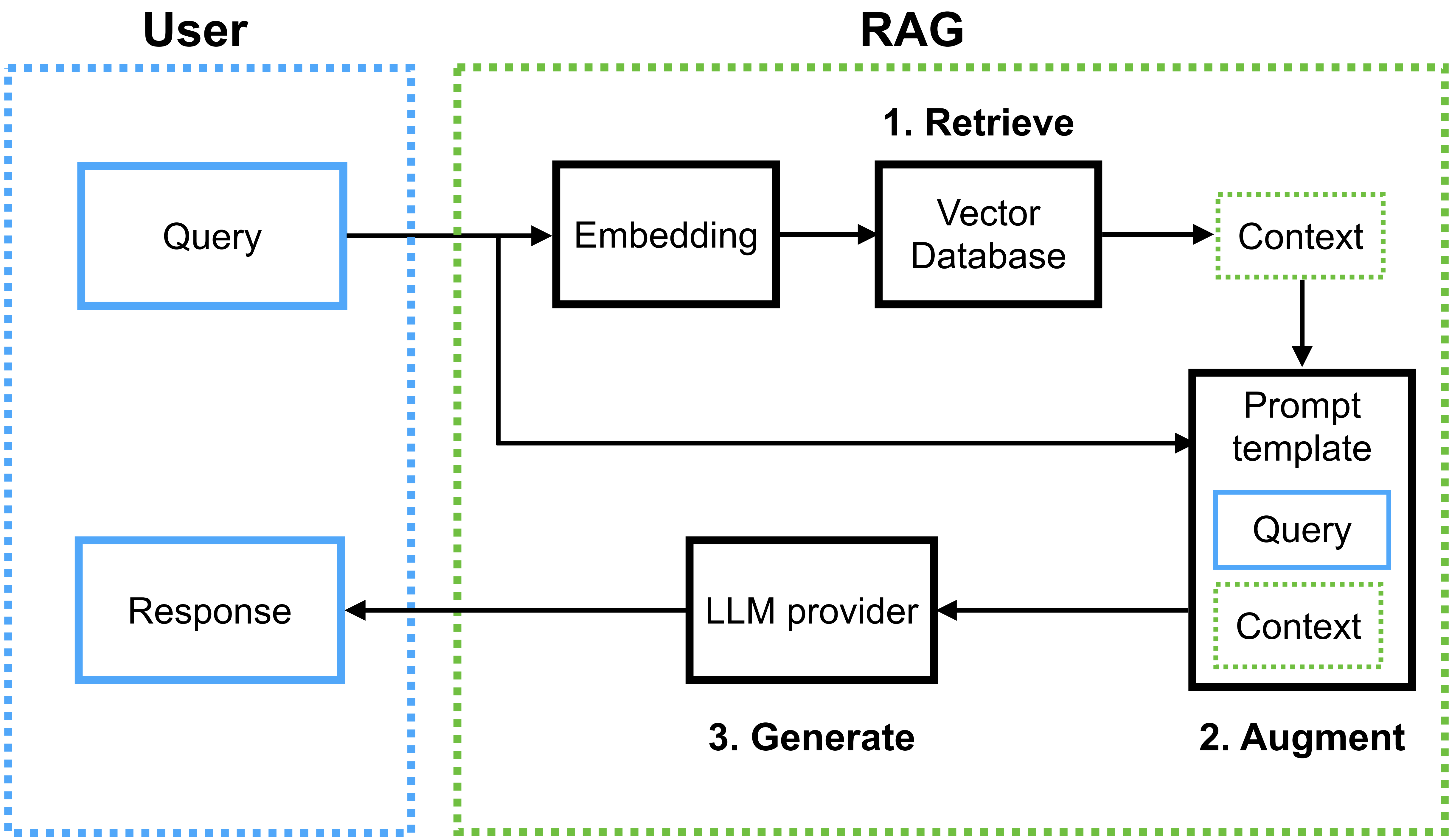
# RAG ?


Fixed-knowledge !!
Prompt
LLM provider
Output
1. Search
2. Relevance data
External
Data source

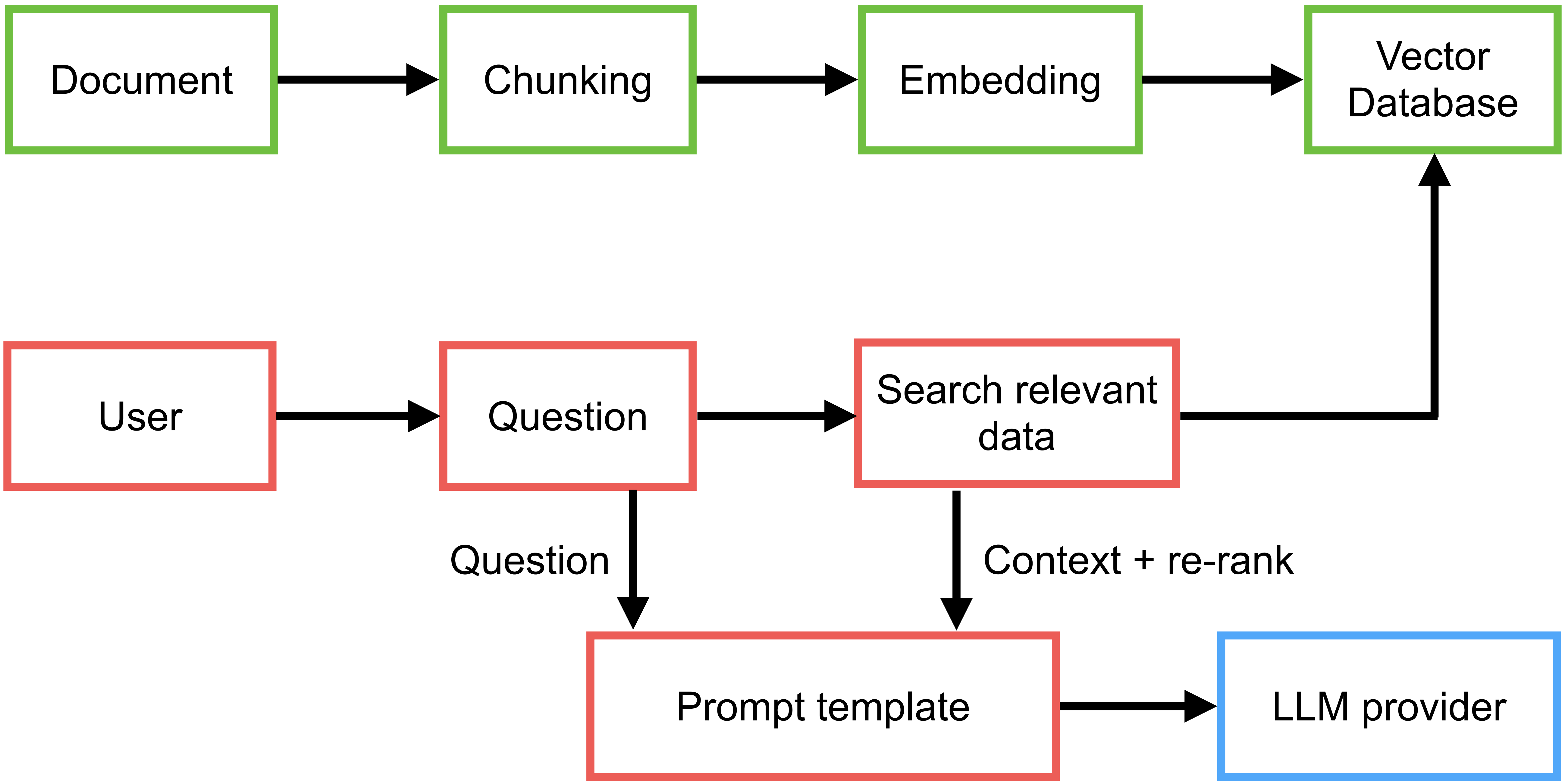
# How to search/retrieve data ?

| | |
|---|---|
| Data source | Data preparation |
| Search | Search result |

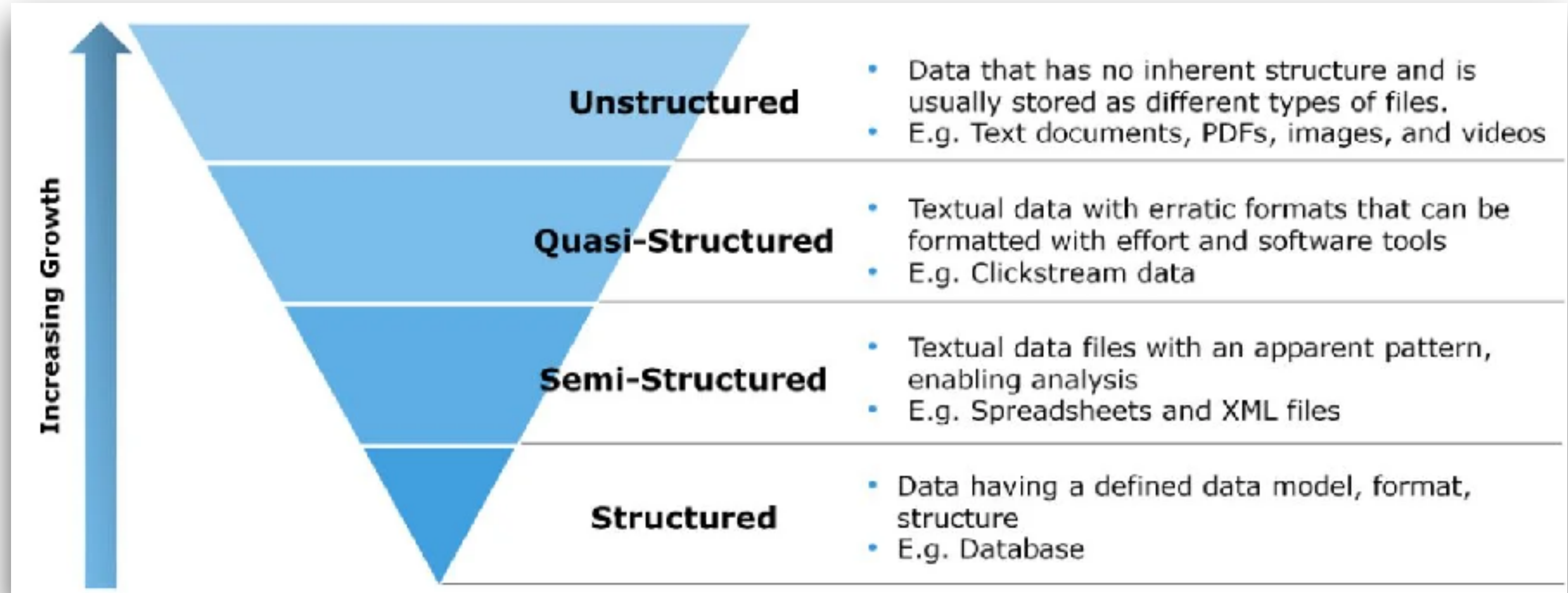
# Basic RAG Architecture



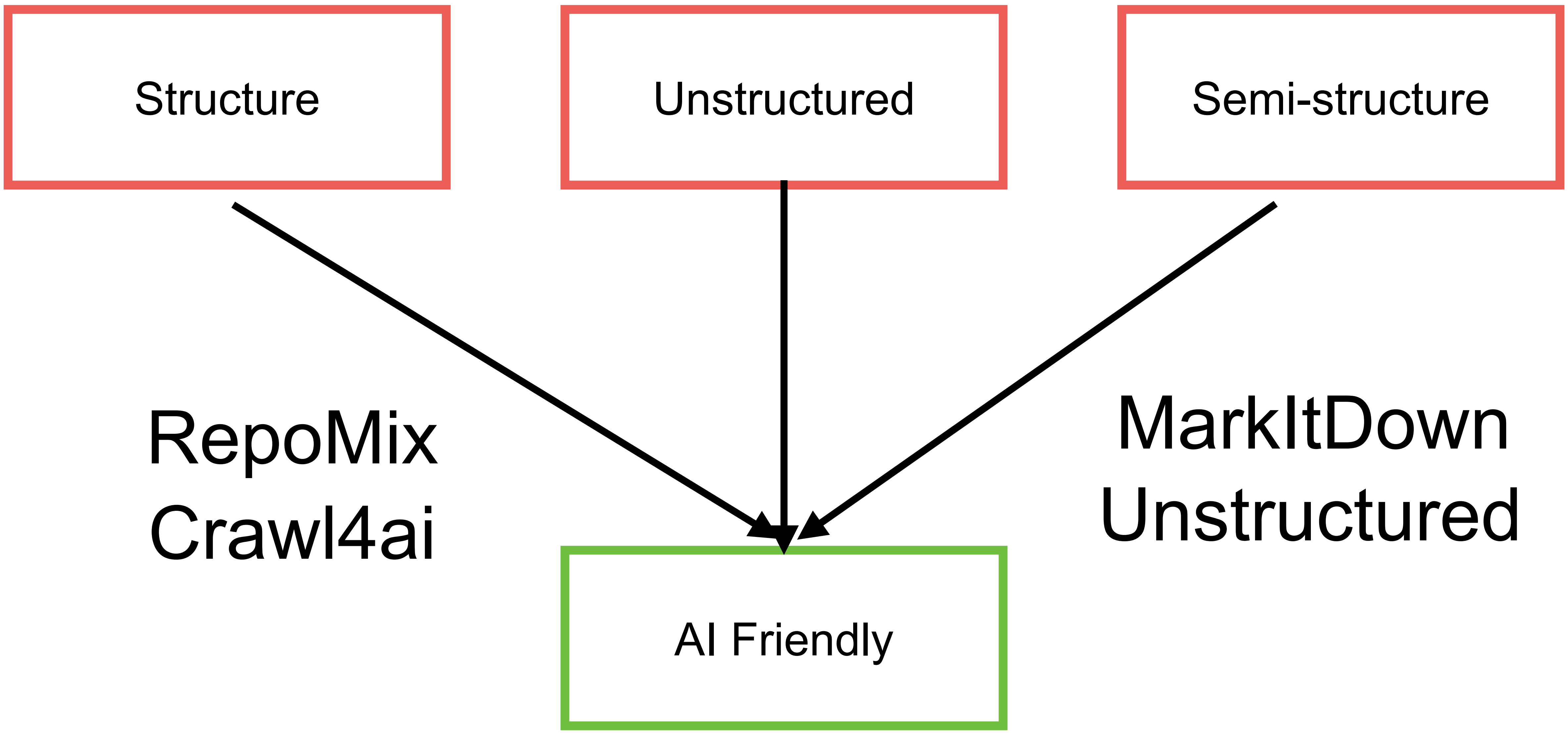
# RAG process


Document
Chunking
Embedding
Vector Database
User
Question
Search relevant data
Question
Context + re-rank
Prompt template
LLM provider

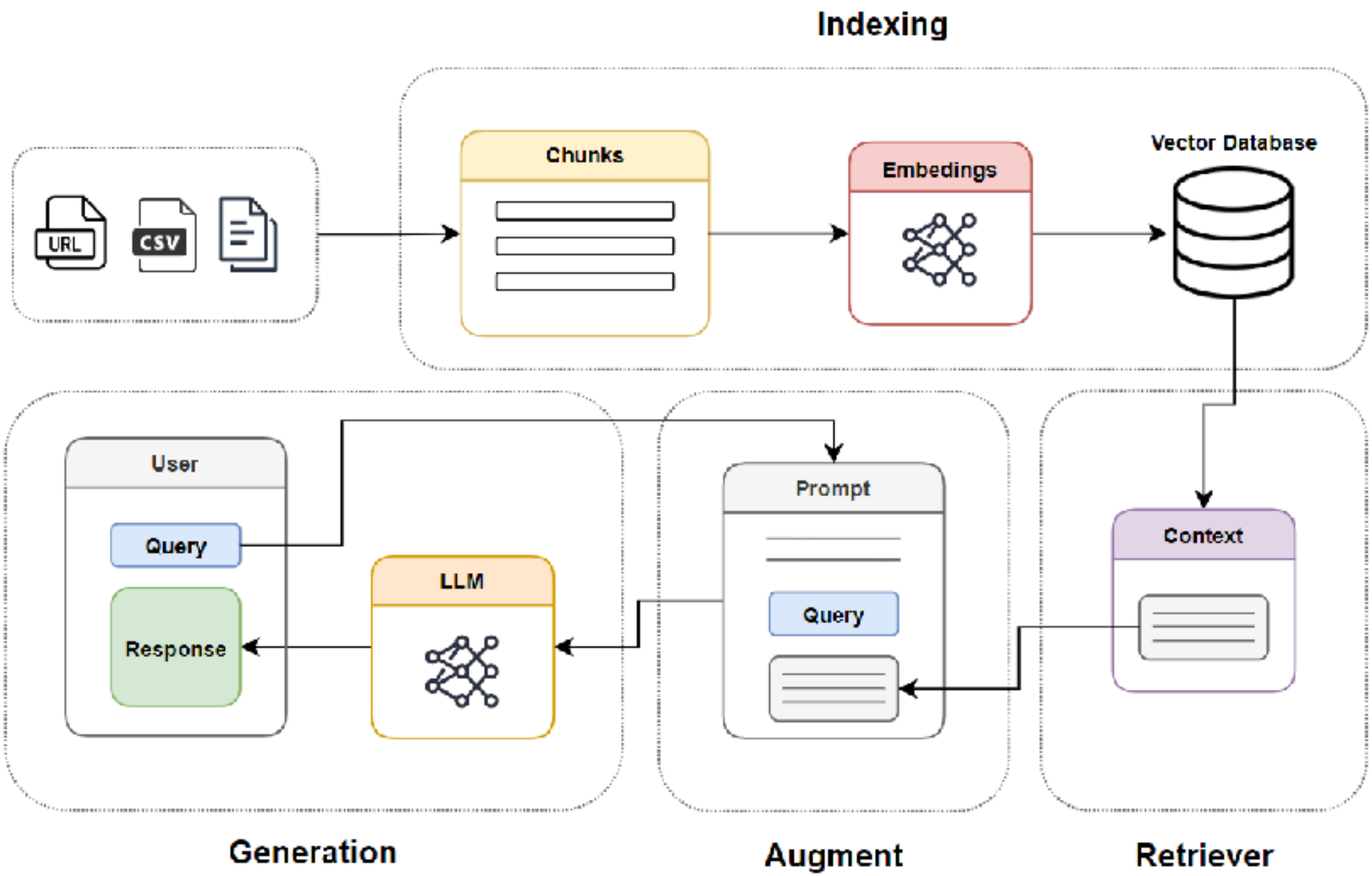
# Structures of Data ?



# Friendly Data for LLM/AI

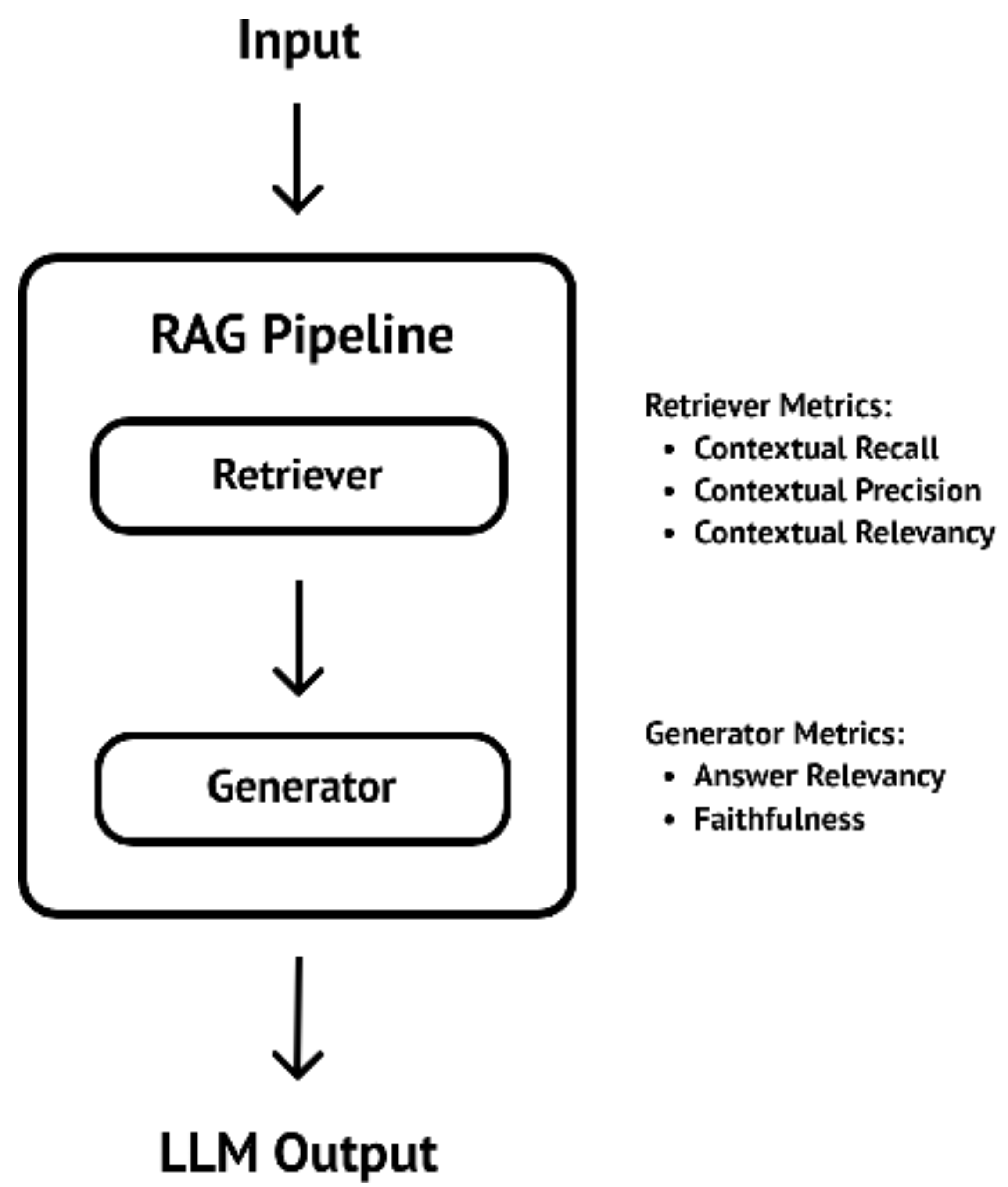


# RAG Cookbook (techniques)

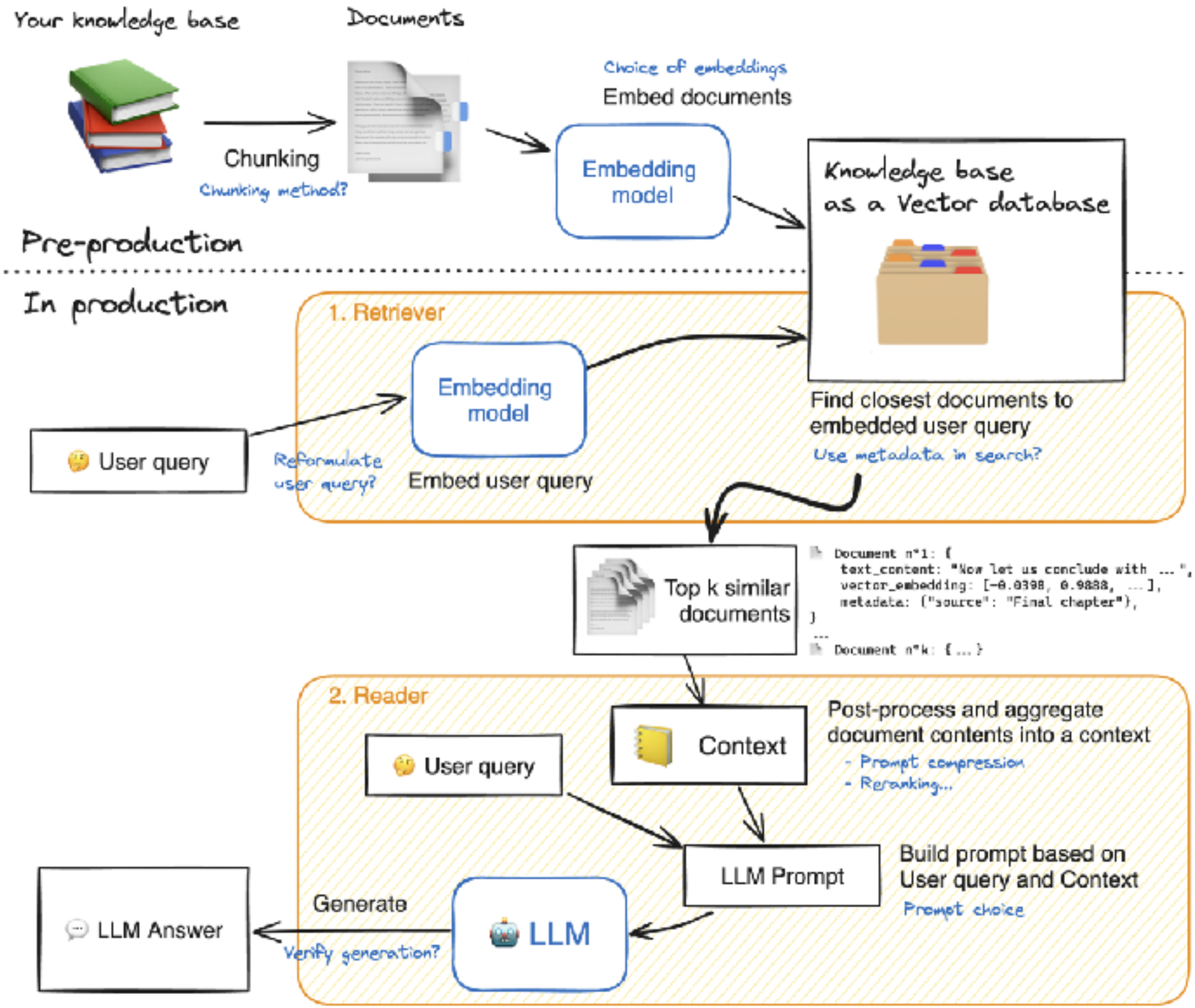



https://github.com/athina-ai/rag-cookbooks

# RAG Evaluation !!




https://www.deepeval.com/guides/guides-rag-evaluation




https://huggingface.co/learn/cookbook/en/rag_evaluation

# RAG Implementation

Document loading

Splitting or Chunking

Embedding

Store in Vector DB

Retrieval and Re-ranking

Send Prompt to LLM

Output

# Failure Points of RAG

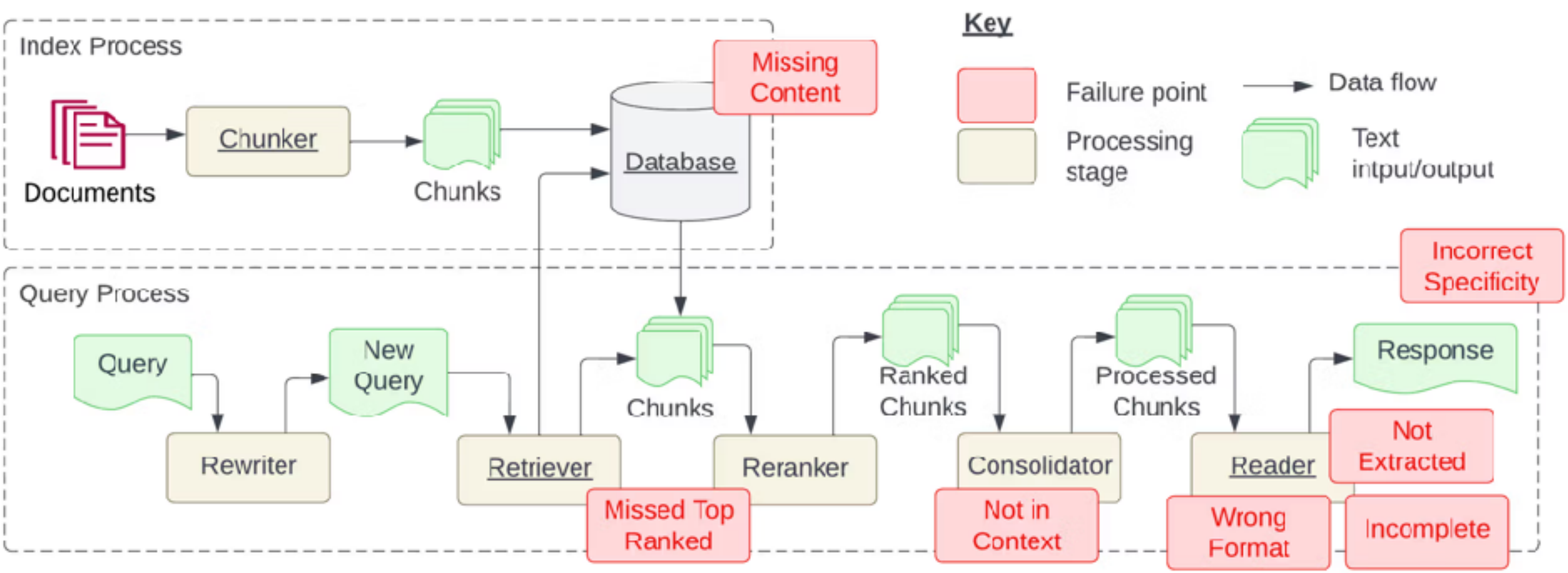



**Figure 1: Indexing and Query processes required for creating a Retrieval Augmented Generation (RAG) system. The indexing process is typically done at development time and queries at runtime. Failure points identified in this study are shown in red boxes. All required stages are underlined. Figure expanded from [19].**

https://www.galileo.ai/blog/mastering-rag-how-to-architect-an-enterprise-rag-system

# RAG is better

## But come with Cost !!

More latency

Retrieval errors

Accuracy !!

More Complexity

Maintenance overhead

# RAG Techniques ?

Semantic chunking

Chunk size selector

Context chunk header

Adaptive RAG

Re-ranking

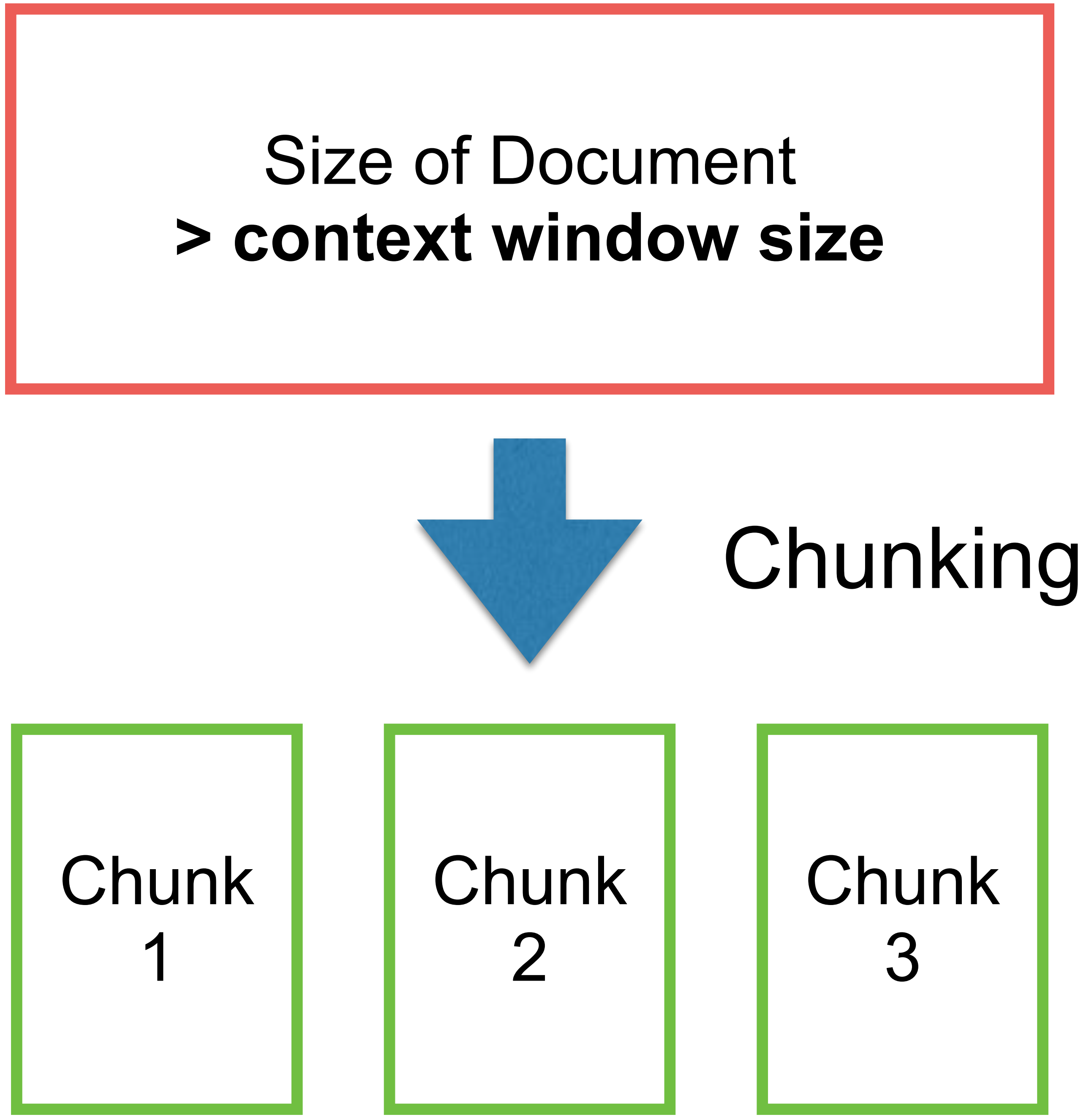
Graph TAG

https://github.com/FareedKhan-dev/all-rag-techniques

# Pre-Retrieval optimization ?

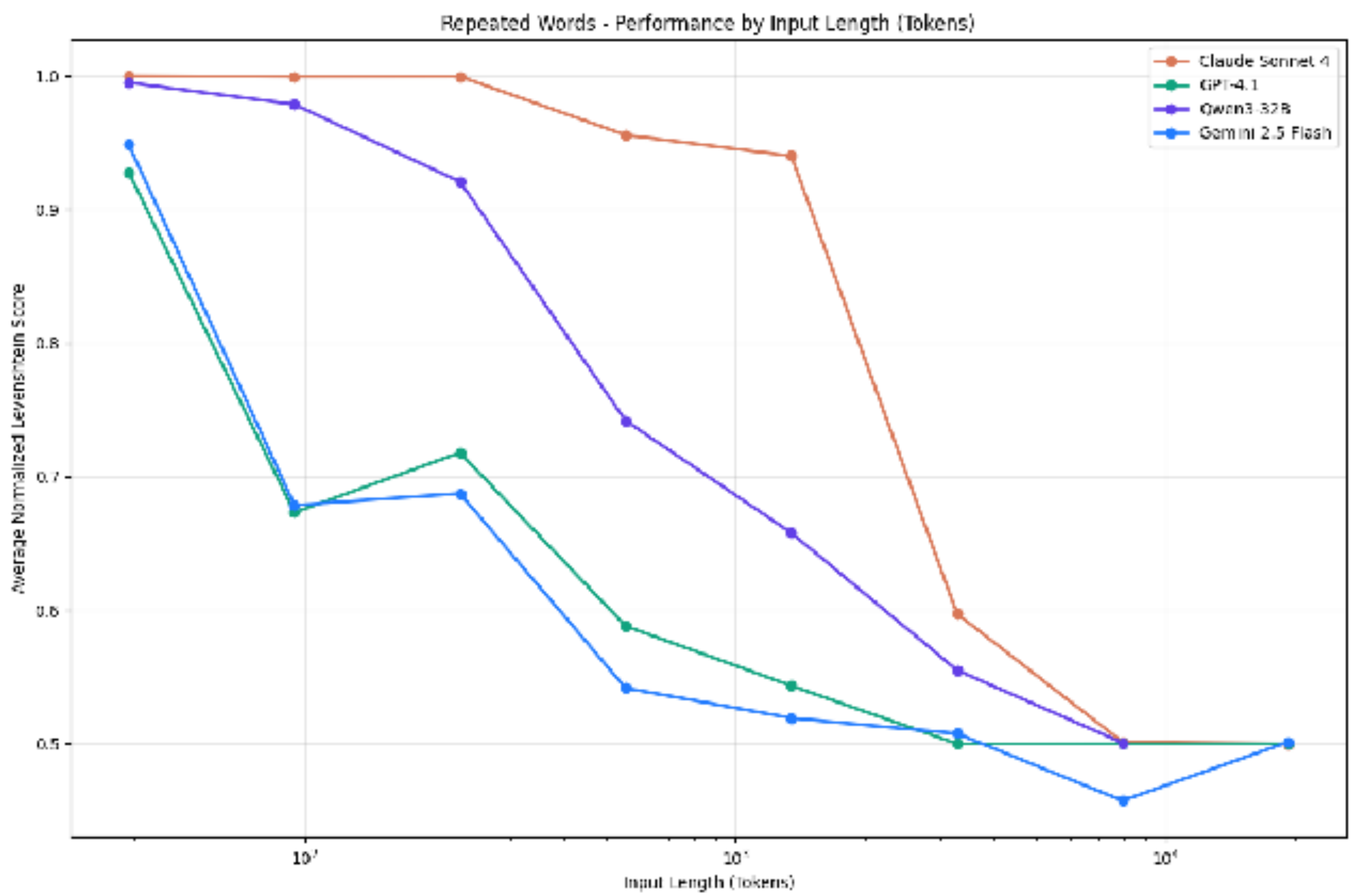
Preprocessing and cleansing data
Chunking strategies
Add metadata
Embedding model selection

# Chunking strategies



# Context Window ?

Number of tokens in the context window increases,
the model’s ability to accurately recall information from that context decreases




https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents

# Chunking Strategies !!

Fixed size
Recursive characters
Document structure-based
Semantic chunking
Agentic chunking
…

https://blog.dailydoseofds.com/p/5-chunking-strategies-for-rag

# Good Chunking

Efficiency

Relevance

Context preservation

Improve content generation

Reduce noise

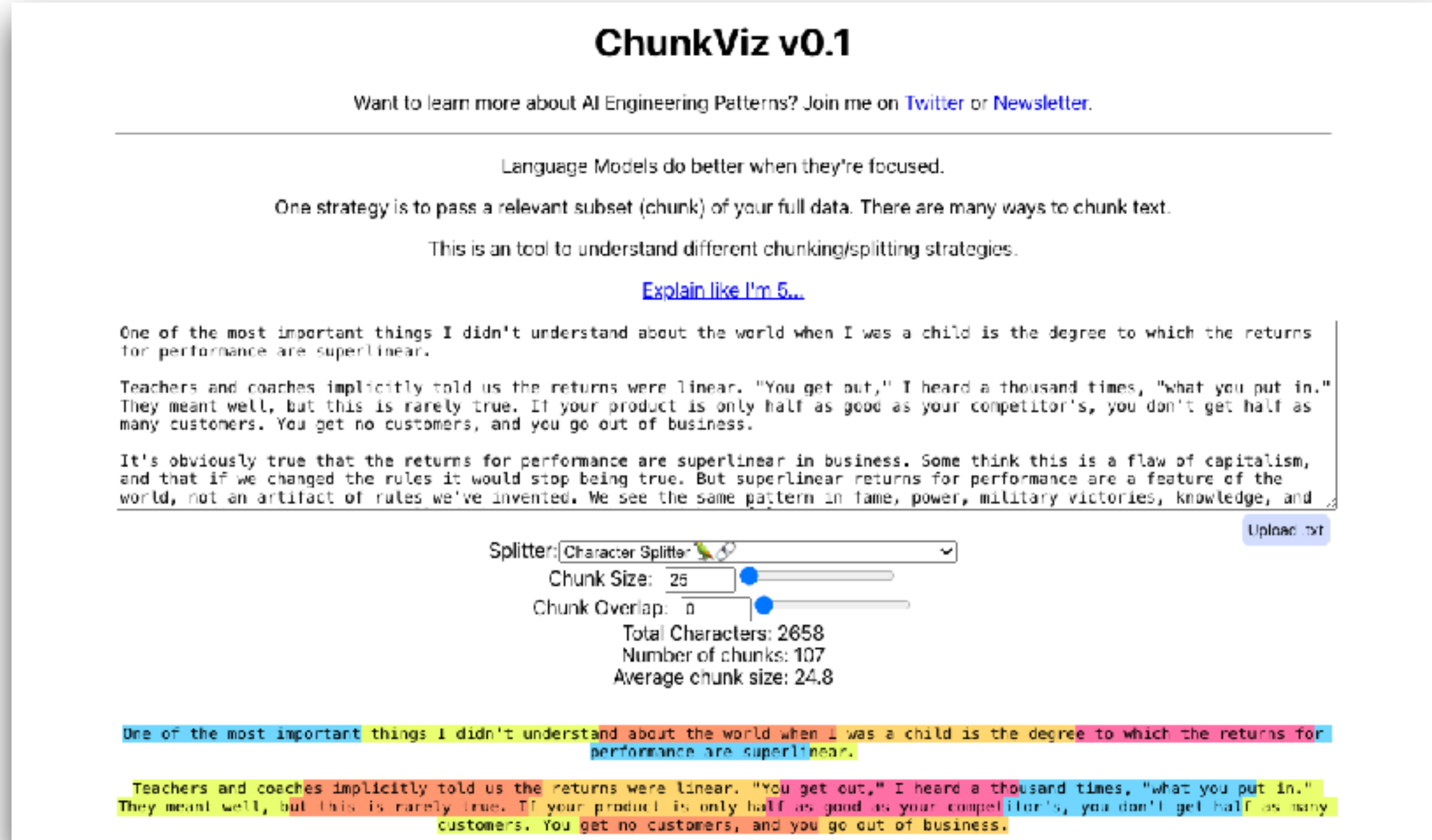
# Bad Chunking

Loss context

Redundancy

Inconsistency

# Chunking Visualization




https://chunkviz.up.railway.app/

# Retrieval optimization ?

Re-ranking

Hybrid search

Query transformation

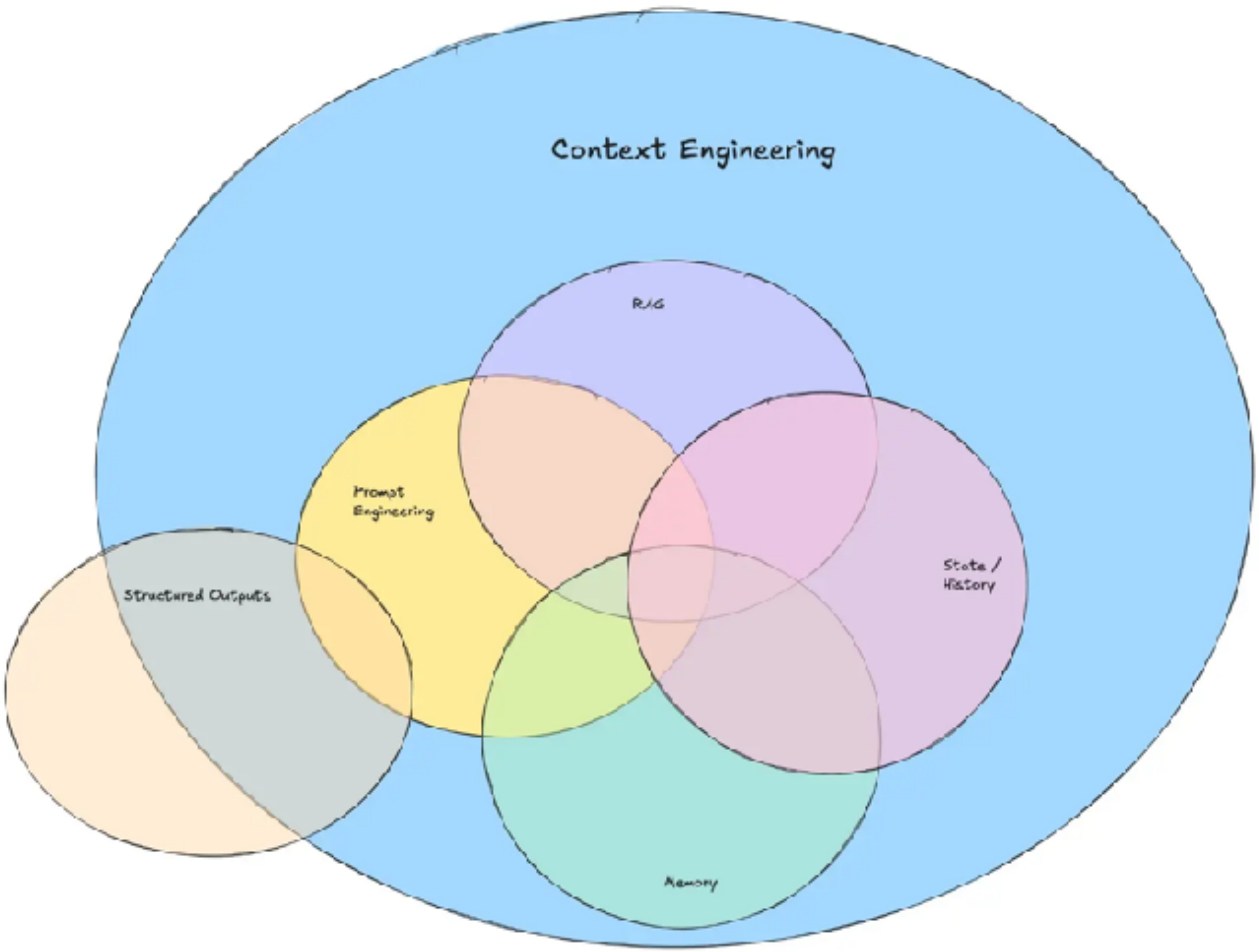
Multi-vector embedding

Contextual retrieval

Vector database Selection

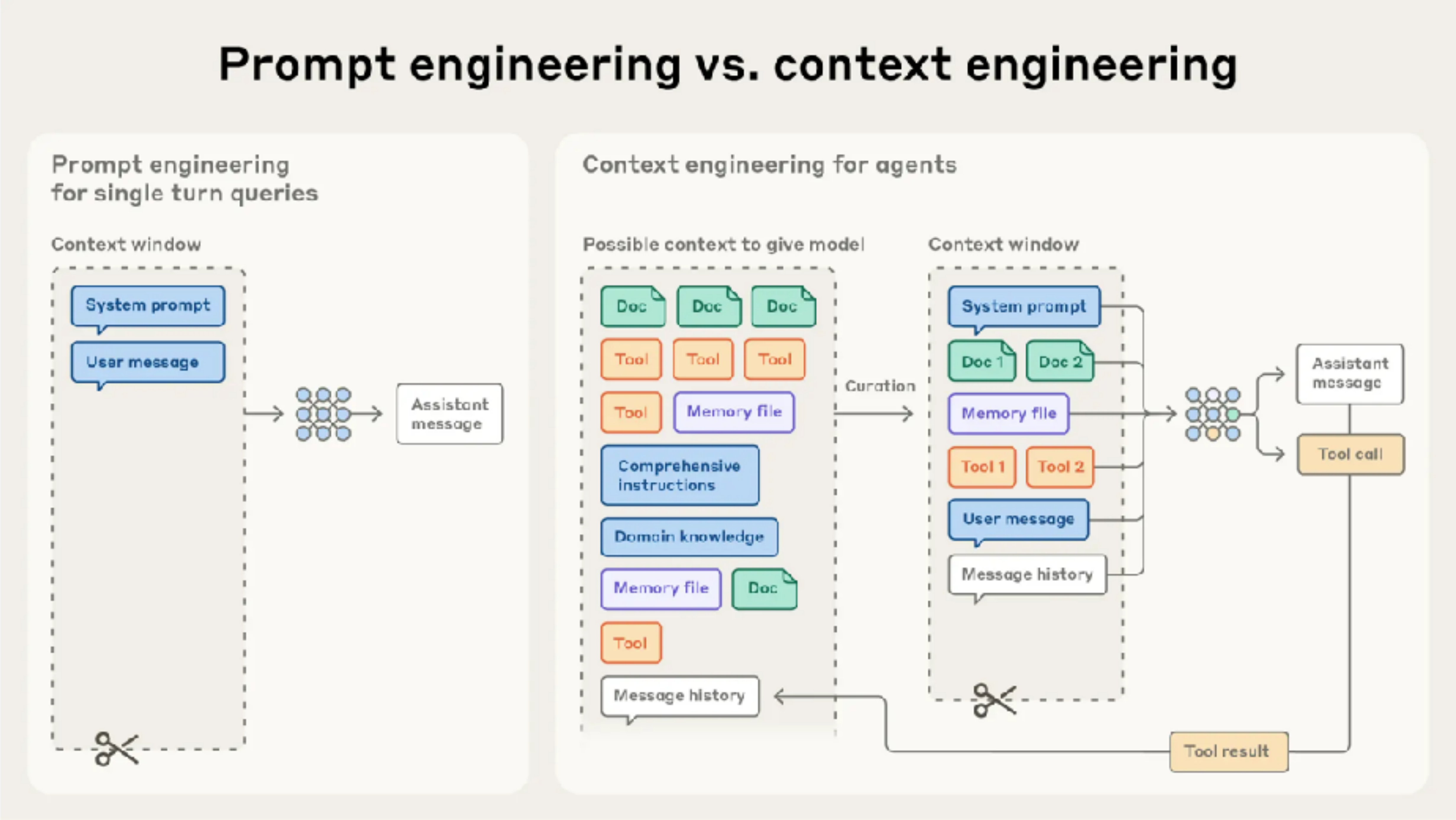

ONE-SIZE FITS ALL
STORE
SALE

# Context Engineering




https://www.promptingguide.ai/guides/context-engineering-guide

# Effective Context Engineering

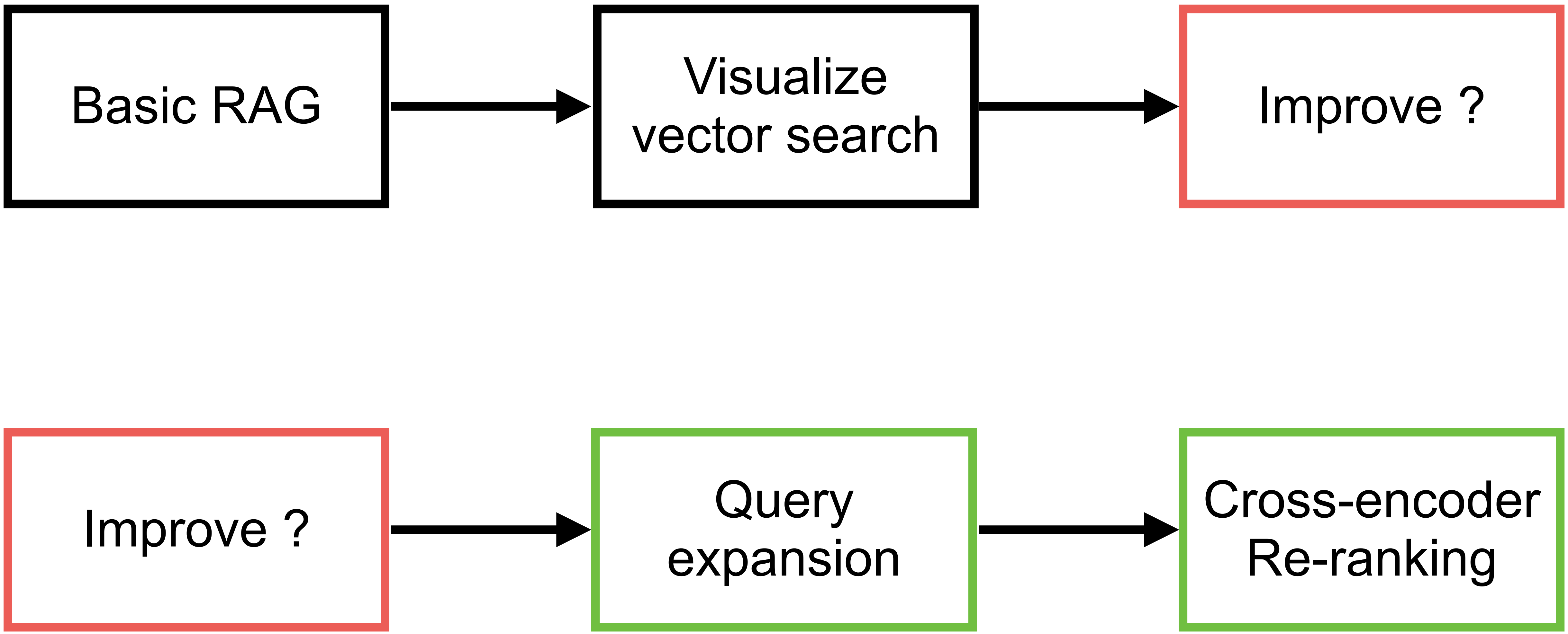



https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents

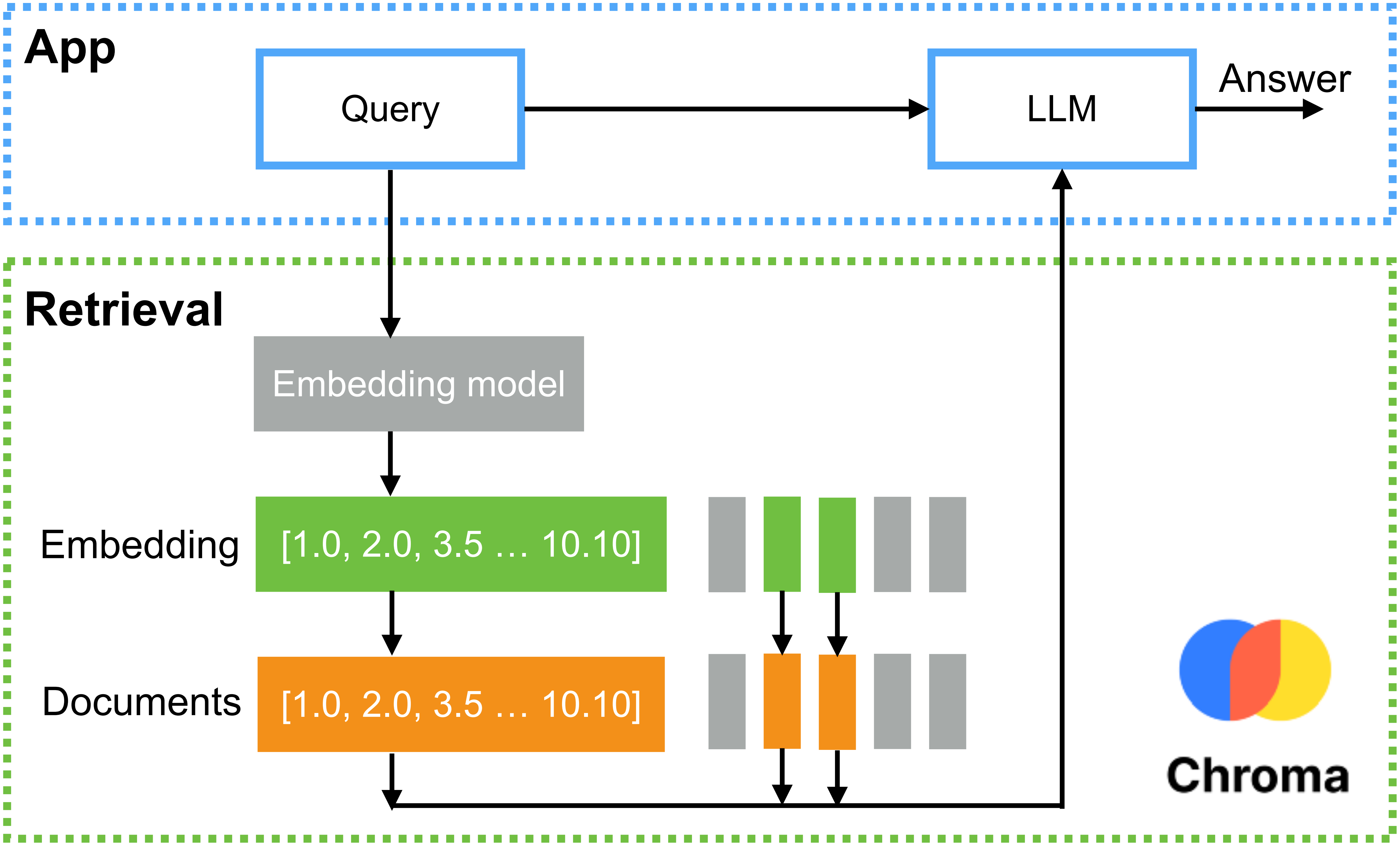
# RAG Workshop

https://github.com/up1/workshop-basic-llm/tree/main/workshop/basic-rag

# RAG workshop



# Basic RAG


App
Query
LLM
Answer
Retrieval
Embedding model
Embedding
[1.0, 2.0, 3.5 … 10.10]
Documents
[1.0, 2.0, 3.5 … 10.10]
Chroma

# Query Expansion

Query expansion is a widely used technique to improve the recall of search systems

Ambiguity

Vocabulary mismatch

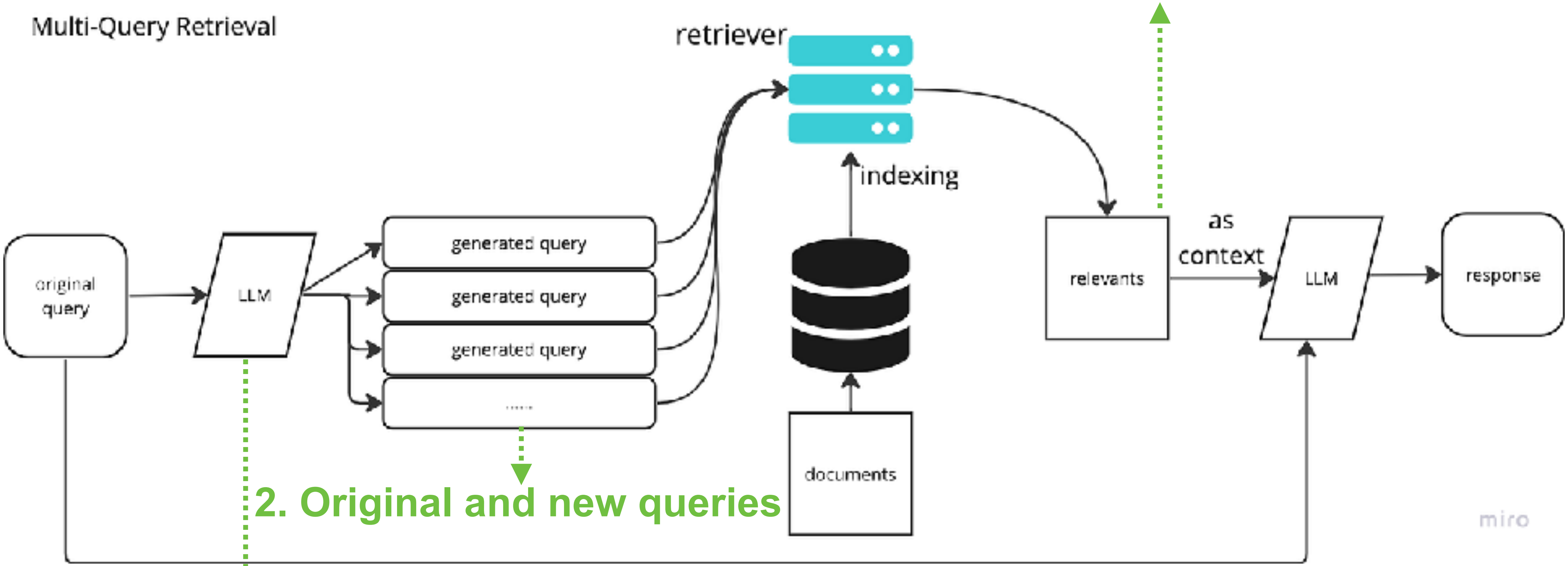
Lack of context

“Add synonyms, related context and contextual terms”

https://arxiv.org/abs/2305.03653

# Query Expansion

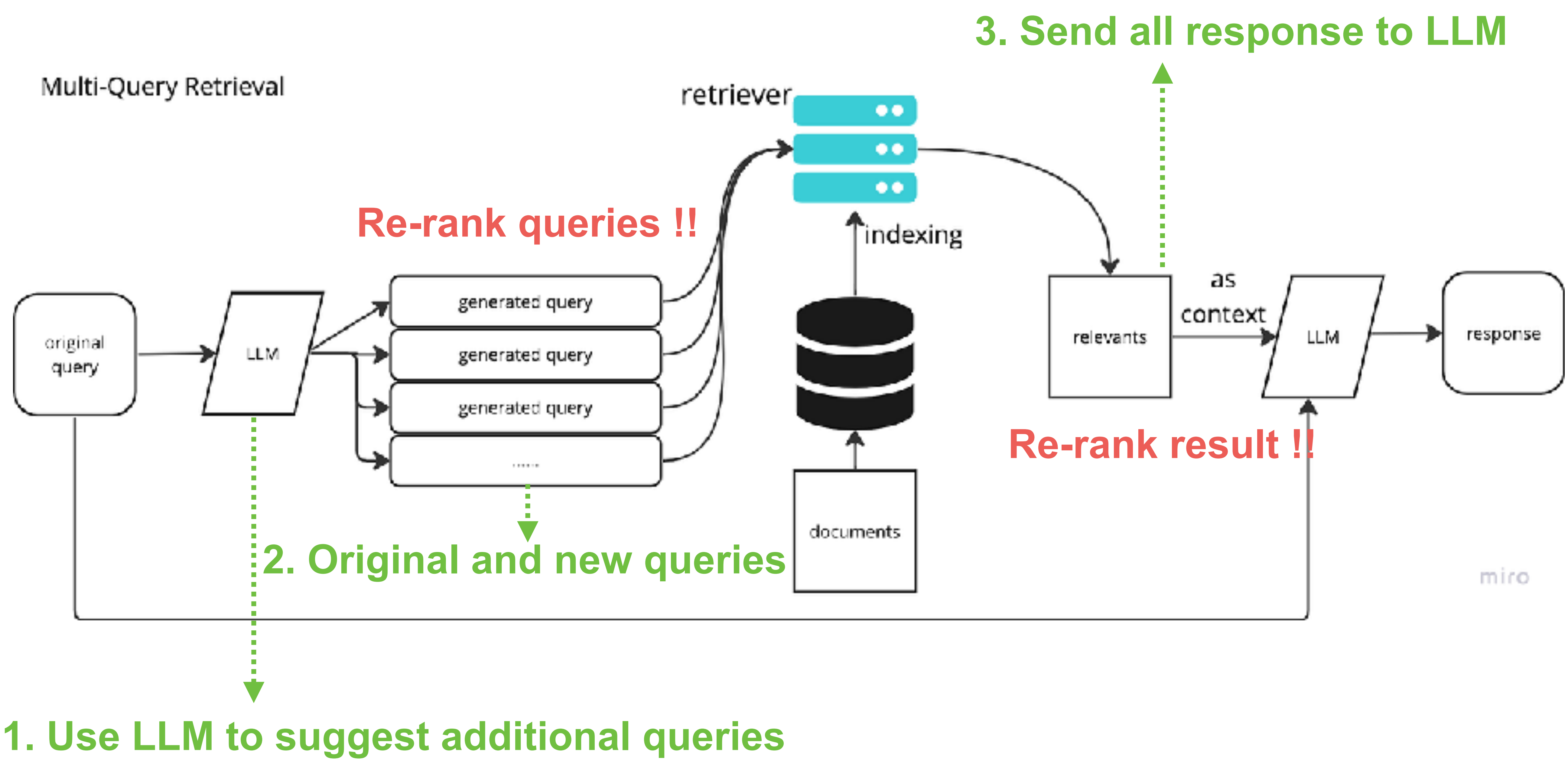
3. Send all response to LLM




2. Original and new queries

1. Use LLM to suggest additional queries

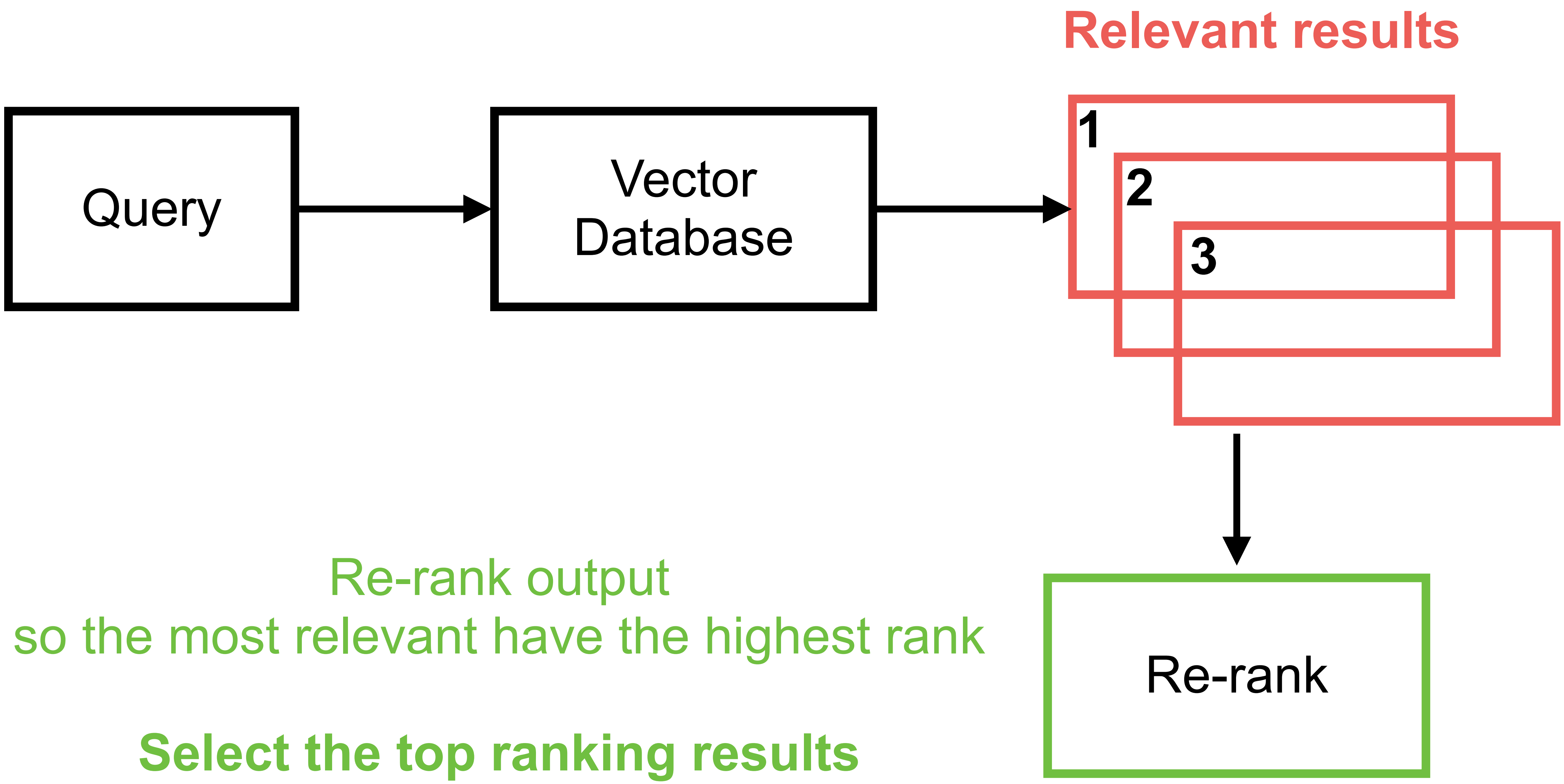
# Re-rank !!



3. Send all response to LLM
Multi-Query Retrieval
retriever
Re-rank queries !!
indexing
original query
LLM
generated query
generated query
generated query
......
relevants
as context
LLM
response
Re-rank result !!
documents
2. Original and new queries
miro
1. Use LLM to suggest additional queries

# Cross-encoder Re-ranking

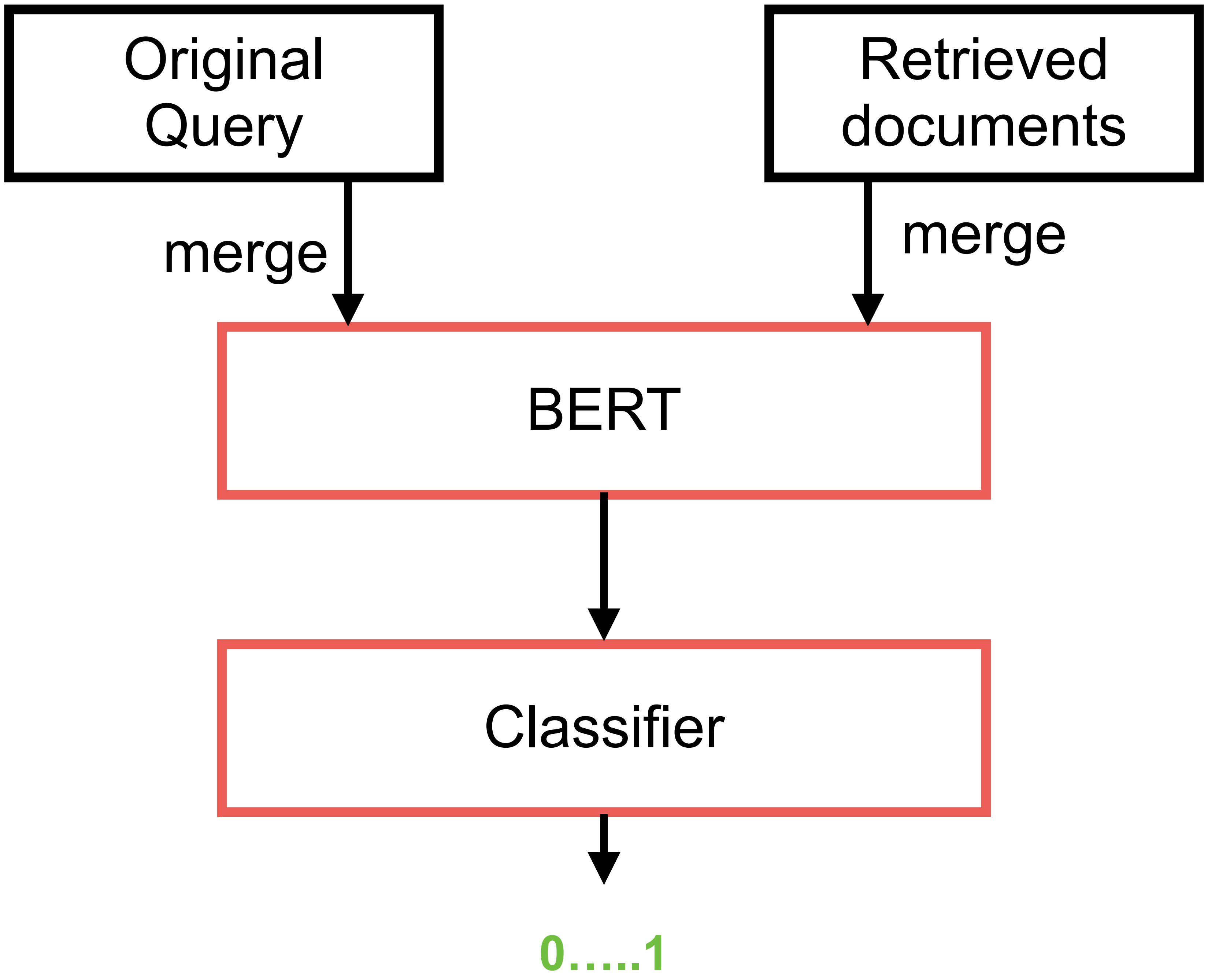
How to ordering relevant results !!



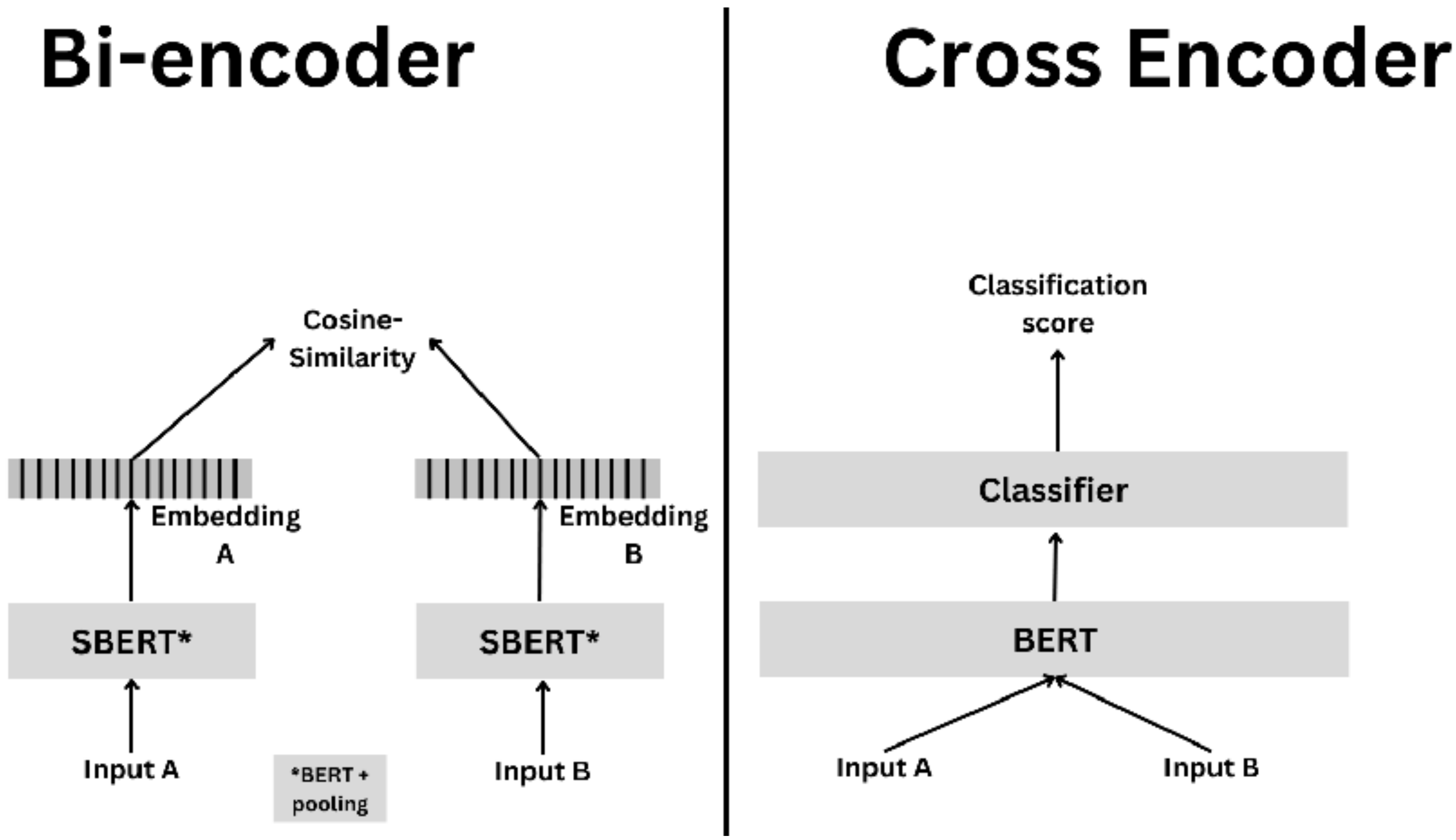

Re-rank output
so the most relevant have the highest rank

**Select the top ranking results**

# Cross-Encoder

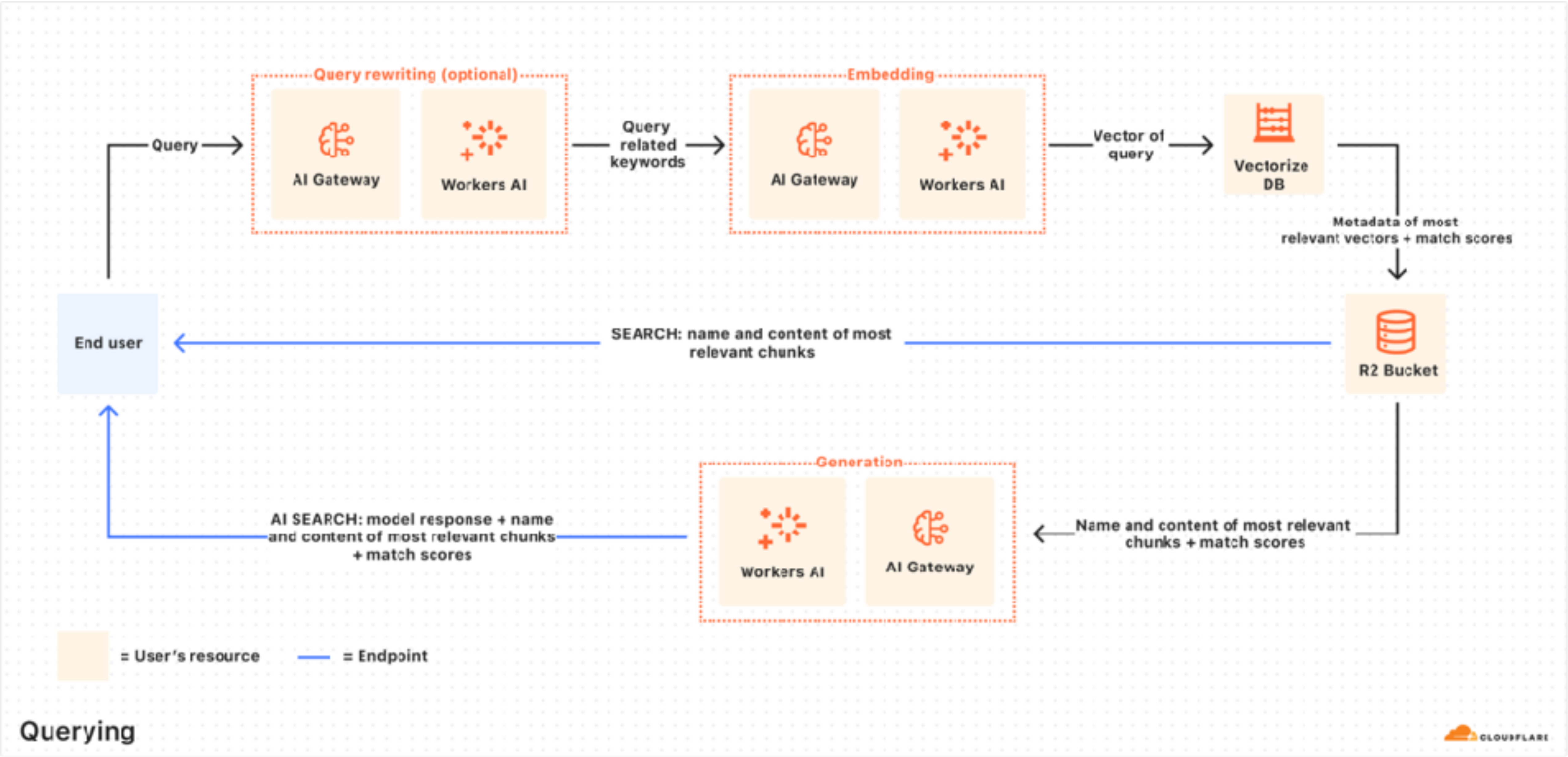


# Cross-Encoder !!




https://sbert.net/examples/cross_encoder/applications/README.html

# Cloudflare AutoRAG




https://developers.cloudflare.com/autorag/

# Guardrails

https://github.com/guardrails-ai/guardrails

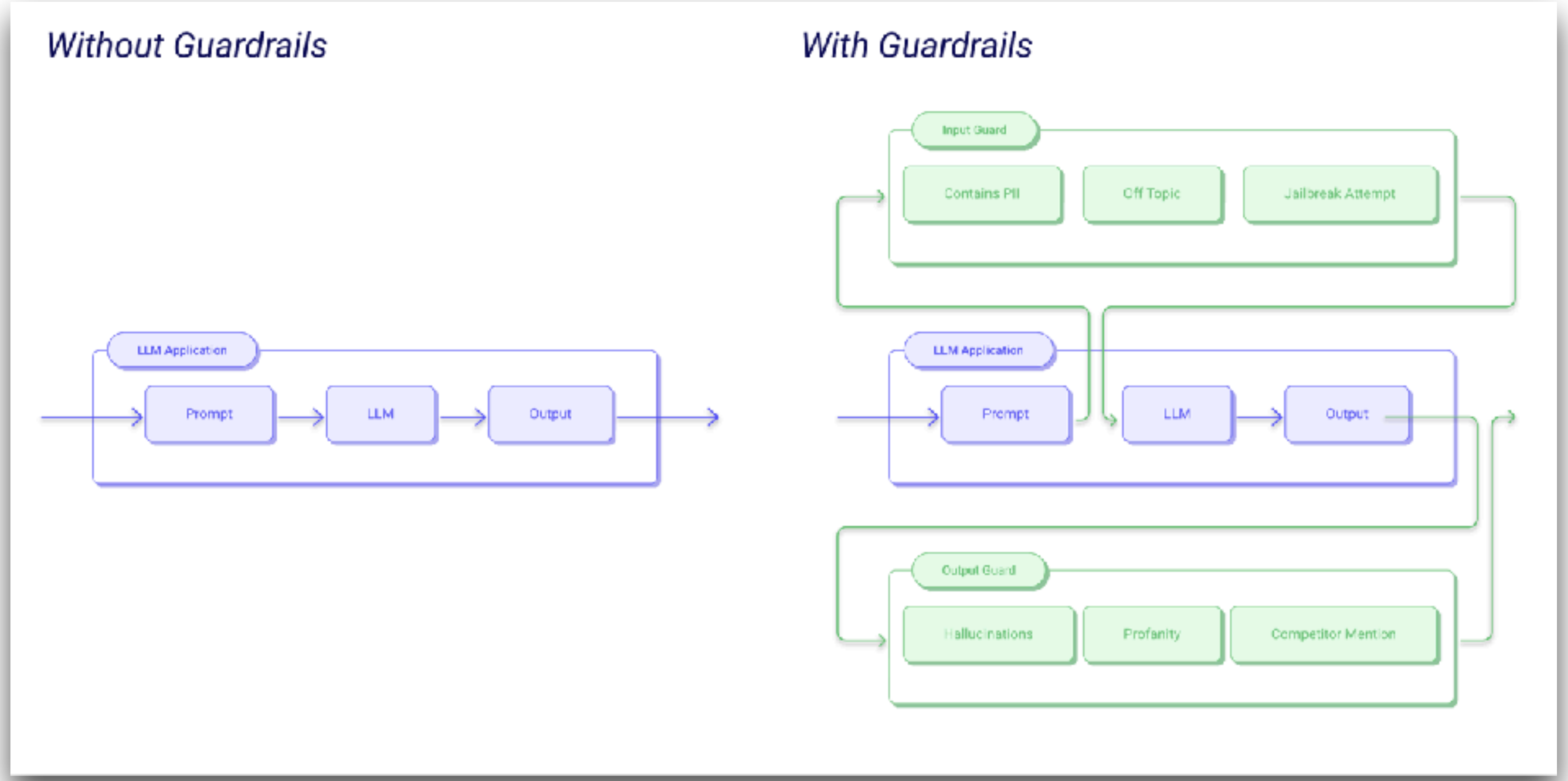
# Guardrails

Help to build reliable AI applications

| | |
|---|---|
| Guard for<br>input and output | Help to generate<br>Structured output |

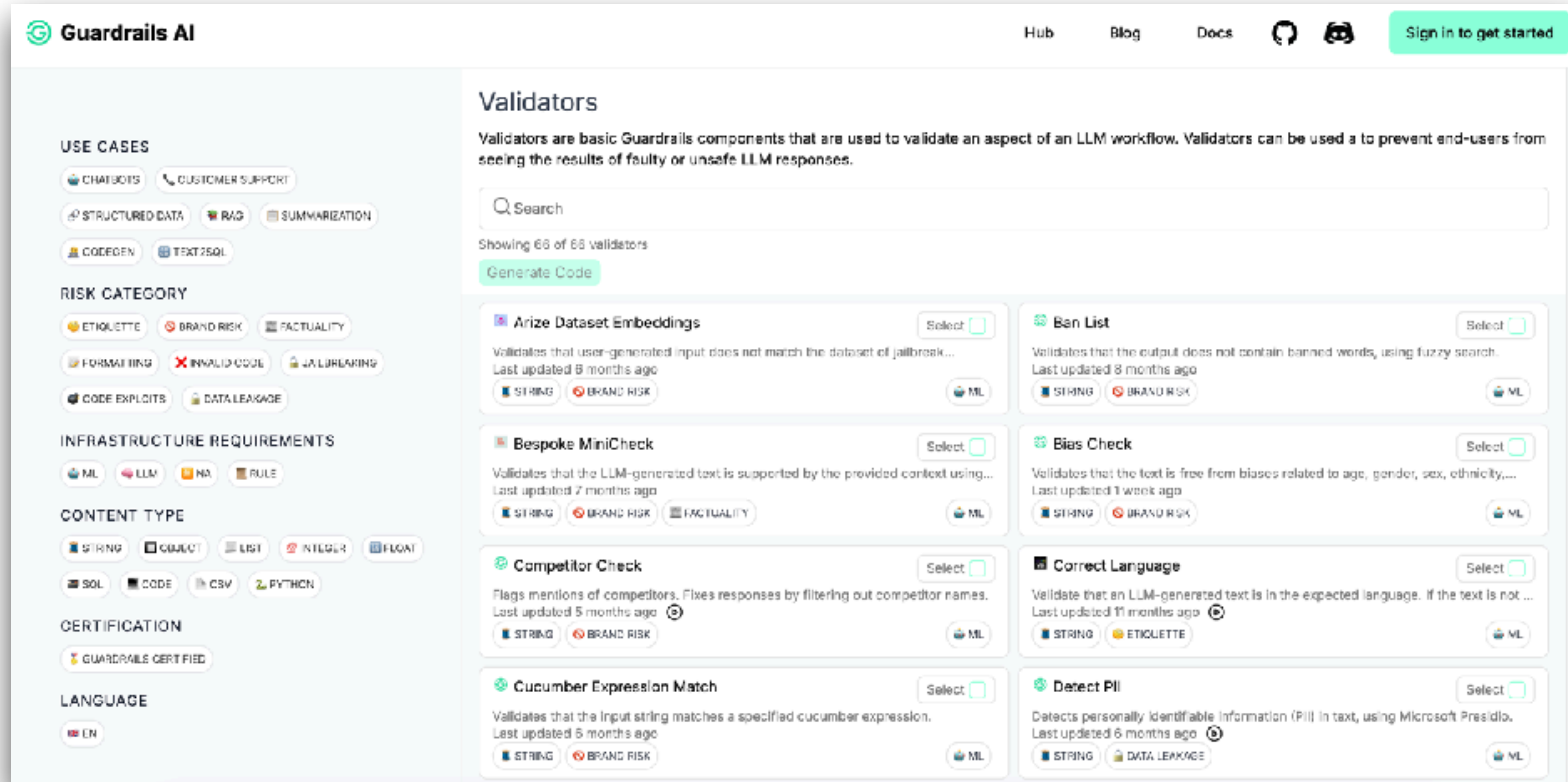
https://github.com/guardrails-ai/guardrails

# Guardrails



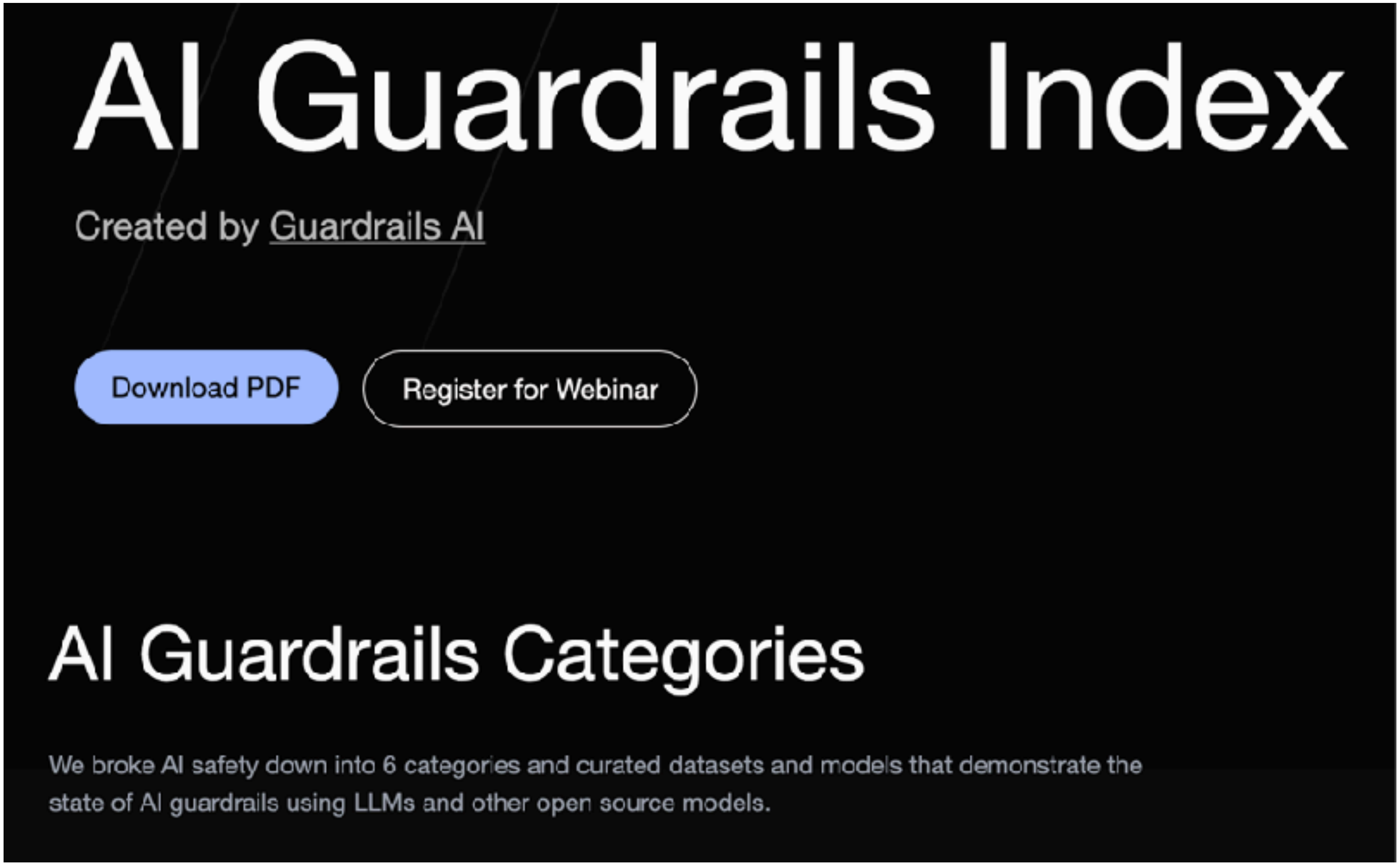
Without Guardrails
LLM Application
Prompt
LLM
Output
With Guardrails
Input Guard
Contains PII
Off Topic
Jailbreak Attempt
LLM Application
Prompt
LLM
Output
Output Guard
Hallucinations
Profanity
Competitor Mention

# Guardrails Hub



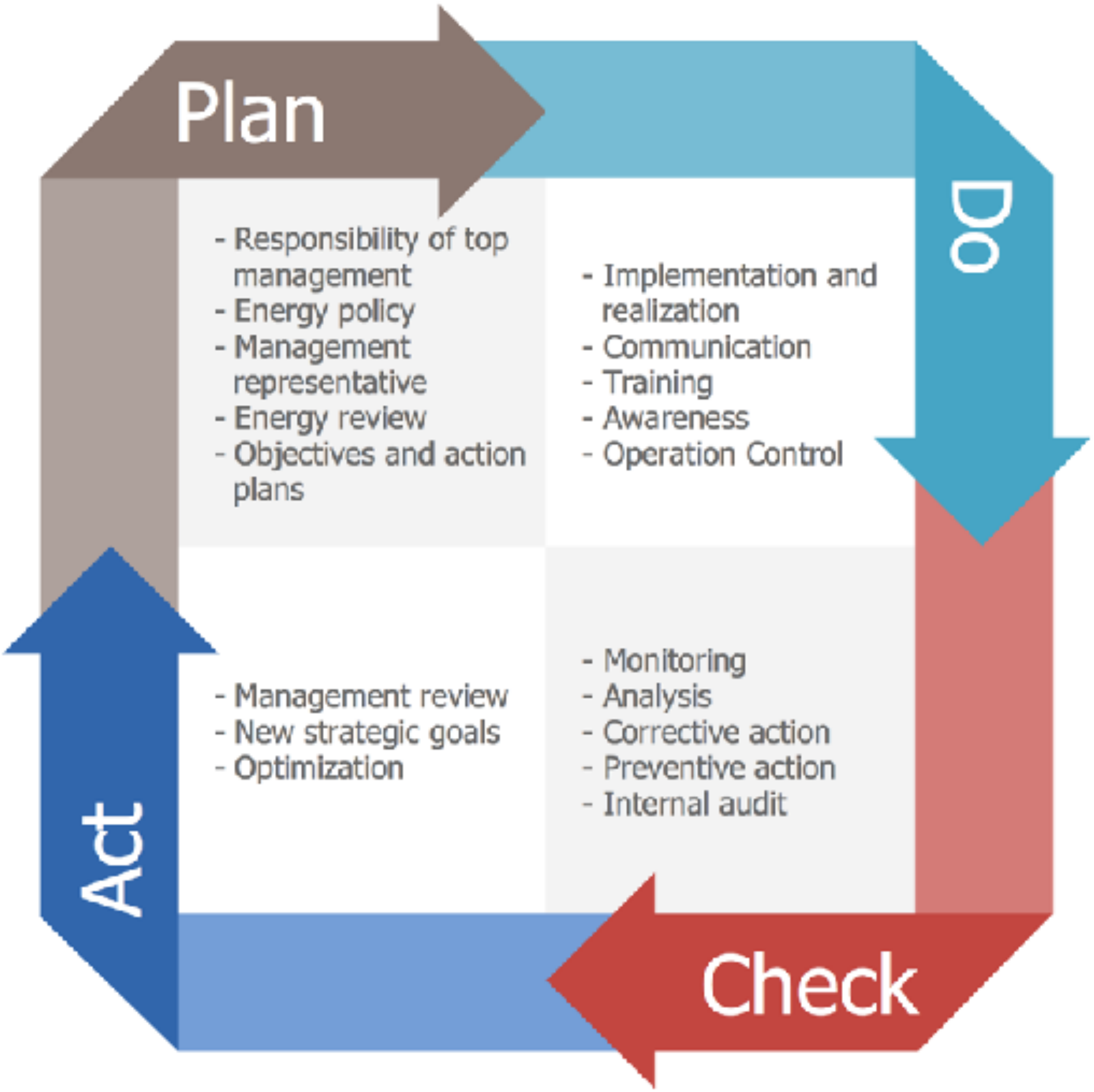

https://hub.guardrailsai.com/

# Guardrails Index



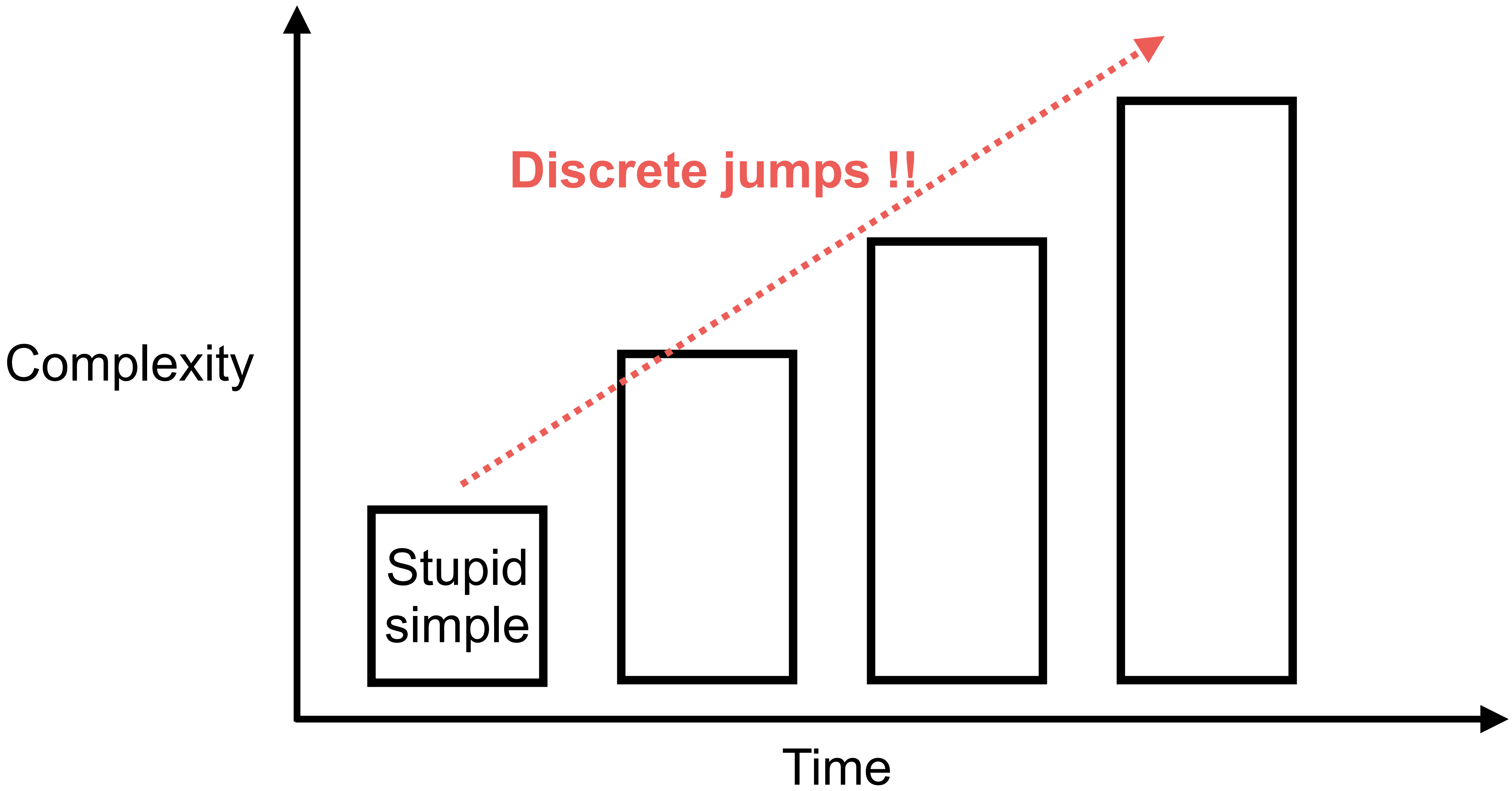

https://index.guardrailsai.com/

# PDCA process



# Staged Complexity

Start small and get a feature working first



# Planning workflow

Requirements → What you're build ?

↓

User stories → What use need ?

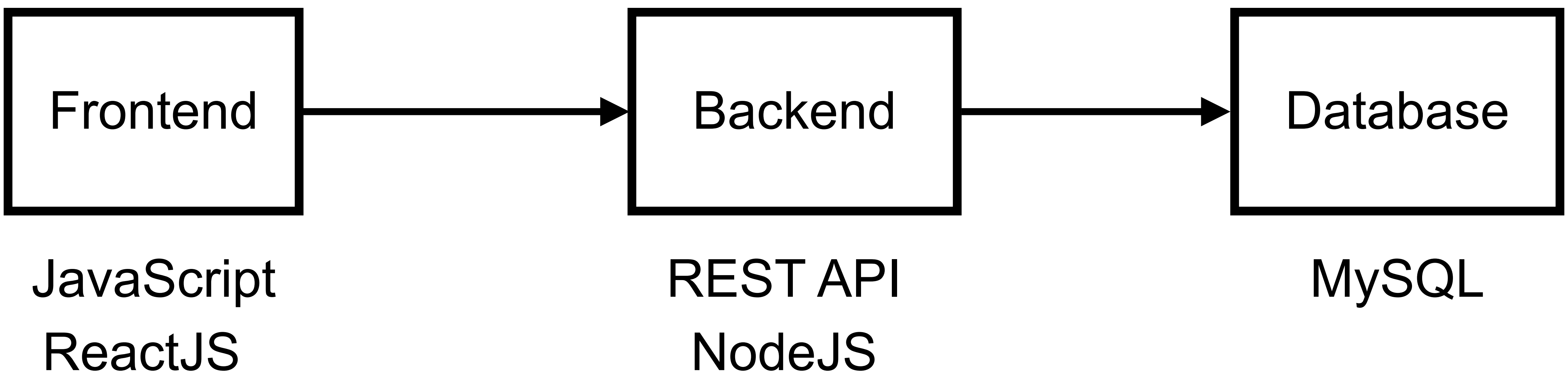
↓

Architecture → Selection ?

↓

Technologies → Selection ?

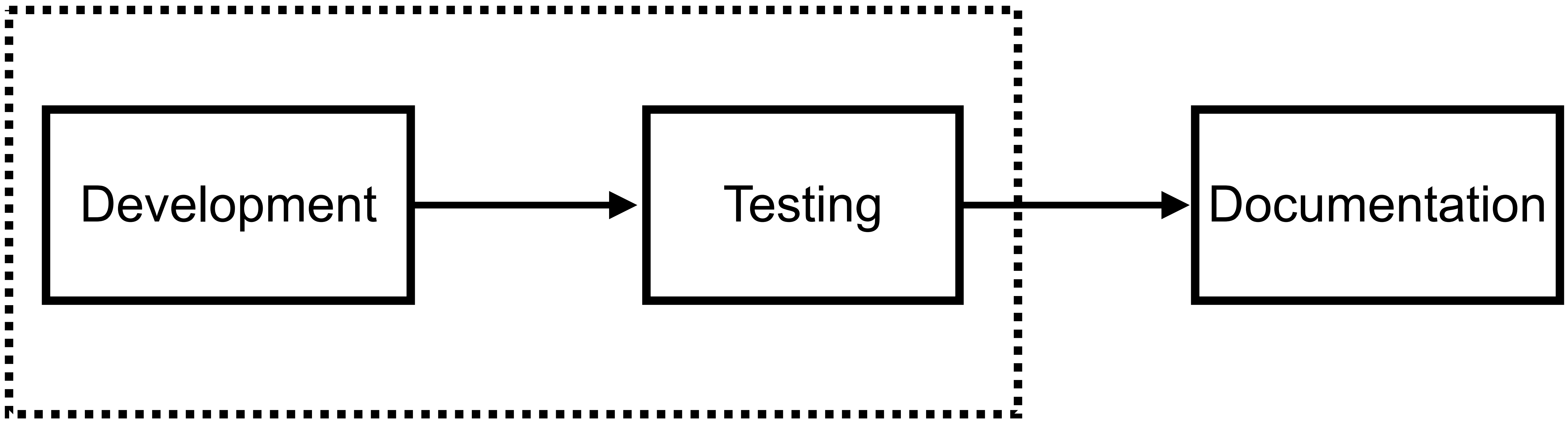
# Architecture of project




Choose your tech stack !!

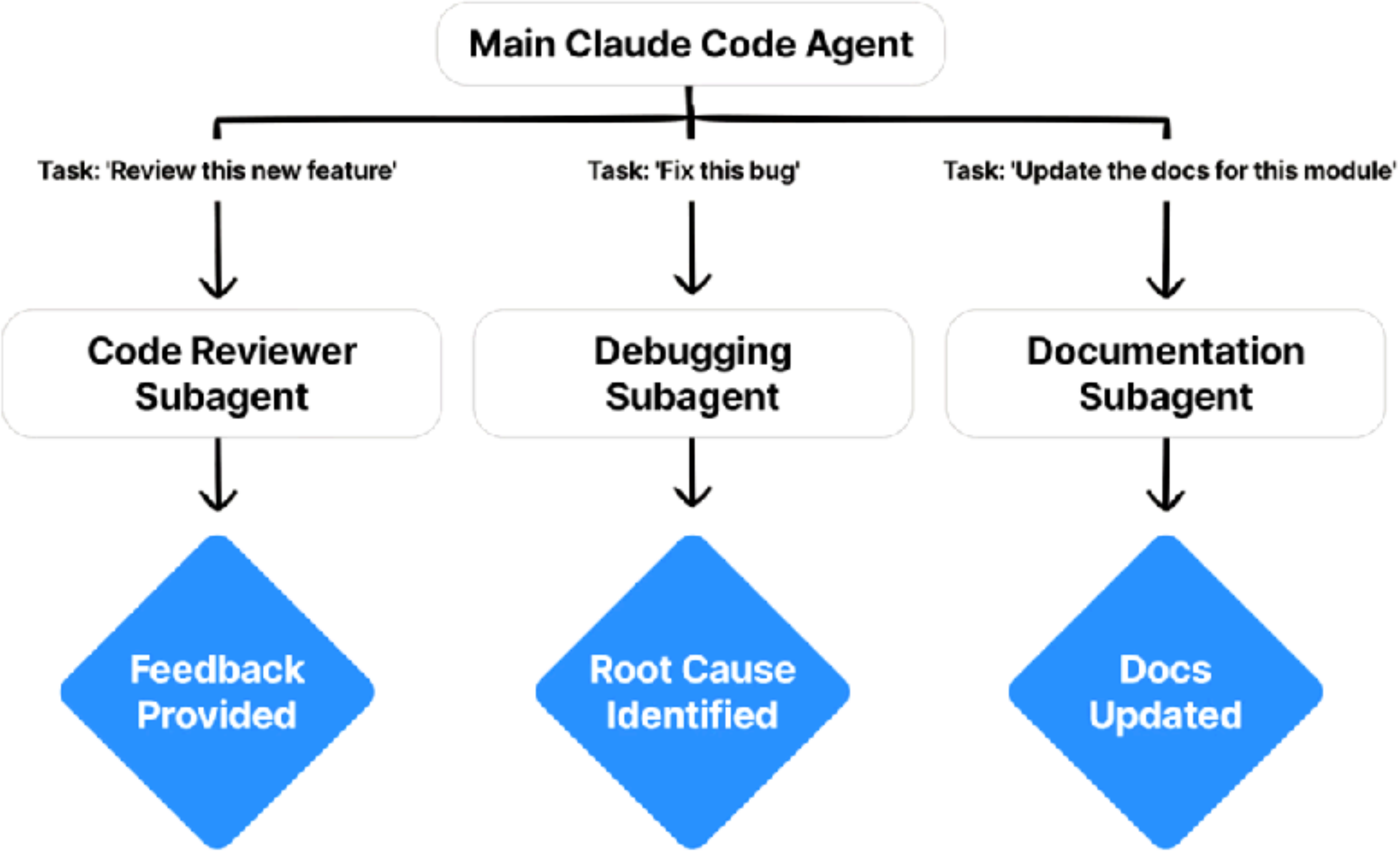
# Processes

## Iterative and incremental process



# AI Subagent

# AI Subagent




https://code.claude.com/docs/en/sub-agents

# AI Subagent



https://github.com/VoltAgent/awesome-claude-code-subagents

# Start your journey